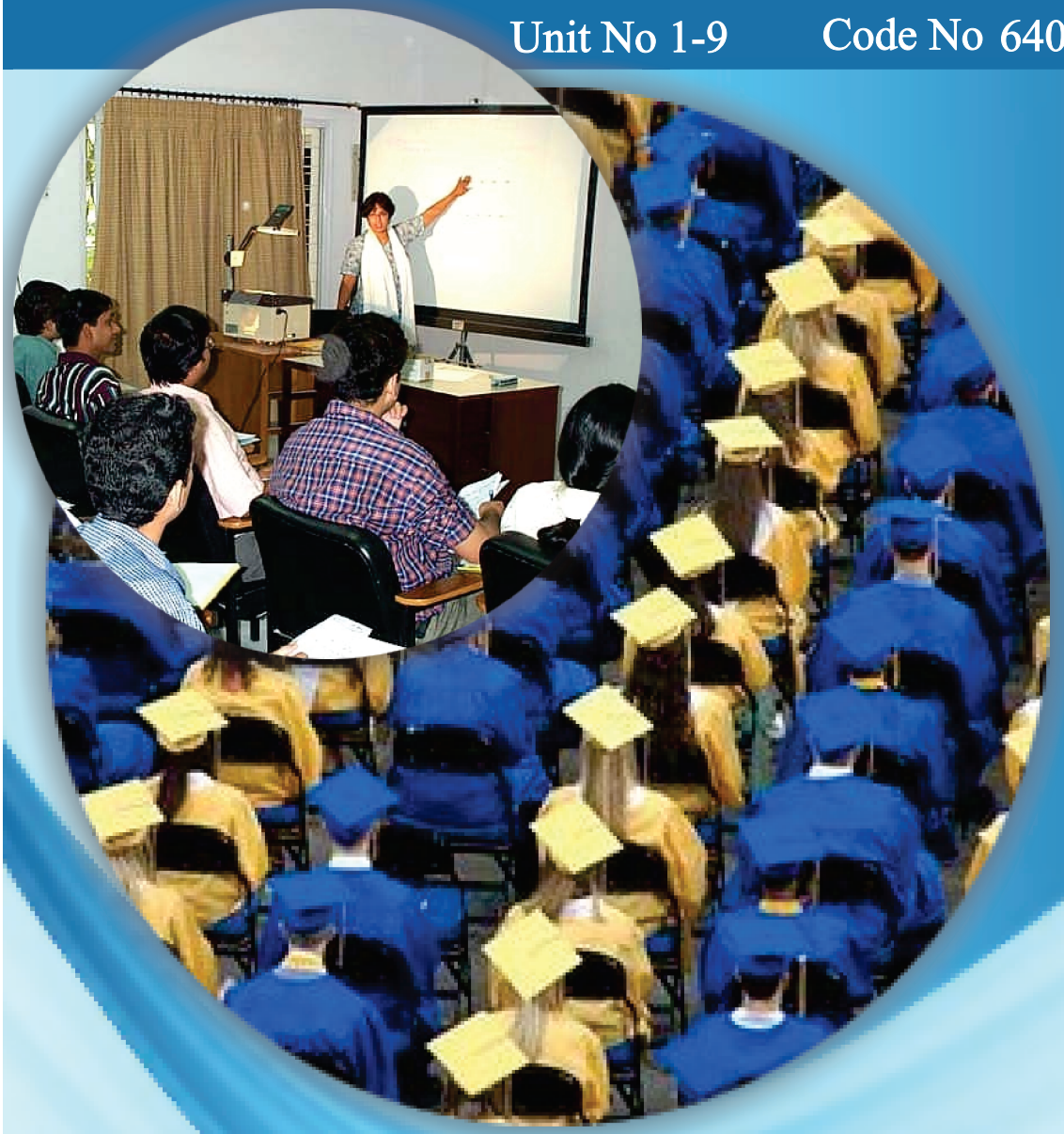
Associate Degree in Education

CURRICULUM DEVELOPMENT

Unit No 1-9 Code No 6406



FACULTY OF EDUCATION

ALLAMA IQBAL OPEN UNIVERSITY
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FORWARD

Education is one of the institutions the human race has created to serve certain needs. Like all human institutions, it responds or should be respond to changes in the environment. The institution of education is activated by a curriculum that it will change in response to forces affecting it.

A major characteristic of any theoretical principle is its capacity for being generalized and applied in more than one situation. Were curriculum theories but one shot solutions to specific problems, it would be difficult to defend the concept of curriculum as a discipline. However, the principles of curriculum theory are often successful efforts to establish rules that can be repeated in similar situations and under similar conditions. Many people will agree, for example, that the concept of balance should be incorporated into every curriculum.

What has led so many people to be dissatisfied with so much of what education is all about? Why is the status quo rarely a satisfactory place to be? In addition, why does it turn out, as will be illustrated, that yesterday's status quo is sometimes tomorrow's innovation? For answers to these questions teachers and specialists who participate in efforts to improve, the curriculum must consider some general principles of curriculum development.

As a discipline, curriculum possesses (1) an organized set of principles (2) a body of knowledge and skills for which training is needed and (3) its theoreticians and practitioners.

If the curriculum is perceived as a plan for the learning experiences that young people encounter under the direction of the school, its purpose is to provide a vehicle for ordering and directing those experiences. This process of providing the vehicle and keeping it running smoothly is known as curriculum development.

Curriculum specialists make a number of distinctive contributions to their field. Specialists know what types of curricula have worked in the past, under what conditions, and with what success. Since the name of the game is improvement, specialists must be well grounded in the historical development of the curriculum and must possess the capacity to use that knowledge to help the schools avoid historical pitfalls. Curriculum specialists generate or help to generate new curriculum concepts.

This book is intended for students in fields such as Curriculum Development, Curriculum Planning, and Curriculum Improvement. It is meant to be especially helpful to Master of Education students of open distance learning system of AIOU.

I congratulate Course Team Chairman, Dr. M. Zafar Iqbal and Course Coordinator, Mrs. S.M. Shahid, for their professionalism and dedicated efforts to procedure this textbook.

Dr. Shahid Siddiqui
Vice-Chancellor

PREFACE

As newer development occur in education, as research adds new insights on teaching and learning, as new ideas are developed, and as times change, beliefs about curriculum and instruction also undergo transformation. The rightness or wrongness of concepts like curriculum and instruction cannot be established by an individual educationist or even by a group of educators. One index of correctness might be the prevailing opinion of most educators at a particular stage in history.

Gainus Julus Caesar and his cohorts of the first century BC had no idea that the oval track upon which the Roman chariots raced would bequeath a word used almost daily by educators twenty-one centuries later. The track the curriculum has become one of the key concerns of today's schools, and its meaning has expanded from a tangible racecourse to an abstract concept.

No education teacher, curriculum coordinator, administrator, or tutor in distance learning system of AIOU would dream of arguing that techniques of coping with the woolly mammoth should be part of the curriculum of schools at the dawn of the 21st century.

The curriculum of the cave dweller, whether informal or non-formal, is quite different from increasingly formal types of schooling that the human race invented over subsequent periods of history. Techniques for coping with the woolly may well have been paramount concern to prehistoric man.

Planning should begin with the programmatic viewpoint that is, with curriculum decisions, rather than with instructional decisions. Appropriate planning begins with the board aims of education and proceeds through a continuum that leads to the most detailed objectives of instruction. A discipline has its theoreticians and its practitioners.

Consequently, we can conclude that the field of curriculum requires the use of an amalgamation of knowledge and skills from many disciplines. That curriculum theory and practice are devised from other disciplines does not in any way diminish the importance of the field. Curriculum's synthesis of elements from many fields in some ways makes it both a demanding and an exciting arena in which to work.

Though it may be vehemently denied, no one has ever seen a curriculum, not a real, total, tangible, visible, entity called a curriculum. The interested observer may have seen a written plan that may have been called a curriculum. Somehow, the observer knows, probably by word of mouth, that in every school in which teachers are instructing students a curriculum exists. A written plan provides the observer with an additional clue to the existence of something called a curriculum. But if by some bit of magic and observer could lift the roof of a school in session and examine the cross section thereof, the curriculum would not be apparent. What the observer would immediately perceive would be many instances of teacher-pupil interaction we call instruction.

This textbook on "Curriculum and Instruction" for Master of Education students of Allama Iqbal Open University is really a product of professionalism and dedication of Course Coordinator Mr. S.M.Shahid and his team and I extend my full gratitude to them.

ACKNOWLEDGEMENTS

In the world of professional education, the word curriculum has taken on an elusive, almost esoteric connotation. The poetic, neutral word does possess an aura of mystery.

Any discipline worthy of study has organized set of theoretical constructs or principles that govern it. Certainly, the field of curriculum has developed a significant set of principles, tried and untried, proved and unproved, many of which are appropriately the subjects of discussion in this text.

Through the process of curriculum development, we can discover new ways for providing more effective pupil learning experiences. The curriculum developer continuously strives to find newer, better, and more efficient means to accomplish the task of educating the young.

One might ask whether the field of curriculum contributes any knowledge of its own to that borrowed from other disciplines. Certainly, a good deal of thinking and research is going on in the field of curriculum. New curricular ideas are being generated continuously.

Curriculum and instruction are viewed as separate but dependent concepts. Curriculum is defined in a variety of ways by theoreticians. This text follows the concept of curriculum as a plan or programme for the learning experiences that the learner encounters under the direction of the tutor of AIOU.

The teacher too, is a curriculum worker who engages in curriculum planning in varying degree, on different occasions, generally under the leadership of a supervisor, be that person a headmaster or principal. How the teacher and the curriculum specialist work together in the two fields of curriculum and instruction is a recurrent theme of this book.

Many people have contributed to the writing and publishing of this next. Through their insights into curriculum and instruction, teachers, reviewers, and editors with who have helped to shape this text I am highly thankful to all of them.

And at the last not the least I feel it my moral obligation to extend my heartfelt thanks to all the Unit Writers, Reviewers, Course Team Chairman and Hakim Safdar Husayn for all the painstaking for this academic endeavor. Any suggestions for improvement from any person will be useful for us while revising this textbook.

COURSE OBJECTIVES

After going through this book, you will be able to:

- Describe models of famous authors.
- Compare the instructional methods on costs, and effectiveness, material and demerits.
- Define various steps of curriculum.
- Differentiate between philosophical categories.
- Explain the functions of conceptual framework for curriculum design.
- Classify aims, goals and objectives and understand their significance.
- Understand the principles and criteria for selecting the curriculum content and learning experiences.
- Mention the main purpose of curriculum evaluation and the extent to which an evaluation can be made.
- Interpret problems and issues in curriculum design.
- Describe dynamics of curriculum change.

UNIT–1

MODELS OF TEACHING

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INTRODUCTION

This unit deals with the models and their importance in the field of teaching. Model is used in several contexts in our daily life. A model is an exact copy of something, which gives detailed information to a person, who works on that plan.

A model of teaching is a plan or pattern that we use to design face-to-face teaching in classrooms, tutorial settings or in shaping instructional material. The use of models is very old. Socrates, the Greek Philosopher, used his own model of question-answer. In ancient India, teachers developed their own models of teaching of desirable changes in the behaviour of the learner. The reader can attain clear concepts about models of teaching. Need for these models are explained in a comprehensive way. There are four families of models of teaching. Two of them are given in detail to help the reader to understand that how a student becomes knowledgeable to his environment. Models of different authors make the reader knowledgeable and know the importance of these models of teaching. It also presents a rich variety of approaches to teach methods with enough illustrations to make a model for real learning situation.

OBJECTIVES

After an intensive study of this unit you will be able to:

1. Define teaching, model and model of teaching.
2. List down the characteristics of model of teaching.
3. Describe functions of models of teaching.
4. Describe models of famous authors.
5. Explain advantages of BTES academic learning time model.

1. ABOUT IS A TEACHING MODEL

There has been a lot of confusion regarding the concept of teaching and how-teaching should be defined. Even in more complex situation called the classroom, a clear and precise definition" of teaching cannot be given. According to Bare (1961) Teaching means many different things: and that the teaching act varies from person to person and from situation to situation".

> "Teaching is the art of assisting another person to learn. It includes provision of *information* (instruction) about appropriate situation, conditions or 'activities designed to facilitate learning". (English & English -1958)
>
> "Teaching is an interaction between teacher and student, under the teacher's responsibility in order to bring about expected change in student's behavior".
>
> "Teaching is an activity which is designed and performed for multiple objectives, in terms of changes in pupil's behavior". (Mujeeb-ul-Hasan Siddiqi 1991)

1.1. Concept of Model

Models are mainly representations, copies or images of real objects. They can be used to represent objectives, systems, concepts or ideas, (Harry Dhand 1990)

> "Models refer, to a set of basic assumptions that-outline the universe of scientific inquiry specifying both the concepts regarded as legitimate and the methods to be used in collecting and interpreting data". (Bhatacharaya-1993)

The model is a pattern of something to he made or reproduced. It is a mean of transferring a relationship or process from its actual setting to another one in which it can be applied more conveniently and effectively. (Mujeeb-ul-Hasan Siddiqi 1991)

1.2. Concept Model of Teaching

Model of teaching "can be defined as an instructional design: which describes the process of specifying and producing particular environmental situations, which cause the students to interact in such a way that a specific change occurs in their behaviour. (S.S. Chohan 1989).

A model of teaching is a plan or pattern that can be used to shape curricula, to design instructional material and to guide instructions in the classroom and other settings. The most important aim of any model of teaching is to improve the instructional effectiveness in an interactive atmosphere and to improve or shape the curriculum. (Mujeeb-ul-Hasan-1991)

1.2.1. Characteristics of a Model of Teaching

All good models of teaching have some common identifiable characteristics, which are described as follows:

1. ***Scientific Procedure:*** A model of teaching is not a haphazard combination of facts but on the other hand, it is a systematic procedure to modify the behaviour of learners. They are based on certain assumptions.

2. ***Specification of Learning Outcomes.*** All models of teaching specify the learning outcomes in detail on observable students' performance and what the students will perform, after completing an instructional sequence.

3. ***Specification of Environment:*** This means that, every model of teaching specifies in definite terms the environmental conditions under which a student's response should be observed.

4. ***Criterion of Performance.*** A model specifies the criterion of acceptable performance, which is expected from the students. A model of teaching delineates: the behavioural outcomes, which the learner would demonstrate after completing specific instructional sequence.

5. ***Specification of Operations:*** All models of teaching specify mechanisms that provide for students' reaction and interaction with the environment.

1.2.2. Functions of a Model of Teaching

The reader may ask the question that why should we develop a model of teaching, what are its specific functions or how does a model, help .the practicing teacher in classroom teaching-learning process.

1. ***Guidance:*** *A* model of teaching serves a useful purpose of providing in specific terms what the teacher has to do. He has a comprehensive design of instructions with him through which he can achieve the objectives of the course. Teaching becomes a scientifically controlled and goal oriented activity. Thus, a model provides guidance to the teachers as well as to students to reach the goal of instructions.

2. ***Developing Curriculum:*** A model of teaching helps in the development of curriculum for different classes at different levels of education.

3. ***Specification of Instructional Material:*** A model specifies a detailed instructional material, which a teacher can use to bring a desirable change in the personality of a learner.

4. ***Improvement of Teaching:*** It helps the teaching-learning process and increases effectiveness of teaching.

1.2.3. Assumptions of Models of Teaching

The models of teaching evolve on following assumptions:

1. The first assumption, underlying all models is that teaching is the creation of proper environment. The various component parts of teaching are interrelated.
2. The second assumption is that contents, skill, instructional rules, social relationship, types of activities, physical facilities and their use, all form an environmental system whose parts interact with each other to constrain the behaviour of all participants, teachers as well as students.
3. The third assumption is that different combinations of these elements create different types of environment and elicit different outcomes.
4. Fourth one is that models of teaching create an environment. They provide rough specification for environment in the classroom teaching-learning process.

1.2.4. Elements of a Model of Teaching

The four elements of a model are focus, syntax, social system and support system. .

Focus: This central aspect refers to the reference frame around which the model is developed. It is the main theme to determine the combination and relationship of various processes, conditions and factors built in a model. Objectives of teaching and aspects of the environment generally constitute the focus of the model.

2. ***Syntax:*** It is the sequence of steps involved in the organization of a complete programme of teaching.

3. ***Social System:*** It refers to two elements: teacher role and Students, particularly hierarchal relationship and student behaviour, which is rewarded. It is an important element of every-model that specific learning's are very much controlled by the kinds of relationship and are structured during the process -of teaching. Models of teaching prescribe system for teaching of attitudes, skills' and understandings.

4. ***Support of System:*** This element means-to provide facilities to teacher and the students to successfully implement the strategy of teaching. For example, if we want implement individualization, we have to provide, a number of audio-visual aids, teaching mechanism and programmed text to eater to the needs of individual learner.

1.3. Need for a Model of Teaching

The theory of teaching in research on teaching is an important aspect for the teachers. It can be utilized in following ways:

1. It is a guide to formation of research hypothesis that may be tested in practical context.

2. It trains investigators to look for facts, which may not be readily apparent.

3: It accumulates research work, allowing old studies to gain support from new studies.

4. It also allows research to proceed systematically.

5. Above all it makes possible the conclusions to take shorthand form and thereby helping communicability among the researchers themselves and also between' the researchers and the practitioners in teaching.

Here raises a question that why the teachers need a theory of instruction, since psychology ahead contains theories of learning and development. According to Bruner (1966) teachers do not need a descriptive theory but they need a prescriptive theory. Theories of learning and development are descriptive rather than prescriptive. They tell us what happens after the fact. A theory of instruction, in short, is concerned with how and what one wishes to teach can best be learned, with improving rather than describing learning.

The Committee on the Criteria of Teacher Effectiveness (1953) concerned itself with theories and paradigms with a view to draw notice to the necessity of more systematic and orderly approaches to research on teaching. The tools and theories allow the research workers to bring clarity into their concepts, order in their procedure and system in their investigation.

Kerlinger (1964) says, "I appreciate the trend of building theories in education". He further says that the development of theory should be the basic aim of research in. education. Theories, he contends, have wider applicability and' usefulness than non-theoretical findings.

The need for theory of instruction may be summarized as follows:

(i) A theory of teaching helps to clarify the description of the leaching situation in ordinary language.

(ii) It summarizes existing knowledge.

(iii) It mediates application of our knowledge of teaching to a new situation.

(iv It leads to fruitful lines of experimental inquiry.

1.4. The Nature of a Model of Teaching

According to the Association for Supervision and Curriculum Development, NEA, Commission on Instructional Theory (1968), the term theory is used-in teaching as "it is used in the natural-science to represent a set of interrelated generalisations derived from data, which permits some degree of prediction or control over the phenomena to which they pertain".

In this monograph the word “teaching” refers mainly to the activity, which takes place during schooling and within classroom setting. The term includes both material variables and human variables. It refers to the interaction between teacher, pupils and situational elements.

Bruner (1966) says that a theory of instruction is prescriptive in the sense that it sets for the rules concerning, the most effective way of achieving knowledge-or skill. By the same token, it provides a yardstick for criticising or evaluating any particular way of teaching or learning.

> “A theory of instruction is a normative theory. It sets up criteria and states the conditions for meeting them. The criteria must have a high degree of generality”.

Bruner (1968) further elaborates the prescriptive and normative characteristics of theory of instruction in the following words: “Such a theory has the aim of achieving particular ends and producing them in ways that we speak of an option. It is not a description on what has happened, when learning has taken place. It is something, which is normative, which gives you something to shoot at and, which in the end must state something about what you do when you put instructions together in the form of a course”.

Travers (1966) classifies the major independent variables of his model into four main classes:

- Pupil variables
- Pupil task variables
- Teacher variables
- Teacher task variables

Bruner (1968) has proposed that there are four aspects of such a theory:

- Predisposition
- Structures
- Sequences
- Consequences

First, a theory of instructions should concern itself with the factors that predispose a child to learn effectively. These factors relate to his earliest childhood experiences and these precede the child's entry into our scholastic care.

The second aspect is that it should concern itself with the optimal structuring of knowledge. By this, he means that for any body of knowledge there is a minimal set of propositions or statements or images from which one can best-generate the rest of what exists within that field.

The third aspect deals with the optimal sequence that is required for learning. In what order do we present things? What exercises do you give to the student to strengthen

the views of his own thinking? What type of representation do you use? How much particular? How much generality?

The fourth and final aspect of a theory of instructions should concern itself with the nature and pacing of-rewards and punishments and successes and failures.

1.5. Approaches in Building Models of Teaching

Basically there are five approaches in building a theory of teaching. These five approaches can be grouped into categories: quantitative and qualitative.

Quantitative: The look at the relationship between observable teacher behaviour and student achievement.

Qualitative: Is a descriptive way, which seeks to understand the reason why teaching is the way it is.

The quantitative approaches are:

- The process product approach
- The aptitude-treatment interaction approach
- The carol model

The Process Product Approach: By this approach the relationship between teacher behaviour and the average amount of learning is found, which is accomplished by a group. A typical process product type would count the frequency with which specific teacher behaviour occurred in many classrooms over a fixed period of time. Then the frequency of teacher behaviour would be statistically correlated with average student achievement scores to determine the cause and effect of relationships. -

The Aptitude-Treatment-Interaction Approach is concerned identifying particular instructional methods that are well suited to particular students. The guiding question of this type of a model is "which teaching method is best for which kinds of students".

The Carol Model focuses on "time" as the most important factor in student achievement. This type of teaching investigates the ways for teachers to optimize the amount of time mat students spend on learning tasks.

Qualitative Approaches:

- Cognitive Information Processing
- Ethnography

Cognitive Information Processing approach deals with the basic psychological processes that organize and direct teacher behavior.

Ethnography, which has its roots in anthropology, looks at a whole social system. Ethnographers frequently participate in the social system they are studying.

1.6. Social Families of Models of Teaching

Joyce and Weil (1992) developed more than 20 models. They organized the alternative models of teaching into four families. These are information, processing, personal, and social and behavioural.

1.6.1. Information Processing Family of Models of Teaching

The teaching models of this family are concerned with the organization, presentation verbal and non-verbal symbols in a way that help in the formation of concept and solution problem and development *of* social relationship and integrated personality. The important models of this family are as follows:

I. ***Instructional Thinking Model of Hilda Taba:*** It proposes to process the information through inductive process.

II. ***Scientific Inquiry Model of J. Schwab:*** It is designed to teach; the method employed by the subject for solving scientific and social problems.

III. ***Concept Attainment Model of J. Bruner:*** It proposes to develop concept inductive reasoning i.e. developing a concept after presenting its examples and non-examples.

V. ***Advance Organizer Model of David Ausubel:*** It proposes to increase the capacity of learner to absorb and relate bodies of knowledge.

V. ***Cognitive Growth Model of Jean Piaget:*** It has been designed to increase general intellectual ability especially logical reasoning.

VI. ***Memory Model of Henry Lorayne:*** It is designed to increase the capacity to memorise concepts, facts, etc.

1.6.2. Social Family of Models of Teaching

The models of this family are concerned with the social relationship of the individual with others in the society. These models aim at the development of' social relationship, democratic processes and work productivity in the society, They are also concerned with the development of mind and the learning of academic subjects. Some of the important models of this family are as follows:

(i) ***Group Investigation Model of Herbert Thelen and John Dewey*** It aims at the development of skills for participation in democratic social process through interaction skills and inquiry skills.

(ii) ***Role Playing Model of Shaftel and Shaftel*** It aims at motivating students to inquire into different personal and social values.

(iii) ***Social Simulation Model of Seren Boocock and Harold Guitzknow*** It is designed to help students to experience various social processes and to examine their own reaction to them and also acquire concept and decision making skills. ,

1.7. Main Points

1. Teaching is the art of assisting another to learn including provision of information, appropriate situations, conditions or activities designed to facilitate learning.
2. Model is a simplified representation of reality, which is often depicted in a diagrammatical form.
3. Model of teaching is an instructional design used to describe the specifying process and to produce particular environmental situation, which brings specific change in student's behaviour.
4. All good models bring change in behaviour and good learning outcomes.
5. Models of teaching provide guidance to the teacher and help in developing curriculum.
6. They also provide instructional material and help in improving teaching-learning process.
7. The four, sources of models are social interaction, source, information processing source, personal source and behaviour modification as a source.
8. The elements of a model are focus, syntax, social system and support system.
9. Models of teaching are needed to have guidance to the formation of research hypothesis.
10. The approaches in building models of teaching are classified as quantitative and qualitative.
11. Families of models of teaching are developed by Joyce and Weil.
12. There are four families of these models: these are information processing family, personal family, social family and behavioural family.
13. The information processing family of teaching model is concerned with organization and presentation.
14. Also a verbal, non-verbal symbols to help in concept formation, problem solving, development of social relationship and integrated personality.

15. The important models of this family are:

(a) Instructional thinking model of Hilda Taba

(b) Scientific inquiry model of J. Schwab

(c) Concept attainment model of J. Bruner

(d) Advance organizer model of David Ausubel

(e) Cognitive growth model of Jean Piaget

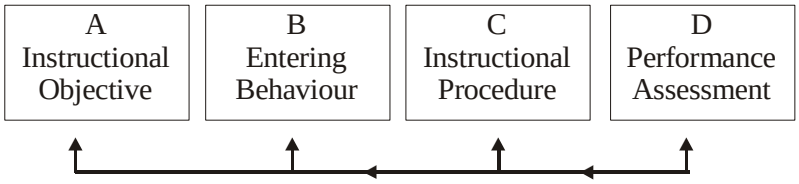
(f) Memory model of Henry Lorayne

2. DESCRIPTION OF MODELS

There are many models- of teaching to guide the teachers in teaching process. Different authors have given different teaching models, but we start from the basic teaching model of Glaser.

2.1. Glaser: A Basic Teaching Model

Robert Glaser (1962) developed a basic psychological model of teaching. In this model the teaching process is divided into four components corresponding to the four major sequences of events in the instructional process.



Basic Teaching Model (Glaser 1962)

Box A: In .box A the instructional objectives show what the students have to attain after the teacher has completed his unit of instructions.

Box B: The box B shows the behavior which indicates the students level before instruction begins. It refers to the students' previous level of knowledge, his intellectual ability and development, .his motivational state and certain social and cultural determinants of his learning abilities.

Box C: Instructional procedure describes the teaching process itself. A teacher has to decide mostly on this procedure. Proper handling of this component will result in changes in students' behaviour such as learning or achievement.

Box D: The box D shows performance assessment. It contains test and observation, which will determine how well students have mastered the instructional objectives. After performance assessment the feedback will be given to C or B or A or any two or all the three of A, B. and C

Some sort of adjustment may be needed in the preceding components if students' achievement is not up to the mark. In this model the teacher is not the person of central importance. With the increasing use of technological devices the model puts a greater emphasis on teacher's competence and skills than on personal art of teaching.

2.2. Herbart: Formal Steps in Teaching

Herbart was a Philosopher and Psychologist, who developed an educational psychology based on the association of ideas. The original formal steps of instruction, according to Herbart, were only four:

(i) Clearness
(ii) Association
(iii) Assimilation
(iv) Application

Later on, Herbart's disciple Ziller divided the step of clearness into two steps viz. preparation and presentation to guide teachers in their efforts. The Herbartians divided a series of instructional steps that became the most widely used aspect of Herbartian teaching method. Its most popular form was in five steps:

(a) Preparation
(b) Presentation
(c) Association
(d) Assimilation
(e) Application

The first task of teacher in preparation was to stir up in pupils' mind the idea upon which he wished-to graft the new material in his lesson. He then presented the material, clearly and attractively, and proceeded to associate it with the ideas previously in the pupils' mind, knitting it skillfully into the existing pattern.

The next step was to examine the fresh pattern formed out of new and old ideas what was the meaning of the new appreciation mass that had been formed.

The final step was that of fixing the new material in the pupils' mind by applying it in a variety of ways in tests, classroom exercise and assignments for homework.

This sequence became a standard approach in the twentieth century for many subjects of the curriculum in primary and secondary schools in many countries. It was systematic, suitable for many different kinds of material, and was based on an easily comprehended theory of learning. In the hands of a skilful teacher it could be both thorough and intellectually exciting. The instructional sequence was associated with the "doctrine of interest". Herbartian method is only applicable to knowledge lesson and not for skill or appreciation lesson. Lastly, Herbart emphasised more on teaching rather than learning. However, interest was as important to the progressive educationalists as to the Herbartians. "Interest", wrote Decorly, is the sluice gate by means of it the reservoir of attention is opened and directed.

Structure of a Basic Lesson for 45 Minutes Class

Time in Min.	Main Phase	Sub-phase	Typical Performance	Herbertian Steps
5	Introduction	Statement of Aim	Linkage Motivation Objectives	Preparation
		Development-1 (Examination Phase)	Attend Classify Label	Presentation
		Review-1	Recall	
	Development	Development-2 (Interpretation Phase)	Identify Explain Conclude	Comparison & Abstraction
		Review-2	Recall	
		Development-3 (Application Phase)	Practice (Encouragement Correction)	Generalisation
		Review-3	Recall	
35	Consolidation	Revision of whole	Retention of whole	
		For mastery	Reinforce Challenge Assess Mastery	Application
45		Any question Next lesson	Linkage	

Sample Forma on Plan (In History)

Name of Institute; Subject:
Class: Topic
Time: lesson Units & Sub-units:

1. Introduction (5 minutes)

(i) Objectives

(a) General — Complete learning of the lessen attainment of desirable behaviours

(b) Specific — To specify in behavioural terms the cognitive psycho motor and effective objectives relating to the lesson.

(ii) Aids and appliances relating to topic:

(iii) Preparation:

(a) Questions and other activities based on other previous knowledge to be linked with the day's lesson for motivating the students.

(b) Announcement of the lesson.

II. Presentation (25 minutes)

	Teacher's Activity	Students Activity
Matter narration	(a) Method leading Developmental questions	(a) Answering questions
	(b) BB Work	(b) Proper use of aids, and appliances
	(c) Use of teaching and appliances	(c) To understand the aids lesson accurately and properly

III. Evaluation (10 minutes)

Asking questions for testing the

(1) Assimilation of the day's lesson — (1) Answering questions to reveal their assimilation.

(2) Providing classroom exercise Etc. — (2) Performing classroom exercise relating to 3 domains.

IV. Assignment (5 minutes)

Assignment to be provided

2.3. Flanders Interaction Analysis Model

Ned A. Flanders (1960) developed this model out of socio-psychological theory. It is a classroom observational technique designed -to test the effect of socio-emotional climate on students' attitude and learning. Flanders developed a technique of recording and analysing the data. The purpose he mentioned as:

> "A theory of teacher influence or more modestly, some, hypothesis of teacher, influence will be developed and tested".

The ultimate goal of the study was to achieve understanding of teacher-pupil interaction and in particular to specify conditions in which learning will be maximised. Flanders classified this influence as 'direct' when a teacher sets restraints upon a student or focuses students' attention upon an idea, and 'indirect' when he reduces a restraint upon student and encourages students' participation. In interaction analysis of Flanders, we use

a 3-second interval time-sampling observation technique for coding the type of teacher pupil talks in 10 categories. By sampling about every 3 seconds, the observer will have made about 400 entries in 20 minutes. The pattern of teacher behaviour can be viewed by the frequency as well as the patterning of categories: 7 assigned to teacher's talk and 2 to students talk and 1 to silence or confusion.

Flanders' study has shown that, by and large, students in classes with higher direct indirect ratio achieve more and possess better attitude than students with lower direct indirect ratio. His work has made a fundamental contribution towards a model of teaching, identifying general patterns of teachers' influence that produce predictable pupils' response. Flanders states the three principles teacher influence to which the interaction model applies.

Restricting students' freedom of participating in the cycle of classroom learning activities does not increase dependence but decreases achievements.

Restricting students' freedom of participation later in the cycle of classroom learning activities do not increase dependence but does increase achievement.

Expanding students' freedom of participation early in the cycle of classroom learning activities decreases dependence and increases achievement.

In short, Flanders' research on classroom interaction has added a new prospect towards building a model of teaching.

Flanders' Interaction Analysis Category System

Teacher Talk	Indirect	1. Accepts feelings 2. Praises or encourages 3. Accepts or uses ideas of students 4. Asks questions
	Direct	5. Lecturing 6. Giving direction 7. Criticises or justifies authority
Student Talk		8. Student talk: Response 9. Student talk: Initiation
		10. Silence or confusion ,

2.4. Concept Attainment Model

The. concept attainment model was developed by Jersome S. Bruner, Jacqueline Good now and George Austin in 1956. The model is developed from their study of thinking process in human being (1977). The concept attainment strategy, as model of teaching is concerned with two separate but related ideas:

- The nature of concepts themselves, and
- Thinking process used by individual to learn concepts.

Nature of Concepts

Concepts are the key building blocks for the structure of knowledge of the various academic disciplines. The theoretical significance of cognitive concept in psychological theory parallels the seminal role of valence in chemistry or energy in physics. Concepts are the distillate of sensory experiences and the vital link between external inputs and overt behaviours. They are the vehicles of thought they are the critical components of an individual's cognitive structure. According to Tennyon and Park, a concept is assumed to be a set of specific characteristics and can be referenced by a particular name or symbol. Concept learning is thus regarded as the identification of concept attributes which can be generalised to newly encountered examples and discriminate examples from non-examples. Concepts can be thought of as information about objects, events and process that allows us to

(a) Differentiate various things;
(b) Know relationship between objects; and
(c) Generate ideas about events, filings and processes.

Brune analyses any concepts as haying five elements: '

(i)	Name	It is the term given to the category
(ii)	Examples	They refer to instances of the concept. There are some positive and negative examples. Positive examples are the instances of the concept.
(iii)	Attributes	These refer to the characteristics of a particular concept that help distinguish instances of the concept from non -instances. These characteristics or attributes may be relevant, irrelevant or criterion-related. Relevant attribute? are common to all examples of concepts while irrelevant attributes vary among examples of a concept, but not with other examples of the same concept.
(iv)	Attribute Value	It refers to the acceptable range for any given relevant attribute. Criterion related attributes are those relevant attributes that distinguish the concept from other super ordinate or coordinate concepts.
(v)	The Rule	It is the definition or statement specifying the special attributes of a concept. To be most facilitative for concept attainment, concept definitions should contain references to both the relevant and criterion related attributes or the concept.

Suppose there is an array of fruits in front of you, mostly apples but one or two-oranges and a pear, each fruit is an example for instance. If my concept is apple the pear and oranges are negative instances and the apples are positive instances. Each example can he described in terms of the basic characteristic. In the case of fruit, these are colour, size, weight, shape and cost. These basic characteristics are called attributes and each

attribute has-an attribute value. The term attribute value refers to-the specific content of that category. Most attributes have a range of acceptable values. The attribute value for an apple is ranged from red to yellow. Purple is out of the acceptable value range of the concept "apple".

Now, the names of the concepts, particularly attributes and examples, play important roles in, student thought process, directed to concept attainment.

Bruner and the associates identify regularities in students', decision making processes that they label thinking strategies. On the basis of their research they have developed concept attainment teaching strategies of two types:

(i) *Selection Strategy:* Here teacher presents unlabelled examples of the concept and the student chooses concept instances. The selection strategies place responsibility for concept attainment and attribute tracking in the hands of the students.

(ii) *Reception Strategies:* The teacher presents example of the concept that arc labelled 'yes' or 'no' and students hypothesise the concept and accept or reject the example.

The Concept Attainment Mode] developed by Bruner and his associates has three variations:

(a) *Reception model* is mere direct in the elements of a concept and their use in concept attainment.

(b) *Selection model* permits students to apply this awareness of conceptual activity in a more active context, one that permits their own initiation. and control.

(c) *Unorganized material model:* The analysis of concepts in unorganized data transfers concepts theory and attainment activity to a real life setting.

Illustration of the Model (From the Reception Strategy)

A primary school teacher decides to teach **a** concept 'domestic animals' to 5th graders. He or she begins the activity by saying that he or she has a category in mind and he or she is going to present certain examples that fit into the category and certain non-examples that do not fit into the category. The positive examples are denoted by "yes' and negative examples are denoted by 'no'. Then the teacher writes on the blackboard the following examples and asks the pupils to determine the category.

Example-1:	Cow	Yes
	'Tiger'	'No'
Student	ST	A- It is an animal that gives us milk.
	ST	B- Whose meat is not eaten by Hindus.

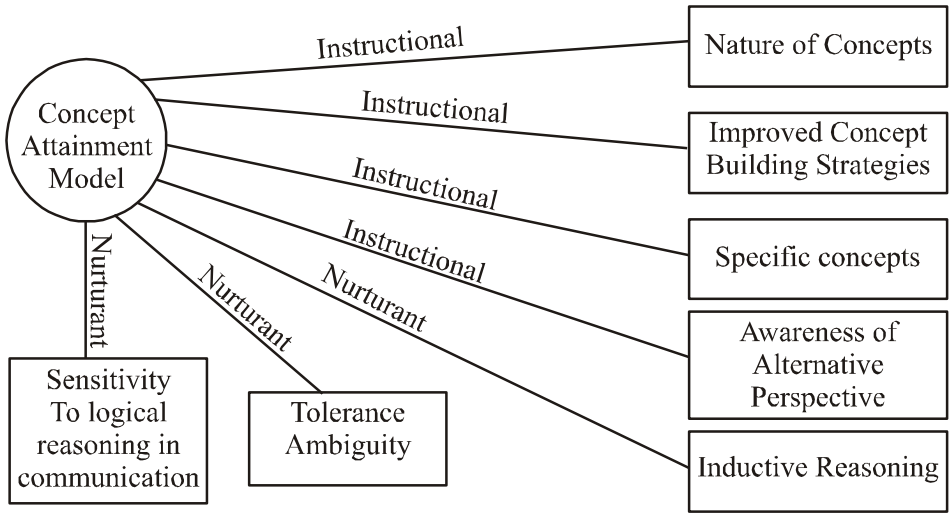
ST C - An animal, which is worshipped by Hindus.

ST D- An animal that people keep in their houses.

Example-2: Goat 'Yes'

ST E - The category is an animal that gives milk. The example of goat has now forced the learners to eliminate the earlier two hypotheses - the animal is not eaten by Hindus; and the animal, which is not worshiped by Hindus.

Concept Attainment Model



2.5. Piaget: Cognitive Growth Model (Increasing the Capacity to Think)

Jean Piaget was a psychologist, who worked on the development of intellectual process. Based on his study, American psychologist Lawrence Kohlberg and Irving Siegel have crafted teaching methods designed to improve intellectual development in general and moral development in particular.

Piaget's studies of intellectual development find their applications to education. His views include organization of the educational environment, curriculum content and sequence, methods of teaching and assessment The role of a teacher is not clearly prescribed anywhere.

Piaget classified intellectual development into different stages. At any given stage one is able to perform certain kinds of thinking. The earlier stages lay the basis for future development. The stages are:

(1)	Sensory-motor stage		(O to 2 years)
(2)	Pre-conceptual stage		(2 to 7 years)
	(a)	Pre-conceptual stage	(2 to 4 years)
	(b)	Intuitive thought	(4 to 7 years)
(3)	Operational Stage		(7 to 16 years)
	(a)	concrete operational though	(7 to 11 years)
	(b)	format operational thought	(11 to 16 years)

Principles of Learning and Teaching

1. Wordsworth summarised Piaget's ideas on teaching and learning. According to him, teaching is the creation of environment where the students' cognitive structure" can emerge and change. Here the students' role is active and self-discovering. The teacher's function is to arrange for learning experience that facilitates stage relevant thinking-and to organize instruction so that students can initiate the activity and discover for themselves.

2. The second principle is based on Piaget's distinction among three types, of knowledge: physical, social and logical. The nature of matter (softness, hardness, etc.) gives the physical knowledge. Social knowledge comes from society in which a student lives. Logical knowledge is mostly related to mathematics and logic.

3. The last principle dealing with the role of social environment specifies the role of teacher as:

   (i) Organizer of the learning environment;

   (ii) Assessor of children's thinking; and

   (iii) Initiator of group activities like play, games and discussion, etc.

   It is clear from all the researches that a good teacher has to undertake constant and continuous assessing process. Piaget descries such a process as "method *Clinique"*.

Based, on the principles of clinical interview we shall now develop the model of teaching. The model requires teachers to give tasks, observe how the students interact with the tasks, and then respond in accordance with the students' action. After making assessment, the teacher must determine the general range of development stages of the students so as to avoid inappropriate tasks being included in the curriculum.

Syntax

This model has three phases:

In the first phase, the students are presented with a situation, which may be puzzling to them as they may confront them with the illogic of their thinking. However, confronting situation must be well matched to the learners', developmental stage. Also the confronting situation should be such as to permit the students to assimilate on the strength of the prior learning but it should be new .enough to require accommodation.

In phase two, the students' responses are called for and their level of reasoning is determined through probing, which generally will consist of asking for justification or offering counter suggestions. To start with, depending on the type of task, the teacher may start with question like what do you think”? Or “what do you see"? The need for probing is an attempt to obtain the correct response. If there is a correct response then \he student is asked to justify the response by indicating his or her reasoning. After the reasoning is persuaded, one or more counter suggestions may be made in order to verify the soundness of a student's reasoning.

Phase three i.e. the transfer phase, has the objectives of seeing whether the student's reasoning will be similar for a related task. Here also the students are given a problem, they make judgments. The teacher requests for reasoning and offer counter suggestions.

Syntax of **Developmental Model**

Phase one	***Phase two***	***Phase three***
Confrontation with Stage-relevant Task	Inquiry	Transfer
Present puzzling situation well matched to learner's developmental stage.	Elicit student responses, and ask for justification, offer counter suggestions, probe student's responses.	Present related task and probe student's reasoning, offer counter suggestions.

Social System

The social system can range from minimally structured to highly structured. However, a fairly structural teaching model with the teacher initiating and guiding the inquiry in a free intellectual and social atmosphere, may be more suitable.

Principles of Reaction

The teacher must create an environment where the students will feel responding spontaneously, and it the same time, he should carefully avoid cuing questions. He has to inquire into both 'correct' and 'incorrect' responses. He should also inquire constantly students thinking with counter suggestions, until he or she is satisfied that the students have attained the particular level of reasoning.

Support System

It is desirable that a teacher is quite knowledgeable in developmental theory. He must also have adequate stock of relevant counter suggestions for Piagetian educational models. A rich and free social environment is essentially important to work out the students' cognitive problems. Such environment is useful for the teacher to act as a facilitator to appropriate comments at the right moment.

Application

This model is useful for both cognitive and social development. In fact, it is useful in areas confronted with illogic, for diagnosis, evaluation and for instructional purposes. The model can be employed to ensure that a child can operate smoothly in his environment or to find out activities that will accelerate the child's cognitive growth.

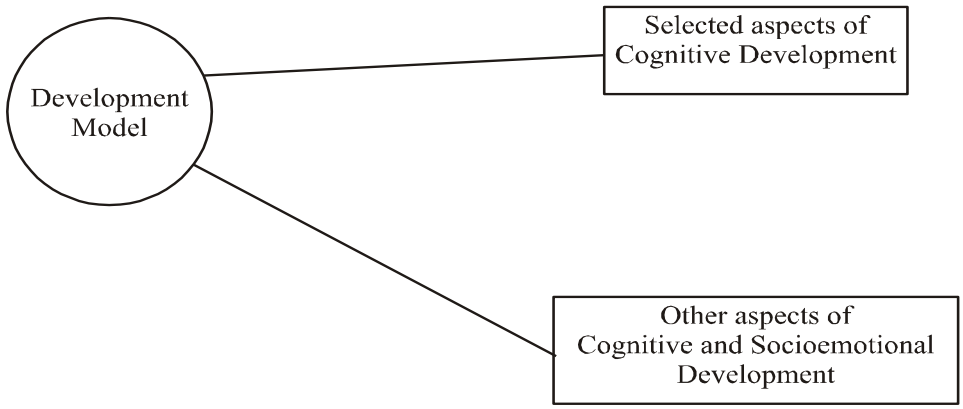
Kohlberg, after Piaget, identifies three major levels of moral development: Pre-conventional, conventional and post-conventional. Each level has two stages.

Instructional and Nurturant effects of Developmental Model

The effects of any model can be categorised as the direct or instructional effect, which are directly achieved by leading the learner in certain directions.

(a) The nurturing effects that come from experiencing the environment created by the model,

(b) To bring change in the cognitive development following Piaget's model, it is necessary to affect all aspects of functioning. Thus, one aspect of cognition will nurture development in all other areas.



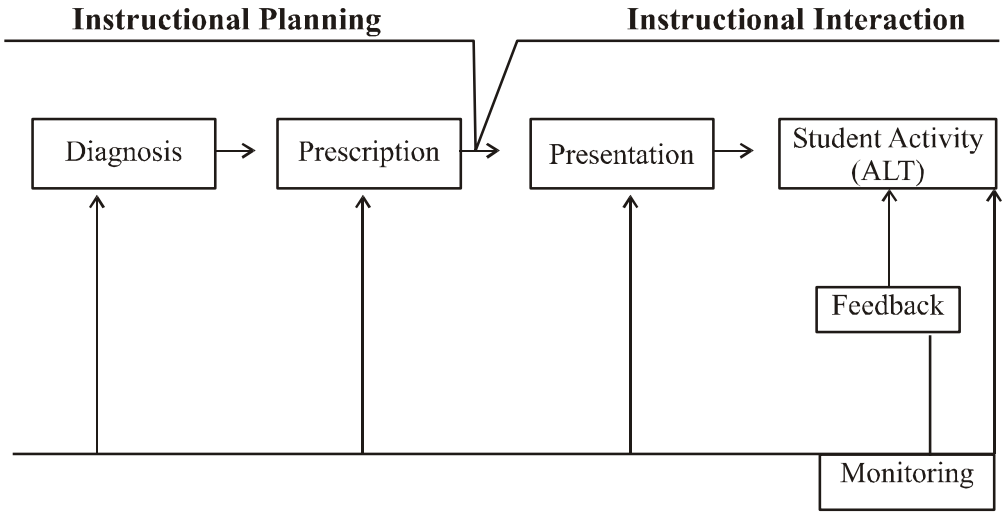
Instructional and Nurturant Effects of Development Model

2.6. BTES: Academic Learning Time Model of Classroom Instruction

Beginning Teacher Evaluation Study (BTES) was carried out by the Far West Laboratory for Educational Research and Development in San Francisco, California, USA, in 1970s, which showed a new path in research on teacher behaviour and pupils' achievement in elementary school classrooms. The study has developed, tested and pitted into operation as the Academic Learning Time Model in Classroom Instruction. Until the 1960s, research on teaching had not received a great deal of attention. By 1975, however, a number of teams and individual research efforts had been completed or were underway. Almost all of these studies were cast in the form of process product analyses, wherein teacher behaviour and classroom characteristics were observed and subsequently related to measures of student achievement and attitudes. Correlation was the frequently used statistics in these studies. The BTES hypothesized that the acquisition of knowledge in the classroom is best conceived as a multiyear process (Fisher and Barliner - 1977). Thus, experimental designs that reflect the process product framework often suffer from problems of ecological validity.

Simple Modification of Process Product Approach

Researchers on the BTES have proposed a simple modification of the process product approach to the study of classroom learning. The link between teacher behaviour and student achievement is the ongoing student behaviour in the classroom learning situation. This complex variable is called the Academic Learning Time(ALT). The investigation of the relations of ALT to teacher behaviour and to student achievement requires a simple flow of events that influence achievement in a particular curriculum area.



Instructional Functions in the Academic Learning Time Mode of Classroom Instruction

The Teacher behaviours, that influence student learning have been conceptualised as serving five interrelated functions. These are diagnosis, prescription, presentation, monitoring and feedback. These functions are broadly of two categories: planning and interaction, and these functions occur through time, in a cyclical fashion. The inajor component of the ALT is the classroom allocation of time from both the teacher and student standpoints. Studying the time variable has led the BTES researchers to observe classroom allocation of time of a teacher. Typically, a teacher in the primary grade allocates 50-100 minutes a day to reading and 30-50 minutes to mathematics. From the analysis of teaching protocols taken in the classes of more and less effective teachers, it has been found that teachers become constrained by their time allocation, starting and ending lessons by clock rather, than on the basis of the cyclical functions of the ALT model. Thus, when teacher rushes students for any reason, skipping one or more functions of the model, they appear to be less effective in helping students achieve in academic areas. Immature learners cannot thrive when disjointed lessons occur within a given instruction period. The ALT is synonymous with engagement, attention, and on task behaviour. Every time, a student is apparently on-task during a teacher's allocated time for a lesson, increase in learning takes place. Even slight increase in ALT would appear to be logically, related .to increased-student acquisition of knowledge. Empirical data from many sources is accumulating to support this position.

2.7. Main Points

1. Basic psychological teaching model was developed by Glaser (1962)
2. Four components of this model are:
   (a) Instructional Objectives
   (b) Entering Behaviour
   (c) Instructional Procedure
   (d) Performance Assessment
3. Formal Teaching Steps of Herbert are clarity, association, assimilation and application.
4. Most widely used instructional steps of Herbartian teaching method are preparation, presentation, association, assimilation and application.
5. The instruction analysis model is a technique of recording and analysing data,
6. The goal of instruction analysis model is to achieve understanding of teacher-pupil interaction and conditions in which learning is maximised.
7. There are three principles of teacher influence to which the interaction model applies.
8. Concepts are the key building blocks for the structure of knowledge of-the various academic disciplines.
9. This model is concerned with two ideas that are the nature of concepts themselves and thinking process used by individual to learn concepts. -
10. Concepts are informations about objects, events and process.
11. They allow us to differentiate, know relationship and generate ideas about events, things and process.
12. The elements of this concept are name, examples, attribute, attribute value and the rule.
13. Cognitive growth model of Piaget is concerned with the development of intellectual process.
14. His views also include organization of the educational environment, curriculum, content and sequence, teaching methods and assessment.
15. Piaget classified this intellectual development in sensory-motor stage, pre operational stage and operational stage.
16. His ideas are summarised into three principles of learning and teaching.
17. The developmental model of teaching is based on these three principles.
18. This model consists of three phases: conformation, inquiry and transfer,

19. There are some principles of reaction, which a teacher must follow while he is with the students.

20. By following cognitive developmental model all areas of development will be nurtured.

21. Chi the basis of BTES research, the students acquire more academic knowledge at primary grades in classrooms.

22. The schools and teachers have allocated more time for academic instruction.

23. The time used for lessons is continuous rather than *disjointed.*

24. Teachers are actively oriented rather than bound by clock as a guide for the length of the lesson.

25. Students are actively involved in instructional episode so that differences between the teacher's and students' allocation of time are minimised.

3. SUMMARY

Teaching is the art of assisting, another to learn. Model is a simplified representation of reality, which is often depicted in diagrammatic form. Model of teaching is an instrumental design used to produce particular environment and to bring specific change in student's behaviour. All good models of teaching provide guidance to the teacher and help in curriculum development. These models are categorised in four families. The information-processing family is concerned with organization and presentation. Social family of these models is concerned with the social relationship of an individual to the society. Different authors have given different teaching model. Basic psychological model of Glaser consists of four components: instructional objectives, entering behaviour, instructional procedure and performance assessment. Most widely used formal teaching steps of Herbart are preparation, presentation, association, assimilation .and application. The instruction analysis model is a technique of recording and analysing data. The concept attainment model is concerned with the nature of concepts themselves and thinking processes used by individual to learn concepts. Piaget explains the development of intellectual process of an individual by giving cognitive growth model. He includes organization of educational environment, curriculum content and sequence, teaching methods and assessment in his views. Beginning teacher evaluation study is based on acquisition of more academic knowledge of students at primary grades in the classroom.

4. SELF-ASSESSMENT QUESTIONS

Q.No.1: Define model, what you mean by a model of teaching?

Q.No.2: List the five main characteristics of a model of teaching?

Q.No.3: Name the four main classes of independent variables given, by Travers(1966)

Q.No.4: The approaches in building, a model are of two main categories. What are these?

Q.No.5: There are five most popular Herbartian steps. What are those steps?

Q.No.6: What are the principles of Flanders interaction model?

Q.No.7: What are the two related ideas concept attainment model is concerned with?

Q.No.8: What are the elements of concepts analysed by Bruner?

Q.No.9: List down die stages of intellectual development classified by Piaget?

Q.No.l0: What do you mean by BTES?

Q.No.11: What is major component of ALT?

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UNIT–2

THEORIETICAL FRAMEWORK OF INSTRUCTIONAL STRATEGIES

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INTRODUCTION

Teachers are blessed persons as they got an opportunity to shape the minds, attitudes and personalities of human beings, the prime creature of Almighty Allah. Therefore in this way a teacher assists the Lord.

Here is a long list of teaching methodologies. In recent years, there has been an increased concern among practitioners and educational research about practicality and effectiveness of teaching. The "HOW" of teaching is now being given as much significance as the "WHAT" and the "WHY" in academic circles?

The Unit in your hands is from the above-mentioned chain. It is organized by addressing individualized, group and class teaching. In the text, you will find the meaning of each of the teaching methods along with its background.

Here principles of teaching are printed, which if followed, guarantee the success rate to the maximum while reading/studying this text you will come to know the techniques and instructional strategies being suggested for each teaching style.

This Unit touches the historical background of methods as well. Mainly, it comprises five parts; (1) Concept of Teaching, (2) Concept of Instructional, (3) Individualized instruction, (4) Meaning of Group Instruction, and (5) Class Instruction.

While preparing this Unit, an attempt has been made to provide readers, learners and teachers of education with comprehensive discussion of main aspects of three instructional strategies. This Unit will serve as "Basic text" at the same time as well.

OBJECTIVES

After reading this Unit, the learner will be able to:

1. Differentiate between styles of instruction.
2. Explain the instructional methods of each style.
3. Identify specific principles applied to each style.
4. Compare the instructional methods on cost, effectiveness, merits and demerits.
5. Follow up the steps of problem-solving and project method in order to their application in the real situation.
6. Discuss, recall and apply the different methods of instruction.

1. CONCEPT OF TEACHING

The word “teach” comes from Old English *taecan* that is in turn derived from the Old Teutonic *taikjan*, the root of which is *teik*, meaning to show and is traceable to Sanskrit die through pre-Teutonic *deik*. The term “teach” is also related to “token” a sign or symbol. “Token” comes from the Old Teutonic word *taiknom*; a cognitive with *taikjan*. Old English *taecan* means to teach. So, “Token” and “teach” are historically related. To teach, according to this derivation, means to show someone something through signs or symbols: to use signs or symbols, to evoke responses about events, persons, and observations, findings and so forth. In this derivation, “teach” is associated with the medium in which teaching is carried on.

A descriptive definition of “teaching” in the 1500s would have been somewhat different from that of today. A descriptive definition then would, have been formulated from such notions as to teach is to give information; to show a person how to do something; to give lessons in a subject. The conventional sense of teaching nowadays is not entirely different. A descriptive definition of teaching can be stated as follows: teaching is imparting knowledge or skill.

The purpose of such a definition' may be to point out the reference of the term, or to show how instances covered by the term differ from others with which it can be confused. Terms such as “imparting,” "knowledge,” and “skills” are often ambiguous. A descriptive definition will remove, as much as possible, this indefiniteness, by resorting to contextual usage. For example, in one context “imparting” means to share, as when someone says to teach is to share experiences. In another context “imparting” signifies the communicating of information by lecture. But even when resorting to context eliminates ambiguities, the term clearly denotes certain objects and just as clearly rules out others, there will be cases to which the term might or might not apply. Is the spreading of propaganda to be counted a teaching? Some persons will say yes .and others no, depending upon their concept of propaganda and how to fit their notion of what it means to be educated. The descriptive definition of teaching as imparting knowledge allows the matter to be settled either way.

A descriptive definition can focus thinking upon a particular course of development (Schefiier 1960 pp. 19-28), It is then said to shade off into a programmatic definition. For example, if “imparting information' is taken to mean that lecturing is the imparting mode. Those, who advocate inquiry as the mode of teaching, will be quick to oppose the descriptive definition. They may admit the accuracy of the historical definition of “teaching” but still turn to their-own programmatic definition of “teaching” as inquiry. Thus, pedagogical discourse is often fraught with semantic issues.

1.1. Nature of Teaching

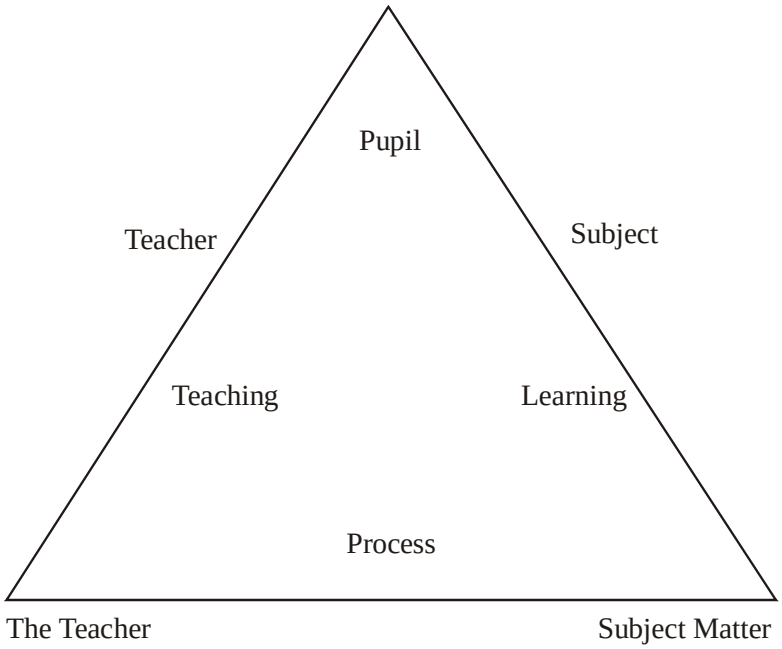
Previously teaching meant nothing more than “giving information and imparting knowledge”. It was the time when teaching was regarded as a bio-polar process: Teacher and the subject being its two poles. The child way altogether ignored. No attention was given to his need and desires. The child was not important and the teacher and the subject

matter were considered more important than him. The child was measured by the adult yardstick. The subject matter was read out, told by the teacher and the child.simply memorized.

Readymade material was given to him, which sopped his energies. Modern education has brought the child into limelight. He stands as the focus of the educative process. Consequently, the concept of teaching has undergone a radical change in the light of this; we shall discuss various aspects of teaching as under.

(1) Teaching a Triangular Process

Modern concept of the nature of teaching is that, it is a triangular or tri-polar process. The three" focal points of this process are the teacher, the child and the subject matter. There is a close relationship among the three in fact. Teaching is a uniting force.



We cannot have teaching unless three are present. In all the three the teacher is the most active agent in the process of establishing the relationship and to bring about effective relationship. The teacher must understand his pupils on one the hand and have a thorough knowledge of his subject on the other. The relationship between the subject and the "child suggests another aspect of teaching.

(2) Teaching is Giving Information

There are so many things, which the child does not know or cannot know. It is therefore, essential to provide some useful knowledge to the child. Keeping in view his needs and capacities, it should be presented in a systematic and interesting way.

(3) Teaching is causing to Learn

Child cannot be treated like an empty vessel into which any type of information can be poured down. We cannot force the child to learn. He will learn only, when he is willing to learn. So the teacher is to make the child-willing worker. He is to help the child to learn for himself. He is a mere instrument in facilitating learning on the part of the child.

(4) Teaching is Stimulating Learning

The child will not automatically become willing to learn, the teacher must think of ways and means of stimulating and encouraging learning on the part of children. He should provoke their interests and motivate them to learn. He should create conditions in which they feel the need to learn.

(5) Teaching is Giving Guidance

Simply arousing his interest and leaving the child to his own efforts is not sufficient. The child may make random movements, resulting in the sheer wastage of time, so he should be guided to do the right things in the right manner and at the right time. He should be able to make correct responses to the various stimulations in his environment. This leads us to another aspect of teaching.

(6) Teaching is helping the Child to Make Effective Adjustments to his Environment

The main aim of teaching is to enable successful social life. This requires the ability in the child to make effective adjustments to his environment. For this, he should be provided simplified environment in the school in which the child learns to overcome difficulties and make effective adjustments on a small scale. Also we should develop his innate powers and train him to use the same in the right direction.

(7) Teaching is helping the Child to Develop Emotional Stability

Teaching will be inadequate if we do not train the emotions of the child and develop his emotional stability. Right actions spring from right feelings. The child should feel a sense of security, which is possible in atmosphere of love and freedom. The teacher should have encouraging-and sympathetic attitude towards his pupils,

Teaching in short, is a process of imparting knowledge, motivating and guiding the pupils to learn through their own activities, training their emotions and developing their powers and capacities so that they are able to make effective adjustments to their environment-and are better prepared for successful social participation.

1.2. Characteristics of Good Teaching

(1) Good Teaching is Stimulating

Simulation and inspiration are of fundamental importance in teaching. A good teacher will try to stimulate his pupils for work, or otherwise they become passive and disinterested in studies. Without adequate stimulation if pupil acts, no worthwhile learning force takes place.

(2) Good Teaching Involves Skill of Guiding, Learning

The pupils need guidance to do the right things, in the right manner and at the right time. But guidance is not to be imposed on the child. It should be given skillfully through suggestions, examples and appropriate environmental situations.

(3) Good Teaching is well planned

A good teacher organizes and plans the subject matter in advance. Without proper planning, effective delivery of the lesson is not possible. However, planning must provide scope for necessary changes due to unexpected changes that may arise in the classroom.

(4) Good Teaching is suggestive and Co-operative

A good teacher will not impose, anything on the child. Rather he will, seek his co-operation and suggest activities, ideas, materials, etc. It is 3 psychological fact that suggestions take deeper root in the minds of children whereas direct advice is generally repulsive to the children. For example, to inculcate his qualities of courage and fortitude, life stories of brave personalities may be suggested to read.

(5) Good Teaching is Kindly and Sympathetic

A good teacher, is 'always kind and attends to his pupils' difficulties and doubts sympathetically. They feel that the teacher is their well wisher and thus whole-heartedly co-operate with him. On the other hand, a harsh teacher can never elicit warm response from his pupils.

(6) Good Teaching is a Democratic

Good teaching should be based on democratic principles. The teacher should have due regard for the rights and individuality of the child. Methods of teaching and subject matter should be selected from this viewpoint.

(7) Good Teaching is both Diagnostic and Remedial

The teacher must study and assess the inborn qualities of the children individually. Their limitations, handicaps and difficulties are to be discovered and remedial measures suggested. Like a Doctor, the Teachers prescribe treatment for it.

(8) Good Teaching is Co-relative

A good teacher does not simply narrate various items of knowledge in watertight compartments. Fie tries to link the new knowledge and experience of pupils. He also tries to co-relate his subject with craft, their physical and social environment.

(9) Good Teaching I iterate the Learner

Good teaching develops in the child habits of initiative in dependent thinking, self-reliance and self-confidence. It enables him to learn for himself and, reduces his dependence upon the teacher. That is to say, the child is liberated from teaching.

1.3. Need of Teaching

M.D. Alcorn, J.S. Kinder & J.R. Schunert (1970) in their book "Teaching in Secondary Schools" quote six patterns of teacher behaviours, which have been developed by Wallen and Travers.

(1) Patterns from Teaching Traditions

The teachers teach generally the same way as they were taught by their teachers and specially by the way of the teacher whom they identify as their ideals.

(2) Patterns from Social Learning

The teachers work generally as a prisoner of their own background and develop the same type of behaviour among their students as seen by them.

(3) Patterns from Philosophical Traditions

Each and every teacher, whether he knows or not, teaches according to one or other philosophy of education.

(4) Patterns from Conditions of School and Community

The teachers generally work in the school in the way, which the school administration and the members of community expect from them.

(5) Patterns from Research

Some teachers conduct' research on learning and teach according to that pattern.

(1) Interpersonal Behaviours Based on Interpersonal Roles

The teacher can know students better if they carefully observe and understand their emotional, intellectual and social needs and interests.

The teachers can demonstrate exemplary values and character if they handle situations firmly, honestly and patiently.

The teachers can work with colleagues, principal, other staff members and parents in proper plan and carry out various inter-school activities and communicate both in formal way and informal way.

The teachers can properly behave if they co-ordinate and assist in for community and hold-and attend seminars, workshops, conferences and discussion groups.

(2) Pedagogical Behaviours Based on Pedagogical Roles.

The teachers can delineate the course of teaching through determining major areas to be covered and scope and sequences of units, topics and lessons to be followed.

The teachers can identify and use basic sources if they enrich their understanding of the subject area and enrich their teaching in classroom.

The teachers can determine the teaching objectives of their subjects if they set objectives concerning content of the subject, concerning growth of students and involving different levels of understanding.

The teachers can carry out classroom teaching if they properly use various types of teaching methods, techniques, devices and activities and integrate various resources into lessons.

The teachers can understand their students if they adapt their teaching according to the spiritual, emotional, social and spiritual needs of-students, relate teaching to students' home and daily environment arid relate teaching to other school subjects.

The teachers can manage classroom climate if they coordinate and organize the physical climate, establish daily routines and maintain disciplined and co-operative behaviour.

The teachers can search theories, ideas and practices if they visit libraries, pursue advance study, obtain higher degrees and conferences and communicate with the teachers on local basis, national basis and international basis.

The teachers can observe and assess classroom teaching through monitoring teaching plans, effects of teaching on students and assessing progress and achievement of students.

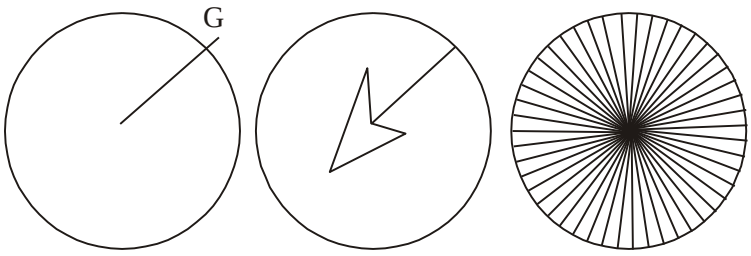
1.4. Teaching Process

The task of teaching is complex and has cyclical nature.

Complexity of Teaching Process

Joel M- Levine presents three models of teaching, which show complex nature of learning.

Basic Model of Teaching (Complexity)



(A)	(B)	(C)
Shortest path to the goal	Movement indirection away from goal	Sixteen major factors to bring into line

Joel M. Levine points the following about complex nature of learning in the above shown three circles.

(a) Point A is beginning of teaching work in all the three circles.

(b) Circle (A) provides shortest path to the goal, which means in other words, it is the most successful path of reaching.

(c) Circle (B) shows that the teacher started into wrong direction from his starting point "A". If he goes either way i.e. AD or AB or ABC or ADB, he is creating problems for the students. In order to be successful in achieving his goal the teacher will have to come back to "A" point to reach "G" point, which is the goal. If he .continues to move in different directions, he cannot reach the goal.

(d) Circle (C) shows that there are sixteen major factors to bring into line. ,

He has identified the following sixteen factors, which have to be managed if the teacher wants to attain his goal of teaching (include P. 642 be here). Each spoke in Circle (C) represents one factor.

1. Delineating course of teaching units, topics and daily lessons.

2. Identifying and using basic sources of the subject.

3. Determining objectives of the course, units and lessons.
4. Procuring and using teaching resources.
5. Getting to know other teachers, administrators and staff:
6. Becoming familiar with school policies and expectations.
7. Maintaining emotional stability and physical health.
3. Getting to know background, needs and interests of students.
9. Establishing roles and standards of conduct.
10. Managing routine duties.
11. Motivating students.
12. Determining teaching strategies and pace of instruction.
13. Utilizing various teaching methods and activities.
14. Assessing readiness of students for learning.
15. Designing and using various assessment techniques.
16. Assessing student's progress and achievement regularly.

The first seven factors he included in the Development phase, the next six in the implementation phase and the last three in the evaluation phase.

- **Cyclical Nature of Teaching Process**

John Dewey (1929) described that teaching is considered of cyclical nature, which have the phases of diagnosis, preparation, guiding teaching and evaluation. It comes back to preparation as the process of teaching repeats. The phases, characteristics and purposes of teaching as a cyclical process are:

Phase of Diagnosis

The characteristics are to determine present state of knowledge and needs of students on the one hand and determine present state of readiness to learn of students on the other. The purpose is to determine what should be done for focus planning and efforts.

Phase of Preparation

The characteristics are to plan teaching, locate resources and carryout research and organize and develop curriculum. The purpose is to master subject matter knowledge and be ready for teaching.

(3) Phase of Guiding Teaching

The characteristics are to select activities and methods appropriate for students, subject and objectives. The purpose is to carry out teaching programme.

(4) Phase of Evaluation

The characteristics are to make plans and assess plans as teaching proceeds to determine success of teaching and learning and modify plans when necessary. The purpose is to assess effectiveness of teaching. 1,5- Maxims of Teaching

1.5 Maxims of Teaching:

(1) Proceed from Known to Unknown

It is said that the old knowledge serves as a book from which the new can be hanged, so that the new knowledge may be acceptable. The teacher must try to win over some part of the child's past experience. To act as it were, as its host ideas, sensations and impressions can, be appreciated or assimilated. When partially familiar they fit into the relevant mass, a foreign idea finds the mind but a glass wall on which it can gain no foothold, so, an attempt has to be made to remind the pupils of what they know already, so as to be ready to pin on to it the new knowledge. While using the previous knowledge the teacher must see that it is .perfect, definite and complete. The teacher must search delight to find which of their experiences has been such that by recalling them to the child, he is likely to ensure a ready welcome for the new knowledge.

(2) Proceed from Analysis to Synthesis

Analysis consists in taking a thing apart into its elements. Synthesis is the complement of analysis. When the child comes to the class his knowledge is incomplete, indefinite and imperfect. The teacher is to round it off. Therefore, the teacher should start with the analysis to view the complex whole through its constituent elements. Analysis is to be employed chiefly to correct and systematize the teachings of every day experience rather than serving and ending in itself. Its purpose is mainly to prepare the way for further synthesis. Thus, analysis must be followed by synthesis otherwise analysis is useless. Analysis makes the things comprehensible and puts it in a clear and broad framework. Synthesis afterwards makes it definite and fixed. The teaching procedures should be neither purely analytic nor purely synthetic but a judicious blend of both.

(3) Proceed from Simple to the Complex

The teacher should keep up the interest of the pupils in the lesson by presenting easier and simpler materials to be followed later by complex and difficult materials. Simplicity and complexity should be determined from the child's point of view. Whatever

material is taught to the child must be authentic.

(4) Proceed from Whole to Part

Whole is more meaningful to the child than to learning of the part: The part approach to learning tends to ignore needs. Learners have to develop some kind of a frame of reference that will help them to relate one aspect of what is to be learned, to its other, aspects as well as to their previous experience. May Seagoe concluded after her research that the ‘Whole’ approach was generally superior to the part approach. ‘Whole’ learning is better than 'part' learning. It is essential to form meaningful wholes.

(5) Proceed from Concrete to Abstract

It is very important for the child to be able to abstract idea. To achieve this purpose, we should approach the child via concrete objects, activity and examples. The imagination is greatly aided by concrete material so while teaching obstruction in any subjects: we should take, help of concrete things as far as possible and then learn to higher levels of thinking.

(6) Proceed from the Particular to the General

Particular facts and examples should be presented to the children before giving them general rules and principles, as they are easier to follow. The process, of induction is easier to understand than the so-called fact. In fact, the study of particular facts should be able to convince the children themselves to arrive at general rules. The process of learning can only become complete if we proceed from particular and finish with the general rule to definitions. So the pupils are to be disengaged from the particular as soon as they can apply the general truths to elucidate further facts. Induction and deduction must go side by side.

(7) Proceed from Empirical to Rational

Empirical knowledge is based on the observation and firsthand experience rational knowledge implies a bit of obstruction and argumentative approach, it is a general feeling that the child feels the rational basis for any knowledge much after he has experienced it in his day-to-day life. For instance, plane geometry would make better sense if it were taught in the practical context of everyday life, instead of in the format of highly abstract theory. It is always good to begin with what we see, feel and experience than with what we argue, generalize and explain.

(8) Proceed from Psychological to Logical

We should proceed teaching in the psychological way, that is, taking into

consideration the child, his interests, needs, recreating and mental makeup. To make a good start we should present the material in a logical way, firstly there should be the selection of the matter in a proper sequence.

(9) Proceed from the Actual to the Representative

The child learns more quickly from the actual and real objects. The real objects from their representations have been taken for granted. Wherever possible the teacher should show the real object. This is particularly, indeed, in the earlier classes, as the information of small children is limited, for them it is always safe to proceed from concrete to abstract, from actual to representative, from near to distant.

2. CONCEPT OF INSTRUCTION

The specific teaching akin to skill training rather than to education in a broader sense is called "“instruction”. May be audio-visual correspondence, individual, etc, from the observation and analysis of forty-nine tape-recorded lessons, the following operational definitions of strategy and tactic, as these terms, apply to instruction, were developed:

Strategy

A generalized plan for a lesson (s), which includes structure, desired learner behaviour in terms of the goals of instruction and an outline of planned tactics necessary to implement the strategy. The lesson strategy is part of a larger development scheme.

Strategies are in a sense, the 'why' of specific teacher behaviour. Some are a function of more explicit 'now' goals, or 'one-lesson accessible' strategies; others, which take more than one lesson to develop and usually continue over a period of several lessons are referred to as ‘overtime’ strategies.

Either one-lesson-accessible or overtime strategies may be changed, modified or discontinued (with another substituted perhaps) as a lesson proceeds and the teacher interprets feedback from the students. One factor, which may give rise to such, change in a lesson strategy, is the nature of that strategy in relation to. the hypothesised and real readiness and rate of progress of the unique group of students (pupils).

Strategy planning is done at a time other than that at which the teacher is. teaching; however, this is not to deny that the teacher may get some ideas for future strategies while in the process of teaching.

Tactic

Goal-linked influenced/influencing behaviour of the teacher - the way a teacher behaves in the instructional situation in working towards the development of the. strategy; units of teacher behaviour through which he fulfils his various instructional roles with the students of his class from moment to moment; the components of teacher behaviour through which the teacher, the students and the subject matter interact.

A tactic may range from no overt teacher behaviour (e.g. using silence for a specific purpose) to one question or statement to a complex of verbal or purposeful non-verbal teacher behaviour interlaced with student behaviors.

In the lessons observed, it was found that one lesson might vary from one tactic; for the whole lesson to a highly complex interweaving of several different tactics. In some cases, while using one tactic with an entire class, a teacher may employ several different tactics at different times or at the same time with different individuals within the total group. Some tactics may be directed towards one child and some towards a group of children; some may be in the form of one teacher behaviour directed towards one learner

response, while others are more complex and dependent upon a sequence of teacher-student interactions; still others may be directed to learner responses at sometime in the future.

It also became evident that a given lesson may illustrate two different , kinds of teacher tactics, planned tactics and responsive tactics. The difference between the two is that planned tactics are those a teacher decides to use to implement his strategy; they are planned before the lesson takes place. Then as the teacher uses these tactics with the class, certain responses by the children may clue him to apply a different tactic with a particular child in terms of the goals established or in order to enhance the child's self-concept. In this case, the teacher draws from his tactic repertory, or he may consciously or subconsciously invent a new (new to the teacher,) tactic to meet the specific situation.

2.1. Instruction

After an operational definition of tactics and strategies had been build, attention was turned to searching for the ways in which tactics and strategies fit into the total picture of instruction; that is, an attempt was made to seek relationships among tactics, strategies, teacher behaviour, goals of education and the learners.

Smith provided the idea that led to the development of the Conceptual Model of Instruction:

Everyone knows that the teacher not only influences student behaviour, but that he is also influenced by student behaviour. The teacher is constantly observing the student and modifying his own behaviour in terms of his observations. We may, therefore, say that instructional behaviour consists of chain of three links - observing, diagnosing, acting, (Smith l963, p.296).

The idea of influencing influenced teacher behaviour seemed most intriguing, as was the picture of the teacher as behaviour observer and diagnoser. By applying these two ideas, the following, four aspects of instruction were identified.

1. Teacher planning - in terms of what the teacher knows of the learner, the curriculum, the situation.
2. Teacher behaviour, initiatory - to create a focus for thinking and working, what the teacher does to get things started.
3. Teacher observation, interpretation and diagnosis of Learner behaviour in terms of the situation, knowledge of prior experiences of the learner, prior observations of learners behaviours, enhancement of child's self-concept the curriculum (affective, cognitive and action dimensions).
4. Teacher behaviour, influenced/influencing influenced by the observations, interpretations and diagnosis of learner behaviour and influencing to the degree that teacher behaviour stimulates further learner behaviour.

Consideration of the flow of those aspects during the process of instruction led to

the development Of the Conceptual Model of Instruction (Fig. 1), from which it can be seen that the potential lesson takes shape as the teacher, making decisions about goals, structure and planned tactics, begins the development of the strategy. Teacher planning (1) is completed.

Teacher behaviour initiatory (2) is a tactic that creates the focus for the lesson and sets the appropriate structure. As a result of the teacher behaviour, initiatory, the children become aware of some of the, goals that will direct their activity as well as their evolving responsibilities in the specific instructional situation: The lesson begins to unfold.

At the same moment the initiatory tactic takes place, teacher observations, interpretation and diagnosis of learner behaviour (3) take place. The teacher observes the consequences of his influencing behaviour. Some observations' may be singular in nature, leading directly to influenced teacher behaviour, some may take place for a longer period of time over, a more complex series of learner-teacher behaviours as related to the purpose of the tactic; some may yield information about the learners that is not directly related to following the teacher behaviour.

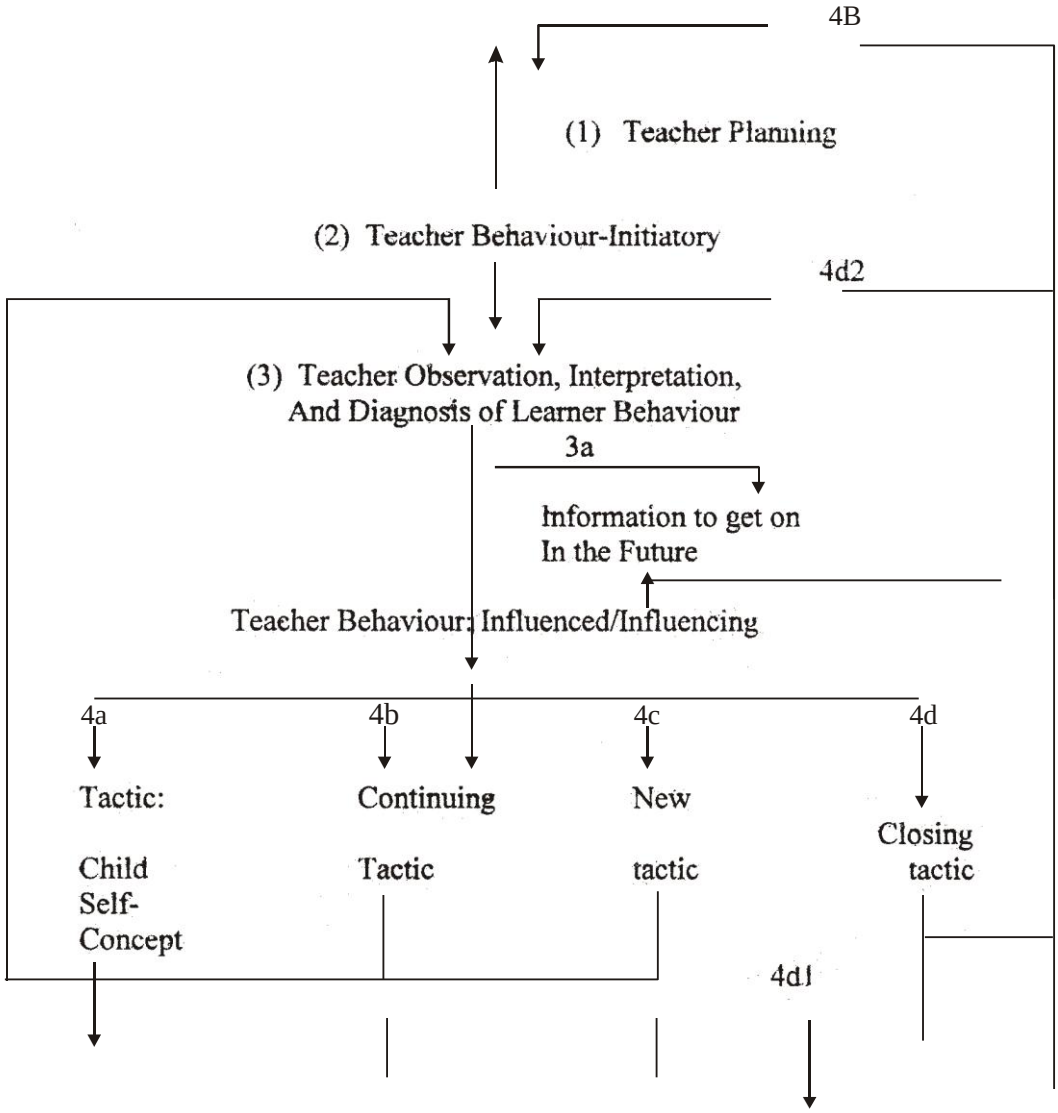


Figure-I A conceptual model of instruction.

For example, while a group of third grade children were working to make a compass, one said. The north part of one magnet will attract the north part of another magnet. Rather than stop the lesson to attempt to correct this misconception, the teacher made a mental note of the response, which became information to act on in the future (3a) and possibly the basis of planning for a future lesson. In this case, the observation interpretation and diagnosis of the learner behaviour did not lead to influence teacher behaviour at that particular moment.

As the lesson continues the teacher's observation and interpretation may lead to

the diagnosis that the initiatory tactic is developing according to plan and the tactic is therefore, continued (4b-Continuing tactic). It may be, however, that the initiatory tactic is not achieving the intended purpose and the planned teacher behaviour is consequently modified or discontinued and another initiatory tactic started (4c-new tactic): or the lesson may he aborted (4d-closing tactic). Thus, teacher behaviour influenced/influencing (4) is initiated. The teacher behaves according to a previous observation, interpretation and diagnosis of learner behaviour (influenced teacher behaviour). And in that the teacher behaviour is intended to feed forward, to new learner behaviour (back to 3), the teacher behaviour is influencing.

Once having achieved the focus and structure for the lesson, the strategy implementation tactic(s) evolves, A 4, 3, 4, 3...behaving-observing cycle develops; this 4, 3 cycle is the interactive heart of instruction. During this sequence, one kind of tactic a teacher might use, which is an exception to be development of the lesson strategy, is teacher behaviour designed to enhance immediately a child's self-concept (4a-tactic: child self-concept).

If the strategy is won or time has run out a closing tactic (4d) is employed. In one case, the topic may be closed off, not to be reconsidered (4d); in another case, the closing tactic may set the stage directly for work next time (4d2). The teacher may say, “We will not be able to complete this work today. Tomorrow, let’s begin just where we left off. When you will come into the room, after recess you may get the materials with which you were working and continue”. The use of 4d implies that the-day's closing tactic provides information, which the teacher will consider in planning the next lesson.

The ways; in which the strategies and tactics relate to the four aspects of instruction; visualized in the model are:

a) The first step, teacher planning is devoted to developing a strategy for a lesson or for a series .of lessons in terms of selected goals. Planned tactics are decided upon.

b) Teacher behaviour in the class situation initiates the tactics in action.

c) Teacher observation, interpretation and diagnosis are made in terms of the purpose for which the tactic was initiated.

d) Teacher behaviour proceeds with tactic as planned or the tactic is modified as a result of prior observation, interpretation and diagnosis. In reaction to students' interaction, teacher may draw on responsive tactics to pull certain students into, class interaction to take advantage of problems question, etc. which arise.

It may be concluded, therefore, that the central directive element of instruction is the lesson strategy and the essence of classroom interactions, the lesson tactic(s).

2.2. The Phases of Instruction

The sequence of transformational events is affected by learning processes.' As these events unfold in the course of an act of learning, they may be influenced to a greater or lesser degree and in a number of different ways, by the external events- of instruction. Some processes (for example, the establishment of expectancy) are obviously affected by instruction whereas others (for example, the internal processes of memory storage) may be only indirectly influenced. It seems reasonable, therefore, to distinguish as successive phases of instruction, those interactions of external stimulation and learning processes that can most clearly be the course of learning. Designating instructional phases in this way helps to emphasise the function of instruction as supportive of learning and thus to suggest the variety of tasks involved in teaching.

Parenthetically, it may be noted that as the learner .develops in Sophistication, many of the events of instruction, initially observed as planned external events, come to be accomplished by the learner himself. In other words, to a greater or lesser degree, he becomes a "self-learner". He may supply his own motivation, develop his own system of coding, initiate strategies of search and retrieval, and supply his own feedback. Naturally, the teacher wishes to encourage this growing tendency towards independent learning. This is done in a number of ways, including the progressive reduction in external "cueing" for coding and retrieval processes as the learner develops.

The phases of instruction as related to the processes of learning may be identified as follows:

(1) Motivation Phase

The preparation for learning is accomplished by instruction, which activates motivation by appealing to student interests. Communications of the teacher during this phase have the additional purpose of relating these interests in an expectancy of "what the student will be able to do" once he has learned

(2) Apprehending Phase

During this phase, teaching is concerned with arousing attention, in a general sense and also with providing stimulation (often verbal), which "directs" attention so that particular features of the stimulus situation are selectively perceived.

(3) Acquisition Phase

This phase of instruction supports the process of entry into long-term store. Coding processes may be provided or suggested. Alternatively, a set to employ a strategy of coding may be activated by communications from the teacher.

(4) Retention Phase

This phase pertaining "storage" is included for the sake of completeness. The manner, in which internal processes of storage (such as interference, simplification and the like) can be directly influenced by instruction, if indeed they can be, is not entirely clear. It seems evident; however, that indirect influence can be brought to bear by suitable arrangement of learning conditions, as, for example, in presenting dissimilar stimuli together rather than highly similar ones, thus reducing the possibility of interference.

(5) Recall Phase

External instructional events during this phase may take the form of providing cues to retrieval or of monitoring the process of retrieval to insure that suitable strategies of search are employed. Teachings also include the conduct of "spaced reviews" providing opportunities for retrieval or occur.

(6) Generalisation Phase

During the Generalisation phase, the teacher provides situations calling for the transfer of learned knowledge and skills in novel ways, and providing cues for application to previously un-encountered situations.

(7) Performing Phase

Instruction oriented to this phase of learning is largely a matter of setting occasions for the student to "show" that he has learned. Obviously, these occasions set the stage for the feedback that comes next.

(8) Feedback Phase

The feedback phase is one in which information is supplied to the student concerning the extent to which his performance has reached or approached a criterion standard reflected in his expectancy. What is accomplished in this phase is the confirmation of the expectancy, affecting the process of enforcement.

2.3. Implications for Instruction

After a look at the Conceptual Model of Instruction, attention is turned to what this model implies about instruction, how it ties together past experiences and opens up some new hypothesis for further consideration.

(1) Instruction is viewed as a Two-way Situation

In one case, students are growing towards the stated curricular goals; in the other, teachers are learning about the children or their classes, the effect of their behaviour upon a specific group of students, ways to behave in terms of certain goals and the limitations of present instructional goals: students grow towards curricular goals and teachers' tactic and strategy repertoire is empirically extended.

(2) One Relationship between Learner Behaviour and Ongoing Instruction is, Indicated

The tactical element loop illustrates the role of the learner behaviours in instruction arid identifies some specific points at which learner behaviours affect the nature of instruction.

(3) The Necessity for Continuous Feedback (completing the tactical element cycle) in Shaping Instruction is Highlighted

According to the model, it may be predicted that if part of the tactical element loop were eliminated the effectiveness of instruction would became limited. When the loop of tactical elements is destroyed, effectiveness in diminished and the cumulative learning effect of instruction for both teacher and student is impeded.

(4) Use of the Model Provides One Basis from which some Aspects of instruction may be observed, discussed and experimented with

The notion of tactics and strategies provides one useful organizer for observing, classifying and experimenting with teacher-learner-behaviour.

(5) Implicit in the Model is the Goal-directed Behaviour of the Teacher; the Control Points at which Goals affect the Behaviour of those involved in Instruction are Suggested

The .model indicates the specific points at which goals affect the instructional process as the bases for strategy decisions, as planned tactics are decided upon as responsive tactics come into the situation and as the tactical element loop proceeds from moment to moment during instruction.

(6) Evaluation in Instruction is not something Unique to the Instructional Situation: Rather, it is a part of the Tactical Element Loop

Viewed from the perspective of the tactical element loop, instructional evaluation proceeds while the students learn, during instruction rather than apart from it (although it

is recognized that for some purposes some evaluation takes place apart from instruction).

(7) One Route to Increasing Teacher Effectiveness through a Purposeful Experimental Approach to Instruction

Implied is a relationship between a teacher's tactic repertory (both planned and responsive) and the variety of instructional goals accessible through his behaviour. Thus, one route to extending one's tactic repertory may be through reflecting on his spontaneous behaviour as well as purposeful experimenting to discover the real effect of a certain hypothesised tactic upon the students.

(8) The Model. Provides a Unifying element through which some of the Studies about Various Aspects of Instruction may be related

The model may serve to place goals, curriculum, pedagogy, teacher,1 student, materials, etc. in perspective so that increased understanding of what such factors are and how or where they relate to the total picture of instruction may be further hypothesised

In. summary, perhaps the essential implication of the model about teaching is that teaching is enquiring and invites a teacher's thinking and behaving in terms of certain goals or enduring purposes. In another sense, teaching is viewed as learning about the learners; learning about the effect of one's behaviour on the learners, based on a continuing stream of feedback from and about the learners and learning as the teacher extends the goals, which direct his activity.

2.4. Need for instruction

The teachers in general rely on their knowledge of the subject matter of their field of specialization alone, in order to teach and thereby try to achieve a high level of competence to become effective teachers. The concentration is more on dispensing of knowledge to the students rather than in finding out if the students have comprehended what the teacher has tried to pass on to them. Invariably, teachers are not concerned about their abilities to communicate effectively, or the pedagogic approaches that they should adapt. There are a number of teachers, who do not even feel the need to improve these. To them teaching centres around one basic issue the subject matter as prescribed in the syllabus. All efforts are made to cover the syllabus rather than uncover it for the students to explore, analyse and critically study in order to assimilate the knowledge and make it their own. Further, the teachers do not feel the need to keep abreast of the latest developments in the field. They do not look upon teaching as a complex process requiring competence in several fields such as ability to understand the students, the different ways by which they learn the skills of teaching, the evaluation measures to be adopted, etc. As such, the concept of professional preparation of teachers, which is an accepted fact at the school level, has not taken roots at the tertiary level.

In what way is the teacher equipped to do his job satisfactorily? He or she has a

post-graduate degree, perhaps with an additional degree, M. Phil, or Ph.D. indicating mastery of the subject matter and/or research experience. If fortunate, he would have had an effective teacher himself and model his teaching accordingly. Limitation appears to be the only method adopted. If the master was an expositor of knowledge, he too is an exponent. On the other hand, if the master has repeated verbatim form the textbook or any other prepared material, the teacher, too, considers that as the best method to follow. The teacher is not aware of the repertoire of method and approaches available, which can be used, judiciously either alone or in combination. How would he know about these unless he is directed and guided in this aspect?

Having knowledge of the methods alone is not sufficient to become a good teacher. Under the guidance of a trained person he should have tried out these methods, remembering to match the methods with the subject taught and the needs of the students. Moreover, methods are likely to differ from subject to subject and from one level of achievement to another. A theoretical knowledge of the methods precedes the application of these methods.

The second aspect of the requirement would be the psychological basis to understand students. Students should be motivated to learn. Unless they are ready to learn, what is being taught is not absorbed by receptive minds? When the inclination to learn is not kindled, an elaborate lecture hardly leaves any effect on the assembled students, A lecturer should be aware of the different ways to prepare and encourage the students to learn. Attitude to learning is more important than learning itself.. Reviewing of the material studied in the previous class, listing the specific objective of a topic to be learnt giving an outline of the material to be dealt with during a period of time, indicating the significant aspects of the topic and introducing the subject especially through any of the audio-visual aids would perhaps create a positive attitude to learning.

Knowledge of certain principles drawn from learning theories will be potentially useful to a new teacher. 'Learning by doing' is still considered as an acceptable slogan. This does not imply that students do not learn anything, when they are passive. They are likely to learn more if, they are actively involved in the learning process. Further, learning-takes place when there is opportunity for frequent practice, repetition in varied contexts and when appropriate learning materials that would encourage generalisation and also the ability to discriminate and critically examine the material studied are provided. Reinforcement when used properly is an important motivator. Reward from a teacher such as nodding, indicating acceptance of a correct response, can be extrinsic or can also be intrinsic reward (self-reward) as in the case of finding satisfaction in learning the material taught. However, there are negative, reinforces such as punishments and failures but generally it is considered that positive controls (rewards and successes) are to be preferred to negative.

2.5. Selection and Implementing Strategies of Instruction

Selecting instructional strategies is one of the final steps in planning for instruction. Instructional strategies are derived “from a number of sources, including the objectives, the subject matter, the pupil, the community, and the teacher.

Teachers vary in their styles, models, and skills. By style we mean the unique, personal qualities that a teacher develops over the years to distinguish himself or herself from all other teachers.

When we speak of models of teaching, we mean a generalized role a pattern of methods - such as discussion leader, television instructor, or foreign language informant. The so-called Socratic method of stimulating thinking is a model. Muhammad (peace be upon him), for example, used both a model (preacher) and a method (sermonizing).

Skills of teaching are those generic and specific competencies necessary to design and carry out instruction. Lesson planning, for example, is a generic skill; that is, it is pertinent to all teachers at all levels. The ability to teach pupils to perform the division of whole numbers is an example of a specific skill. Both the, models and skills must be compatible with the teacher's style. Instructional strategies must be appropriate to the teacher's style, model, and skill.

Instructional strategies, styles of teaching, and teaching skills are all selected, adopted, and implemented to successfully fulfill instructional goals and objectives.

The ultimate purpose of all strategies, styles, models, and skills is the fostering of student achievement.

The various instructional components should be organized into, among other types of plans, short-term units and daily lesson plans. Although teachers may design their own formats for unit and lesson plans, generic outlines are suggested in next sections. As teachers gain experience, less detail in planning is possible. However, some planning is always necessary. The reader is referred to selections from the new large body of research on effective presentation of instruction.

3. INDIVIDUALIZED INSTRUCTION

Education and growth, both are two individual affairs, and can be well shaped only in healthy and free atmosphere. Arty form of pressure or stress stops a child's natural growth and learning. So we recognize the child, and child-centred education is .responsible to give rise to the thinking of individual freedom.

In other words that is "to consider one's level and desire". Thus individual method provides for an individual child, facilities of instructions, keeping in view his needs, intelligence, and motivation, rate of learning, capacity and aim in life.

In short we can say that it would bring confidence, self-respect and self-reliance in children. In modern era, a child-centred education system is thought as the best. Therefore, it is needed to prefer the individual brought-up to make teaching and-training meaningful.

3.1. Merits of Individualized Teaching

We can list its benefits as below:

- This kind of teaching is imparted in accordance with individual conflicts of a child so it would satisfy him individually. Though an individual is a member of a society but one possesses one's own personality or individuality. For instance, two students cannot hold same position at one time; there will always be a difference even in points. So, a good teacher must keep in practice, to develop personalities at individual level.
- With the help of individualized teaching, a particular subject can be well acknowledged and, therefore, can be given due, complete/full attention to each and every aspect of that particular subject.
- Individualized teaching method builds the character and brings insight. It develops qualities of self-confidence, self-respect and self reliance in individuals.
- This kind of instruction assists the students to develop the habit of self-study that takes them towards their natural aptitude. It becomes a natural way for them to find a particular aim.
- It evokes one's curiosity and creativity. Therefore, it satisfies natural desires or instincts. According to Froebel, "the ideas for things) are hidden inside the mind and heart of the child, which come into light" so individual attention is a. source to probe hidden potential and. qualities of a child, to enlighten and polish them. In return, a child (Karner) feels satisfied and solaced.
- It trains students for practical life because it lays emphasis on objectivity or practicability. His well-built confidence helps him greatly to face the hardships and to solve problems.
- It greatly affects the literature, geography, mathematics and writing ability of a learner, so it is best to teach children by applying this method. Whereas, music,

art and physical education can be well taught in groups especially in primary classes. The individualized teaching is very effective and brings good results in the long run.

- If we do not teach our learners how to read, write, do or pronounce individually, it will not satisfy them and, therefore, the next educational process will be difficult for them.
- It is fruitful especially in case of weak students. It provides opportunity to express their difficulty without any hesitation whereas, in a group they cannot express well. Any absentee cannot recover missing lesson but in case of this method he may recover before new lesson is taught.
- In this method it is convenient for teacher' to identify weaknesses and problems of learners and solve them accordingly. But it is not possible in a group to pay special attention to each and every individual.
- In individualized teaching, one is not stuffed or forced rather one learns according to his capacity. In this way, a learner is also responsible for his own progress or position.
- It connects teacher and learner closer. Close relationship enables teacher to guide and train his learner more appropriately.
- Teacher plays vital role in individualized teaching. At a time he performs as a guide as well as a friend. So his accountability increases to a great extent.

3.2. Principles

If we follow these principles as guidelines this method would be more fruitful and effective.

(i) Individualized instruction should be in accordance with child's aptitude, interest, demand and learning rate.

(ii) Educational purposes must be integrated with ethics and morals to make this method meaningful and helpful for building stronger foundations of one's own standards.

(iii) It must be adopted as a stimulant or motive so that educational motivation can be propagated into our learners.

(iv) A learner should get opportunities to flourish in stress-free environment. Freedom helps to mould a child's capabilities in the framework of nature.

(v) It is crucial to teach fundamental skills to our learners like reading, writing, and speaking. Once he has learnt, it would facilitate him towards study, exercise, assignments, etc.

(vi) A child should be provided with opportunities to experiment with what he learns. It would inculcate the habit of self-study along with

developing a sense of responsibility.

(vii) Teachers should guide and assist only when needed. They should eliminate the obstacles and let learner work independently so that they may grow and shape their natural abilities well.

(viii) A teacher should blend individual teaching with group teaching occasionally to add a new flavour, in groups students compete with each other, learn from others mistakes and accelerate their learning pace.

(ix) Co-related activities enable a child to be more confident. It is also a mean for teacher to assess one's aptitude, which is expressed in child's activity.

(x) The level and of assignments should be rational so that learners may be able do the job well and get positive feedback for their achievements as token of encouragement.

3.3. Procedure:

While individualized instruction is to be practiced, the following two procedures are suggested to be kept in mind.

i. Assignment

ii. Study.

Here we will briefly discuss each procedure that will further be adopted to make our instructional method more comprehensive and purposeful.

3.3.1. Assignment:

We may define an assignment, as "An assignment is a job, a piece of work or a task accepted by an individual pupil or the class" Philosophy behind assignment method is the utilization of the merits of lecture, demonstration and laboratory work.

Sound planning of Assignments requires:

- Time, work and depth of subject matter should be prescribed
- Learning process, objectives, methods and techniques
- Age, intelligence, interest, abilities of the pupils
- Reference material and laboratory facilities
- Time allotted should clearly be explained
- Questions/task given must be relevant and clear

Type of Assignment

There are two types of assignments:

- Class Assignment
- Home Assignment

Class Assignment

Students are asked to answer the questions given, to them or they have, to perform some experiment, the teacher corrects the answer and maintains a record of progress on regular basis in his diary. Pupils are supplied with a set of instructions to be followed.

Home Assignment

The teacher gives a topic with list of references. The assignments are completed and returned to teacher for evaluation the teacher checks these assignments and if he finds some of them not up to the mark, he returns them for further clarification, reading and resubmission.

Assignment, may be done/assigned in the form of:

- Exercise
- Research
- Skill
- Experiment
- Projects
- Problems Solving

Now we will look at Merits of Assignment

- Related to subject under study
- Enhance understanding
- Brings satisfaction and confidence too
- Purposeful activity
- It creates curiosity, thinking and challenging
- It encourages habits of extra study and self help
- It maintains progress records of evaluations

- It is economical, as shortage of apparatus is not felt
- It is a training in scientific method

Let us look at its Demerits

- It needs more work on. the part of art already overloaded teacher
- It may suffer due to lack of reference and textbooks, etc.
- It requires up-to-date laboratory and library
- It may prove to be a bottleneck for syllabus.
- It allows weaker students to copy from others.

3.3.2. Study

Second important component of individualized instruction, is study. It includes observation, exercise, experiment and reading the books and magazines/ l! is conditioned with one's conscious willingness, interest and motivation. Effective study not only requires ability, interest and mental health but also better and successful techniques. For effective teaching, teachers should supervise thoroughly. Only those teachers can play their role appropriately, who themselves are fond of studies.

Educationists have suggested three types of methods,

(1) Supervised Study

(2) Non-Supervised study

(3) Liberal/Free study

Supervised Study

This type requires constant supervision of a teacher or an individual.

Usually it is done/practiced in classroom, hall or library, where a great number of students study under a teacher's guidance, It is also named Study-hall Method, It is common in the West, not in Eastern countries. Here the golden rule of silence is practiced. This method develops study habits and manners among the pupils. They learn to study in groups as well as individually. It develops concentration.

Non-Supervised Study

It is the opposite of supervised study method. No textbook study is included and teachers only guide the pupils where they need. Here a pupil is directed towards a subject or topic. Learners take benefit from resource centres and practical books too,

This study greatly benefits intelligent and hard-working types of students and moves them to a vast field of knowledge.

It requires great knowledge of teacher so he may guide his pupils On a number of various topics.

Free Study

It is third category of study method it is usually carried out of school boundary and timings. It aims at individual efforts, interest, responsibility and persona attention. Few experts insist that it is just wastage of time of our learners, as it disturbs home activities. Therefore, it should be so planned that they may be able to do it after completing their school or home assignments. However, evening time is considered the best for its application. This study depends on teacher, curriculum and parents: so it would require participation of parents along with teachers, so pupils may get necessary Hews for study.

3.4. Instructional Methods

3.4.1- Kindergarten Method

In German language, kindergarten stands for garden for children. Basic Philosophy of this method is that children grow like plants or flowers in a garden.

Froebel along with Pistalozzi used this method. Froebel considered a school as a garden, and a teacher as a gardener, who has to look after the little human kids as plants and waters them to grow their beauty and perfection.

- Froebel believed that life is always evolving and developing. This dynamic growth is a sign of eternal unity.
- Self-activity is innate. A child's intellectual, emotional and physical growth should be developed through play/sports /games.
- He describes the connecting force as:

  It is different from all other forms, yet it contains all of them, it is formless yet it contains all of them, and it is formless yet the most perfect form.
- The plan works on freedom, play joy. There are no books, no formal storing of knowledge. Exercise is the key to this method. Learning takes place through activity.

  So, there are three ways in which it works.

  1. Songs
  2. Movements
  3. Construction

All three go as a unit. Play is pivot. Songs provide experiences. Creative activities provide expression to children. Children are trained intellectually, physically and emotionally through concrete means of activities. -

A teacher selects songs, plays and games. He shows pictures and arranges activities like construction.

Songs

Every song has three parts:

- A motto for mother or teacher
- A verse together with music.
- A picture showing the illustration of song

- A child is able to use his limbs, muscles and senses;
- Each song provides an opportunity of some nursery game;
- Songs aim at meeting the child's needs of a different kinds; and
- An activity to exercise; so it helps in education.

Gifts and Activities

Gifts are graded and provide some activity. Occupations are activities suggested by gifts. We can thoroughly understand that how it moves forward a child's step from, one to another stage.

First Gift

There are six different balls in a box. The child has to play with them by rolling them. It gives him the idea of colour, material, motion and direction.

Second Gift

A sphere, a cube and a cylinder are in a box. During play a child notices that cube is static, sphere is mobile and cylinder possesses both properties. It harmonises the qualities of both a cube and a sphere.

T**hird Gift**

It comprises a building box, with a number of construction materials like bridge, steps, and benches.

Fourth Gift

A big cube divided into oblong prisms in each of which the length is twice the breadth, and breadth is twice the thickness. A child can make various kinds of building designs when combined with the third gift.

Fifth Gift

This gift is a big cube, divided into twenty even cubes, three of which are again divided diagonally into twos and threes into fours: By combing third, fourth and fifth gift, many beautiful shapes and designs of pattern can be made.

Sixth Gift

It is very important in teaching a child numbers. It is helpful to learn, making more new and different patterns. It is a big cube divided into eighteen whole and nine small oblong blocks.

Seventh Gift

It comprises a set of square and triangular tablets of a very fine wood made up of two colours. It is in geometrical form and mosaic

Firstly, a demonstration of gift must be presented. Teachers must carefully select the play most suitable for child's learning.

Practice should be given in plays and activities.

Second Stage of Scheme

At second stage manual work is expected. Manual work is a necessary condition for the realization-of a pupil's personality.

Other subjects

According to Froebel the main aim of education is an all-round development of a child's personality. He has divided curriculum as below:

- Religion and religious instructions
- Natural sciences and mathematics
- Language
- Art and objects of art

Children would develop a sense of observation, analysis and. later on of appreciation.

3.4.2. Montessori Method

Doctor Maria Montessori was a scientist, an anthropologist, educationist and a physician. She desired to teach children through real experiences by giving them concrete material and maintaining learning situation for them.

Now we will look at her philosophy and principles, which she framed for her method of teaching the children.

- Full freedom
- Education for growth
- Education for self
- Individuality as a psychological process
- Sense training (sensations of colour, sound, weight, touch and temperature etc.
- Motor efficiency (muscular activities)

Procedure and plan of method

1. An atmosphere of complete freedom is to be provided to the students.
2. Supply of suitable furniture and equipments to students.
3. The didactic sensory apparatus is provided for various types of exercises, intellectual sensory and motor.

Children's House

It is a multi-roomed school with a well laid out garden. Rooms are for lunch, normal work, laboratory, rest and bath. Rooms are equipped in accordance with the spirit and needs of this method.

Now we will discuss the activities of Montessori Method

- Exercise in practical life
- Exercise in sensory training
- Didactic exercises for language and arithmetic

Let these points be discussed briefly.

Exercise in Practical Life

In a children's house real activities of everyday life are carried out, in which all household is entrusted to little children. They get full training here for their future lives. They do activities like sweeping, cleaning the floors, dusting cupboards and shelves, learning how to dress-up accordingly, blowing their noses, coughing, walking, how to take their meals, offering prayers, taking gymnastic exercises, doing clay modeling, making tiles and bricks, etc.

Exercise in Sensory Training.

The "Didactic Apparatus" is very finely graded. It provides exercises in sensory training. Various types of apparatus are provided for perception of size, form, weight, touch, hearing and colour etc.

Didactic Exercise For Language and Arithmetic

Children are taught third reading, writing and arithmetic. Writing comes before reading in this method. The movements, which a child makes, are more important than what is written.

Principles for Writing

- Recognition of forms of letter
- Learning of phonetic songs
- Control of pen

Reading Rules

- Interpretation of an idea from written signs.
- Word/phrases are written on a card in bold letters
- Ask to name familiar objects
- Ask to translate the writing into sound
- Himself selects a card/slip and reads them

Rules for Teaching Arithmetic

- To teach numbers a long stair used
- A set of ten rods with varying length is used
- Ask to arrange rods in length

- Ask to arrange weight/diameter-wise
- Ask to arrange blue ones and red ones separately

Now we will list the advantages of System

- It lays emphasis on self-discipline
- Teacher plays an important role in it
- Method of teaching 3Rs is different in it
- Learning by doing is the basis here
- Complete-freedom is given in learning
- Sensory training helps in a child's training of education
- Individual attention is paid in the plan
- Respect the child and his education
- Scientific method in nature

Its Limitation may be listed as following:

- It places limited emphasis on play
- Self-expression does not find sufficient scope
- Sensory training is not always transferable
- Senses can be trained in isolation
- Social training is very limited
- It lacks moral and emotional development
- Too much emphasis on self-education
- Economically this system is expressive
- It requires suitably trained teachers

Thus, keeping in mind these facts and figures, we can say that it is costly and in Pakistan we are not in a position to afford it, but, on the other hand, we do accept that it plays a vital role in basic training and upbringing of children.

3.4.3. Dalton plan

Firstly, we will look at its historical background.

The plan was experimented by Miss Helen Parkhurst at Daltori, Massachusetts. She, being acquainted with Montessori's method, discovered that children might be

taught better if some scheme based on individual differences was introduced. Dalton's plan was, thus, the fruit of educational experiment in teaching methods in the Dalton High School by Miss Helen in 1920.

Dalton's plan is not a particular method of teaching any particular subject rather a method of organizing the work of a school to promote purposive self-activity.

It aims at solving the problems of children even when they are classified by ability. In the words of J.H. Panton, "among other things, the plan is an attempt and a highly successful one under certain circumstances, to provide an education in self-development and social cooperation of pupils". In this plan, independence of the pupils to work at their own rate and the consultative capacity of the teacher to guide them has been stressed. The traditional curriculum is preserved but timetable is discarded. Let us look at the underlying Principles of Plan.

1. Principle of individual work
2. Principle of freedom
3. Principle of self-effort
4. Principle of group interaction
5. The psychology of viewpoint.

Features of Dalton's Plan

Assignments or Contracts and Checkups:

In Dalton's Plan, work of each grade for each subject is laid-out in a series of related jobs and contracts. Teachers outline monthly contracts or weekly assignments for the whole year supplies, guide sheets and guidance for work to be done.

Assignments may be self-corrected, peer review, corrected by other teachers or are handled in pupil-teacher conference.

Subject Teachers

Instead of class teachers, there are subject teachers as specialist. Teachers completely handle their subjects and confine themselves to five duties:

(1) To preserve an atmosphere of study in the classroom

(2) To explain any detail of the assignment

(3) To give information for the use of departmental equipment

(4) To give suggestions with regard to the methods of attacking particular problem

(5) To explain a point and its relation to the general principles of the subjects whenever need arises

Subject Laboratories

Instead of formal old-fashioned classrooms, there are subject laboratories each one equipped for a particular study subject to enable a child to have access to vast literature on a subject. Under Dalton scheme, the pupil, not the teacher, asks most of the questions.

Records

Pupils' progress in each subject is recorded by means of graphs and diagrams. Pupil maintains two records himself and one by teacher in the subject room. This graph links the teacher and pupil to bring them closer for effective teaching and learning.

Conferences

Under this plan, in addition to assignments and contracts, some provision has been made for exchange of ideas in the form of conferences. These are held by teachers and last for 15 to 45 minutes. Teachers and class hold these from 45 minutes duration. Class conferences are held to discuss difficulties or overview purpose.

Suitability for Middle and High Classes

This system demands command of the subject before pupils work under contract. It cannot, therefore, be applied in primary classes.

Merits of Dalton Plan

There are many more advantages of this system, which are listed as below:

- Individualized instruction
- Continuity of work is assured
- It fosters initiatives, self-reliance and self-discipline.
- Graph records are useful
- Problems of formal class discipline much simplified.
- Better and quicker learning is possible
- Compulsory homework does not remain a necessity
- Good for low and high ability pupil

The plan has following limitations as well:

- Smaller classes and generous staffing are required to make the plan

successful

- Preparation of graded assignments is also quite difficult;
- Oral work is not given adequate attention;
- Teachers must be excellent workers;
- Expensive interns of bearing specialist teachers and up-to-date quipped laboratory.

3.4.4. Programmed Instruction

Programmed instruction emerges out of experimental researches on operant conditioning. It incorporates the principles of operant conditioning in human learning.

Chronologically, the Greek philosopher, Socrates is said to be the first programmer, who developed a programme in geometry.

In this method the material to be learned is arranged in graded units, according to levels of difficulty.

It is presented in such an order that it results in the best understanding and retention.

It is based on the, "principle of reinforcement". The really big milestone in development of programmed instruction came in mid fifties with the work of Prof. B.F. Skinner of Harvard University. The operations involved in programmed learning instructions are as follows:

(1) The subject matter is logically sequenced and broken into small steps.

(2) A small bit of meaningful segment of information (frame) is presented to learner.

(3) The learner reads the frame and is required to make a response to the information in writing.

(4) By immediate feedback system he is informed about the correctness of his response.

(5) And the learner is presented with a second frame and the stimulus response, i.e. reinforcement cycle, is repeated until the series of frames presents a complete programme.

Forms of Programme

Text-Books
Cards
Teaching Machines

Classification of Programmes

Mostly there are two varieties of programming:

- Linear
- Branching

Linear programming

It consists of strict, unchanging and carefully thought out series of questions and answers in each unit or frame. It may be presented in the following way:



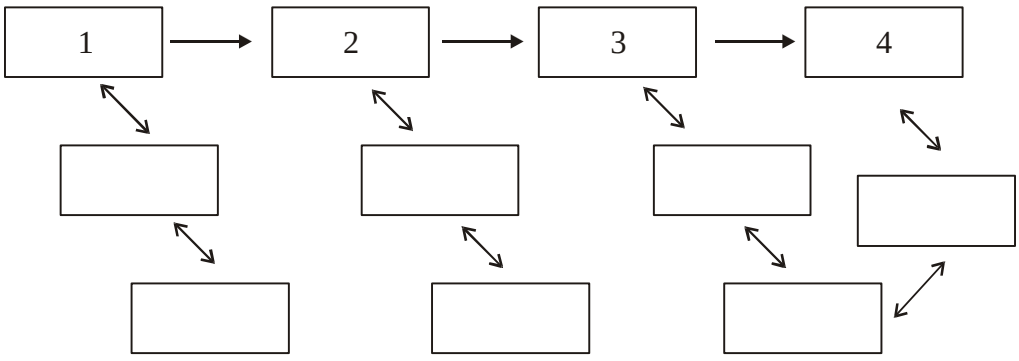
In this method of programming, a pupil has to master step one before proceeding on to step two and so on. Teachers beforehand arrange these steps and pupils find it easier to learn them as compared to in other methods.

Branched Programming

S.L. Pressey and N.A. Crowder have given this process, which consists of many branches and offshoots.

The information sheet at every step guides a student. Sometimes the learner has to retrace an earlier step to ensure that he is proceeding on right lines. This technique may be presented as below:

Branched Programming



N.A. Crowder employed the technique of intrinsic programming, with multiple-choice items. In this method every mistake made by a student is further utilized to build up new' knowledge and skill so mistake opens a new avenue for learning.

Now we will count merits of programmed instruction.

- The use of programmed learning material helps in the improvement of the quality of teaching.
- It may be utilized as a supplementary process along with regular textbooks.
- A teacher is helped by application of this technique.
- Apart from printed material in the form of books and cards, used under this technique, there is a device named teaching machine.
- It is different from an ordinary audio-visual device in form of a workbook, card, film, and TV etc.
- In teaching machines, answers are fed for the* benefit of learners.

Thus, it is a self-study method.

Functions of Teaching Machine

- It is used for individual teaching.
- It contains and delivers programmed contents in steps.
- It works as psychological reinforcement.
- The whole process is foolproof.
- Learning takes place in quicker way.
- It enables a learner to judge his own work.

3.4.5. Computer

In the age of "information" and computer literacy, computers have found their way into classrooms faster than most of us thought would be possible.

Computers and software are referred to as CAI (Computer Assisted Instruction) when students are directly dealt with them. Computers function as an aid to instruction.

The most common uses of CAI are: drill, practice, tutorial, simulation games and problem solving.

Merits and Uses of Computer

- The computer can store and retrieve information, conduct statistical analysis of data, making graphs, tables, and display them effectively.
- It assists in composing, editing, and formatting compositions, and communicating with other informational networks.

- CAI can provide individualized, self-spaced instruction with easy access to enriched exercises.
- CAI can raise student's performance in both cognitive and affective domains. As a motivational device, CAI can enhance interest levels and general attitude towards school.
- CAI can provide immediate feedback.
- CAI can be used to develop' inquiry skills, problem solving skills, critical thinking and decision-making skills.
- Students can write their own programmes.
- CAI does not have human weaknesses, which may interfere with excellent teaching techniques. For example, a teacher will not take a bad day out on students; a computer is not impatient and it cannot get tired.
- Content material is easily conveyed and its learning is easily evaluated.
- CAI can be used for interviews and practice, tutorial work, exploration of a problem, simulating and gaming for providing entertainment from time to time.

Limitation

Although it has a great number of merits yet it must be kept in mind that machine does not possess any emotions or feelings, therefore, it has to be supervised to determine as to what extent the learner can benefit from it.

4. GROUP INSTRUCTION

Group instruction is based on the principle that no student is particularly different from any group or class of students. All students are equally capable and efficient. It is a balanced instruction for individuals and groups.

Thus, we may define group instruction as, "this instruction is imparted collectively to a group of students at the same time and place" so we can say that it is not specified/restricted to a single student rather it focuses on all students equally.

In modern and democratic system of education it is defined as, "this is a process whereby a group endeavours to solve a problem through mutual discussion, thinking, planning, determination and analysis". Through co-operation and creative thinking, the members of such group achieve their goal.

Latest research findings have proved that individual attention is important and effective in teaching-learning process. Such larger groups are divided into sub-groups of 5-6, keeping in view the equality of capability, efficiency and aptitudes of students.

Every group learns and makes progress in uniform manner. This type of instruction yields both individual and collective benefits. It promotes intimacy and affection among the students. Kurts Lewin made first attempt towards group instruction in Lowa University on experimental basis. This system of education is prevalent in Pakistan.

4.1. Benefits

We may list its benefits as follows:

- Group instruction promotes social wisdom in a very natural way among students. Shy students gradually come up and develop a spirit of competition.
- Under this system, the students' strengthen undertakes the learning process through exchange of views. It promotes debating and discussion.
- Weaker/less efficient students can make up through competition and with the help of fellow students.
- It promotes social development and consciousness among the students for greater cooperation, sympathy, sacrifice, affection and loving behaviour.
- The students learn to perform social work. It builds a sense of responsibility in individuals.
- This system saves time and money. In other words within limited/lesser resources instruction is imparted to a great number of students.
- Certain disciplines like physical education, music, art, poems and stories can be well taught in groups.

- Students get opportunity to become social and develop leadership skills, too.
- In-group appreciation of students promotes their dedication, interest and devotion to work more hard.
- It promotes and develops group dynamics.
- Shortage of educational accessories can be overcome through this method. Our educational system is based on it.

4.2. Principles of Group Instruction

The success .of group instruction is conditioned with proper organization based on the following principles:

Equality

While organizing a group, the competence and intelligence of students, must be well considered. They should be almost equally capable and efficient in each group.

Classification and Blending

Equality must be observed in classification of group. Every group should bea well blend of intelligent, mediocre and weak students.

Participation

In running teaching-learning process all group members should participate actively. All affairs should be run with cooperation of all students. '

Evaluation

Group work must be evaluated against the set standards with both strategic i.e.. formative and summative evaluation. It would help a lot towards keepings them on track.

Period of Existence

Strength of students in a group should be fixed according to the subject and nature of assignment. However, freedom of choice should be maintained.

Group Size

Size of a group should be restricted to 5-6 members otherwise it would become

very difficult to pay individual attention.

Emphasis

As the method itself is competition oriented, too much emphasis should not be laid on competition.

Assignments

Different assignments should be assigned to each group, as variant activities and tasks promote a sense of collective participation and social harmony among the students.

Capability

Task assigned must be in accordance with the capabilities, need and choice of students. Discussions and debates should be encouraged. Tasks should include problems of day-to-day life.

Supervision

The responsibility of teacher does not end after assigning tasks to the students. He must be available for supervision, help and guidance without too much interference.

Appreciation

The teacher should perform his role well and encourage and appreciate his students for their performance and achievements.

4.3. Instructional Methods

4.3.1. Problem solving method

We all know that education is through the life and for the life. In our life, we face many good and bad times. There are problems in various situations. We learn solving them, without which life is incomplete. We train and educate our learners how to deal with situations, how to tackle and settle things in context to school learning.

We may define problem solving as; "A manner of dealing with that, which is problematic. A method involving clear definition of problems confronted formation of hypothetical solutions, hunch or suggestion, deliberate test of hypothesis until evidence warrants its acceptance".

It may be defined as a planned attack upon a difficulty or perplexity for the purpose of finding a satisfactory solution. So, it is a method in which the felt hurdle to act

in an educational situation is realized and an attempt is made in a conscious, planned and purposeful way to find its solution.

Location

Students should be directed to locate some problem and their interest in its solution should be aroused. Questions and discussions are a very important tool at this stage. They should take a problem as challenge and find a solution.

Explanation

Either a teacher has to explain a problem or else pupils know of it after a discussion. Student's age level and their understanding are important in determining a problem.

Collection of data

After discussion, data is to be procured. Sources are to be given by a teacher. Pupils getting data, start evaluation.

Evaluation

Only pertinent data should be included in evaluation, which should be based on careful analysis.

Tentative Solutions

All inferences are tentative in the beginning. Here a teacher's guidance is very important and his patience is also tested. He should suggest inferences and solutions in the right way without discouraging the pupils.

Verification of results

After the best conclusion has been drawn, it should be proved to be right, for acceptance. This can be achieved as below:

- By using hypothesis in new contexts;
- By further experimentation; and
- By collecting new information data through study and investigation.

Methods

There are two methods of problem solving:

1. Inductive
2. Deductive

Inductive

In this method generalizations are drawn from particulars. Principles are framed from observations and rules are made from instances or examples.

The pupils define their problem.

After analysis, relevant information is found out. Books help after study, pursuing references and visits to places etc. are some of the ways.

After collecting information, the pupils put it in order, under the guidance of their teacher.

Pupils find out the probable solutions, while the teacher remains in background. Here we may say that tentative solutions and its alternatives are developed and framed by pupils independently.

Learning the most probable solution, others are excluded.

Solutions are applied to the situation and results are checked Repetition of this step is carried out until correct solution is found out.

Deductive

In this process, rules, principles and conclusions are applied to particular cases. Following steps are preceded:

Pupils follow it. They define the problem and formulate it.

They gather information for study later.

Principles and generalizations are reviewed to find as to which may be applicable to find a suitable solution.

Generalizations, principles or rules are applied to a case and inference is drawn that a problem falls under a particular principle.

The principle is applied to a case, if it solves the problem; otherwise the method is repeated to find the correct one.

Problem solving method do possess benefits that are listed as below:

- Pupils learn facts themselves. This is real education/ learning.
- Discovery takes place. Acting in new situation is learnt. .
- Confidence to face the problem is developed.

- Defining the problem and verification of result is learnt.
- Curiosities of pupils are satisfied and devices are found out.
- Ideas are born and originality is developed.

4.3.2. Project Method

We may define it as, "A problematic act carried out to its completion in its natural setting".

This method builds a unit around an activity that is carried out in education setting, might be formal or informal.

Its main aim is to accomplish a useful task in-group setting, whereby all Work is done in a cooperative manner. In this method students are given freedom to select their projects after careful thinking. Two important laws of a project are:

- Learning by Doing
- Learning by Living

New we will look at the steps of a project..

Steps in a Project

Provision of a situation

A situation is identified or arranged in which some problems are available with, interesting ingredient for the pupils. .

Selection of Objectives

A teacher guides and facilitates his pupils in this phase too, to select and find out aims and objectives of the project being selected, keeping in mind real needs of pupils.

Planning

This is a very important step, so one must be careful in drawing a blueprint for a project. Planning should include a few alternatives. A discussion to exchange group ideas and views is very crucial. Suggestions must be critically examined and utilized.

Execution

Among group members are assigned duties in a way that all get activities of their choice and capability: No doubt, it heeds patience of contributors to make it a success. In one project many activities can be developed. They all help in knowledge growth.

Evaluation

Work done on project must be evaluated by themselves (student group) and supervisor (teacher) as well, to locate the follies. Self-criticism is worth training giving. In this step objectives (pre-set) of that project are the standard and performance is

measured against this standard.

Recording

Group/individuals must maintain a full record of various steps

Planning, discussions, assignment of duties, criticisms and such like other points are noted for future reference and guidance.

Qualities of a good Project

To make this method a fruitful and purposeful one it must possess following qualities:

- Clear and solid aim.
- Practicable.
- Interesting and relevant to pupils' need.
- Knowledge based to bring some development.
- Encouraging ingredient and have pupils' level.
- Clear and well defined scope.
- Economical in terms of money and time.
- Time oriented neither too short nor too long.
- Challenging.
- Attainable. (Availability of material information, tools, etc. should be kept in mind before starting work on a project)

A Project Aims to Bring a Child out of Bewilderment

After looking at qualities, now we will learn about merits and demerits of this method. Like other methods, it possesses benefits and drawbacks. So keeping in mind these aspects we can use this method in the best way.

Merits

This method is dependent upon the following laws of learning:

Demerits

- Time consuming.
- Teacher is overloaded with work.
- Teacher does forced correlation, which is much harmful.

- Syllabus of advance classes cannot be covered with this method, as it requires greater time.
- Teacher is expected to have full knowledge of all subjects, which is ideal but not practicable.
- Students gain outward ideas of subjects only.
- Text and reference materials are found with scarcity.
- Expensive, as well as fully equipped laboratory and library is required.
- Practice and drill for skill are not provided in project.
- Subject development is not in order.
- While working on one project pupils do not pay due time and attention to other subjects.

5. CLASS INSTRUCTION

Class instruction has always been important in every period of time. We look back into history, education was the sole right of nobles and elites of Europe and few people could receive education. When education was opened to general public, it created lots of problems for school. In our sub- continent, a Christian Priest stayed in schools for a short time. He observed that students were arranged in classes for teaching-learning session.

Now we will look at the system[4], what does it mean and how it works?

In this system, the class is a unit of almost equally proficient students. They can benefit with a proper pace of time. Europeans adopted an inelastic approach within this system that yielded better results. Keeping in view the individual differences, the education is imparted on the basis of .identical qualities of students. Learning is a social process and it preserves social values. It is the focal point of this system. In this system students are grouped in class (one) and taught in a way that they can benefit from education and* also learn to respect the values. As such, a greater number of students are imparted education with common interest and qualities. Despite individual differences various aspects of personality are groomed.

Under class instruction task is a unit and one teacher teaches one subject according to curriculum. Every teacher, therefore, fulfils the requirements of education. This further fulfils and promotes students' interest in education process. Along with education, emotional aspects like sympathy, brotherhood, affection, cooperation, hardworking and competition are developed.

It leads to character building. Today, class instruction is inevitable in every respect. According to Pope Pious, "Education is a social process." And according to Michael Lee, "In fact, class is a model Social environment with a fundamental aim to make conscious attempt and effort for pupils learning.

5.1. Principles of Class Instruction

Each teaching method is based on a few principles, which make it a more fruitful activity. Here we would look at principles of class instruction.

Organization

A class organization should be done on the basis of common characteristics. In other words, we can say that pupils of similar aptitude and proficiencies should be taken.

Class Size

Reasonable number of students suits this method so that the teacher may be able to pay attention to the whole class properly.

Uniformity

Effective teaching requires a uniform social background; therefore, all students should be given equal and justified attention.

Appreciation

Due appraisal must be given to deserve students. It would promote a spirit of healthy competition.

Individual Differences

Although in this method the teacher is concerned with whole class but one must keep in mind that individual differences must be kept in view to have-good results and satisfaction of the students.

Promotion of Reaction

Since me students are at the learning stage, try to develop emotional reactions and avoid negative ones.

Additional Ingredient

A teacher should take interest in students' hobbies and activities in addition to curricular guidance.

Discipline

It affects a lot on teaching-learning process.

Style

The teacher should communicate to his class in a soft but firm tone and avoid shouting.

Punishments

Extreme punishments should be avoided. Positive reinforcement should be preferred to maintain discipline.

Expectation

Rate of learning may be higher or lower against teachers' expectations. Let the students learn; do not force too much considering that their potentials and capabilities are to grow steadily.

Privacy

The teacher must withhold all personal talks and matters of students to him and should not disclose until unavoidable. It develops intimacy among teachers and students.

Openness

Never develop enmity and hostility towards your learners. Always encourage and boost up their qualities.

5.2. Instructional Method

After learning meaning and essential principles of class teaching, now we will discuss instructional method. In this module you will find three methods:

- Demonstration
- Storytelling
- Lecture

5.2.1. Demonstration

- The word demonstration is derived from word demonstrate, which means, “show how to do”.
- Talking about abstract ideas is not easy to grasp. While concrete things get fixed up in minds of students. Thus, from “concrete to abstract” is followed in this

technique.

- In this method the teacher explains the details of the experiments, which he performs before the students.
- Pupils see the apparatus and operation, help the teacher on his asking and at the end; they perform reverse demonstration by themselves, which enhances their confidence and knowledge.
- Demonstrations can prove useful if leaned through rehearsals. If demonstration fails once, repeatedly it effects badly, but successful work maintains enthusiasm and interest of learners in the task and learning.

To make this method successful the teacher must possess following qualities:

- Well planned.
- Purposeful.
- Organized.
- Relevant.
- Challenging for students.
- Mastery over what he demonstrates.
- In students access:
- Its site.
- Operational View
- Clear sounds,
- Quick and clear.
- Students' involvement.
- Connected with simple life situations.

Let us look at merits of this method:

- Demonstrations are economical in view of both times money.
- Psychologically sound, as students see concrete objects.
- Interests and enthusiasm are aroused.
- Useful when devices are costly.
- Creative and acquisitive instincts get satisfied.
- Useful when revision of a principle or project is desired in short time.

- Repeated when and where desired. ,

Demerits

- Generally pupils don not get a chance to do.
- They only observe and remain passive.
- In big class situation, it may not be possible to allow handling of demonstration devices by the pupils.

5.2.2. Story Telling Method

Dear readers:

As you know, among the methods of teaching, there are some, which are used only at times. Such methods are important for their significance' from academic point of view.

One such technique is, "story telling". It means teaching a topic in the' form of a story. Usually the subjects like, Islamiyat, Social Studies and History are well taught through this method.

By adding events and tales, these subjects are made more charming and interesting for pupils. It attracts the children. Even carefree children also take interest. According to Khawaja Ghulam-us-Sayyedain, it is a very useful technique for initial/primary classes.

By applying this method, we can make boring subjects/topics, interesting.

Salient Features

- Students feel charmed and drawn towards the stories of inventions.
- If a teacher has to introduce the principles of Archimedes, he can narrate the story of the king, who announced reward for finding purity of his golden crown without breaking into parts for analysis.
- He should narrate how Archimedes, while taking his bath in a tub, experienced his body thrust upwards by water.
- He discovered that like his body everything lost its weight in water and in that way he could find out the purity of gold in the crown.
- A teacher can use this method in hundred and one ways while teaching Radio, T.V., Telescope and similar other modern discoveries to sketch each detail.
- It motivates students towards learning
- They would move to read or take interest in adventure of the histories of scientists

and religious scholars.

- This method is more suitable for elementary classes.
- This method is more suitable for elementary classes.
- A Teacher cannot use this technique as a method of teaching in general but its judicious use is allowed.

Demerits

These are fewer and are mentioned as follows:

- It is more or less, lecture method.
- The students do not take interest if it does not hold them. '
- Subjects like Literature, Social Science, Islamiyat, and Humanities can only be taught through this method.
- For grown up pupils it may be used sparingly and in lower classes can be used extensively.

Let us learn principles of storytelling method:

- It should be used when felt necessary otherwise children are fascinated and become impractical.
- It should be short and carrying moral values. Detailed/long tales/become boring for the children.
- Keep silence in classroom.
- Teacher must be skilled and knowing how to control his tone, volume and pitch of sound.
- Story must be chosen in accordance with the age and mental level of children.
- During story session ask related questions from children to keep their involved.
- After completing story, children should be asked by teacher .to revise the tale in their own words.
- Summarise important points on blackboard with the help of audience.
- Opportunities of writing and telling a story must be provided to pupils so that they may develop a habit how to express their own ideas.

5.2.3. Lecture Method

This method is next to discussion method, It is an old method and being used

from the time of Socrates.

This is a lecture technique whereby teacher talks while students listen and take notes. It benefits larger groups as well. Though this method is not child centred and keeps lecturer busy and active, yet it is more common in colleges and universities.

This method is not according to the principles of teaching. The learners remain passive and there is no problem solving.

International standard time is 40 minutes for one lecture but mostly teachers take double the time, which brings boredom, anxiety, dullness and exhaustion.

This method can be-used in the following ways:

- To motivate towards specific points.
- To clarify, a unit, lesson, problem.
- To revise, whatever previously learned.
- To expand the knowledge and contents.

The lecture is also a method of exposition. According to James Michael Lee, the lecture is a pedagogical method whereby the teacher formally delivers a carefully planned expository address on some particular topic or problem.

You can show its significance in above paragraph. Now we will see the merits of Lecture Method,

- It is economical.
- Saves time and covers the syllabus in short period of time.
- The work of teacher becomes simple, as he has nothing to be bothered about experiment or demonstration preparations.
- It can be based on the rules of education and psychology and not on scheme of textbook material.
- It provides better scope for clarification and for lying stress on significant ideas.
- It brings a persona contact and touch to impress or influence the pupils.

Demerits

- Does not encourage students' activity.
- Pupils remain passive.
- Teacher/subject centred activity.
- No way to check the reactions of students.

- Teacher may be fast due to time constrains and learners do not grasp the idea. .
- Average student may not be able to fix up his attention to a lecture of 40 minutes duration.
- Students may not follow lectures properly, facing difficulty to take notes.
- Teacher may cover more courses but less learning takes place.
- A lecture is frequently monotonous to the students, as after a while, very few teachers can keep interest up to the end.

Guideline

- Decide on specific topics.
- Research on the topic, if necessary.
- Prepare a lesson plan.
- Before beginning draw attention of pupils.
- Outline main points at the beginning as well as at the end.
- Periodically ask students for any questions
- Speak loudly and clearly, controlling tone volume correlating with body action and expression too.
- Make reasonable eye contact with audience.
- Do not be afraid to stay away from the lesson plan if an unrelated topic or issue arises.

6. SUMMARY

Now let us go through the summary of the unit.

In teaching/educational strategy, we have learnt about three styles of, instruction i.e. individual, group and class instructions. Individualised instruction is a child-centered activity or to consider ones level and desires, it focuses on an individual personality development, to build the character well and to bring insight. It follows through assignment and study methods. Its instructional methods are: Kindergarten, .Montessori, Dalton Plan, Programmed instruction and Computer use.

Froebel and Pistatozzi introduced kindergarten methods. Plan works on freedom, play and joy. Three ways are used in form of songs, movements and construction. Froebel arranges a total of twenty gifts and activities out of which seven are most important. Occupations are activities suggested by gifts, froebel divides curriculum in religion, science, language and arts.

Dr. Maria Motessori used Montessori Method. She taught children through real experiences by giving them concrete material. She insisted on freedom, education for growth, self-sense and motor training for children, in this system, training focuses on bearing certain responsibilities. Basis of this method of teaching is 3Rs, which is quite different from other methods.

Dalton Plan was experimented by Miss Helen Parkhurst. It is not precise for any subject rather organises schoolwork to promote purposive self activity. In this system, specialist teachers supervise pupils through assignments, contracts, and subject laboratories, which are well equipped, and are used to make more access of pupils to vast literature on the subject.

Programmed Instruction' is based on principles of operant conditioning (reinforcement). Real milestone of this system is Prof. B.F. Skinner. In this system, subject matter is arranged in bits, one after the other and presented to pupils and stimulus 'response - reinforcement cycle is repeated. Form of Programme is textbooks, cards, and teaching machines. It comprises branched programming. Teaching machine device requires answers fed by learner -- a self- study method.

Computers and software are referred to as CAI, Computer functions as an aid to instruction, commonly used to drill, practice, tutorial, simulation and problem solving.

Group instruction is imparted to a group of students at the same time and place. Kurts Lewin was the first, who attempted this style in Lowa University. Disciplines like physical education, music, arts, are taught through this method by addressing certain principles. Problem solving and project methods fall in this category, which is proceeded through inductive and deductive approach. Both methods require passing through certain steps. Class instructor considers the class as a unit. The very same method is prevalent throughout the subcontinent. Preservation of social values and continuity of social process are the focal points in this method.

Instructional Method includes demonstration, storytelling and lectures.

Demonstration reveals, “How”, i.e. a practice oriented activity with certain merits and demerits.

Story telling is useful for elementary groups. Topics like Islamiyat, social studies and history may be taught well. It cannot be used as a method. However, a judicious (careful) use is allowed. It motivates students but over fascination may lead to impracticality. It requires the following certain principles.

Lecture method is next to discussion. Usually, it is a one-way and teacher-centred activity but it saves, time and resources. Pupils remain passive and, especially, average students find it difficult and may not be able to take notes properly. However, following the guidelines it may be used in a better way.

7. SELF ASSESSMENT QUESTIONS

Q.1: Explain the kinds of instructions with examples?

Q. 2: Describe the main principles of individualised, Group and Class instruction?

Q. 3: Narrate procedures adopted to practice individualised instruction?

Q. 4: Illustrate methods (instructional) of individualised instruction?

Q. 5: Discuss background of Kindergarten, Montessori, Dalton Plan, Programmed Teaching and Computer?

Q. 6: Narrate merits and limitation of each method?

Q. 7: Define problem-solving method and outline its steps briefly?

Q. 8: Give two examples of problem solving applying deductive and inductive method in daily life?

Q. 9: Define project and list the steps to be followed in this method?

Q. 10: Explain meaning of Class Instruction?

Q. 11: Prepare the checklist to arrange Demonstration?

Q. 12: Write merits and demerits of demonstration, storytelling and lecture method?

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UNIT–3

INTRODUCING CURRICULUM

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INTRODUCTION

Curriculum is an area of vital importance to the professional teacher. Over the past two decades, the study of curriculum has become an established part of teacher education programmes and all serving teachers have become familiar with the concepts of curriculum in some way. Similarly, teachers undertaking postgraduate studies and professional development activities have been exposed to the concepts associated with curriculum, and in recent years, the term is being used quite frequently in the media and the community in general.

Such a development is highly appropriate for professional teachers. Curriculum is. after all, the very substance of schooling for teachers in school. Therefore, teachers need to be knowledgeable about curriculum and understand the processes by which curricula may be developed. When teachers consider curriculum issues, for example, they tackle the substantive matter of schooling, which may be expressed in terms of the fundamental questions of curriculum.

In this Unit, different concepts associated with curriculum, its nature, important definitions and characterises, its need and scope, its planning and development including characteristics involved and effectiveness, its major tasks, and so forth, have been discussed.

OBJECTIVES

After studying this Unit, you should be able to:

1. Define various concepts of curriculum.
2. Describe the desirable characteristics of the curriculum.
3. Describe the broad components of the curriculum.
4. State the need and scope of curriculum.
5. List the steps and characteristics in curriculum planning.
6. State and analyse the principles of curriculum development.

1. THE NATURE OF CURRICULUM

Curriculum plays a vital role in attaining the aims and objectives of education tend to. It reflects the curricular and co-curricular trends in our institutions i.e. the courses of study, the objectives of education, the methodology of teaching including teaching aids, and evaluation methods.

Curriculum comes from the Latin root, Currere which means "to run". Which later come to stand as the "course of Study".

Curriculum i.e the sum total of all learning content, experience, and resources that one purposely selected, organized and implemented by the school in pursuit of his peculiar mandate as a distinct institution of learning and human development.

1.1. Important Definitions of Curriculum Compared.

The term 'Curriculum' is defined in many ways by educators. Some use the term in very limited and specific contexts while others attach very broad and general meanings. Some define it in descriptive terms, i.e., what curriculum is and others in prescribed terms, i.e. What curriculum ought to be., Again curriculum is defined in terms of subjects, activities and experiences.

Following are some of the important definitions of the curriculum:

Alberty A, and Alberty E. (1959) define curriculum' as the sum total of student activities which the school sponsors for the purpose of achieving its objectives".

In the words of H. Robert Beck, and W. Walter Cook, "Curriculum is this sum of the educational experiences that children have in school".

Blond's Encyclopedia (1969) of Education defines "Curriculum as all the experiences a pupil has under the guidance of the school".

F. Bobbit in ""The Curriculum" (1918) has observed that "Curriculum is that series of things which children and youth must do an experience by way of developing abilities to do things well that make up the affairs of adult life; and to be in all respects what adults should do".

Derek Rowntree in A Dictionary of Education (1981) has defined 'curriculum in these words. "Curriculum can refer to the total structure of ideas and activities, developed, by an educational institution to meet the needs of students and to achieve desired educational aims".

R. Doll, in Curriculum Improvement: Decision Making (1982) has stated: "Curriculum embodies all the experiences which are offered to learners under the auspices or direction of the school".

The first definition lays emphasis on the world "sponsors' and the second definition tends to give importance to spontaneous as well as 'sponsored'. The third definition stresses the word "guidance' in providing experiences. The fourth, definition

gives importance to 'adult activities'. The fifth definition denotes that is something, "fixed'. The sixth definition like the fourth one highlights the significance of guidance in providing activities.

1.2. Characteristics of Curriculum

1. ***Totality of activities:*** By Curriculum it is meant all the school activities, which are used to promote the development of the pupils. It refers to the totality of subject matter, activities and experiences, which constitute a pupil's school life. Pragmatists have also included the entire range of learner's activities in the curriculum because according to them the child learns by doing.

2. ***A means to an end:*** It is known that Curriculum is not an end in itself, but a means to an end. Therefore, it is created so as to achieve the aims of education. That is the reason why different educationists have suggested different kinds of curricula to conform to the aims and objectives ascribed to education. It follows that the curriculum will change with every change in the aims of education,

3. ***Total school environment:*** The total environment of the school influences Curriculum. It is made up of everything that surrounds the learner in all his working hours. It is "the environment in motion". It refers to the total educational programme of the school (school environment) including all experiences, activities and learning, in which the learner is expected to progress and attain the goals of education.

4. ***Totality of experiences:*** Curriculum refers to the totality of experiences that a pupil gets in the school (i.e. the class-room, library, laboratory; workshop, play-ground and in the numerous informal contacts between the teacher and the pupils) as well as outside the school. These experiences help him in the development of personality: Curriculum includes not merely syllabus and books but all those experiences and relationships. Which get indulged in by the student both inside and outside the school. Thus, the syllabus specified by the authority should not be taken to mean curriculum.

5. ***Mirror of curricular and co-curricular trends:*** Curriculum forms the mirror of curricular and co-curricular trends and is able to reflect the curricular and co-curricular trends in our educational institutions i.e. the courses of study, the aims and objectives of education, the methodology of teaching including teaching aids and evaluation techniques.

6. ***Mirror of Educational Trends:*** Curriculum is the mirror of educational trends. It depicts the total picture about the prevailing educational system. The objectives behind the educational system highlighted through the series of experiences, which are provided by the curriculum.

7. ***Development of balanced personality:*** Curriculum is quite helpful in the development of balanced personality. The activities in curriculum concerning physical, intellectual, emotional, social, economic, aesthetic and cultural development play their role for developing balanced personality.

8. ***Process of living:*** Curriculum is a process of living in which interaction between the individual and his environment takes place. Curriculum is concerned both with the life of the individual and his environment.

9. ***Dynamic:*** A good curriculum is dynamic. The needs and interests of the pupils go on changing with the passage of time. For varying needs different types of activities are required. This tends to necessitate some changes in the curriculum. Curriculum is never static. It has to be different, for different students, different classes and different schools. It has to be kept dynamic in order to keep with the needs, interests, abilities, attitudes and lives of the pupils.

10. **Mirror of Philosophy of life**: Curriculum is regarded as the mirror, of philosophy of life. It depicts philosophy of life. Democratic or autocratic way of life is reflected in, the curriculum. Each way of life is having different philosophy and hence different goals of education.

11. ***Achievement of goals:*** Curriculum is prepared to achieve some set goals and objectives of education, which are set by society. Curriculum helps in achieving the aims and objectives of education.

1.4. Broad Components of Curriculum

The components of curriculum, according to Agnes. S. Robinson (5971). are the goals, objectives, content, processes, recourses and means of evaluation of all the learning experiences planned for pupils both in and out of school and community through class room instruction and related programmes (for example: field trips, library programmes, work experience education, guidance and extra classroom activities.)

According to K.A. Leithwood (1981), curriculum encompasses educational philosophy, values, objectives, organizational structures, materials, teaching strategies, student experiences,assessment and learning outcomes.

Three main components of the curriculum are as under:

- Programme of studies
- Programme of Activities
- Programme of Guidance

Programme of Studies: This refers to the various subjects like History, Languages, Mathematics, and Science etc. Emphasis on the study of a subject/subjects has changed from time to time in accordance with the philosophical and sociological ideals. Conservation and promotion of culture has been an important determinant in the

selection of the contents of the subjects. In view of the vastness of culture, principle of selection is followed. The level of information to be imparted at a particular stage or class is graded suitably.

The methods of imparting knowledge are determined on the basis of psychological findings especially regarding learning.

Programme of Activities: With the changing concepts of education and consequently curriculum an increasing emphasis of being laid on the organization of various activities in the schools. In view of the importance of activities in the promotion of ideals of citizenship, cooperative living and democracy, many educators advocate that curriculum should be envisaged in terms of activities rather than subjects,. The principles of learning emphasize that participation in activities goes a long way in sublimating the instincts of children and making teaching-learning more enjoyable as well as-effective.

Programme of Guidance: A comprehensive programme of guidance includes helping students solve their educational, vocational and personal problems. With the rapid changes in the society in various fields, it has become very much necessary to include the guidance programme in curriculum.

Curriculum and Syllabus

A UNESCO publication entitled "Preparing Textbook Manuscripts' (1970) has differentiated the curriculum and syllabus as: ‘‘"The curriculum sets out the subjects to be studied, their order and sequence, and so ensures some balance, between humanities and science and consistency in the study of subjects, thus facilitating, inter-subject links. It follows that the curriculum determines the amount of school-times allotted to each subject, the aim of teaching each subject, the pace of the psychomotor domain, whichtakes time, to acquire, and possibly, the variations between rural and urban school teaching. The curriculum in the schools of developing countries is often directly related to the requirements for development. The syllabus determines the basic content of instructions in a given subject and the range of knowledge and skills which the pupils must acquire and establish in detail the themes and individual points to be studied in each school year the syllabus is a refined detail of the curriculum at a particular stage of learning for a particular subject”.

1.5. Need of Curriculum

The following points demonstrate the importance of curriculum:

1. ***Achievement of educational aims***: Curriculum renders help in achieving the aims of education. Without suitable curriculum aims of education cannot be achieved. In the absence of curriculum it is not possible to do anything systematically.
2. ***Fixing- limits:*** Curriculum is quite helpful in fixing limits of teaching and learning. It helps in determining the work of the teacher as well that

of the pupil.

3. ***Development of democratic values:*** Curriculum is helpful to the students in developing democratic values such as liberty, equality, fraternity, justice, respect for dignity of the individual and group living.

4. ***Development of citizenship:*** Development of citizenship is one of the major responsibilities of education. Suitable curriculum is helpful in. achieving this aim of education.

5. ***Development of character:*** There is no system of education, which does not aim at developing character in the youth. Character can be developed through suitable curriculum.

6. ***Satisfaction of needs:*** Curriculum is able to satisfy educational, vocational and psychological needs of students. There is a great variety of interests, skills, abilities, attitudes, aptitudes, and requirements of students.

7. ***Criteria of suitable teachers:*** The curriculum mainly shows what type of teachers is needed in the schools. We should know what type of work they are. required to do and this is to be in accordance with the requirement of the curriculum.

8. ***Selection of suitable methods:*** Curriculum makes the teacher to select suitable methods of teaching. 'How to teach' will be determined by what to teach i.e. the curriculum.

9. ***Acquisition of knowledge:*** -Curriculum helps the student in the getting knowledge. By studying various subjects laid down in the curriculum the student gets knowledge in conformity with his abilities and level of intelligence.

10. ***Development of personality:*** Curriculum is useful in developing physical intellectual, aesthetic, social, cultural, moral, spiritual religious and vocational abilities of the student. Suitable curriculum is necessary for the complete and balanced development of personality.

11. ***Reflects trends in education:*** Curriculum is meant to achieve the end i.e., the changing aims of education with the changing social requirements. Hence curriculum reflects the trends in education and changes in philosophy. Modern education expects following demands on the curriculum:

    (i) ***Providing suitable knowledge****:* The curriculum should provide suitable knowledge, which will be quite helpful in the achievement of aims of education.

    (ii) ***Providing suitable activities and experiences:*** The curriculum includes well-selected activities and experiences required for development of pupils according to social requirements.

(iii) ***Providing wholesome influences:*** The curriculum should provide wholesome school programme for developing the desirable behaviour patterns in the pupils.

12. ***Discoveries and inventions***: At the university or higher level of education the main aim of curriculum is to encourage research and inventions.

Thus curriculum acts as a pivot in organizing educational effort on some manageable basis and is the heart of the school.

1.6. Scope of Curriculum

The scope of the content concerns question about what to include and what to exclude as for as the selection of subject matter is concerned. It includes the broad form, which is the range and extent of each area to be covered. Determining the scope must include some reference to aims and objectives and range of the learning experiences. Counseling what is to be included in the content is sometimes referred to as determining the scope and it can be seen by operating at four levels.

a) First there must be decisions regarding what to include as a whole in the major area within which the curriculum operates.

b) Should the concerns be with certain subjects that are basic to the understanding of the human caring, such as those within the behavioral science and humanistic?

c) Should the selection drawn from the life science in that these may help the student understand physical factors of care?

d) Should the material be developed that addresses the learning?

e) Is there a need to include a study of the exceptional in terms of psychological considerations that are appropriate for the study of pedagogy? Thus, sort of questioning is directed at the identification of the subjects, which are thought to be eclectically desirable for the underpinning of the base of teaching practice. These have been considered at length before and viewed as the essential antecedents to distinctive instructional knowledge.

Second there is level of scope, which looks at the area of a distinctive subject and considers what might be borrowed and used. This eclectic use of knowledge draws and extracts from the particular established discipline concerned without necessity of accepting the total area of ideas and beliefs within the totality of the discipline.

Third level of scope determination concerns the individual teachers working from curriculum at classroom level and involves independent decisions, about how much material can be developed within a particular period of time and where the emphasis should be placed within certain subjects and topics.

A fourth level of scope relates to individual lessons concerning the areas to be dealt with and the extent they are to be covered and the level of intellectual difficulty involved that the patterns of the curriculum should adapt a certain idea of scope, which focuses on certain centres of organization. Such centers of organizations can be seen to operate through themes or modular theme.

In dealing, with the scope of the contents certain basic principles can be usefully followed and these can also act as a general guide from making decisions.

1. There is a need to work from basic generalizations or universal thinking about the content progressively towards specific or particular elements.
2. The initiated decision-making should be concerned with the identification of the major areas of knowledge to be included. .
3. The content to be included must be educationally worthwhile.
4. The content must occupational be relevant for teaching.
5. The content should be professionally relevant for teaching.
6. The scope must determine the appropriate skills, attitudes and beliefs, which the students need to be aware of.
7. The scope should determine the appropriate intellectual and practical level of difficulty in the content.
8. The scope needs to be taken, into account whether additional, external expertise is needed from outside the field of teaching
9. The scope of the content should reflect the necessary educational technology to be included and the curriculum materials.
10. The scope ought to include consideration of the basic form of assessment and examinations to be used.

Inevitably when the scope of curriculum content has been determined there will be concern for the tradition approaches of the past, which may well be thought to have stood the test of time.

2. CURRICULUM PLANNING AND DEVELOPMENT

Like curriculum, curriculum planning has been defined in various ways:

1. Curriculum planning is a process in which participants at many levels make decisions about what the purposes of learning ought to be, how those purposes mighty be carried out through teaching-learning situations and whether the purposes and means are both appropriate and effective.
2. Curriculum planning consists of all the processes necessary to plan for and write a curriculum.
3. Curriculum planning is the process of gathering, sorting, selecting, balancing, and synthesizing relevant information from many sources in order to design those experiences that will assist learners in attaining the goals of the curriculum.
4. Curriculum planning is the orderly study and improvement of schooling in the light of stated objectives.
5. Curriculum planning is the preliminary phase of curriculum development when the curriculum workers make decisions and take action to establish the plan that teachers and students carry out.
6. Curriculum planning is the process whereby arrangements of learning opportunities or curriculum plans are created.

2.1. Characteristics of Curriculum Planning

These may be stated as:

1. Curriculum planning is ultimately concerned with the experiences of the learners.
2. Curriculum planning involves decisions about both content and planning.
3. Curriculum-planning involves decisions about a variety of issues.
4. Curriculum planning involves many groups.
5. Curriculum planning takes place at many levels.
6. Curriculum planning is a continuous process.

2.2. Process of Curriculum Development

The process of curriculum development begins with the following:

1. Formulation of objectives of education, which are based on the ultimate, aims of life, and also on the political and social philosophy and development needs of a country.
2. Specification of objectives of education in respect of different stages of school

education and also in respect of different subjects and activities and experiences at different stages of education.

3. Determination of the scheme of studies, syllabi, etc.
4. Preparation of instruction materials like textbooks, supplementary readers workbooks, teacher guides, etc.
5. Implementation of curriculum in the school.
6. Evaluation of curriculum to ensure quality control for effecting suitable modifications in the curriculum.

2.4. Effective Curriculum Planning

Following are some crucial factors, which guide effective curriculum planning:

1. Curriculum planning must be based upon a clear conception of what makes a good life.
2. Curriculum planning must take into account the characteristics of past, contemporary and future society.
3. Curriculum goals should address a broad range of needs, interests of both individual and society.
4. Curriculum planning should recognize and encourage diversity among learners.
5. Curriculum planning must make provision for all aspects of teaching-learning situations. They should include suggestive activities, content, resources, and measuring devices, etc.
6. Curriculum planning should be done in the light of the characteristics of the learners for whom they are intended.
7. Curriculum planning should also make use of approaches other than subject-area approach. .
8. Curriculum planning must provide flexibility to allow teacher pupil planning.
9. Curriculum planning must provide for the infusion of spontaneous ideas, which emerge during the interaction of the learners and teachers.
10. Curriculum planning should reflect a balance among cognitive, affective and psychomotor needs of the learners.
11. Curriculum planning should include provision for reflective thinking, values and valuing, enhancement of self-concept and self-esteem.
12. Cooperative planning and development are most effectively done in cooperative settings.
13. Curriculum planning should take into consideration an integrated set of

experiences rather than a disjointed conglomeration of activities.

14. Curriculum planning must provide for continuous evaluation of all aspects of curriculum.

2.4. Basic Principles of Curriculum Development

Curriculum is a tool in the hands of the teachers, to give training to children in the art of living together in the community. It is a tool, which, considerably helps to inculcate those standards of moral action, which are essential for successful living in society and for getting true satisfaction out of life. It is therefore; very essential that the curriculum should be, based on sound principles.

The following principles should be kept in mind when framing curriculum;

l. ***The Conservative Principle:*** It has been stated that nations live in the present, on the past and for the future. This means that the present, the past and the future needs of the community should be taken into consideration. The past is a great guide for the present as it helps us to decide what has been useful to those, who have gone before and what will be useful to those, who are going through now. The point has been explained in the Handbook of Suggestions for Teachers as "It is, in fact, being increasingly recognized that the various subject of the curriculum represent certain form of skills and certain branches of knowledge, which have proved to be of importance in the experience of the race, and which have to be taught to each succeeding generation. From this point of view it is the function of the school to preserve and transmit the traditions, knowledge and standards of conduct on which our civilization depends.

This principle will be of help only when we carefully select as to what things of the past are likely to help us in the present. All the things of the past may not suit us. It is, therefore, essential that we should select only those subjects and activities, which are required by the present generation.

This principle has been criticized by some educationists on the ground that it looks to subjects and not to the pupils. It is contended by its antagonists that in these days when it is accepted that education and schools, and therefore curriculum, should be child-centred, this principle does not take us very far. It may be replied to such critics that to condemn everything of the past is not a sound policy and especially in a country, which has had a glorious past and, which showed the path of knowledge to other countries. Another point to be stressed is that there must be a base to stand upon and if the base is durable it must be accepted. Thirdly, it would be wrong to say that in the past the child was entirely ignored. However, as stated above, the principle of selectivity should be adapted in the selection of the curriculum on the basis of the past.

2. ***The Forward-looking Principle:*** While discussing the first principle, we have pointed out that in the present, future needs and requirements of the community should be given their due place. Children of today are the citizens of tomorrow. Therefore, their education should be such as it enables them to be progressive

minded persons. Education should give them a foundation of knowledge and feeling that will enable them to change the environment where change is needed.

3. ***The Creative Principle:*** In the curriculum those activities should be included, which enable the child to exercise his creative and constructive powers. The objective of education is to discover and to develop special interests, tastes and aptitudes. Weeler says, "The promptings to anyone of the intellectual interests may fade away and become practically irrevocable unless opportunities be then and there forthcoming. None of the child's native gifts should escape our notice and so perish of inanition. In a curriculum that is suited to the needs of today and of the future, there must be a definite bias towards definitely creative activities.

4. ***The Activity Principle:*** The curriculum should be thought in terms of activity and experience, rather than of knowledge to be acquired and facts to be stored growth and learning take place only where there is activity. 'Experience' rather than 'instruction' is to meet the needs of the various stages of growth. In the words of John Dewey. "The general movement is away from the old over-reliance and verbal instruction, formulated subject and learning from books, and in the direction of more varied and many sided individual activities in keeping with child's real interests".

   Playgrounds, shops, workrooms and laboratories not only direct the natural active tendencies of youth, but they also involve intercourse, communication and co-operation.

   Abbott and Wood report has also stressed the activity principle, "It is vitally important that young children should not be required to sit still for long periods at a time. A young child needs rest, it is true, but he must play, he must explore, and he must be physically active if he is to derive a daily satisfaction out of his attendance at school. In short, he needs experience more than instruction".

   The curriculum must ensure the activity of body and mind. It should be the centre of the curriculum. All modern methods of teaching, i.e. Kindergarten, Montessori, Project, Basic, etc., are based on this principle of activity.

5. ***Principle of Preparation for Life:*** This is most important principle in the construction of the curriculum. Education must equip an individual for life. Hence, curriculum must include those activities, which enable the child to take his part effectively and amicably in the activities of the community when he becomes an adult. We have to prepare him in such a way, as he is capable of facing the various challenges of the complex problems of the future.

6. ***Child-Centred Curriculum:*** It is true that the child is to be prepared for life. But this does not mean that his immediate interests should be sacrificed for the sake of the future, which is indefinite. The best preparation for life that we can give a child is to help him to live fully and richly his life at that stage at which he is. The child automatically prepares himself for the next stage by living well and true life at one stage. Smith and Harrison observe, 'Education.... regards the child as an individual growing by his own activity, living in his own environment, and

preparing himself for adult life, not by imitating the adult, but by living as fully as possible in the environment of childhood"

7. ***Principle of Maturity:*** Curriculum should be adapted to the grade of the pupils and to their stage of mental and physical development. In the early childhood; 'wonder' and 'romance' predominate. So subjects and activities, which present the elements of 'wonder' and 'romance, should be included at this stage. At a later stage they are interested in practical things. So at the Elementary Stage the curriculum should provide for practical problem. At the next stage, that is the Secondary Stage, students are interested in generalizations and accordingly curriculum should provide such activates. The child at this stage is keen to discover, to find out and discuss new facts. The curriculum should harness the adventurous spirit of the growing child.

   The experience provided should be within the comprehension of the students.

8. ***Principle of Individual Differences:*** Individuals differ in taste, temperament, skill, experience, aptitude, and innate ability and in sex. Therefore, the curriculum should be adapted to individual differences. It should not be rigid.

9. ***Vertical and Horizontal Articulation:*** On the one hand, each year's course should be built on what has been done in previous years and at the same time should serve as basis for subsequent work. It is absolutely essential that the entire curriculum should be coordinated.

10. ***Principle of Linking with Life:*** The community needs and characteristics should be kept in view while framing the curriculum.

11. ***Principle of Comprehensiveness and Balance:*** The curriculum .should be framed in such a way as every aspect of life i.e. economic relationships, social activities, occupations and spiritual life, is given due emphasis.

12. ***Principle of Loyalties:*** The curriculum should be planned in such a manner that it teaches a true sense of loyalty to the family, the school, the community, the town, the province, the country and the world at large. It should enable the child to understand that there is unity in diversity.

13. ***Flexibility:*** Curriculum should take into consideration the special need and circumstances of the pupils. Curriculum of the girls may not always he identical with the hoys. The special needs of both the sexes should be given their due consideration.

    In general the curriculum of the village and the urban school will be the same but there might be variation according to the specific needs of the locality.

14. ***Principle of Core or Common Subjects:*** There are certain broad areas of knowledge, skill and appreciation with which all the children must be made conversant and these should find a place in the curriculum. This is more important at higher secondary stage where there are diversified courses. These subjects are to be common to all groups. They are known as core subjects.

Mother tongue or regional language, special studies (general course), general science including mathematics and one craft are expected to be the core subject.

15. ***Principle of Leisure:*** The curriculum should prepare the child for the use of leisure time. According to Herbert Spencer, Literature, Music and Art occupy the leisure part of life and should, therefore, occupy the leisure part of education. The capacity to enjoy leisure greatly determines a man's capacity to work. If leisure is spent in gambling, drinking and reading obscene literature, it will hamper progress not only of an individual but also the nation as a whole. The school curriculum should therefore, prepare the would-be-citizens to use effectively their leisure time.

16. ***Principle of All-round Development of Body, Mind and Spirit:*** All kinds of experiences should be provided to the students so that they may develop their all powers.

2.5. Elements of Curriculum

Curriculum development process consists of various phases or elements. The curriculum planners have to consider all these phases while working on such process. According to J.S, Farranta (1990; p-24) the elements of curriculum development are philosophy, society, aims, educational administrators, school organization, teaching material teacher's role and teaching methods.

In the words of Lawton D. (1976; p-21) the elements of curriculum are the goals, objectives, content, processes, resources, and means of evaluation.

Weeler (1974; p-30) has given five elements of curriculum as:

1. The selection of aims, goals and objectives.
2. The selection of learning experiences calculated to help in the attainment of these aims, goals and objectives.
3. The selection of content (subject matter) through which certain types of experience may be offered.
4. The organization and integration of learning experiences and content with respect to the teaching learning process within school and classroom.
5. Evaluation of the effectiveness of all aspects of phases 2,3, and 4 in attaining the goals detailed in phase I.

In the light of above mentioned elements and taking into consideration the educational system in Pakistan, curriculum must have the following important basic factors:

(a) Situational analysis
(b) Setting the objectives.
(c) Content

(d) Learning experiences
(e) Evaluation

(a) Situational Analysis:

Geographical condition of the country, society, its religion and culture, all influence the process of curriculum development, as one of the functions of curriculum is to preserve the cultural heritage of society and the school programmes are responsible to transmit this heritage to the coming generation.

The curriculum development process is also influenced by the geographical Conditions especially when general decisions are made for school timings, duration of working time, summer and winter vacations, availability of natural sources, animals, plants, etc.

In the same way, sociological conditions are also important to be kept in view. According to Murray, Print (1993; p-39), "Curriculum developers serve the function of translating traditional assumptions, ideas values, knowledge and attitudes into curriculum objectives, content, learning activities and evaluation. Of these elements sociological sources have their greatest impact on education system of a country,

When curriculum is planned, the cultural background of the concerned society is also considered. The objectives, which are set, the content, which is selected, the methods, which are followed, and the evaluation procedures, which are adopted, all can only be effectively done if they are according to the cultural and religious background of the country.

Curriculum planners have to give due importance to the society, beliefs, attitudes, values, likeness, dis-likeness, etc. during the process of curriculum development.

(b) Setting the Objectives:

The objectives play pivot role in the curriculum design. They are the starting point. Nothing can be done without having any target to be achieved.

Hughes (1967) indicated that there are normally four sources or inputs through which-curriculum developers go for objectives. These arc society. knowledge, learners and learning process. Curriculum planners use research findings and opinions of philosophers, educationists. Psychologists, subject specialists and sociologists as inputs for their designs. The selection of objectives is based on their collective value position, which is supposed to be the representative of society because the composition of the committee is meant to-reflect this and the opinion of the committee is focused on the main targets or objectives to be achieved.

Perhaps you tend to use the terms 'goals' 'aims' and 'objectives interchangeable. But, in fact, these terms express different concepts. Therefore, we should be clear about the main distinctions at this stage, although they are treated in much more detail in coming units.

Educational aims refer to the overall purposes of education, which represent the needs, and aspirations of a society as manifested in its educational policy.

Goals are the general purposes of education that are normally prepared for particular stage or level of the education system; whereas objectives refer more precisely to the changes in behaviour, which we hope will result from studying particular courses. While goals are less general than aim but less specific than objectives.

The very first question one must face in developing a curriculum-is "what are its aims and objectives". There, must be some point to planning activities, some intended educational outcomes.

When consider the nature of aims and objectives, we come across many conflicting views about issues like "Why do we teach"? What should be the purposes of education? For example, what is the major purpose of education and what should we select as a teaching material to reach and achieve the main purpose.

(c) Subject Matter or Content:

This is the element, which has been emphasized mainly in the past. At the time of subject matter selection, the following factors are to be considered:

- Type of society/culture
- Type of curriculum
- Number of diversified courses offered
- Scope of the subject itself
- Level age of the learners
- Number and qualification of the teachers
- Available sources and resources
- Strength in the classrooms
- Demands of the society
- International needs
- Methods of content organization
- System, of examination, etc.

(d) Methods of Teaching/Activities:

This is another important element of curriculum development. It includes the methods adopted by the teacher during instruction and the learning experiences or activities performed by the students. This element has been ignored too often in the past and even it is being ignored at present. It is not fair to ask teachers to achieve certain

goals without giving them guidelines how to go to the achievement of desired goals. In Pakistan the curriculum development process is centralized. Teachers are not directly involved in this process. So most of them are ignorant of the process of achievement of desired goals. They may be subject specialists with full command on subject matter but still they may have the problem of selecting appropriate methods of teaching and engaging the learners in various activities accordingly after we have determined the goals, aims and objectives, the next important problem is the selection of the subject matter. Out of the huge amount of ever expanding knowledge, what should we impart to our pupils? What knowledge is of greatest value to them? Further, should we emphasize process or" information? Should a curriculum be fixed or flexible, constant or common to all or differentiated, practical or liberal?

(e) Evaluation:

Curriculum development is a dynamic process, which needs a continuous research and evaluation for its betterment.

In order to cope with the society demands and bring about desirable changes, evaluation is made by the curriculum planners/researchers and teachers.

Curriculum evaluation must not be misinterpreted as student evaluation. It is a broad term used to make judgment about the set curriculum and to make decisions about the worth or effectiveness of it on the basis of evaluation the experts can modify the curriculum by bringing about desirable changes. Keeping into account the process of dynamic nature of the curriculum development, evaluation becomes necessary for decision making at all stages in curriculum planning. Guba and Stufflebeam (1970) identified four types of decisions, which are involved in-curriculum evaluation.

(a) Planning (objectives to be selected).

(b) Planning procedure (personnel, method and material to employ).

(c) Implementing procedures (whether to continue, modify or abandon a procedural plant plan).

(d) Outcomes (extent of achievements of targets or objectives).

3. SUMMARY

Curriculum is a Latin word which means the "race course" or a run-way, which one takes to reach a goal. Thus, a curriculum is the instructional and educative programme through which the pupils achieve their goals, ideals and aspirations of life. Cunningham says, "Curriculum is a tool in the hands of the artist (the teacher) to mould his material (the pupil) in accordance with his ideals in his studio (the, school)." The curriculum however cannot be considered synonymous with courses of study, although they form an important part of curriculum.

The term curriculum in recent years has come to mean all the planned activities and experiences, which are available to students under the direction of the school. It "includes the totality of experiences that a pupil receives through the manifold activities that go on in the school, in the classroom, library, laboratory, workshop, playground and in the numerous informal contacts between teachers and pupils. In this sense, the whole life of the school becomes the curriculum, which can touch the life of the students at all, points and help in the evolution of a balanced personality.

For constructing school curriculum, these principles should be borne in mind:

1. Curriculum should be child centred
2. Curriculum should provide fullness of experience.
3. Curriculum should make provision for flexibility within framework of acceptable principles and values.
4. Curriculum should be related to the life, needs and aspirations of the people.
5. Curriculum should be able to develop a rational outlook.
6. It should lay emphasis on learning to live rather than on living to learn.
7. Curriculum should make provision for socially useful productive work.
8. Curriculum should be able to promote social justice, democratic values and national integration.
9. Curriculum should make adequate provision for the study of languages.
10. Curriculum should make provision for artistic experiences and expression
11. Curriculum should make provision for physical development
12. It should have provision for character building and human values.
13. Curriculum should provide for uniformity and variety.
14. Curriculum should provide for continuity in the whole programme.
15. It should be well integrated.

4. SELF-ASSESSMENT QUESTIONS

Q.1 Write down different definitions of 'curriculum' quoting at least three educational theorists.

Q. 2 After Considering various definitions and views on curriculum, write down your own ideas about the meaning of ‘curriculum’.

Q.3 What are the main principles for curriculum construction? Explain.

Q.4 Explain the meaning and characteristics of curriculum.

Q.5 What is the importance of curriculum?

Q.6 What are the defects in the existing curriculum? Suggest principles of curriculum construction.

Q.7 Explain the scope of curriculum in the light of four levels.

Q.8 Write short notes of the following

i. Effective curriculum planning

ii. Basic principles of curriculum development

iii. Elements of curriculum

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UNIT–4

FOUNDATIONS OF CURRICULUM

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INTRODUCTION

The curriculum foundations may be defined as those basic forces that influence and shape the minds of curriculum developers and enhance the content and structure of the subsequent curriculum.

Ideas about curriculum do not arise in a vacuum. Such ideas are actually based upon views about human nature, source of values, worthwhile knowledge, and the role of teachers and the school. Therefore, the development of curriculum depends largely on the ideas that grow out of the fields of philosophy, psychology and sociology. These three ideas contribute to curriculum development plans and help learners to grow and develop into unique personalities for accomplishing satisfactory lives within the framework of acceptable norms of society.

The foundations of curriculum development can be analysed in terms of philosophical, psychological and sociological foundations.

These sources of curriculum foundations constitute the principal areas of curriculum development and affect the ways developers think and conceive about curricula.

Philosophical work can aid curriculum developers to understand the nature of objectives, structure and interrelationship of objectives, nature of curriculum activities, the structure of curriculum plan and the contents and methods.

Psychological foundations aid curriculum developers to understand the nature of the learners, learning process, learning experiences, interest of learners and the conditions facilitating optimum learning.

According to sociologists, schools are social institutions especially set up for the preservation and transmission of culture by the society. The curriculum, therefore, includes learning experiences based on the way of life, kind of knowledge and attitudes and beliefs Considered important by the society.

Curriculum must take into account the philosophical, psychological and sociological considerations otherwise; it will remain bookish and divorced from life. A curriculum that ignores these foundations does not serve any purpose. A sound curriculum must be based on the needs and aspirations of the learners as well as of the society.

OBJECTIVES

At the end of this Unit, the learner will be able to:

1. Define curriculum foundations.
2. Differentiate between the various philosophical categories.
3. Relate growth and development of learners with curriculum.
4. Define sociology culture and values.
5. Describe role of sociology in curriculum development.
6. Discuss the influences and biases of the society and .culture on curriculum.

1. PHILOSOPHICAL FOUNDATIONS

Philosophy is the pursuit of wisdom and knowledge. It is the study of realities and general principles. It concerns with the research of internal truths.

Philosophy provides systematic procedure for clarifying issues and problems and making decisions on critical points of curriculum development.

Philosophical work can aid curriculum development in many ways but it is particularly useful in helping us to understand:-

- Nature of educational objectives:
- The structure or inter-relationship of the objectives:
- and Nature of curriculum activities.

Defined literally, philosophy is "pursuit of rational thinking and sound judgment". "Aristippus (435-359 BC) wrote, "Philosophy is the ability to feel at ease in any society." Kabir, Huyaun (1902-1969) stated, "Philosophy seeks to give knowledge of the whole".

1.1. Philosophy and Curriculum

Every society is held together by a common faith or "Philosophy", which serves its members as a guide for living a good life. It is, therefore quite natural for the adults of the society to pass on this philosophy or "knowledge of good" to their children. In primitive societies knowledge of the good life was passed of informally, from father to son and from mother to daughter. But in developing and developed societies, schools are established to induct the young into the ways of living that adults consider well. Thus the curriculum of the schools, whatever else it may do, is first and foremost designed to win the hearts and minds of the young to those principles and ideals that will direct them to wise decisions; i.e. decisions whose consequences lead to the concepts of good life. Indeed, the curriculum is so thoroughly permeated with the cultural aspects of philosophy of life that a certain philosopher of education was prompted to write, "what a man really, believes is frequently more clearly revealed in what he teaches to his students than in what he professes in his public statements (Thut 1957).

1.2. Philosophical Categories

There are three philosophical categories that have particular relevance for curriculum development.

(1) ONTOLOGY (The Nature of Reality)

(2) ESPISTEMOLOGY (The Nature of the Knowledge)

(3) AXIOLOGY (The Nature of Value)

1.2.1. Ontology

Ontology is concerned with the nature of reality, and by asking the question "what is real"? A number of queries become clear. While this question may appear simple at the first glance, it deserves greater consideration. Different societies, for example, perceive reality in quite different ways, as do the individuals', who constitute these societies. In primitive societies, it was accepted as real that the earth was flat, yet today we would regard this as nonsense. Similarly, one society may regard the use of chemical fertilizers as essential, while another may consider biological wastes more useful than the chemicals.

Thus what is real to a society is very important and must be taken into account while constructing curriculum. Indeed, some curriculum developers see, their role as a vehicle for change: Thus a new social studies curriculum may depict "Kelly Gang" as a symbol of working-class resistance to authoritative oppression and so re-create reality for a new generation of school students. Thus, some recent curriculum developments like multiculturalism gender equality and environmental education, etc. have sought to achieve just that.

1.2-2. Epistemology

The philosophical problem that deals with the nature of knowledge and of knowing is called epistemology. For Waller and Evers (1988), "epistemology is the study of the nature, scope and applicability of knowledge'". In curriculum, what we advocate becomes the basis for student learning. We are actually concerned with the nature of knowledge, its basis, how we know and what we know.

When studying epistemology, we ask:

What is true?

How do we know the truth?'

How do we know what we know?

These are obviously vital questions for curriculum developers to consider, particularly in a society, which purportedly values truth and seeks to pass the truth to subsequent generations.

Ultimately our position becomes a statement of faith, a stand on those questions, which we believe, and are prepared to accept, as true. As such we rely heavily upon our fundamental ontological beliefs (what is real?)

In this way the close relationship between epistemology and ontology .is consolidated.

Thus in any curriculum development activity, but particularly in relation to schools, the epistemological stance taken by those developers involved is of vital importance. Will they include the accepted truth? What does that constitute? To what degree is there a consensus accepting that truth, or is that "consensus" a fallacy too?

And so the epistemological questions continue. At the very least curriculum developers should be aware of epistemology and be prepared to pose the fundamental questions involved in such a study.

1.2.3. Axiology

Axiology is that aspect of philosophy that is concerned with the nature of value. Axiology questions are a fundamental feature of our life in that the resulting decisions have a profound effect upon our behaviour. Questions such as:

what is good? and

what is attributable to humans? etc.

are both fundamental to our very existence and constantly present in our daily lives. Thus the axiological considerations are important in one's development of a curriculum for future generations.

Zais (1976) contends that axiological questions are usually divided into two main categories.

(a) Ethics
(b) Aesthetics

(a) Ethics:

Is concerned with concepts of good and bad. right and wrong as they apply to human behaviour. When constructing the curricula, developers should be aware of both their own ethical positions and the ethical basis (hopefully not biases) that they are integrating into the curriculum. Thus, developers will select objectives and contents that in their minds are more ethical both in terms of knowledge and process.

Robert Zais summarized the situation succinctly" Education, after all, is a process of deliberately influencing. Children and youth in such a way that they become what they would not otherwise become. And the curriculum is the master plan by which this purpose is accomplished. At this point it is important to raise these issues and questions in the minds of curriculum developers. There is increasing evidence in recent years that saver elements of Pakistani society want a greater and more purposeful input of ethical aspects into school curricula.

(b) Aesthetics:

Is concerned with such values and issues as beauty and enjoyment of human experience. Aesthetics questions: What is beautiful? What aspects of the senses produce enjoyment? And what aesthetic experiences yield "higher order" enjoyment?

The issues involving aesthetics produce particular difficulties for curriculum developers because individuals answer the above questions in very different ways. What is beautiful to one person may be ugly to another, particularly if they come from different cultures. And what produces aesthetic enjoyment to one individual, may produce hay fever in another'! The sensory pleasure associated with a bottle of quality cold drink may be difficult for the patient/individual who suffers from allergies.

In answer to these questions, curricula developers in the part have opted for a more conservative, accepted view of what is beautiful and what is enjoyment. In more recent times, this position has weakened and we have witnessed the emergence of more "popular" aesthetics within the school curriculum.

For the curriculum developers the value of philosophical considerations is abundantly clear. Ontology, epistemology and axiology provide a useful structure for examining one's own philosophical position as well as how philosophical stances affect the development of curricula. In these considerations, one might pose some typical philosophical questions that are useful to curriculum developer. The questions are:

- On what grounds should contents be selected or rejected?
- How different is instruction from conditioning?
- Are there distinct forms of knowledge?
- How can specific curricula be justified?
- How should content be structured within a curriculum?
- Should a curriculum be differentiated for different students?
- What is fact?
- To what degree should "new" reality be included within the curriculum?

2. PSYCHOLOGICAL FOUNDATIONS

The word psychology is derived from two Greek words “psyche” means soul and “logos” means study. Psychology is the scientific study: of human behaviour.

2.1. Role of Psychology in Curriculum Development

Psychology attempts to describe, explain and predict human behaviour. Psychology gives us an insight into the child's development and learning and provides various techniques of inquiry for use in the curriculum area.

The contribution of psychological basis to curriculum is significant and is growing. As this is a relatively young discipline, the scope for applying its concepts, principles, processes and values to curriculum development is gradually increasing.

2.2. Psychological Sources

The purpose of psychology is the study of human behaviour. The psychologists are concerned with:

Describing
Explaining
Predicting

Evaluating (investigating) the behaviour of human being, curriculum developers, therefore, can draw upon psychology, particularly educational psychology, for at least five areas of information.

(1) Educational objectives
(2) Student characteristics
(3) Learning process
(4) Teaching methods
(5) Evaluation procedures

The study of psychology does not provide a source of contents a school curriculum (other than for a few subjects on psychological studies).

Let us briefly examine the psychological sources that the curriculum developers can employ.

2.2.1. Educational objectives

Knowledge of the psychology of learning helps the curriculum developers to devise and phrase appropriate goals and objectives. The curriculum developers can determine whether goals and objectives are suitable for various developmental levels and

ages of learners and that whether they are attainable or not. Subsequently, the formulation of curriculum goals and objectives has profound influence upon the selection of contents for the curriculum.

2.2.2. Student/Learner Characteristics

An understanding of the nature of learners particularly of individual differences and personalities will assist, the curriculum developers to-make more choices in curriculum decision-making. The study of personality can tell us whether different personalities respond to learning experiences in different ways, Indeed, this is something the experienced teacher has long known and some teachers have, endeavoured to accommodate these differences within their classrooms. Similarly, an understanding of individual differences is most useful to the curriculum developers, An effective curriculum is able to accommodate .differences in student skills and abilities.

2.2.3. Learning Processes

Perhaps the greatest contribution that psychology makes to curriculum is an understanding of how people learn. The curriculum developers, who have a sound grasp of learning and learning theory, are in a commanding position to devise an appropriate curriculum for learners. In particular, an understanding of learning is essential to the effective selection of appropriate learning/teaching strategies.

Whether or not one supports a theory- of operant conditioning such as that of B.F. Skinner, some forms of Gestalt theory (K. Lewin). Jean Piaget's approach to growth and development, or some other form of explaining how learning occurs, the final outcome speaks how the curriculum is shaped. Indeed, one of the difficulties encountered by curriculum developers is the vast array of theories, paradigms and algorithms that support to explain the process of learning.

2.2.4. Teaching Methods

Psychology makes a significant contribution to both the selection of learning experiences and the way teaching is conducted in the classroom. In the school curriculum an understanding of psychology is essential to the curriculum developers in devising appropriate learning experiences and conditions for learning, In selecting learning experiences, the curriculum developers should take an account of: -

- Learning theories
- Individual differences amongst students
- Motivational strategies
- Personality
- Cognitive and affective development
- Teaching style

- Group dynamics
- Teaching methodology and
- Learning styles

This extensive list of psychological factors suggests that, the curriculum developers can make substantial use of psychological sources while selecting learning/teaching experiences.

2.2.5. Evaluation Procedures

Psychology can also provide curriculum developers with directions for undertaking the evaluation of students and teacher's performance. Educational psychologists have developed a vast array of techniques for measuring the degree of students learning, students' attitudes towards learning/ teaching and so forth, as well as the extent of teacher's effectiveness.

Educational psychologists, as well as other educators, have been concerned with such evaluation issues as:

- Norm-referenced assessment or criterion referenced assessment
- The role of formative evaluation
- Appropriate instruments to measure students' performance.
- Determination of teacher's effectiveness.

These aspects of psychology indicate the range of influence that psychology has upon the development of curriculum.

2.2.6. Human Growth and Development

Knowledge about the growth and development of the child has a great bearing on what to teach at a given level. Learning outcomes have to be determined with reference to the characteristic thought forms at the various age-levels, with a view to orienting curriculum to child/learners needs and capacities. We must consider the functioning of intelligence and development of capacities. The processes of human development and nature of learning have special significance for curriculum development.

The relationship between psychological foundations and the curriculum are given below:

- Curriculum to be child-centered, must take into account the psychological make-up the learners i.e. nervous system, has a great bearing on the curriculum development for different age groups.
- Learning experiences should be provided according to the mental development of the learner. On this account, learners are divided into ability grouping.
- The effectiveness of the curriculum depends on the interests of the learners. So the curriculum planning, must take into account the interests

of the learners.

Human growth and development are very important elements in curriculum development. Curriculum decisions are not to be made arbitrarily but on the factors determining individual's growth and development. A fixed and rigid curriculum is hardly suitable as it fails to accommodate the needs of slow learners, late starters as well fast learner.

School curriculum on the whole, should aim at enabling the learners to acquire knowledge, develop concepts and inculcate skills, attitudes, values and habits conductive to the all-round development of their personality and commensurate with the social, cultural, economic and environmental realities at national and international levels.

3. SOCIOLOGICAL FOUNDATIONS

Curriculum must take into account the sociological considerations otherwise it will remain bookish and divorced from life. A curriculum that ignores sociological foundations does not serve any purpose. It results in waste of time, energy and resources. It will produce individuals, who cannot play their role effectively as enlightened members of a society.,

A sound curriculum must be based on the needs and aspirations of a society. An unrelated curriculum may lead to individuals, who can neither find employment nor engage themselves in fruitful occupations and consequently remain dissatisfied, maladjusted and frustrated.

3.1. Etymological Meaning of Sociology

Sociology as defined in dictionaries is "the science or study of society",

The term was coined by COMTE (1830) linking the Latin "socius" (originally a people, tribe or city allied to Rome, but later a society) to the Greek "logos" (study). The term spread rapidly and is now used in virtually all languages to denote any real rigorous, reasoned study of society.

3.2. Definitions of Sociology

"Sociology is the systematic study of the groups and societies, human beings build and the way these alliances effects our behaviour".

"Sociology is the study of social life and social causes and consequences of human behaviour".

"Social life" encompasses all interpersonal relationships. The "causes and consequences of human behaviour encompasses how these relationships,' groups, and organizations are inter-related and how they influence personal and interpersonal behaviour.

"Sociology is the study of social relationships, social institutions and society".

3.3. Role of Sociology in Curriculum Development

According to sociologists, schools are social institutions especially set up for the preservation of culture and transmission of culture by society. School seeks to discharge this function through the curriculum. The curriculum, therefore, includes learning experiences based on the ways of life, kinds of knowledge, attitudes and beliefs that are considered important by the society. Some kinds of selection are needed, as all the aspects of culture cannot be included in the curriculum planning thus becomes a way for the selection of various elements.

There are various sociological foundations like cultural, economic and political that deeply influence the school curriculum, its conception, content and organization.

Among the most important sociological considerations that should guide those engaged in the task of curriculum development, may be listed as below:

- Core values of society
- Changing values of the people
- Demands of modernization
- Criterion of a good family life
- New forms of cooperation
- Media explosion
- Population explosion
- Regional and national imbalances
- Economic efficiency
- Education for fellowship and leadership creative and purposeful activities

3.4. Curriculum for Modernizing the Society

The curriculum for modernizing the society stressed the following:

- Restructuring contents of the various subjects in the light of modern development in science and technology.
- Adopting new methods of teaching.
- Encouraging activities for awakening curiosity and developmental interests, attitudes and values and the building up of such essential skills as independent study and capacity to think and judge for oneself.

3.4.1. Culture

Culture defines an accepted way of life. One implication of this statement, of course, is that the "accepted" way of life is the preferred way that is the "accepted" way of life has more "value" then other alternative ways.

We can see that culture is a "value loaded" enterprise, so to speak. It provides the members of society with the "goods" and the "bads", the "beautifuls" and the "uglies," the "shoulds" and the "should nots".

3.4.2. Society and Culture

A society is a collection of individuals, who have organized themselves into a distinct group, to be a society, however, a distinct group and not just a collection of individuals, the members of the group must perceive themselves as "having things in common", which enable them to "belong". These "things in common" are the stuff of which culture is made. Culture, then, may roughly be viewed as a kind of social cement

that consists of the characteristics, habits, ideals, attitudes, beliefs and ways of thinking of a particular group of people. Even from these sketchy definitions, it is clear that while society and culture is certainly not the same thing, "without a culture there could be no society, and without a society there could be no culture".

(Smith, Stanley, and Shores 1957)

3.4.3. Complexity of the Concept "Culture"

Culture is a highly complex concept that requires a great deal of considerations and study then we are able to allow here; it is similar to such concepts as "democracy" "morality" and " love", which have a multiplicity of meanings.

In relatively broad term we might say that culture defines an accepted way of life. It includes a vast array of easily observed facets of living such as material products, political and social organizations, characteristic vocations, modes of dress, fads, foods, games, music, child bearing and rearing practices, and religious and patriotic rituals. .

3.4.4. The Structure of Culture

A structural framework was proposed for the purpose of facilitating the Study of curriculum. This theoretical model was based upon a classification of the total curricular phenomenon into a complex of eight internal factors, which are:

- Epistemology
- Society/Culture
- The individual
- Learning theory
- Aims
- Contents
- Learning activities
- Evaluation

In much the same way, the study of culture will be facilitated if we are able to proceed upon the basis of a hypothetical structural framework. One useful framework, proposed by Ralph Linton (1936), is presented here because of its simplicity and its congruence with the theoretical constituents of society and culture that we have been developing.

Linton has proposed that all elements of culture can be classified into three principal categories.

- The universals
- The specialties
- The alternatives

The Universals

The universals comprise those values, beliefs and customs that are generally held by the entire adult population. For example, in a wide variety of instances, behayiour in such areas as language, food, religion and economics tends rather circumscribed in our society.

The Specialties

The second category identified by Linton, includes those elements of the culture to be found only within sub-groups of the society. Among the most common of these are the vocational subgroups; in our society certain behaviour is expected of professors, for example, that be quite different from those expected of businessmen. Thus, pr6fessors are expected to be shabbily dressed: thinkers leftist in their politics, absentminded, and impractical in worldly affairs; businessmen, by contrast, tend to be viewed as smartly dressed doers, conservative in their politics, mentally alert, well organized, efficient, and practical. .

The Alternatives

The alternatives are those beliefs and practices that violate culturally accepted norms (universals and specialties) in their attempt to fulfill a need, solve a problem, or simply to allow a more congruent perception of reality. Alternatives are like specialties; however, all members of the society may not share them. Unlike specialties, however, any sub-group may not share them. A simple, tangible example of an alternative might be the introduction of pizza in place of the traditional food as an afternoon (lunch) or dinner.

Since curriculum workers and teachers are in the business of intervening. in the lives of young people for the purpose of making them something that they would not otherwise become, it is necessary not only that they have full and conscious knowledge of cultural universals and specialties, but that they evaluate and compare them with all manners of alternatives. Current practice, however, consists mainly of accepting cultural directives and transmitting them through curriculum as efficient as possible. This procedure, of course, constitutes little more than mindless indoctrination. To educate, on the other hand, calls for a curriculum, that promotes the illumination, examination, and evaluation of cultural universal and specialties in the light of projected desirable alternatives. As we shall see in the following sections, breaking out of the cultural trap is a long, demanding process requiring substantial quantities of both wisdom and courage.

3.5. Values Based Curriculum:

Like the concept of “culture”, value is a misleading complex idea/opinion.

Following types of experiences and activities may be planned for developing desirable values among pupils:

- Exemplary behaviour of teachers.
- Value formation through various types of co-curricular activities i.e., student participation in school management, social service programmes, labour Weeks, visits to hospitals, etc.
- Lectures or discourses.
- Creation of an environment of psychological safety and security for the students in the school.
- Value oriented content in various subjects studied in school.

Classification of Values

Values have been classified in a number of ways and their meanings also vary:

Instrumental Value

A subject is said to have instrumental value when it is pursued, not for its own sake, but for some ends beyond itself. Instrumental values include preparatory or introductory, practical or utilitarian, socializing and conventional values.

Preparatory Value

A subject is said to have a preparatory value when it prepares the way for other studies. Arithmetic prepares the way for Algebra.

Introductory Value

Since a particular study introduces us to a number of subjects, it has an introductory value. It is identical with the preparatory value. To illustrate we may select physical Geography which introduces us to a little of Botany, Zoology, Physics and Chemistry.

Practical or Utilitarian Value

These values lie in a study of subject whose pursuit is individually and directly useful as it satisfies many wants and needs. The subject is individually and directly practical when the knowledge of the subject is applied directly by the person himself. A person may use his knowledge of hygiene to avoid smallpox. The same knowledge can, however, be used through other agencies, through society, for example. A subject is said to possess a socializing value when it creates socially desirable habits and reactions, or when it enables us to understand society in its complexities. The socializing value includes moral values too. These moral values refer to certain moral habits that society wants individuals to develop through education.

Conventional Value

Conventional values implies to a value, which is customarily described and desirable in a subject. Certain subjects may be studied because their knowledge is expected of people of a certain class. Their knowledge is indeed, for proper social enjoyment. In the time of Locke, a gentleman was expected to learn Latin, Greek and German because they were necessary in that social set up.

Intrinsic Value

Dewey has described these as the appreciative studies. The resulting experiences in these subjects are worthwhile on their account. The values that occur from their studies are in the form of pleasures or intellectual joys.

Liberalizing and Sentimental Values

The liberalizing values consist of the pleasure that accompanies intellectual insight. The sentimental value, on the other hand, refers to pleasure that emerges when our feelings are exercised. The aesthetics, the comics, the social and the moral situations may arouse our feelings. Drama, literature, music, painting, sculpture and religion have, in varying degrees, these values.

Essential Values

These values involve the basic nature of man himself and include elementary physical skills, basic social skills, ability to use symbols such as language and numbers.

Personal Values

These values make a person good for himself.

Social Values

These values are good for the society and form the basis of relationship of an individual with other people in society.

Institutional Values

These include values established by institutions.

Global Values

These values are determined completely outside the schools.

Values may differ from place to place and time to time but values like truth and love remain constant.

3.6. Influences of Society and Culture on Curriculum

The social and cultural influences that affect curriculum developers are evident in both conscious and unconscious ways and their impact is certainly profound. Education manifests through the curriculum and reflects society and culture, that reflection is a result of curriculum developers being an integral part ‘‘of that society and culture in both of the above ways. In this way the curriculum more ‘‘reflects’’ society and than society leads to change.

Indirectly society and culture influence curriculum developers simply because they are members of a particular society, cultural values, attitudes and beliefs are acquired by individuals unaware of that process, yet, once acquired, these cultural traits become consolidated and affect our social behaviour. And when the process of curriculum development takes place, the cultural traits within developers influence the very selection of objectives, contents, methods and evaluation that constitute the curriculum they are devising.

Take, for example, a group of primary school teachers, who decide to, enhance the literacy component of an existing curriculum on completion one could analyze the result to determine why they undertook the task, what objectives were formulated, what content was employed, how it was taught and how it was evaluated. In many instances, when probed deeply enough, the teachers would be unaware of the basis of their decisions, if story reading were a component of the revised curriculum, what proportion would be of oral reading? Why? And how would that be assessed? What stories were selected for students? Why? These are typical of the questions that should be asked of curriculum developers when they construct curricula, and which reveal indirect influences of society and culture.

Alternatively, curriculum developers may be well aware of social and cultural influences and have the deliberate intention in mind (or not) of reproducing aspects of that culture m the curriculum.

3.7. Culturally Induced Bias and the Curriculum

One particular aspect of the social and cultural influences on the curriculum, which deserves specific attention, is that of culturally induced bias. As societies perpetuate themselves through implementing values in the young through institutions such as schools, it is distinctly probable that some of these values will be culturally biased. Indeed, these values may be so effectively integrated within schools and society

that they are perceived not as biased but as accepted components of the very fabric of society.

It can be argued that until recently the perception of the traditional occupational role for women was that, of child rearing and domestic duties. A small range of their typical occupations such as nursing, teaching, secretarial duties and so forth were also condoned with society. But to imagine, even before 30 years, women lawyers, engineers, politicians, pilots, judges and senior business executives would have been almost unthinkable.

Today the former view is perceived largely as ludicrous. But to achieve this change in values and attitudes many barriers have had to be surmounted, not the least being pervasively held sex-stereotyped beliefs. These beliefs were so tightly woven into the fabric of society that they were perceived as natural and essential. In changing these stereotypes, the school curriculum was seen as an important vehicle in promoting and consolidating the new values and attitudes.

In fact, it can be seen that social and cultural forces have a profound effect upon the curriculum in both direct and indirect ways. Curriculum developers whether at systemic, local or school level within educational enterprise, should not forget that they are a product of their culture and that every decision that they make will be culturally related.

Lastly, curriculum developers serve the function of translating traditional ideas, assumptions, values, knowledge and attitudes into curriculum objectives, contents, learning activities and evaluation. Of these curriculum elements sociological sources have their greatest impact on contents.

Thus it is not possible to talk about a culture free curriculum.

4. SUMMARY

The word curriculum comes from the Latin work *"currere"* which means. "to run", it is a runway, a course on which one runs to reach a goal.

Curriculum foundations are those forces that influence and shape the minds of curriculum developers and enhance the content and structure of the subsequent curriculum. They influence developer's thinking about curriculum.

Three foundations of curriculum are: philosophical foundations, psychological foundations and sociological foundations.

Philosophy is the pursuit of wisdom and knowledge, the study of realities and general principles. It concerns with the search of internal truths. Philosophies foundations give understanding of nature of educational objectives, structure or interrelatedness of objectives, nature of curriculum activities and the structure of curriculum plan.

Philosophical categories have particular relevance for curriculum development and "these categories include: Ontology (the nature of reality). Epistemology (the nature of knowledge) and Axiology (the nature of value).

The word psychology is derived from two Greek words *"psyche"* (soul) and *"logos"* (study). Therefore, psychology is the study of Human behaviour. Psychological foundations give us an insight into child development and learning and provide various techniques of inquiry for use in the curriculum area.

The particular areas of information in psychology includes; educational objectives, studies characteristics, learning processes, teaching methods and evaluation procedures. .

Knowledge about growth and development of the child has a great bearing as on what to teach at a given level. The process of human growth, development and nature of learning have special significance for curriculum development.

The word sociology is derived from Latin word *"socins"* (society) and *"logos"* (study). Sociology is the systematic study of social relationships, social institutions and society. Sociological foundations deeply influence the school curriculum, its concepts, content and organization.

Sociological considerations engaged in the task of curriculum development include: core values of society, changing values of people, demands of modernisation, criterion of a good family life, democratic temper of the society, new forms of cooperation, media explosion, population explosion, regional and national imbalance and economic efficiency.

Curriculum for modernising society includes restructured contents of various subjects in the light of modern development in science and technology, adopting new teaching methods and encouraging activities for awakening curiosity, developmental interests, attitudes, values and essential skills.

Culture is a “value loaded” enterprise. It provides the members of the society with the “goods” and the “bads”, the “beautifuls” and the “uglies”, the “shoulds” and the “should nots”.

Society and culture are certainly not the same things while, “without a culture there could be no society and without society there could be no culture”.

Litton has proposed that all elements of culture can be classified into three principal categories: universals, specialties and alternatives.

Society and culture influence curriculum developer because they are members of a particular society. When the process of curriculum development takes place, the cultural traits influence developers in selection of objective, content, methods and evaluation procedure.

In fact, it can be seen that social and cultural forces have a profound effects upon curriculum.

5. SELF-ASSESSMENT QUESTIONS

Q. No.1: Define Curriculum?

Q. No.2: Define curriculum foundations?

Q. No.3: Enlist three terms in which curriculum foundation can be identified. '

Q. No.4: What is the role of curriculum foundations in curriculum development?

Q.No.5: What is the relationship of philosophy and curriculum?

Q.No.6: Write components of philosophical foundations, which give understanding in curriculum development?

Q. No.7: Enlist philosophical categories that have particular relevance for curriculum development?

Q. No.8: Differentiate between the old and new concepts of psychology?

Q. No.9: What is the role of psychology in curriculum development?

Q. No.10: How is the growth and development related to psychological foundations of curriculum?

Q. No.11: How does sociology affect curriculum development?

Q. No.12: Enlist the types of values needed for curriculum?

Q. No.13: How does culture and society influence curriculum?

Q.No.14: What are the culturally induced biases and how do they affect curriculum?

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UNIT–5

CURRICULUM DESIGN

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INTRODUCTION

Curriculum planning and design are essential preliminary ingredients in the curriculum development process. This Unit examines the concepts of curriculum planning and design, as to how they function as an integral part of the curriculum development process and what this means to curriculum developers. It also examines the nature of the design process and considers the principal forms or types of common place curriculum designs found in schools and educational systems. It then suggests some ways by which curriculum designs are created and lastly addresses the important issue of curriculum planning.

Before the detailed construction of a curriculum document occurs, curriculum developers are involved in planning and designing their proposed curriculum. Indeed, it is not possible to develop a curriculum without some form of curriculum design, though some curriculum developers address this issue more systematically and substantively-than others. Moreover, while curriculum planning and design may be essentially conceptual in nature, they may be quite overt acts of groups of curriculum developers who, together, resolve issues relating to the design and planning of the proposed curriculum.

This unit argues that a well-conceptualized approach to the process of curriculum development is an effective use of resources and hence suggests that time will be profitably spent on substantive curriculum planning and design. As Karen Zumwalt suggests, give the view that professional teachers should have the knowledge to enable them to create sound educational programmes, it is essential that teachers have knowledge of some planning process that enables them to think about curriculum beyond the individual lesson.

In the unit, concept of curriculum design, categories of curriculum designs, conceptual framework of curriculum design and its functions, critical problems of curriculum design, criteria for a desirable curriculum, major patterns of curriculum designs, different steps in curriculum design and benefits of curriculum design have been elaborated. Besides, questions likely to arise with regard to curriculum design have also been given to provoke the students to study the subject matter deeply.

OBJECTIVES

After studying this Unit intensively, the students may be able to:

1. Define the curriculum design.
2. Explain categories of curriculum design.
3. Explain the functions of conceptual framework for curriculum design.
4. Explain critical problems of curriculum design.
5. Explain the criteria for a desirable curriculum design.
6. Explain the major patterns of curriculum design.
7. Describe the steps in curriculum design.
8. Describe the benefits of curriculum design.

1. CONCEPT OF CURRICULUM DESIGN

(i) A curriculum design is a framework or plan of action for preparing a course of study or a set of students' experiences. It is a deliberate process of devising, planning and selecting the elements, techniques and procedures of curriculum. Curriculum design is a method of thinking.

(ii) A curriculum design is the result of curriculum decision.

(iii) Curriculum design has been explained in the "Dictionary of Education" (1982) edited by PJ. Hills as, "a curriculum design is a set of abstract relationships embodied in the materials and learning activities of a course in use. The critical variables of subject matter, pupils, teachers, and milieu are brought the way together. Thus, the progress of curriculum design is distinct from the broader social process, extending over time of curriculum planning. The feasibility and educative potential of a finished curriculum design can be relationally analysed, but its formation or synthesis is an art".

(iv) Curriculum design may be defined as the substance and organization of goal and culture content so arranged as to reveal potential progression through levels of schooling.

(v) Curriculum design is a statement, which identifies the elements of the curriculum, states their relationships to each other, and indicates the principles of organization and the requirement of the organization for the administrative conditions under which it is to operate.

5.1.1. Categories of Curriculum Designs

Over the years, it has been possible to distil a number of different approaches to curriculum design. Broadly, they may be classified into four groups, though different curriculum writers suggest differing patterns of designs (Smith, Stanley & Shores, 1950; Taba, 1962; Saylor, Alexander & Lewis, 1981; McNeil, 1985; Gress & Purpel, 1988; Murray Print, 1988; Wiles & Bondi, 1989). The four recognized groups of designs are subject-centred designs, learner-centred designs, problem-centred designs and core designs. Different curriculum designs largely reflect differences in the organization and consequential focus of subject matter.

School curricula designs have been distinguished because of how the subject matter is delivered (instructional emphasis) while most emphasize what subject matter is delivered (curriculum emphasis). In this unit, the curriculum approach has been emphasized. Indeed, this is largely the most common and fruitful way of conceptualising curriculum design (Smith, Stanley & Shores, 1950; Taba, 1962; Saylor, Alexander & Lewis, 1981; Print, 1988).

Thus, curriculum developers, drawing upon their personal experiences; their preferred conception of curriculum and their understanding of curriculum drawn from the

curriculum foundations, have constructed curricula according to designs, which may be categorized as:

1. Subject-centred designs
2. Learner-centred designs
3. Problem-centred designs
4. Core designs.

1. Subject Centred Designs

This group of designs revolve around the teaching of an established body of content that has been derived from the accumulated wisdom of the academic disciplines. Content considerations are the basis for decisions about the horizontal and vertical dimensions of the curriculum while other curriculum elements have considerably less impact. Three related, though distinct, designs have emerged.

(A) Subject Designs

This is probably the oldest and most widely used form of curriculum organization found in schools and educational systems, enjoying a long tradition at least in the Western world. In essence, this design is based on the classification and organization of subject matter into discrete groups, which we have called subjects. These groupings, which have become known as school subjects, were initially based on evolving divisions of labour in research that produced physics, history, literature, geography, mathematics, and so forth. In more recent times practical areas such as typing, home economics and industrial arts have become accepted as subjects.

Emphasis in subject designs is placed on the acquisition of subject matter knowledge, and content is structured sequentially, such as in mathematics or history. The subject approach is commonly employed in schools, particularly at secondary levels, though not without some concern. Because these subjects have been studied in isolation from each other, a factor reflecting the development of the subject design, it has been extensively criticized over the years.

(B) Academic Disciplines Design

This approach to organizing curriculum is essentially a post-Second World War phenomenon, gaining greatest support in the inherent organization of content, as is the subject design, the academic disciplines design emphasizes the role played by those distinct entities called academic disciplines. These can, it is claimed, be clearly delineated in terms of knowledge, skills and values.

Thus, in a school setting, the content of this design would focus on what an academician does, that is how a biologist, historian or a mathematician actually works at that discipline. That is, how the disciplinarian thinks, what research is done, how that research

is carried out, how data are analysed, how research is reported, and so forth. The result, it is hoped, is that the schools would produce mini versions of academic disciplinarians.

To assist students to acquire the knowledge, skills and values of the academic disciplines, numerous curriculum projects were developed. Among the more well know are:

(1) Biological Sciences Curriculum Study
(2) Social Studies
(3) Pakistan Studies
(4) Pakistan Science Education
(5) School Mathematics Study
(6) Individualised Science Instruction System

(C) Broad Fields Design

This third design was developed to overcome a perceived weakness in the subject design that was evident in the nineteenth and early twentieth centuries. Essentially, the broad fields design attempts to rectify the fragmentation and compartmentalisation evident in the subject design by combing two or more related subjects into a single broader and integrated field of study.

Furthermore, the broad fields design was deemed more suited to younger learners and to those less able to cope with the rigours of subjects and academic disciplines. Through the process of subject amalgamation; not only were subjects to be made more integrated, but also pedagogically modified to suit the needs of learners.

The broad fields design is commonly found in primary and lower-secondary schools. Attempts to introduce broad field subjects into higher-secondary schools have been resisted largely because they are inadequately prepared for tertiary education. With the changing nature of higher-secondary school enrolments in recent years, we are witnessing some change to this resistance. The vocational training system may well produce a modified version of broad field subjects based on the concept of 'key competencies', though strictly speaking these competencies are an instance of a core curriculum design. What we do not know at this stage is what form the competencies will take when implemented into the school curriculum.

Features common to the three subject-centred designs include:

(1) Classification and organization of all contents into subjects or subject-like groupings.

(2) Subjects are clearly defined and distinguished.

(3) A hierarchy of subjects is commonly found according to their perceived value.

(4) Methodology applied and practiced is largely teacher-centred and expository in nature.

2. Learner-Centred Designs

Supporters of these curriculum designs generally view society in democratic terms and perceive individuals as being 'naturally good'. Hence, learner-centred designs emphasize individual development and their approach to organizing the curriculum emerges from needs, interest and purposes of students. As a result, there are two essential differences between this and the subject-centred designs.

Learner-centred designs take their organizational cues from individual students' needs rather than a body of subject matter.

Consequently, learner-centred designs are usually not as pre-planned as other designs, for they evolve from teacher-student interaction in relation to learning tasks. In some cases, the curriculum may have no preplanning at all and may emerge as a group of students reveal their concerns, interests and needs. More commonly, however, curriculum developers have purposeful intentions, which are learner oriented and these may be modified later upon interaction with students.

(A) Activity/Experience Design

This approach is based on determining the genuine needs and interests of learners, which in turn form the basis of the curriculum (Smith, Stanley & Shores. 1950). An important claim of this approach is that 'people only learn what they experience.... learning in its true sense is an active transaction' (Taba, 1962; 401). Consequently, it is very difficult to pre-plan, although with experience certain trends emerge. An important role for the teachers, therefore, is to develop the ability to ascertain genuine students' interests and then create an appropriate curriculum around them. To do so implies competence with the curriculum development process.

Other features of this design include a lack of formal subjects, joint planning of learning experiences between curriculum developers and students and, finally, skills and knowledge taught as the individual requires or needs them. It should be noted that this approach to organizing curricula is found mostly in the literature, with few examples in reality. In some alternative schools, such as summer hill, examples of activity/experience curriculum may be found and most preschools/kindergartens use the activity design as a basis for their curriculum.

(B) Humanistic Design

Similar in approach to the experience design, the humanistic design emphasizes the meeting of individual's needs in a conductive, supportive learning environment. The humanistic approach to curriculum design may well incorporate all the features of the experience design with the additional factor providing a supportive environment for the individual learner. Alternatively, it may offer some subject structure as a guide to learners.

The design has emerged from the humanistic conception of curriculum, which seeks to provide learners with intrinsically rewarding experiences for self-development, that is, to enhance an individual's self-concept through a supportive learning experience. As a further guide, the writings of Carl Rogers, Arthur Combs, John Holt, Mario Fantini, Jonathon Kqzol and C.H. Patterson should be consulted.

Alternative schools, particularly of the open variety, provide examples of the humanistic curriculum design in operation. You may be able to identify specific examples of schools that have a humanistic design as the basis of their curriculum or have components of a humanistic curriculum within a broader curriculum design.

The essential difference between the two designs is that the curriculum developer, in the humanistic designs, has some preconceived views, based on the intents of humanistic psychology, on what is of value to learners and that those ideas are integrated into a curriculum based upon the learner's needs. The intent of this design then is to deliberately provide a curriculum, which purposively reflects those supportive features from humanistic psychology such as caring, support, enhanced self-concept and the like.

3. Problem- Centred Design

These curriculum designs direct learners to focus their attention on and attempt to resolve, problems of living that are both individual and social in nature. As this is such an enormous task, considerable variations in the nature of problems to be studied may be found. Themes might include persistent life situations contemporary social problems, and personal concerns of youth, major social functions and worldwide problems such as peace and environment.

Generally a pre planned approach, the problem-centred curriculum design seek to provide learning's with opportunities to resolve problems they are likely to encounter or already are, in their life. Unlike the learner-centred designs, emphasis is placed on group welfare, group activities and the group solution of problems. At least two variations of problem-centred design may be found in the literature on curriculum design.

(A) Thematic Design

The argument of thematic designs is based on the artificiality of other curriculum designs in providing appropriate learning experiences, which reflect what we experience in real life. A curriculum design should, it is suggested, reflect the types of experience we encounter in our lives in order to be useful and meaningful. Designs based on non-realistic bases, such as subject, academic disciplines, broad fields and so forth, are remote from reality and from student's perceptions of what they need to learn A useful thematic approach, by contract, can meet those needs.

Thematic approaches to curriculum design are employed commonly preschools and primary schools. In many primary schools teachers have deliberately programmed their curriculum across subjects in order to produce a more useful thematic approach.

(B) Problem Design:

This Curriculum approach argues that learners should, encounter substantial real life problems in order to understand the world. Like the thematic approach, the problem-centred design argues for the more meaningful basis to curriculum design in order to involve learners in what they perceive, as relevant. However, distinct emphasis is placed upon the concepts of identifying, addressing and resolving multifarious problems. Through this process, it is argued: students will obtain 'meaningful learning and be able to play a more purposeful role in society.

Here the curriculum would be designed so that students might study problems experienced by people that are of both an individual and group nature. The emphasis- in this curriculum design is upon resolving the problem to the satisfaction of learners. Examples of typical problems would be:

(1) What can we do about air pollution in our cities?

(2) How can adolescents live harmoniously at home?

(3) Should a new freeway be constructed to solve the traffic problem?

(4) Can the Kashmir issue be resolved?

(5) How far should genetic research be allowed to proceed?

(6) How can nuclear waste be stored effectively?

(7) What should one aspire to in life?

(8) Where should a new airport be located in our city/town?

This curriculum design appears to be most appropriate to the various life skills curricula that are now emerging certainly those students, for whom the traditional academic curriculum designs are inappropriate, would benefit from a problem-oriented-curriculum. This situation is particularly applicable to the increasing numbers of non-.tertiary-bound students remaining in higher-secondary schools.

4. Core Learning Deigns:

Many forms of core learning design may be distinguished and there is evidence of this design having been used for some time (Smith. Stanley & Shores1950: Zais. 1976, CDC, 1980, Kirk, 1986. Marsh & Stafford 1988).

Core Design

The notion behind a core design, usually called a core curriculum, is that there exists a set of common learning's (knowledge, skills and values) that should be provided to all learners in order to function effectively in a society. The core concept, however, does vary considerably in interpretation and one writer has suggested that it is possible to distinguish no less than six forms of the core design (Zais. 1976; 42-3). For our purposes, it is sufficient to understand that a curriculum may be organized around the idea of a core as a set of learnings essential for all students.

The key issues in any core design for a curriculum are:

(1) What should be included in the core?

(2) How large should be core be, i.e. what percentage of the total contents within the written curriculum should be?

(3) What should be excluded from the core?

(4) Is a core required of all learners?

The notion of a core design is usually referred to a small, separate group of subjects that are compulsory to study, while the curriculum is topped with options or elective subjects. Typically, the 'core' consisted of English, mathematics, science, social studies (or equivalent subjects such as history and geography) and physical education for these were the subjects studied by all students. To these were then added a number of optional subjects such as fine art, home economics, music, languages, and so forth.

The emphasis of this approach to core curriculum was that all students would experience a set of common and essential learnings that were necessary for learners to function effectively in society.

5.1.2. The Functions of a Conceptual Framework for Curriculum Design

Generally speaking, a conceptual system for the curriculum or a theory of curriculum is a way of organized thinking about all matters that are important to curriculum development: what the curriculum consists of, what its important decisions are, and how the information and criteria from these sources are translated into curriculum decisions?

Only a few scattered remarks are available in the curriculum literature about the functions of a curriculum theory or a conceptual framework in designing curriculum. Herrick uses the term "curriculum theory" and suggests three chief functions of such a theory:

1. T identify the critical issues or points in curriculum, development, and the generalisations, which underlie them.

2. To point out the relationship, which exists between these critical points and their supporting structure.

3. To suggest the approaches that need to be made to resolve these critical issues.

(Tylel and Herrkk, 1950, p.!)

Goodlad uses the term “conceptual system” and indicates that this system must identify the methods of inquiry by which to answer some basic questions about curriculum development (1988, pp. 391-396). This volume has identified certain points at which crucial curriculum decisions are made, such as the choice of objectives, the selection and organization of content and of learning experiences, and evaluation. A theory should help to identify what these points of crucial decisions are, and the bases on which these decisions are made.

To establish a systematic set of issues, with which a theory of curriculum should deal, might yet be premature.

2. CRITICAL PROBLEMS OF CURRICULUM DESIGN

Critical problems of the curriculum design are:

(1) General versus special education design.

(2) Maintaining relationship among curriculum component.

(3) Problems of scope and sequence.

(4) Identification of organizing centres.

(5) The problem of balance.

5.2.1. General Versus Special Education Designs

The initial consideration in curriculum design is the determination of whether the focus is to be on general or special education, or both. We have indicated elsewhere that general and special education are not discrete, that elements of both-occur in all curricula, regardless of their intent. However, it is important for design purposes to establish clearly if the curriculum is chiefly to train for specific skills and knowledge or to educate for community participation and human development, if the former design will tend to be determined mainly which these are most efficiently transmitted to learners, in driver training or film developing for example, curriculum objectives are, relatively speaking, clear and specific enough so that content, learning activities, scope, and even sequence can be developed from, regularised, explicit procedures. The design, in fact, reflects a conceptualisation based on logic emperical and technical considerations; In contrast, a design intended to emphasize general education must take into account such factors as cultural norms, human ideals, ethics, and life style. These value issues are so deep-rooted and complex that a general education design can property he articulated only after long and difficult inquiry into substantive foundational issues. Even after such at;' inquiry, problems of scope, sequence, and integration arc far from easily handled, in short, the fact that designs for general educational represent commitments to preferred life styles needs constantly to be kept in the foreground so that the full moral importance of the curriculum is never underestimated.

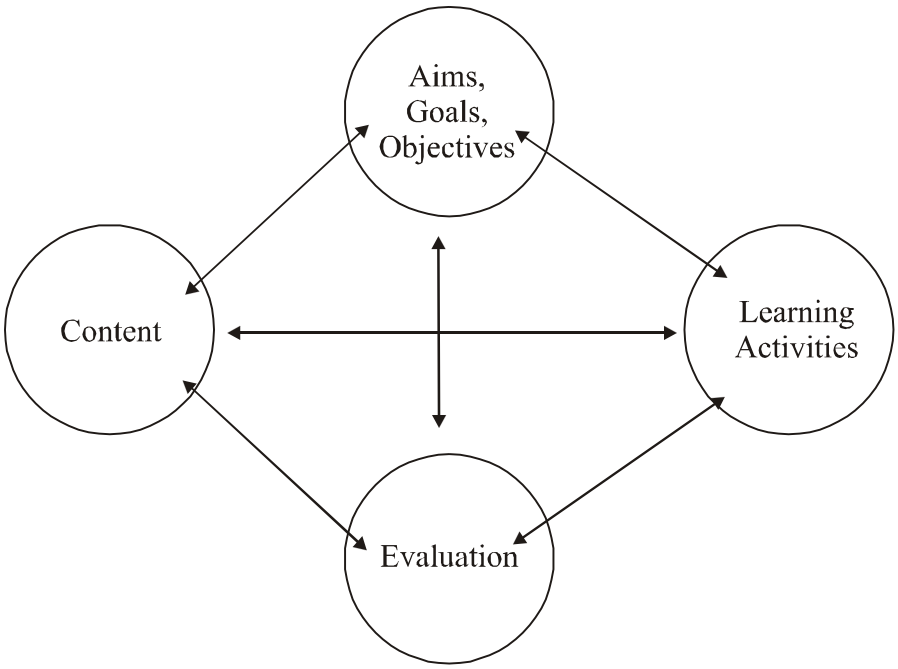
Because of the distinction between general and special education and the n.eed for both, it seems appropriate to suggest that all public school curricula be formulated in terms .of the basic core design. This is not to say that all curricula should have a core pattern on the general education component of a core design but that public school curriculum design should include provisions for a general education core (whatever in design) as well as an appropriate complement of special education alternates.

5.2.2. Maintaining Relationship among Curriculum Components

Another major problem area in curriculum design has to do with maintaining a relationship among the four components of curriculum, aims goals and objectives,

content, learning activities, and evaluation. To begin with a takes a good deal of keen observation and clear thinking just to maintain a good correspondence between long range aims or goals and the more immediate contributory objectives to be stipulated in the design. For example, consider the difficulty of formidating the optional schedule of specific objectives contributing to the attainment of the long-term goal, "ability to employ inductive reasoning in problem solving situation". Second, it is even more difficult to maintain (at the same time) a clear-cut relationship between curriculum goals and content learning activities, and evaluation. For example, given a curriculum goal in literature that calls for the development of empathy and the capacity to identify with human values, problems, and dilemmas, care must be taken to devise a variety of learning activities with potential for attaining the goals and not (as is often done) to overemphasize activities that centre on the intellectual analysis of the literature read. In the same way, the evaluation of students achievements on this literature objective with most likely require quite unconventional devices in place of or in addition to usual paper and pencil test that require only recall of literary material

One way of graphically demonstrating the necessary relationships of curriculum components is suggested in the figure given below. In additions to indicating that each of these curriculum components is related all of the others, it suggested that decisions about any one will be hearing on decisions about the others.



5.2.3. Problems and Scope of Sequence

A third critical problem of curriculum design centres on the two basic dimensions introduced at the beginning of the unit: horizonal and vertical organizations, or, in the

terminology of the curriculum field, scope and sequence. Scope, it will be recalled, refers to the extent and arrangement of curriculum elements that occur at the same time, while sequence describes their progressive, level-to-level organization over a period of time. . ,

The principal design problem associated with sequence is the difficulty of developing cumulative and continuous learning as students move through the curriculum. According to Taba (1962, p.429), this problem has been particularly resistant to solution because sequence has traditionally been considered only in terms of content, with little or no attention being given to the sequence of processes i.e. the skills and competencies that students should development in their interaction with content. For example, in spite of many "logical" arguments, in support of the vertical arrangement of its content, the conventional science sequence of first biology, then chemistry, then physics mandated by most secondary schools makes very little sense in terms of cumulative student development from a process (or even from a content) standpoint. This failure to pay serious attention to sequence in terms of the cumulative development of intellectual and affective process has resulted in innumerable curriculum difficulties, among which many identify poor articulation among the levels of schooling, perennial complaints at each level of poor preparation at the preceding one, misplaced expectations, and generally reduced student development.

The principal design problem associated with scope is the difficulty of integrating the variety of learning that students undergo at a particular level of the curriculum Enough has already been said about fragmentation and the compartmentalisation of learning's from English, mathematics, history, and science, for example to preclude the need for solution in part for the same reason that the problem of cumulative development in sequence has failed to pay sufficient attention to the interrelationship of intellectual and affective processes. While specific, nontransferable skills can certainly be claimed, for each of the separate curricular areas, a broad range of reading, thinking, expression, aesthetic, and interpersonal processes are common to all of them.-These common processes needed to be indentified and utilized as integrating elements to bolster the horizontal coherence of the curriculum.

Another problem associated with scope has to do with the breadth-or-depth dilemma that was raised in the section describing the broad fields design. Broader the scope, the lesser possibility exists for depth. This dilemma, it should be noted, applies whether we are dealing with scope in terms of content or process learning's. It is also related in many significant ways to our basic epistemology and learning theory. But given the chronological framework, in which curriculum must be set it is difficult to avoid consideration of the breadth-or-depth problem of design.

2.4. Identification of Organizing Centres

A fourth problem of curriculum design, i.e. the problem of determining the organizing centres of the curriculum is related to scope and sequence problems insofar as it defines to some extent scope and sequence possibilities. We are already somewhat familiar with a variety of macro design organizing centres as a result of our discussion of

representative curriculum designs. Among others, we should briefly recall the separate subject, the broad field, the learner's need, and the contemporary social problem as typical organizing centres.

Goodlad (1963, pp.25-50) has proposed that two organizing concepts be used to deal with the scope and sequence problems. The first of these is the organizing centre. This may be a book, event, problem, epoch, experiment, or field trip selected to achieve one or more curriculum objectives with a specific group of learners. Either narrow or broad in scope, the organizing centre defines the substance of the learning that is to occur at a given level of schooling.

Goodlad calls his second organizing concept, the organizing element. This is the thread running through a series of organzing centres, "holding them together like beads on a string". An organizing element may be a concept, generalisation, skill, mode of inquiry, or other developmental theme, about which specific organizing centres are longitudinally culstered. For example, the concept of energy might be an organizing element, which is developed through organizing centres on magnets in the primary years, and further expanded through a study of a solar system in high school. It is fair to say that the organizing element controls sequence by guiding the upward progression of students, while the organizing centre defines scope by specifying content, materials, procedures, etc., at each level of the curriculum.

As we have seen earlier in this unit,-determining what the organizing centres of a-curriculum design are to be is a process involving assumptions and beliefs in the foundational areas. Subject, learner, and problem centredness are conclusions that evolve from basic value orientations. Inasmuch as the nature of organizing centres profoundly influences design possibilities, it is most difficult to deal with the technical problem of manipulating centres of organization without prior consideration of their specific nature.

2.5-The Problem of Balance

A .fifth problem of curriculum design is the one that has been alluded to above: the problem of maintaining balance in the curriculum. Balance, of course, means ensuring that .appropriate weight is given to each aspect of the design so that distortions due to overemphasis and under emphasis do not occur. It is noteworthy that most designs are remarkably weak in this respect.

A great deal of controversy in curriculum design has traditionally centred on the question of how much emphasis, respectively, should be accorded to subject matter on the one hand, and learners on the other. This is of course, a question of balance. As we have seen, some curriculum designs are heavily subject centred, while others are fundamentally learner centred, in the former case, emphasis on content usually results, in the conclusion that arithmetic for example, is an important subject that can be learned by first grades therefore, that subject is sympathetically taught, beginning in the first grade.

The result is frequently frustration and failure-with the victims of an inflexible Curriculum being charged with responsibility by reason of stupidity indolence

recalcitrance, or worse. In the latter instance, where primary emphasis is placed on learners, first graders are customarily taught what they are interest in learning and consequently may not be introduced to arithmetic until much, later on. Even then, the subject will probably not be taught systematically, but rather functionally, in accordance with learners needs and interests. The result here is frequently number illiteracy and other basic education inadequacies.

Reasonable consideration of the balance factor in design suggests that neither of these "either-or" positions are very satisfactory, and that renewed; non-ideological inquiry into the problem is long overdue. Some previously discussed "problem-centred" designs, of course, represent efforts in this direction but their practical results have been somewhat less than spectacular. The reason is that balance in this case is not a simple matter, of more or less ernphasis but rather a very complex process involving the integration of two diverse factors; subject matter and learners. A great deal of work still remains to be done in the development of designs that respect the logic and authority of knowledge and at the same time provide for the human development of learners.

Balance is also a factor in the .development of the four curriculum components. For example, subject-centred designs tend to overemphasize content to the, neglect of objectives, learning activities, and evaluation. When objectives are defined, in these designs, they are usually quite narrow, mainly stressing ingestion of content. Also, learning activities centre on .the memorization of information, and evaluation follows suit by measuring only the quantities of data that students retain. In contrast core curricula often overemphasize learning activities but are deficient in the specification of content (Taba.1962. p.422). Objectives and evaluation in these designs also tend to be under emphasized probably because their breadth of scope makes them difficult to handle. The main difference in emphasis between the subject and the core designs, however, probably derives from the former's propensity to view learning in terms, of content, and the latter's tendency to view it as process. We may note, finally, that the fascination with curriculum outcomes the dominates many current "compenrency-based" designs have resulted in curricula that are mainly comprised of objectives and evaluation, with extremely superficial consideration given to the components of content and learning activities.

The distinction between content and process learnings mentioned above represents another point, which requires special attention if balance is to be maintained in curriculum design. As we have noted, earlier in connection with the problem of scope and sequence, traditional designs have tended to overemphasize, content learning to-the neglect of many important intellectual and affective process: Content and process are inextricably linked, of course, but inadvertent, overemphasis of either results in distortions that permeate the entire design structure.

The five problems of curriculum design-described above reflect a writer's conception of the major issues in curriculum organization.

2.6. Criteria for a Desirable Curriculum

The problems of curriculum, change become the criteria for determining the desirability of a curriculum. A list of the problems of curriculum development. recapped and stated as criteria, follows.

The curriculum, that is the pattern of courses and the selected learning activities, must do the following:

(1) Be consistent with the conceptual framework and implement the conceptual framework commitments.
(2) Derive and test its concepts and theories in teaching process
(3) Respond to the educational needs of society and the immediate concerns of students:

(4) Cope with the knowledge explosion and the short "half-life" of scientific knowledge.

(5) Use the logical, precise, effective, and efficient educational technology that is currently available.

(6) Use teaching personnel in the most economical and efficient way (time, energy and money).

(7) Enable utilization of cognitive teaching input.

(8) Provide for student testing of learned behaviour in real situations.

(9) Produce a graduate capable of delivering creative teaching care for the next fifteen to twenty years.

(10) Spend a reasonable length of time accomplishing the goals of the curriculum.

A "good" or desirable course pattern and implementation strategy will meet the foregoing criteria and will utilize all data generated in the previous stages of curriculum development. There are many ways the curriculum commitments can be implemented that will meet the foregoing criteria and, therefore, be "desirable" curriculum. All the effective patterns have one common element - the capacity to continue to change and grow without the necessity of rebuilding the entire structure. Curriculum verification is not the final step in curriculum building; there is no final step. Curriculum verification is only a beginning or trial Validation of ideas. From the first day of the first class of the implementation of the "new" or "revised" curriculum, feedback about implementation begins and enables plans to be made for continuing changes. Courses must change, grow, and improve; teachers must develop new teaching skills, learn new teaching strategies and tactics; and students must acquire new patterns of learning. All this is a continuing process-a process of trial, feedback, alternation - trial into infinity. The curriculum is like the limerick of the lady and tiger;

There was a young lady from Niger

Who smiled as she rode on a tiger?

They returned from the ride

With the lady inside

And the smile on the face of the tiger

The curriculum "tiger" consumes the "lady" or those who ride, since the process of change, when built into the curriculum, continue to operate through the graduation of class after class of students and the attrition of faculty member after faculty member. This tiger with a smile on his face is the antithesis of another curriculum tiger, the "sabertoothed curriculum tiger". The sabertoothed curriculum was allowed to continue to decimate the brain power, initiative, and creativity of students and faculty long after it was extinct because somehow it had become sacred, ritualistic, and frozen into a time and space that no longer existed. The tiger with a smile on his face is an ever-changing beast, who is never to become extinct because he adapts to his ever-changing environment.

2.6.1. Selecting a Format

Selecting a curriculum pattern, format, or plan is a task following an examination of the several patterns that are consistent with the conceptual framework and appeal to the preferences of the faculty. To select a pattern that will provide optimal usefulness, consistency, and appropriateness, faculties take two or more plans and weigh the advantages and disadvantages. One maxim in this stage of curriculum development is never to offer the faculty only one choice of a plan. The more choices offered the better the faculty will be to synthesize the best features of the alternatives and to generate a plan pleasing to most faculty members. Task groups working independently to devise alternative plans usually produce two or more different plans. If the faculty wishes more plans, more task groups will be utilized.

5.3. MAJOR TYPES/PATTERNS OF CURRICULUM DESIGNS

Perhaps the most useful existing classification available today is the one developed by Luces. In this classification, School forms are separated by function. Six major types of school design and their rational are presented.

(1) Conservative liberal arts

(2) Educational technology

(3) Humanistic

(4) Vocational

(5) Social reconstruction

(6) Deschooling

5.3.1. Conservative Liberal Arts

This traditional form or design is based on the belief that a human being's unique and distinctive quality is intellect, and that the quest for knowledge is the natural fulfillment of such an intellect. In short, the highest purpose in life is to engage in the process of inquiry; to move "from ignorance to truth, from confusion to enlightenment.

Historically, this quest for knowledge was seen as a reflection of a world whose law and physical order were fixed properties. The process of education was concerned simply with the pursuit of objective knowledge for its *own* sake. A liberal education was suitable to free people, who possessed the legal opportunity and means to devote themselves to cultural attainment

In later times, after scientific, revolutions and the loss of a shared cultural had diminished the concept of *paldeta,* the cultured man the liberal arts approach to educating became, a perspective. Liberal arts were not so much a mastery of subject matter as it was a way of looking at things. The human mind was trained so that the individual might live fully.

This notion of education was translated into a public education. In the words of R. Freeman Butts; "....the prime purpose of the public schools is to serve the general welfare of a democratic society, by assuring the knowledge and understanding necessary to exercise the responsibilities of citizenship are not only made, available but are inculcated.

The curriculum of the conservative liberal arts is familiar to most of us, as the curriculum we studied in high school and in college was also similar in nature. Usually dominated by a standard text or set of materials conducted according to time to master units and leading to formal recognition of learning in a graduation from study, the liberal

arts model is familiar to many people. In recent years, Mortimer Adier's Paideia Proposal has served as a guide to this form of learning in public schools.

3.2. Educational Technology

Education in the current decade will experience a technological atmosphere. The technological instruments – personal computers, fax machines, versatile copy units; offer the possibility of significant alternations in the teaching-learning act as we know it. Research, communication, training all are targets of the proliferation of gadgets and hardware. More important for planners such technology also suggests an exit point for the spiraling costs of financing public education.

Historically, technological designs stressed objectivity, precision, and efficiency: As major proponent B.F. Skinner stated, "The traditional distinction comes down to this: when we know what we are doing, we are training.....any behaviour that can be specified can be programmed". Usually, in the past, learner behaviours were described overtly, infrequently lasting beyond the immediate treatment.

In the 1970s, all of the objectivity, of the technology of behaviourism began to break down as it was recognized that students were being impacted by unplanned technology. Technology is powerful in that it allows the learner new powers.

Technology in the 2012 be of a different scale than previous decades even though each decade has seen its own miracle technology. Televisions in the 1950's, transistors .and computers in 1960, video cams and compact discs in the 1970s, and facsimile machines and CD-ROMS in the 1980s all have impacted learning. However, the 2000s will be the decade of integration of these tools, and more with the result that the learner will possess world-wide power from home, Schools if they are to survive in a viable form, will be hard pressed to ignore the new technologies, and in some Technologically unadvanced districts, it will be a difficult transition; Technology in 1990s already has the power to reform or destroy public education.

In the 1980s, the lack of meaningful use of technology depressed its impact on classroom learning. In most districts, there appeared to be little integration of hardware or software in traditional courses, and true visions of how the new technologies could be utilized were nearly nonexistent. In addition to the computer, schools grappled with interactive television, various electronic nets, interactive cable and satellite-beamed video, CD-ROM technology, and other technological advancements.

Now in 2012 curriculum specialists must do better if they are to maintain control of the programming function. Not only is knowledge of how technology work (literacy) important, but also how technology can be used to improve communication and transmission of knowledge. Five questions should focus this effort to regain control:

(1) What are the implications of techno logy in the 2012?

(2) How can schools receive technology in a meaningful way in the 2012?

(3) What technology is most effective for learning in schools?

(4) How can we prevent technology from creating an intellectual elite in schools?

(5) How can we confront the trend towards home schooling from the society deschooling the learning process?

Curriculum leaders in the 1990 played a very large role in defining technological designs by the answers given to these questions. Education in public schools was redefined by technology, only the form remained to be determined. Had the educators not been aggressive students of technology, the forces that control the medium would have programmed the curriculum.

3.3. Humanistic

Such designs generally feature student-centred curriculums and instructional patterns and a dcentralization of authority and organization.

Human curriculum designs have deep roots in American education and have taken numerous forms in this century. In such programmes there is a shift in atmosphere towards understanding, compassion, encouragement, and trust. Physical settings usually encourage freedom in the form of student mobility, increased choice of curricular activities, and a learning-by-doing format.

An early example of this design in the United States was the Dalton Plan, which was implemented in the Dalton. Massachusetts Schools in the 1920s. The programme featured freedom of movement and choice of materials by students, cooperation and interaction of student group life through a house plan, and subject matter laboratories in the classrooms.

Another early version of a humanistic curriculum design was the organic method of education developed at the Fairhope. Alabama, school around 1910. This programme held that children were best prepared for adult life by fully experiencing childhood. Children were led naturally into more traditional areas of schooling only after experiencing a curriculum of physical exercise, nature study, music field geography, storytelling and fundamental conception of numbers, drama and games. General development rather than an amount of information controlled the classification of students.

Contemporary versions of the humanistic design are to be found in open elementary schools, emerging middle schools, and student-centred programmes. such as Outward Bound. In such programmes, the instruction is human personalized, and individualised. Curriculum is geared to the maturationa levels of students, and teachers serve a guides to learning rather than authority figures or purveyors of knowledge. The problem-solving process of the instructional format borrows heavily from another humanistic design, the core curriculum,

The core curriculum, developed in the 1930s in schools, such, as the Denver public schools, attempted to present learning from a human and holistic perspective. The following excerpt from an-evaluation report outlines the programme objectives:

"it is so named because it represents an attack upon those problems, which are relatively common to the young people in the school and, because it carries the chief responsibility for guidance, for general testing, and for record keeping. It is the part of the total school programme, which is planned for the development in boys and girls of the ability to solve common problems and of the power to think together and to carry on the democratic process of discussion and group decisions".

Core curriculum used a ten-point plan in organizing for instruction:

(1) Continuity of teacher-pupil relationships.

(2) Greater teacher participation in formulating policies of the programme.

(3) Elimination of barriers to learning experiences through the attack of problems rather than through reliance upon the logical organization of such matter in isolated courses.

(4) Development of core courses based on student concerns.

(5) Relating school activities to the community.

(6) Pupil-teacher planning emphasising choice and responsibility.

(7) Guidance by a teacher, who knows the student in an intimate classroom testing.

(8) Using a wide variety of sources of information.

(9) Using a wide variety of means of expression - Words, art, music.

(10) Teacher-to-teacher planning.

Humanistic designs generally are characterized by highly flexible instructional areas, high degrees of student involvement, and an emphasis on the process of learning as opposed to a product orientation or a preparation for life outlook.

5.3.4. Vocational

A fourth curriculum design present in the last century has been one concerned with vocation and economic aspects of living. Such programmes generally go under either the traditional term *vocational* education, or the broader and newer term, *career education.*

In the early years of the last century, vocational programmes were perceived as separate from the academic tracks and focusing on the non-college-bound student. The curriculum consisted of crafts and labour skills and such programmes were strongest in areas within an industrial or agricultural community. More recently, there have been efforts to connect or combine the vocational and academic areas. In the words of Sidney Marland, former Commissioner of the US Office of Education:

"It is flatly necessary to begin to construct a sound and systematic relationship between education and work. This system will make it a standard practice to teach every

student about occupations and economic enterprise. A system that will markedly increase options' open to each individual and enable us to do a better job than we have been doing of meeting manpower needs of this country".

Vocational education programmes generally study eight areas

(1) Trade and industrial education;

(2) Business education;

(3) Agriculture;

(4) Home economics;

(5) Marketing education;

(6) Technical education;

(7) Technology education: and

(8) Health education.

Vocational curriculum designs either try to integrate academics (interdisciplinary) and use vocational applications to illustrate the utilization of knowledge, or they create, new tech-prep alternatives, where students branch off from the pre-college curriculum while in high school.

Efforts to implement a comprehensive vocational design have been increased during the past two decades by a number of factors. First, there has been a growing recognition that the schools are an essential piece of the national economic condition. Welfare, unemployment, large segment of the population without useful skills, and the fact that only 40 to 45 percent of all high school graduates attend college, have been given as reasons for an increased vocational emphasis in the schools.

Second, the entire relevancy movement of the 1960s revealed a condition of students, who are bored and listless in senior high school, resentful of the holding pattern of formal schooling.

Third, vocational/career education has been promoted as a means of assisting minority groups and other disenchancted members of the society in breaking out the cycle of poverty. Students experiencing such programmes can escape the containment of environments and firmly backgrounds.

Finally, the whole concept of utilitarian education and no-frills curriculum has increased the awareness and demand for vocational designs. There is a growing opinion among the public that insufficient attention has been paid to the hard social reality that everyone must eventually seek, gainful employment. Technological and political conditions demand a change in the basic definition of an education.

Proponents of vocational designs in the twentieth century have pictured them as a necessity: a means of serving all students in the public schools, a vehicle for making

school useful and relevant, a contributor, to the well being of the American society. More recently, American business has stressed economic education as stimulus to a high school curriculum that incorprates more training activities. Vocational designs are practical. Critics of vocational designs, including career education, see them as static conceptions of life in the American society and insufficient preparation of life in an unknown future.

3.5. Social Reconstruction

A fifth curriculum design found in the United States in the last century had as its main theme, social reconstruction. The conception of the school, as a vehicle for social improvement was not new. Arguments for this type of school were made in the 1930s by members, of the social reconstruction wing of the Progressive Education Association. Harold Pugg, for example, spoke of the changes impending in the American society and encouraged the schools to influence social changes. He outlined characteristics of a needed curriculum in the 26th National Society for the Study of the Education Yearbook.

The social reconstruction designs sought to equip students with tools (skills) for dealing with changes about them. So equipped, the student can meet an unknown future with attitudes and habits of action. In the 1970s and early 1980s, special schools for minority youth illustrated this special kind of intellectual "arming" of the student. Such programs continue into the 1990s.

More recently, the American curriculum has seen an increase in the teaching of thinking skills, skills that teach students how to use information. Only recently has it been acknowledged that the way in which information is treated causes the student to use information in certain ways. In other words, how we teach is how students learn.

Another example of social reconstruction designs is the use of schooling to encourage certain social tends. For example, in the Duval Country School in district Jacksonville, Florida, magnet programmes are offered to encourage voluntary racial integration. Here, the curriculum offerings are multiple and secondary to the primary objectives of meeting court, order compliance to desegregate under local initiative.

The major assumption of social reconstruction designs is that the future is not fixed, but rather is amenable to modification and improvement. The school, as an institution, cannot remain neutral in a changing world and can influence and direct social change.

Some recent applications of the social reconstruction design have used "futurism" to justify the necessity of social intervention. Since the future will be like the present, it is necessary to be flexible and develop the ability to make value decisions. In the words of Kirschenbum and Simon:

> "Social reconstruction designs generally combine classroom learning with application in the outer world. Teachers and students are partners in inquiry, and instruction is usually carried on in. a problem-solving or inquiry format".

5.3.6. Deschooling

As strange as it sounds, it is possible to design the de-schooling of public schools. Through purposeful organization, or lack of it, it is possible to de-emphasise or disestablish the public school, programmes and the formality of education by redirecting resources to alternatives. While early efforts in such designs sought to free the learner from bureaucracy and control by the institution of the school more recent attempts have attempted to de-school in order to replace public education with economic or political alternatives,

According to its chief spokesman, Ivan Illich, schools are social tools that actually operate to deprive individuals of an education and real learning. Schools are not the panacea for social ills, but rather are rigid, authoritarian institutions that perpetuate the social order through a number of functions; Illich sees de-schoolling as an alternative design.

Problems of institutionalised education revolve around questions of power, leadership, and structure. Schools, by dominating the values and focus of organization, control the learner. Such control is often racist and sexist and is always oppressive. 'Further, schools are undemocratic in their method of converting knowledge into power.

Opposition to formal schooling and its structure has been a continuous phenomenon of the twentieth century, but the free school movement of the late 1960s presented the best examples of the de-schooling design. Glatthorn outlines the emergence of the free school movement during that period.

Glatthorn identifies a number of ways in which free schools and alternative schools attempted to release the individual student from the institutional oppression of the school: travel-learn programmes, work and apprenticeship programmes, volunteer service, informal study in the community, and affective experiences. Collectively, these curriculum arrangements sought to define education as a personal act.

In the 1990s, new forces fell within the de-schooling design. On the one hand, there was a significant *home schooling movement* in America where parents refused to enroll their children in school and served as a surrogate teacher at home. While the variety of laws governing these phenomena precluded any real generalisation of motivation and practices it can be stated that such parents did not support the concept of organized schooling for their children.

A second force that impacted public schools in the 1990s was the political push to give parents vouchers and the right to enroll their children in a school of choice. While this ten-year-old movement was seen as a sincere effort to upgrade the public schools, it was more probable that this theme was more about privatising formal education. Using either legislation or incentives, or both, proponents of this educational movement eliminated foundational funding of public education.

Efforts to break the monopoly of formal education and de-school learning continue today. They seek to downgrade the importance of accepting the functions of

formal schooling, and to break the myth of a need for education. While on the decline, such curriculum designs are likely to emerge again in the future.

Together, these six curriculum designs outlined the diversity of educational programmes during the twenty-century. Curriculum leaders need to be aware that such diversity has always been present in education and will continue to be present in the future.

Hopefully, these designs will suggest other possible curriculum designs. Curriculum leaders need to develop the intellectual freedom that will allow them to design the best possible school programmes for children. As Lawrence Cremin has so eloquently stated in *The Genius of American Education,* "Education is too significant and dynamic an enterprise to be left to mere technicians".

5.3.7. Steps in Curriculum Design

Feyereisen, Fiorno and Nowak 0970) suggest the following steps in curriculum design:

(1) Identification of the problem.

(2) Diagnosis of the problem.

(3) Search for alternative solutions.

(4) Selection for the best solutions.

(5) Ratification of the solution by the organization.

(6) Authorization of the solution.

(7) Use of the solution on a trial basis.

(8) Preparation for adoption of the solution.

(9) Adoption of the solution.

(10) Direction and guidance of the staff.

(11) Evaluation of the effectiveness of the solution.

3.8. Benefits of Curriculum Design

(1) It focuses attention on goals.

(2) It improves the probability of success.

(3) It improves economy of time and efforts.

(4) It facilitates communication and coordination of projects.

(5) It reduces stress.

3.9. Questions in Curriculum Design

(1) How can balanced development be assured? What guarantees can there be that important areas of life in which students or teachers are insensitive to problems will not be neglected?

(2) How can there be continuous growth from year to year without undesirable repletion or undesirable gaps in learning?

(3) How can desirable depth of knowledge be assured? How is it possible to possible to prevent learners from ending up with a smattering of superficial knowledge about many areas and little of the depth they need for genuine understanding?

(4) How will the depth of command of special subject areas important to individual learners be assured? Where will the persons with specialized talents come from the scientists, historians, linguists, artists, teachers, and political leaders, philosophers of the future?

(5) How can it be guaranteed that children and youth will become acquainted with the broad cultural resources, which arc part of our heritage and become skilled in drawing upon and' using these resources in meeting life situations?

(6) How can there be guarantees that choices of problems are not trivial and do not represent transitory interest rather than basic concerns?

(7) How can there be genuine group problems? Will there not actually be many times when extrinsic motivation will be needed if group study is desired? Does the point of view not logically lead to work that is completely individualised (Stratemeyeret al. 1957, pp.111)?

(8) What is a curriculum: what does it include and what differences are there between the issues of a curriculum and those of a method of teaching?

(9) What are the chief elements of the curriculum and what principles govern the decisions regarding their selection and the roles that they play in the total curriculum?

(10) What should the relationship among these elements and their supporting principles be, and what criteria and principles apply in establishing these relationships?

(11) What problems and issues are involved in organizing a curriculum and what criteria need to be applied in making decisions about the parents and methods of organizing it?

(12) What is the relationship of a curriculum pattern or a design to the practical and administrative conditions under which it functions?

(13) What is the order of making curriculum decisions and how does one move from one to another?

SUMMARY

(1) Curriculum design may be defined as the arrangement of the elements of a curriculum. The different parts (intention, content, learning experiences, evaluation) of any curriculum are known as curriculum, which may be arranged in different ways to produce different designs.

(2) By curriculum planning, we mean that process whereby curriculum developers conceptualised and organize the features of the curriculum they wish to construct. This involves a broad analysis of the curriculum intent and context (what it will look like), organizing the sequencing of developmental task (how to construct the curriculum) and arranging for the process of implementation and evaluation.

(3) Curriculum design is an example of the application of curriculum presage in practice. In the curriculum presage phase, developers are influenced by various forces and conceptions, which help account for the different curriculum designs that emerge.

(4) Design forces in curriculum are referred to as horizontal and vertical organization.

(5) Horizontal organization, often referred to as scope of horizontal integration is concerned with the arrangement of curriculum components at any one point in time.

(6) Vertical organization, sometimes known as sequence or vertical integration, addresses the relationship between curriculum components over the entire duration of the curriculum's application.

(7) The main categories of curriculum designs are:

Subject-centred designs
- subject designs
- academic discipline designs
- broad field designs

Learner-centred designs
- activity/experience designs
- humanistic designs

Problem-centred designs
- thematic designs
- problem designs

Core-designs

— core designs
— national core designs

Throughout the last century, divergent opinions about education have led to efforts to reform the public school because these efforts continue today, curriculum leaders should be aware of the multiple curriculum designs in existence and to open to new thinking about the way schools and educational programmes are organized.

Six curriculum designs are prominent: educational conservationism, technological design, humanistic design, vocational design, social reconstruction design, and de-schooling design. Such diversity has unquestionably enriched the programmes of the public school education.

5. SELF-ASSESSMENT QUESTIONS

Q. 1: Write a note on concept and categories of curriculum designs?

Q. 2: Discuss the subject-centred and learner-centred designs?

Q. 3: Describe the problem-centred and core designs? '

Q. 4: Explain the functions and problems of curriculum design?

Q. 5: Explain the criteria for a desirable curriculum?

Q. 6: Write short notes on the following:

(i) Conservative Liberal Arts

(ii) Educational Technology

(iii) Vocational Curriculum Design

(iv) Humanistic

(v) Deschooling

(vi) Steps in Curriculum Design

(vii) Benefits of Curriculum Design

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UNIT–6

SITUATIONAL ANALYSIS AND GURRICULUM INTENT

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INTRODUCTION

In this Unit, the concept of situational analysis, its need, process method the main factors involved, have been examined. Besides, a needs-assessment technique which may be used as an important part of the situational analysis procedure, or additionally it may be employed as an independent diagnostic tool at one or more later phases in the curriculum development process has also been examined.

Understanding aims, goals and objectives of curricular is essential for curriculum planners and teachers. Failure to gain this understanding is likely to produce a variety of problems for the various groups participating in educational programmes and activities. This Unit is, therefore, concerned with the technical explanation of these terms.

OBJECTIVES

After studying this Unit, you should be able to:

1. Define the situational analysis.
2. Describe the phases of need-assessment.
3. Classify aims, goals and objectives and understand their significance.
4. Show the difference as well as the relationship between aims, goals and objectives.
4. Show the difference as well as the relationship between aims, goals and objectives.
5. Understand the List the major sources of aims, goals and objectives.
6. Explain taxonomies of educational objectives and mention important points in their favour and against them.

1. SITUATIONAL ANALYSIS

Situational analysis is defined as the process of examining the factors and content for- which a curriculum is developed, such analysis is used in curriculum planning. The application of that analysis to curriculum planning, it involves a detailed analysis of several 'factors that relate to that content albeit the learning experiences inside and outside a school. These, the curriculum continents structure and development depend upon situational analysis a school group of schools, region or system. The analysis then incorporated within the subsequent planning for and development of curriculum-intent, curriculum content, learning, activities and evaluation. Nicholls & Nichotls mention the factors involved in the process of situational analysis as:

> a situation which is made up of a number of factor such as pupils, pupils' homes and background school, its climate its staff facilities and equipment. Analysis of these factors, together with a self-analysis, followed by study of their implications for curriculum planning constitutes one-step towards the rational approach… (of curriculum) (1978:22).

A situational analysis is a basic element for the construction of a curriculum. Curriculum developer's reed to be aware of the factors affecting curriculum, to bring a reasoned, rational approach to the development of curricula. Above all, situational points out the local factors developing curriculum to meet students needs. The argument, for undertaking a situational analysis is therefore summarized as under:

1. Identifying local needs of students, .parents, teachers and the community.
2. Understanding the local curriculum contents.
3. Facilitating planning and subsequent curriculum development.
4. Providing a data for determining curriculum goals and objectives.

1.1. Need Assessment

Need assessment is a starting point of a situational analysis, for collecting data within a situational analysis, for undertaking, reviews of aspects of curricula at other times. John McNeil defines need assessment as:

It means before going into the process, curriculum developer should assess the needs which are focused.

> "Need analysis is process by which one defines educational needs and decides what their priorities are. In the content of curriculum, a need is defined as a condition in which there is a discrepancy between an acceptable state of learner behaviour or attitude and an observed learner state". (McNeil. 198:74)

Need assessment provides the guidelines for further action a means for reaching consensus over future directions for a curriculum by determining the discrepancy between current and preferred situations. In turn this requires procedure to ascertain the nature of the discrepancy, which is outlined below:

Keeping the difference for current and prefaced situation in view. The curriculum

developers use need assessment in facilitating a situational analysis and for laying the foundations for an effective set of curriculum aim, goals and objective.

1.2. Phases of Needs-Assessment

While there are numerous ways available for curriculum developers to undertake a needs-assessment, the following procedure is simple and useful. The Algorithm suggests using a five-phase approach employing a; discrepancy concept to determine needs. It assumes the needs assessment activity is being conducted at a school level though it may be conducted at other levels such as regions and system.

(i) Phase-I: Formulate Goal Statements

.Those involved in conducting the needs-assessment (teachers, perhaps parents and students) make statements of acceptable and preferred, student behaviour; These goal statements may relate to literacy levels, abundant competencies, personal skills, content knowledge, subject skills, self- concept, physical development and so forth. Stated briefly, the goal statements represent positions that the curriculum decision-makers believe of value to students. For example:

(a) Students should be literate in .English to an acceptable level.
(b) Students-should develop a positive self-concept.
(c) Students should understand the nature of government.

(ii) Phase 2: Rate Goal-Statements

The goal-statements developed in phase-I have priorities assigned to them be a group such as those from phase-I or a broader group of teachers, parents and students. Goals are rated on a scale (say 1 to 5) by individuals and then averaged amongst the group. For example:

Goal statement	1=4.51
Goal statement	2=3.27
Goal statement	3=3.77
Goal statement	4=4.21

(iii) Phase 3: Rank Goal-Statements

The averaged ratings are then ranked in order to determine a priority of importance for the preferred goals A ranked list now exists of preferred learning goals. Ranking the above goals, the result is:

Goal-Statement	1
Goal Statement	4
Goal-Statement	3
Goal-Statement	2

(iv) Phase 4: Determine Goal-Statements

By examining each goal-statement, either subjectively or objectively, in terms-of what is happening within the institution, a discrepancy factor may be determined, which represents the difference between the current situation and the preferred learner state.

Subjectively, using teachers to act as judges and then to rate the goals in terms of what is happening in reality may achieve this end. Objective data may be collected through tests, examinations and so forth to determine levels of learners' proficiency and their rating.

Where a discrepancy exists between what is preferred (ranked goals) and what exists (observed learner state), a need is indicated. It is possible that the goals ranked highest may not produce the greatest need, although this obviously, depends upon the level of discrepancy evident between the observed and preferred learners states. However, those goals ranked highest usually tend to 'have high discrepancy factors simply because they are judged to be important.

Extending the example above we find different levels where discrepancy has resulted.

Goal-1	=	moderate discrepancy
Goal-2	=	high discrepancy
Goal-3	=	negligible discrepancy
Gaol-4	=	high discrepancy

(v) Phase 5: Developing plans of Action

Goals-Statements, with high discrepancy (the difference that affects) factors and priority demand immediate attention. Using the discrepancy as a basis for change curriculum developers then create or adjust curricula that resolve the observed and perceived problems. New curriculum objectives are required and from these subsequent curriculum element developed or altered.

An example of this technique may be a school that decides to examine the relevancy of its curriculum on a regular basis. Before it can develop a new curriculum staff needs to reflect upon the existing situation: or rather than, react intuitively, a intuitively needs-assessment is recommended and a small group of staff, together with a small group of parents creates a list of goal statements of preferred behavior for the students of the school. Curriculum developers may develop an integrated approach to enhancing literacy as well as specified programme of student’s self-concept. In this way, curriculum developers react systematically to areas of perceived needs. In a somewhat different way, Wiles and Bondi (1989) advocate a needs assessment of a far more substantive nature. They contend that a need assessment is basic to all curriculum development and should be like an inquiry into school programmes. As such, they appear to argue for a situational analysis of through nature.

1.3. Conducting Situational Analysis

While the need for conducting situational analysis is considered in most of the curriculum literature, uninitiated curriculum developers. Who think it as difficult time-consuming task, may nevertheless, resist it. Some Curriculum developers may argue that they do not need a situational analysis to understand the context of their curriculum. After all the have an unconscious understanding of student needs and necessary curricula to meet those needs. However, we should question curriculum developers as to where this unconscious response comes from and to what degree is it valid?

Amongst other things, a situational analysis provides a clear vision of the curriculum context as well as a systematically divided database, both of which provide a more reliable and valid source for later curriculum decisions. This is not only a more systematic way of concept making and conducting curriculum development .than an unconscious understanding approach but one may gain substantially greater support from other participants in the curriculum development process. In this regard, the following factors are to be considered as:

(1) External Factors to the School

External Factors means those things (valuables) that can affect curriculum developments implementation.

(i) Cultural and Social Changes and Expectations

This includes major changes to society, such as unemployment, societal values economic growth and family relationships, partial employer and community expectations of schools.

(ii) Educational System Requirements and Challenges

This includes systemic influences, such as policy requirements, inquiry reports, external examinations, major curriculum projects and significant educational research. Major inquires, such as, McGowan (NSW 1981) Beazley (WA. 1984). McGraw (WA. 1984) and Blackburn (Vic. 1985) have all been very influential at the systemic level.

(iii) Changing Nature of Content

The subject matter taught in schools requires constant revision to update it with developments in the outside World. Examples include new knowledge acquired, technological- developments and new literature.

(iv) Teacher Support Systems

A variety 'of external systems can contribute to enhancing teaching/learning strategies, content updates, evaluation techniques, audio-visual material and other resources. Support may come from tertiary institutions, educational institutes (ACER. CDC), local leaching centres, curriculum consultants, advisory teachers, in-service courses and subject associations (e.g. Sciences Teachers Association).

(v) Resources

Curriculum developers need to be aware of the availability and flow of resources into the school. These may come from Commonwealth Source (priority schools project, projects of national significance) state education departments, the community and business organization.

(2) Internal. Factors to the School

(i) Pupils

Significant data that may be gathered on students include abilities physical and psychological development, aptitudes, emotional and social development and educational needs. An accurate understanding of the nature of students allows for effective curriculum planning.

(ii) Teachers

What are the skills, experiences, teaching style, values and special strengths and weaknesses of a school teaching staff? Special strengths may broaden curriculum offerings (e.g. aeronautics, horticulture, meditation) and allows for curriculum enrichment and extension.

(iii) School Ethos

The school climate/environment is a significant factor influencing curriculum and it includes principal involvement, sharing of responsibility, social and professional cohesiveness and operational procedures.

(iv) Material Resources

What exactly does the school possess in terms of buildings, equipment resources (books, curriculum materials), land and vehicles as well as financial sources for-future purchases, knowledge of resources, facilities, curriculum planning, etc. (e.g. can we offer gardening, Book fair Farming? Kitchen Gardens etc.)

(v) Perceived Problems

Curriculum Change is basically evolved from needs and problem. Curriculum planners communicate these needs and problems with parents, teachers, students and the community. Needs-assessment techniques may be used.

1.4. Situational Analysis Procedure

A desirable process of conducting a situational analysis involves four steps:

1. Identify problems in context.
2. Select appropriate factors.
3. Data collection and analysis.
4. Make recommendations.

Curriculum developers, while beginning the development phase of the curriculum model outline, should devote considerable time and effort to a situational analysis effectively it will contribute to the development of the curriculum elements.

1. The Problems

The problems that initiate a situational analysis may be specific (prepare a course for non-tertiary-bound upper-school students), general (improve student literacy abilities at primary school) or fundamental (create a curriculum for a new school). The need to address these problems could well be so obvious that the school would face significant difficulties if they were ignored. The increasing number of non-tertiary-bound students in years 11 and 12 are a case in point. Should school offer a traditional community design to those students continually? If it is, there will be great problems to face. A thorough, analysis of student needs and expectations is required as a preliminary stage to later curriculum development. Indeed, this has been the brief of the Finn Report (1991), the Mayer (1992) discussion paper and the Carmichael (1992) Recommendations for a Vocational Training Certificate.

It may be that teachers are not well aware of the problems and feel needs of students and parents. Regular investigation of these attitudes is a useful activity by teachers for keeping the curriculum relevant and realistic. A needs-assessment technique may be helpful here as seen earlier in this unit. In many instances feedback to students and parents will also provide a beneficial, public relations function. Skilback and Saliman have suggested various factors to be used in a situational analysis at the individual school level, the Nicholls factors are appropriate for collecting data relating to the nature of the curriculum problem. This approach requires an analysis of (1) the pupil abilities and backgrounds; (2) teacher strengths and weaknesses; (3) school environment, (4) school resources; and (5) school climate (after Nicholls & Nicholls. 1978).

2. Appropriate Factors

Upon determining, which needs and problems are to be addressed in the situational analysis, the curriculum developer can then relate to the appropriate factors suggested by Skilbeck. It may be that the problems are so obvious that progress direct to data collection is warranted, but often that is not the case. A careful consideration of Skilback's factors will illuminate the nature of/the-problems concerned. It should be noted, however, that the curriculum developer is not expected to collect and analyse data from all of the factors outlined by Skilbeck rather those that are relevant to the problem/need. Should the need be "extremely general, it would then be appropriate to collect and analyse data on all factors, but in most cases some selection of factors is sensible.

3. Data Collection and Analysis

How are data to be collected, which will throw light on the selected factors being analysed? Many teachers will adopt an intuitive approach, relying upon their experience and intuition to explain the situation. Others, to their peril, may simply ignore this stage. However, a more systematic approach to data collection provides useful information that after conflicts with intuitive understanding given below are a few suggestions. Some techniques are useful in data collection to the types of data that may be obtained. Curriculum developers then analyse these data to determine their significance in the contents of a specific situation. It **is** not possible to provide an analysis of the data-collection techniques mentioned below. Nevertheless it is important for curriculum developers to be knowledgeable about these forms of data collection. The use of the above technique will vary with the nature of the situational analysis being undertaken. A small-scale review of a physical education programme might only require informal interviews with students and a brief survey of parental attitudes. A more realistic comprehensive example might involve an entire school staff in an analysis of student numeric skills. Such an analysis may require special student ability tests; students self-reporting Scales; questionnaires to parents; inventory of school mathematical resources: analysis of school records and external examination results, as well as interviews with staff to, elicit teaching strengths and interests. The latter is obviously a time consuming task, although clearly the data collected would provide a sound foundation for developing appropriate curricula.

Another example may revolve around a problem of student attitudes towards schooling. An examination of the school-ethos is recommended and several forms of[1] data can be collected. Employing instruments to measure the psychological environment of classrooms (Fraser & Fisher. 1983) curriculum developers can ascertain student perception^ of that situation. These instruments are basically determined according to a number of dimensions and student differences between actual and preferred classroom environment. Once data have been collected they must be analysed to determine .trends, and areas of consistency. This does not usually require sophisticated statistical treatment but rather a systematic, analysis and synthesis of the collected data to determine what patterns are revealed.. From these patterns, developers are able to recommend directions for action.

Data Collection Techniques

Factors	***Techniques***	***Data Collected***
Pupils	Interviews School records Systematic observation Questionnaires External examinations Psychological environment Self reporting scales	Student information and attitudes. Background and achievement data Student behaviour patterns. Student attitudes (large scale). Comparative student performance. Student perceptions of room climate Student attitudes
Teachers	Anecdotal records Staff profiles Questionnaires	Information on teacher behavior and attitudes. Records of staff skills and abilities. Teachers' attitudes.
School ethos	Systematic observation Psychological environment interviews Inventory check list Systematic observation	Impression of school climate Aggregated classroom climate student/ teacher/parent attitudes Listing of school resources Impression of school resources

1.5. Recommendations from a Situational Analysis

The final step in the situational analysis model requires curriculum developers to make recommendations based on the analysed data. This would typically consist of a list of recommended actions based on the previous goal-statements. Examples of recommendations may be:

i. Students require substantial literacy development to establish secure English communication skills.
ii. An advanced course in Japanese culture is needed to support Japanese language course for upper school students.
iii. Staff has expertise to offer introductory units in aeronautics, marine biology and yachting.
iv. The school and staff have appropriate resources to address locally the needs of gifted and talented students.

The value of undertaking a situational analysis has diminished considerably without the translation of results into recommendations or plans of action. The resulting recommendations will serve to guide the curriculum team with development of goals and objectives, content, learning activities and strategies for evaluation.

1.6. Situational Analysis Link

Developers while having sufficient data from a situational analysis, are able to continue with the effective preparation of the, curriculum utilizing able situational analysis data. For example, developers are better able to address the problematic task of devising a purposeful curriculum. When considering the curriculum intent for learners, developers will know about contextual factors teaching resources, the nature of learners, school facilities, school ethos and so forth, this information will provide a substantively sound basis for their future deliberations.

Similarly, a soundly conducted situational analysis will facilitate curriculum developers with their deliberations, on suitable curriculum content, appropriate learning activities, useful forms of student assessment and even how to facilitate effective implementation of the curriculum document.

Thus a situational analysis provides very basic information about:

- Individual & group needs;
- Socio-cultural perspectives;
- Core issues and basic problems
- Religious and ethnic information;
- Trends and attitudes adopted by the people
- Contextual approach & meaning to facts.

2. FORMULATION OF CURRICULUM OBJECTIVES

If the formulation of aims, goals and objectives is undertaken at this point in the development process, a clear direction is provided for subsequent steps. Instead of relying upon intuition and experience for the subsequent selection of content, learning activities and evaluation strategies, the curriculum -- developer has a sound foundation for later decision-making by using the stated goals and objectives to provide direction. Thus from objectives comes content, from objectives and content together comes learning activities and the three elements together then help to direct evaluation strategies.

This has far more value than constructing goals and objectives after content, as would appear to be the case in some practice, the latter approach requires objectives appropriate for the content, father than the other way around, and questions the very purpose of writing objectives at all. It is both logical and useful, therefore, to use a situational; analysis as a starting point for; formulating curriculum intent and for facilitating the formulation of other curriculum elements. The subsequent aims goals and objectives would then be-used to develop appropriate content, learning activities and evaluation procedures. While this may appear a hard task, it is rewarding when the process of developing curriculum unfolds.

2.1. The Nature of Aims

Curriculum aims, are statements of general expectations based on some value scheme borrowed from philosophy, consciously or unconsciously. Their distinctive quality is that they are not directly related to school or classroom outcomes. “Human Survival”, ““self-realization” and “ethical character” are examples of a few curriculum aims. Aims are thus remote and long-range and have to be converted into more immediate and specific school' outcomes if they are to be realized in actual practice (Zains. 1976. p.306).

An aim gives a direction to a set of more detailed intentions for the future. They are just a starting point - and represent an ideal, an aspiration and a direction, which education system should take in general. Thus they act as a guide to action and provide a general framework for the overall education process (Davies 1976 p.12). Since they are principally concerned with larger ends and purposes that act as binding forces for different programmes and activities, they are relatively few in number but are broad in scope and use.

Aims are inspirational and visionary in character and are therefore, vague and permanently open-ended (Richmond. 1976. p.175). They have to he clearly defined from age to age as have take on new meanings, for example, the concepts of a “good life” or “educated citizens“ admit of several interpretations that vary from age-to-age and from nation-to-nation even during the same period. Thus an aim needs to be clearly defined, interpreted and explained with reference to the national ideology and socio-political climate obtaining in a country (Sharif. 1964. pp.40-45). Besides, an aim has to be analysed and broken down-into its constituent parts for its practical realization and accomplishment.

In brief, statements of aims just provide overall direction and guidance to a school

system; but they are not always helpful to teachers in classroom instruction or evaluation. They are basically meant to provide direction to policymakers at different levels national, provincial and local. Hence they are not valid, for specific and concrete action'(Bloom, 1971, p.21).

Examples of Aims

Harry S. Broudy classified aims into four categories:

(a) Value pattern
(b) Social organization;
(c) Social roles; and
(d) Lifestyle. (Zais, 1976, pp.307-308)

All these categories must be inter-related with each other to produce the desired results. For instance, it is necessary that aims in other categories should be consistent with the aims and requirements of the central value patterns and display the same spirit. Any inconsistencies in this regard will be harmful to dominant value pattern.

(a) Value Pattern

This is the general category, which actively influences the character of aims in the other three categories. Aims in this category represent a philosophical viewpoint eg: If the central aim were the development of Islamic character, all other categories will be directed toward the realization of this basic value.

The students[7] of different educational systems bear the stamp of their typical value pattern. For example, the products of Aligarh; Deoband, Nadva and public schools in Pakistan reflect the different value patterns and characteristics of their institutions. Sir Syed's main concern was to promote western sciences and help the Muslim youth to get jobs in the Government. Deoband aimed at the preservation of a classical religious orientation, while Nadva attempted to modernize religious knowledge and bring it in line with the new demands of modern times (Ahmad! 1977', pp.72). These diverse value patterns shaped the institutions and hence the attitudes and behaviour of their students.

(b) Social Organization

A social organization acts as a social habitat for the development and promotion of a preferred value pattern whether Islamic, democratic of socialist. Given the above value pattern, social organization would be focused on Islamic social outcomes such as unity and social integration, brotherhood, concern for the collective good and a deep sense of respect for all individuals and groups on the basis of intrinsic personal qualities rather than surface qualifications, i.e. race, creed, colour, status or geographical location.

(c) Social Roles

A social role refers to pattern of behaviour associated with a distinctive social

position “such as the position of a mother, teacher, administrator, employer or student etc. It indicates what a person ought to do in his or her typical position. Social roles are complementary as well as complex (Broom and Selznick, 1968, pp. 12-12). Aims specifying preferred social roles would offer a particular set of qualities to be developed in parents, teachers, family members, neighbours, citizens and officials, which are in consonance with the national ideology. The Prophet of Islam (peace is upon him) is the best model for the Muslims. The Quran exhorts believers to assimilate this model into their personality.

In Islam, several roles are combined within a single individual and he or she is expected to behave likewise in a given situation during peace or war, a ruler is not simply a head of the state; he is also a guardian of public morality, a friend, neighbour and a citizen. This is how Islamic values would influence the social role of individuals.

(d) Life Style

This refers to the way in which one lives one's life. It is the practical manifestation of one's preferred value pattern. You can see a variety of life styles in any, society for example, that of a businessman, a gypsy or a hippy. The Eastern life style is very different from that in the West. However, each ideology attempts to, develop, on the whole a typical life style with reasonable scope for variation and spontaneity within its cultural framework. In brief, the life style must be in consonance with the spirit of the central value pattern.

Examples of Aims

Statements of aims may be short or long but detailed statements are generally preferred for curriculum building. The examples that follow are taken for the National Education Policy (1979, pp. 1-2).

(1) To foster in the hearts and minds of the people of Pakistan in general, and the students in particular, a deep and abiding loyalty to Islam and a living consciousness of Muslim Nationhood.

(2) To develop and inculcate in accordance with the Quran and Sunnah, the character, conduct and motivation expected of a true Muslim through effective elimination of gaps and contradictions between the professing and practice of Islam.

2.2. The Nature of Goals

Curriculum goals refer to school outcomes as a whole, and they are somewhat removed from immediate classroom assessment (Zais 1976.p.306). They lie in the middle of the aims-objectives continuum and goals actually represent different aspects or major constituents of an aim and thus prove helpful in identifying its principal parts. They clarify and explain what a particular aim is directed at the intent as well as the content.

Goals are derived from aims and must be consistent with them. They attempt to operationally the thinking represented by an aim making it relatively clear and practical (Davies. 1976. p. 14): They are more explicit than aims and thus indicate the broad pathways to the attainment of over-arching and all-inclusive aims. In short, goals simply facilitate the achievement of aims.

An aim indicates the direction, while a goal points to the actual destination. Rather than being visionary in character, a goal takes on a concrete form and becomes the focus of an activity. It serves as a basis for action and helps us in bridging the gap between the 'ideal' and the 'real'. However, goals are less special than objectives and are no more than inferred descriptions or hypotheses about the things learners will be able to do at the conclusion of a learning sequence (Davies 1976. p. l4). They do not specify the expected behaviour of learners with precision.

Several goals may be derived from an aim. These goals are then ordered so that priorities can be determined and allocated. Some of the goals may be realized at an early stage in the learning process: others may be taken up in the middle and still others may be accomplished very late (Davies. 1976. p-14).

These goals serve two main purposes. Firstly, they help us put concepts into writing and indicate what the learners must know at the end of a course (terminal goals). Secondly, they help the teachers and others concerned to bridge the gap between an aim and a specific objective (Leonard and Utz, 1974. p. 88).

Example of Goals

Examples of goals may be as follows:

1. To became a proficient and competent researcher in the field of education:
2. To get research certification of graduation in the field of research.
3. To become economically self reliant by utilizing the research degree
4. To successfully adopt the new kills and expertise required for social psychological and educational research
5. To get through the research programme organized by an educational insinuation.

Goals are the statements of individual's expectations from the given programme or descriptive.

2.3- The Nature of Behavioural Objectives

Behavioural objectives go by several names - specific objectives performance objectives and instructional objectives, increasing interest has been shown in them during the last two decades. The educational community is however, divided with a great many outstanding-scholars like Bloom, Tyler Gagne and Taba considering them a virtual renaissance, while-other prominent figures regard them as too! mechanistic and dehumanising. The debate continues but with current rethinking, some sort of compromise may be reached in the near future.

The current interest in behavioural objectives has arisen from several sources. Among these are the famous works of educational theorists like Tyler, Bloom. Mager and Krathwohl, which lay heavy stress on the need for accurate assessment, and measurement of learners knowledge in terms of observable and specific human behaviours. Several, educational taxonomies have been offered and a considerable literature is available on the subject. Bloom's taxonomy stands out to be of great value written about in the realm of education. Another source is industrial economics with its input—output models and techniques such as cost benefit analysis, operational research, systems engineering and job evaluation, all of which require precise, detailed and clear-cut objectives, prior to the formulation of a problem and .strategy of attack.

According to Davies behavioural objectives are called behavioural because they are stalled in terms of overt human behaviour, which is path demonstrable and measurable. They describe in unambiguous terms the expected behaviour of a learner at the end of a learning experience. Objectives are very specific the high explicit with no ambiguity about, their meaning, focus and intention. They are short-range, time bound, quantifiable and operational in form and spirit.

They refer to the most immediate specific outcomes of classroom instruction, the most important thing to note about the objectives that they act as a clear guide and provide immediate direction to curriculum planning, classroom instruction and evaluation. They are sharply focused on intended learning outcomes and leave no scope for misinterpretation. Precision clarity, specificity and quantify-ability are the principal features of a truly behavioural statement of an objective.

Generally Speaking behavioural objectives are detailed specifications of desired outcomes at the end of a lesson, teaching unit. And a large number of objectives can be developed from a given set of goals. This actual number depends upon several factors such as purpose, grade level, teacher's convenience and the generative capacity of the goals.

Examples of Behavioural Objectives

Now look at the examples below and consider their adequacy as behavioural statement:

1. "To be able to write a summary"
2. "To explain the theory of relativity"
3. "To be able to complete a 100 item multiple-choice examination on the topic of Muslim contribution to Science within one hour, with 70 correct answers as the lowest limit of acceptable performance".

You will probably have noted that (1) is an example of an incomplete behavioural objective. It does not mention the related content to be summarised which could, for example, be an English poem, an historical event or a scientific theory. As regards (2), both the behaviour and the content are mentioned. However, by far the most comprehensive behavioural statement is (3) which contains all the necessary elements of a complete

behavioural objective.

2.4- Taxonomies of Educational Objectives

A number of writers and teams of experts have offered several taxonomies of educational objectives. The ones presented by Bloom. KrathwonJ and Harrow for the cognitive, affective and psychomotor domains respectively are very popular and have been extensively used by curriculum planners, teachers and learners. The taxonomies represent classifications of objectives in hierarchical order in each domain and thus provide guidance in curriculum development, in test construction and evaluation and in teaching and learning.

The taxonomies serve several useful purposes. They help in specifying aims and goals in clearer terms, with the result that they no longer remain vague and hazy. Taxonomies also serve as tools in test construction and evaluation, and enlighten teachers and curriculum planners on several possible categories of objectives for different courses, lessons or tests. Moreover, they can also be helpful in the analysis of tests and examinations and can aid teachers and learners to focus their attention on the relative emphasis they have placed on different categories or levels of objectives.

Let us now look more closely at the work done on the three domains - (a) cognitive, (b) affective, and (c) psychomotor.

(a) The Cognitive Domain

Bloom and his associates developed this taxonomy, which includes six levels or categories of objectives ranging from simple to complex cognitive processes. The levels and their developmental sequence are arbitrary, based on the assumptions of the authors. The same levels may be defined and interpreted differently by other writers. However. Bloom's Categories are as follows.

(i) ***Knowing****:* this is the lowest category of objective in cognitive domain dealing with remembering facts and information. It treats knowledge as product and includes facts, concepts, principles, generalizations, trends, classifications, theories and structures (Michaelis et. al. 1975p.78). This level does not pre-suppose any understanding and is confined simply to the memorization of the elements of knowledge. Corresponding pupil behaviours included recalling, stating, identifying, listings describing an example of an objective at this level would be to describe the battle of Badr.

(ii) ***Comprehensioning****:* this is more complex than the first category. It involves interpretation and explanation of what was memorized and stored in the mind; and prediction based on material learned earlier. Corresponding pupil behaviours include paraphrasing, summarizing, illustrating, interpreting, explaining, interpolating, extrapolating predicating, etc.

Examples: To illustrate the concept of Jihad with reference to Muslim conquests

in the early days of Islam.

To interpret a population graph.

(iii) ***Applying:*** is the ability to solve new problems; using previously learned material, concepts, laws, principles and theories in new settings. Corresponding pupil behaviours include solving, modifying, applying, computing, demonstrating, constructing, performing.

Examples: To apply two concepts of civil liberties by stating how they are involved in a recent action taken by the court to protect the rights of minority groups..

(iv) *Analysing:* This refers to breaking up the whole into its different components, determining their relationships, distinguishing relevant features from irrelevant ones and recognizing the underlying theory involved. It is a process of reasoning in its highest form. Corresponding pupil behaviours include distinguishing, discriminating, analysing, categorizing.

Examples: To analyse the components of the cognitive domain.

(v) ***Synthesising:*** This refers to the ability to join together different discrete parts to form a new and complete whole. This is a process of reorganization and rearrangement, -culminating ultimately in something new and fresh. It involves some sort of creative activity. Corresponding pupil behaviours are designing, creating, composing, formulating, concluding, generalizing, etc.

Example: To draw a generalization from the data collected in an experiment.

(vi) ***Evaluating:*** this is the ability to make judgments based on a given set of criteria. Judgments involve a high level of thinking. This is the lightest level of cognitive domain. Corresponding pupil behaviours include criticising, justifying, concluding, evaluating, defending, comparing, contrasting, assessing, etc.

Example: to evaluate the achievement level resents of the learners a reading is class 10th.

(b) The Affective Domain

Krathwohl, Bloom and Masia elaborated on this domain that includes five categories of objectives dealing-with values, attitudes, feeling appreciations and other affective dimensions of learning. The categories are arranged in hierarchical order according to the degree of internalisation (Michaelis et, al 1975 p.79). Unlike the cognitive domain, which entails both product and process, the affective domain is mainly process oriented (Davies, 1976 p. 1-51). The categories are:-

(i) ***Receiving****:* This is the lowest level. It rejects to awareness or attention on the part of a learner towards the material being presented. It implies that the communication will be attended to Corresponding pupil behaviours include listening, attending, describing, identifying, etc.

***Example**:* to demonstrate an interest in music by listening to it.

(ii) ***Responding:*** *The* category goes beyond awareness and results in some sort to response on the part of a student, accompanied by a feeling of satisfaction. The student not only attends but also reacts. Corresponding pupil behaviours include answering, reading, greeting, performing, complying, following.

***Example**:* To enter the playground to play football.

(iii) ***Valuing:*** This suggests commitment and the worth a student places on a particular object, material or behaviour. Values of students are difficult but not impossible to ascertain: clues to one's values may be obtained from ones behaviour. Corresponding pupil behaviours are inviting, justifying, supporting; advocating, and preaching.

Example: To demonstrate a conviction by writing a letter to the editor of a newspaper on the need to educate the adult population.

(iv) ***Organization:*** This involves conceptualisation of values clarification of' relationships among them and organization of a value system. It involves the ability to bring together different values and an attempt to produce a coherent value system. Corresponding pupil behaviours include altering, combining, generalizing, formulating, and organizing.

***Example**:* To make a judgment on women's freedom in Islam.

(v) ***Characterization of a value complex:*** this is the highest level, which is combined with developing a life style that covers a broad range of activities consistently related with *one* another. This level may be achieved only in adulthood and is indicative of one's philosophy of life. Beliefs, attitudes, and ideas are fused into one integrated whole.

Corresponding pupil behaviours include performing, believing, qualifying and questioning, influencing i.e.

Example: To support these activities in the world

(c) The Psychomotor Domain

Anita Harrow developed this taxonomy dealing with psychomotor activities in 1972. It includes six levels ranging from the simplest to the most complex motor activities. These are as follows:

(i) ***Reflex Movements:*** There are involuntary motion responses to stimuli. They are functional at birth and develop throughout life and include, for example, stretching stiffening, relaxing etc.

(ii) ***Fundamental Movement**:* These refer to inherent body movement patients such as walking, running, jumping, pushing, pulling, and manipulating objects.

(ii) ***Perceptual Abilities**:* they involve kinesthetic discrimination visual

discrimination, adultery discrimination and coordination of eye and hand: eye and foot. They help learners to interpret stimuli in order that they can adjust themselves to their environment.' Corresponding pupil behaviours include bending, bouncing, eating, writing, etc.

Example: To demonstrate the abilities copy letters of the alphabet.

(iv) ***Physical Abilities:*** They are concerned with the vigour of the person and are, therefore, related to strength, endurance, flexibility, agility and dexterity. Corresponding pupil behaviours include enduring strenuous activity, moving quickly and precisely, touching toes, stopping and starting immediately.

Example: To demonstrate the ability to carry 120 Kilograms.

(v) ***Skilled Movement:*** These refer to efficiently performed complex movements as in games, sports, dance and the arts. Corresponding pupil behaviours include typing, skating, filing, juggling, playing musical instrument etc.

Example: To demonstrate the ability to juggle with four balls at a time.

(vi) ***Non-Discursive Communication:*** This refers to behaviours that are involved in movement communication, ranging from facial expressions to highly sophisticated communications. Responses in non-discursive communication come more from intuition than from reason. Objectives at this level are related to posture: gestures, facial expression and interpretive movement through creative expression.

Example: To move expressively so as to communicate emotions.

Although the three principal domains of educational objectives have been treated separately, they are fundamentally inter-related to each other. "When a student reviews a poem, he not only uses cognitive analytical skills but also gives his valuing and organization of values by the selection of certain skills and the insistence upon a set of criteria in evaluating and ranking the poem and poet. Besides, the differences between various levels in the domains and the examples of corresponding pupil behaviours are not inclusive. An overlapping repetition of terms between and across the domains is possible and real (Leonard and Utz. 1974 p.83)

Inspite of several limitations, the taxonomic approach is useful in many ways. As indicated before, it is helpful in planning the curriculum and in teaching and evaluation. The various domains make us attentive to the different levels or categories of objectives, some of which might otherwise be ignored in the educative process. These taxonomies have provided a great impetus to thinking in the field, of learning and education. However, they should be handled intelligently, with due regard to the nature of the subject, its methodological treatment and other genuine concerns.

2.5. Writing Behavioural Objectives

Writing behavioural objectives is a highly technical and complex task. It involves

several aspects such as deciding on the elements of objectives, seeing, taxonomic relevance, choosing proper language and sequencing. Each of these requires understanding and experience. Tyler. Mager, Michaelis, MaAshan, Gronlund. Leonard, Utz and several other writers have developed typical forms; it is neither possible nor desirable to survey all these different formats however a few significant viewpoints are mentioned for your consideration and use.

Authors differ over the necessary elements of a behavioural objective. Tyler proposes two, Mager three, Leonard and Utz four. Kibler and associates five and Michael is, Grossman and Scott as many as six elements for really complete behavioural objectives.

Tyler (1949, p.46) insists that both the intended pupils' behaviour and the related content or area of life must be included. Mere indication of behaviour is not enough: the content related to that particular behaviour must also be specified.

> ***Example:*** To illustrate the law of supply and demand. To develop critical thinking in mathematics.

'Illustrating' and 'developing critical thinking' indicate behaviours, while the law of supply and demand' and 'mathematics' point to the related content. This is the general pattern that is adopted in most curricular materials.

However, Robert Mager (1962, p. 12) mentions three elements of a specific objective. His approach is also very popular and has greatly influenced the thinking and practice of writers and educational practitioners. The three elements are as follows:

(a) The desired terminal behaviour that we want the student to perform should be clearly specified.

(b) The objective should state the important conditions (materials, procedures, activities, etc.) under which the students are to perform the desired behaviour.

(c.) The criteria of acceptable performance should be specified.

However, Mager is of the view that all three elements need not necessarily be present in each objective statement. The central purpose of writing behavioural objectives, he clarifies, is to communicate clearly the intent of a lesson course or unit. That's all. Of course, Mager lays special stress on the "doing" aspect of behavioural objectives. "An objective is useful to the extent that it specifies what the learner must be able to do or perform when he is demonstrating the mastery of the objective". (Mager, 1962 p. 13).

> ***Example:*** The student will list (behaviour) at least three causes of the revival of Islam (criteria) as given in the class textbook (conditions).

It is interesting to note that Leonard and Utz recommend four elements: Kibler, Barker and Miles, five elements and Michaelis, Grossman and Scott as many as six elements for complete performance objectives. These six elements are concerned with time, person behaviour, object, performance level and conditions. The format is reproduced below:

(a) *Time:* Indicate the time by which a student should meet the objective, e.g. "after completing this lesson at the end of the unit'… by the end of the year;…

(b) *Who:* Note whether all or only certain students should attain the objective, e.g. students who have complete 80% of the students'… all students should…

(c) *Behaviour:* Note the behaviour to be observed, e.g. state.... describe… list… name..... arrange…

(d) *Object:* Note the criteria of performance that are desired

e.g. at least three...100% accuracy.....

(e) *Conditions:* Indicate related activities or materials that the essential. e,g.. "given a ruler and. pencil'...'using a dictionary".... When given three choices... (Michaehs et, Al. 1975. pp.85-86).

***Example**:* By the end of the term, all the students of the course of 'Islamic Culture" will be able to complete a 100 item multiple-choice examination on the topic of Muslim contributions to science within one hour duration, with 70 correct answers as a lower limit of acceptable performance.

In this example, the desired behaviour is the completion of examination. The conditions are the 100 time multiple-choice tests and the one-hour time limit. Muslim contributions to science represent the object and the criterion in 70 correct answers. Other elements of this format are obvious.

The above format is exhaustive, covering almost all the possible aspects of an ideal objective. It must, however, be mentioned that most curricular materials do not generally contain all these elements. They are mainly confined to the specification of 'behaviour' and the related 'object' or the content. The most important, point in behaviour specification is effectiveness of communication. And it is "either necessary nor desirable to follow completely any one format. These different approaches are simply meant for general guidance. Therefore, Pakistani teachers and planners should feel free in adapting these approaches to their typical conditions and working environments.

Relevant educational taxonomies may be used for guidance. These will help considerably in selecting the desired learning level and matching it to the appropriate word that describes the behaviour. If the teacher wishes a student develop his or her cognitive powers, he can have a look at various levels of the cognitive domain, select the suitable level and proceed methodically in objectives formulation. Thus he can get to know the various aspects of cognition, its progressive development and the related action-verb pertaining to each level or aspect. This is how taxonomy may be used as a tool for writing objectives.

The language .of objectives should be very simple and clear. Although it looks plausible that behavipural statements should be stated in a single, sentence, it is not necessary that they should always be in this form. More than one sentence may be used to write a behavioural objective (Leonard and Utz, 1962, p.90). Clear, operational and action verbs should be used and ambiguous verbs avoided, as should be clear from the following

examples: ,

Action Verbs	***Ambiguous Verb***
To define	to know
To classify	to understand
To analyse	to appreciate
To compare	to enlighten
To compute	to learn
To list	to respect

In addition, behavioural objectives should be properly sequenced in progressive order to facilitate the achievement of the intended educational goal. Thus, they should move from simple to complex categories. Sequencing objectives helps teachers to focus their attention on several dimensions of knowledge, values and skills. They no longer remain confused to a few limited learnings, as is often the case in traditional instruction. Many new avenues are thrown open to them by way of this analytical approach.

In conclusion, besides the above consideration, several others should also be kept in view while writing behavioural objectives, e.g., the nature of the subject, the grade level, teachers' competencies, future learning's and the distinction between closed and open objectives. (An open objective is meant to develop creative behaviour, while a closed objective is designed for repetitive performance).

3. SUMMARY

Situational analysis is defined as the detailed evaluation of the content (situational) into which a curriculum is to be placed on one hand and to the application of that analysis to curriculum planning on other.

A needs-assessment is a useful technique for defining needs and determining their priorities. It is particularly helpful in schools when conducting a situational analysis as a base for curriculum planning. It involves:

(a) Formulating goals statements.

(b) Rating goal statements.

(c) Ranking goal statements.

(d) Determining discrepancies.

(e) Developing plans of action.

In a school setting, a situational analysis covers many factors, which are both external and internal to that setting, having a profound effect upon the school.

To undertake a situational analysis, a curriculum developer is required to:

(a) Identify problems in the context;

(b) Select appropriate factors;

(c) Collect and analyse date; and

(d) Make recommendations

After undertaking data collection and analysis in a situational analysis, curriculum developers make recommendations for the proposed direction and structure of the curriculum.

Curriculum intent provides direction for subsequent development by specifying the ends for learners to achieve. A statement of curriculum intent includes the aims, goals and objectives to be achieved by learners.

Aims are broadly phrased statements indicating what is hoped to be achieved by the curriculum in terms of general student behaviour. Goals are more specific, precisely worded statements of curriculum intent and are derived from aims. Objectives are specific statements of curriculum intent, are derived from goals and are expressed in precise terms, including specific student behaviour.

The sources for aims, goals and objectives include curriculum foundations (philosophy, society, learners), subject matter, curriculum conceptions, situation) analysis and educational forces such a government reports.

Objectives may be written as;

(a) general or unit objectives;

(b) specific or instructional objectives; and

(c) behavioural objectives.

Effective objectives are characterised by the presence of features such as comprehensiveness, consistency, attainability, suitability, validity and specificity.

Curriculum outcomes are related to curriculum intent and, as the intended results of teaching and learning, are expressed as general statements of expected student achievement.

4. SELF-ASSESSMENT QUESTIONS

Q. No.1: Define & explain the situational analysis?

Q. No.2: Explain the conducting situational analysis and discuss situation analysis links.

Q. No.3: Write short notes on the following steps of situational analysis procedure respectively:

1. Identify problems in context.
2. Select appropriate factors.
3. Data collection and analysis.
4. Make recommendations.

Q. No.4: Aims, goals and objectives are relative terms. Explain this statement with the help of two suitable examples?

Q. No.5: Interrelate aims, goals and objectives in a page?

Q. No.6: Explain the taxonomies of educational objectives?

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UNIT–7

CURRICULUM CONTENT AND LEARNING EXPERIENCES

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INTRODUCTION

Education is a purposeful activity, which schools organize and arrange to achieve some pre-determined objectives specifically assigned to them. Such educational activities, which take place inside or outside the school under the direction and control of the teachers, are known as “the curriculum”. How to develop a curriculum is a question of great importance and concern. Conceptual frameworks help determine relationships among various elements of the curriculum. The conceptual frameworks also help in evaluating the outcomes of the curriculum, and make it possible to modify or change the curriculum in a systematic way.

In this Unit we shall discuss, the various conceptual frameworks of curriculum development and then develop a model of our own.

OBJECTIVES

After studying this Unit, you should be able to:

1. Define the meaning of the term "curriculum content” and “learning experiences”.
2. Understand the need for content selection and know the different levels of content and their functions.
3. Understand the principles and criteria for selecting the curriculum content and learning experiences.
4. Know the main procedures for content selection and learning experiences.
5. Understand what is meant by 'curriculum organization' and be familiar with criteria for assessing its effectiveness.
6. Outline a procedure for content selection in a field of your choice.
7. State the bases for selecting instructional methods.

1. SELECTION AND ORGANIZATION OF CONTENT

Curriculum content is defined as the subject matter of the teaching-learning process and it includes the knowledge (facts, concepts, generalizations, principles and so forth), processes or skills associated with that knowledge base and the values associated with subjects or whatever is being learnt. R. Hyman, for example, has defined content as consisting of 'knowledge (i.e. facts, explanation, principles, definitions), skills and processes (i.e. reading, writing, calculating, dancing, critical thinking, decision-making, communicating) and values (i.e. the beliefs about matters concerned with good and bad right and wrong, beautiful and ugly).

Smith, Stanley and Shores in their book. Fundamentals of Curriculum Development discussed the issue of auricular content in detail. According to them, "In the process of teaching, one must teach something to someone, the someone being the pupil and the something the content. Content might be described as the knowledge, skills, attitudes and values to be learned". It includes what men know and believe, their ideals and loyalties, but not everything they have created. For example, an institution such as the family is not subject matter, but what is known and believed about the family and the ideals we hold for family life, are subject matter. Similarly, tools and machines are not subject matter, but knowledge about tools and machines about how they are made, their operations, and their uses ~ is subject matter that can be included in the curriculum.

Ralph Tyler has, however, differentiated between the terms "learning experience" and "content", According to him, the term "learning experience" refers to the interaction between the learner and the external conditions in the environment to which he can react. Learning takes place through the active behaviour of the student; it is what he does that he learns, not what the teacher does. It is possible for two students to be in the same class but it is just possible that they will be having very different learning experiences. This definition of experience as involving the interaction of the student and his environment implies that the student is an active participant The problem of selecting learning experiences involves, therefore, determining the experiences likely to fulfill given educational objectives; and also setting up situations, which will evoke the kinds of learning experiences, needed.

1.1. Levels of Content and their functions

Hilda Taba has pointed out the following three levels of curriculum content, along with their functions:

(1) Specific facts and processes

One can view school subjects as consisting of knowledge on several different levels. One level is that of specific facts, descriptive ideas at a low level of abstraction and specific processes and skills. For instance, a description of the branches of government, of the characteristics of the digestive system, dates of events, and the computational processes

in arithmetic and algebra belong to this category. This kind of knowledge is described as static. Its mastery does not produce new ideas although specific facts do constitute the raw material for the development of ideas. All students must master precisely the same content items therefore, a careful choice of the contents to study is as important as ever.

(2) Basic ideas and principles

Basic ideas and principles represent another level of knowledge.-For example, idea about casual relationships between human culture and the natural environment are of this sort. Such ideas and principles constitute what current is referred to as the 'structure'' of the subject ideas, which describe facts of generality, facts that, once understood, will explain many specific phenomena.

(3) Concepts

The third level of content is composed of what might be called concepts, such as the concept of democracy or interdependence, of social change, or that of a ‘‘set’’ in mathematics. Concepts are complex systems of highly abstract ideas on which context are built. They cannot be isolated into specific units but must be woven into the whole fabric of the curriculum.

Smith, Stanley and Shores also discussed this issue of the nature of content. According to them, there are two kinds of subject-matter, one descriptive, the other normative. A brief account of the same is however, given in the following:

(i) Descriptive Subject matter

Descriptive subject matter consists of facts and principles. Books and other instructional materials are heavily loaded with factual content. Courses in history, for example, consist largely of factual statements about what happened, when and there it happened, and who was involved in the events. Biology courses contain facts about the structure of plants and animals, their classification, and their functions etc. Descriptive principles are laws, rules and theories, such as scientific laws and theories. In general, these principles do not involve values and resemble facts. Just the objective description of principles and facts is called descriptive subject matter.

(ii) Normative subject-matter

Individuals make moral and aesthetic choices. They consist of the norms or standards, which represent the values/standards that men have built up through generations. The objectives of education, the social-moral controls of economies, the ideals operating in and through-political institutions, the legal system, the rules of acceptable family conduct, the standards, for judging the merits of artistic works-all these would be included in such a list of normative item, and it would still be incomplete.

Normative content and descriptive content may be further distinguished by relating them to the kinds of questions that might be asked about an event. What happened?. Only factual statements can answer these questions. Why did it happen? What would cause theories and general ideas, using descriptive principles rather than facts to provide the answers? Was its occurrence good or bad? Should it have happened? Should it happen again? Was it beautiful? Answers to these questions require more aesthetic judgments, and hence involve normative content. Thus, what constitutes normalize science be included in normative subject on matter.

1.2 The Principles Selecting of Curriculum Content

(1) The Need for Selection

The concept of education has broadened and, as the percentage of school-age children in schools increases, there is a growing need for a careful selection of curricular content, enabling the learners to cope effectively with the complexities of modern living. Similarly, finding of latest research studies about the nature of the learner and the education process bear on the selection of materials and experiences. The study of youth provides many clues, which are useful, and an understanding of the learner is indispensable. As changes take place in our concept of learning, they operate to produce changes in curricular content. Finally, for social continuity and national integration it is important that the Islamic values and the ideology of Pakistan are provided for boys and girls through selected content which prepares them to be true Pakistanis an competent persons in their respective fields.

(2) Principles for Selecting Curriculum

According to our socio-cultural framework the following constitute some of the most important general principles for selecting the curriculum content.

(a) Promoting Islamic Ideology

The content should reflect and promote the Islamic ideology of Pakistan.

(b) Developing Social Understanding

A common body of educative experiences should be selected which develops social understanding in judging different problems.

(c) Promoting Maximum Personal Development

The learning situations must be related to the unique characteristics of each learner. Desirable personality traits and a wide range of special interest

opportunities should be promoted.

(d) Promoting Continuity of Experience

This principle relates/to the maturity and learning sequence of each learner.

(e) Providing for educational Aims

The content should reflect the stated aims of the curriculum.

(f) Maintaining a Balance Among all the Goals

This is done through a balanced allocation of time and judicious requirements.

(g) Involving the Pupil in the Learning experiences

There should be flexibility and democratic procedures. Pupil participation and cooperation should be emphasized.

(h) Using Effective Learning experiences

The experiences chosen must utilize the most appropriate situations for promoting effective learning.

Although the particular learning experiences appropriate for attaining the objectives will vary with the kind of objectives aimed at, there are certain general principles, which, according to Ralph Tyler, apply to the selection of learning experiences, whatever the activities may be.

These general principles of content selection are briefly described as under:

(a) For a given objective to be attained, a student must have experiences that give him an opportunity to practice the kind of behavior showed by the objective. For example, if one of the objectives is to develop skill in problem solving, this cannot be attained unless the learning experiences give the student many opportunities to solve problems.

(b) The learning experiences must be such that the student obtains satisfaction in carrying out the kind of behaviour implied by the objectives. For example, in the case of learning experiences intended to develop skill in solving health problems. It is important that the experience not only gives the student an opportunity to solve health problems, but also that effective solution of these problems is satisfying to him. If the experiences are unsatisfying or distasteful, the desired learning is not likely to take place.

(c) The content should be such that reactions desired to be produced are within the

range of possibility for the students. That is to say, the experiences should be appropriate to the student present attainments his predispositions.

(d) The proposed content should help in achieving the specified objectives. No part of the content should prove to be a hindrance in the achievement of an objective while helping in the achievement of some other objectives.

(e) The content selected should ensure that the same learning experience would usually bring about several outcomes. For example, while the student is solving problems about education, he is also acquiring certain information about the field of education. He is also likely to be developing certain attitudes toward the importance of public education procedures. In addition, he may be developing an interest in or a dislike for work in the field of education. Every experience is likely to result in the achievement of more than one learning objective.

1.3. Criteria for Content Selection

It is generally agreed that content selection should be based primarily on the stated objectives. Changes in objectives require changes in the curriculum content. It is, therefore, necessary to have valid criteria for content selection, so that, the final evaluation of the curriculum content can be carried out in a proper context, Below, six standards for content selection are presented in question form:

(1) Does the curriculum content, reflect the basic principles of Islamic values and teachings?

(2) Does the content .contribute to the growth and development of an Islamic society?

(3) Is the content significant to an organized field of knowledge?

(4) Does the content stand the test of survival?

(5) Is the content useful?

(6) Is the content interesting to the leaner?

Perhaps you can suggest some other criteria. In thinking about this, look back at the principles outlined earlier.

In addition to the above, the following criteria, according to Nicholls and Nicholls, are also externally relevant and important:

(A) Significance

The criterion of significance applies where content is judged in terms of .how essential or basic it is to the discipline or theme under study. Where content is considered to be of value to the subject area, it is deemed to be significant and thus worthy of inclusion in a curriculum. For most curriculum developers this criterion involves an appropriate balance between concepts, ideas and facts. As Nicholls & Nicholls state:

> "If study were to be based on a number of carefully selected principles, concepts, ideas and facts would be learned to illustrate these and would be included only in so far as they contributed to an understanding of these. This would reduce the problem of learning the large bodies of facts, which seems to be the bugbear of so many courses at the secondary level". (Nicholls & Nicholls, 1978:52)

The significance criterion may be applied to any body of content considered for inclusion in a curriculum. For example, it may be used where content is based upon themes, problems, activities, such as in primary schools, or the most common from-subjects and disciplines, Certainly the criterion has greatest applicability where content is... perceived as a logical structure and finds its widest application in curriculum development situations that involve experts and scholars in those disciplines which the curriculum includes (Zais,. 1976:344).

Nevertheless, this criterion is somewhat problematic: when we raise the question 'Significant for whom? At this point it is useful to reflect upon the discussion made in the unit on curriculum design. Certainly those who are concerned with the curriculum development process will bring differing perspectives to that task. Many argue that nowhere is the impact of that decision more important that in deciding what is significant content for inclusion into a curriculum.

(B) Validity

An important criterion to apply when selecting content is that of validity. Content may be regarded as valid when it is authentic or true, and to a large measure this means whether the content is accurate. Accurate or true information says what it is supposed to say. For example, content that is meant to cover the geography of Pakistan should do just that and not included Pakistani economics.

A significant test of the validity of content is to determine the degree of its obsolescence. In this rapidly changing world, the obsolescence of' content is a continual problem faced by curriculum developers and those who implement curricula. Some school subjects, such as mathematics, science and social studies, appear to be in a state of almost constant flux.

The criterion of validity of content may also be measured in terms of the relationship between content and objectives. For content to be valid it must reflect the stated objectives. If objectives claim one thing while the content selected for the curriculum teaches something different then it is regarded as invalid. For example, if an objective seeks a student to achieve understanding of Pakistan's political structure, and the ensuring content deals only with political party only, then the content would be invalid.

This use of the validity criterion is particularly important for practicing teachers who implement syllabuses developed by those external to the school such as state educational authorities. Exact accord/bond between the stated intent and evident content may always be possible. Situations may arise where a subject-syllabus committee agrees

on the intent of the curriculum and then proceeds to develop content more in line with the results of political pressures amongst committee members rather than the agreed objectives.

(C) Social relevance

It is a little bit controversial to select contents for a social setting. This criterion suggests that content for inclusion in a curriculum should be selected on the grounds of its relevance to the social needs, but within a community. Thus this criterion is concerned with content relating to moral values, ideals, social problems issues and so forth that would assist students' to become more effective members of their society.

But what content would be included on these grounds? In a thinking, responsible society, curriculum developers may well incorporate content that reflects:

1. Democratic principles and values
2. Understanding of cultural
3. Social awareness and criticism
4. The facilitation of societal change

(D) Utility

A related criterion to consider when selecting content is that of utility. This criterion appears similar to the criteria of significance and social relevance but the term is defined rather in a rather specific manner, in terms of individual learners. When employed to select content for a school curriculum, the criterion of utility applies to the usefulness of content in preparing students for adult life. As such, it implies a very directly relevant and functional approach to the selection of content that will lead, to a desired outcome on behalf of the learner.

This criterion is also individually oriented, reflecting the concept of the value or usefulness of the content to individual learners who are experiencing the proposed curriculum. This is compared with the notion of value to learners in a social sense as discussed in the social relevance criterion. Together with the notion of usefulness for adult life, the utility criterion is an important consideration to be made when content is included in a curriculum.

As a criterion, utility is extremely valuable to those who helpful favour a more functional school curriculum' because it can be helpful in maintaining real world relevance in schools. Nevertheless, some schools and colleges, and many universities and other educational institutions, applying the criterion of utility to the content selection process, developers can expect a curriculum to be more relevant and hence valuable to the real world.

The rigid ultitarian approach is a static one and does not prove fruitful. At an extreme position, however, rigidly applied utilitarian criterion could well perpetuate the status quo or provide a limited perspective, on available content. This approach may cause

difficulty within departmentalised institutions (such as secondary schools) as the utilitarian needs of adulthood may suggest that family life, career education, interpersonal skills, personal health, personal finances and so forth be included in the curriculum. Such topics, however, tend to be thematic or problem oriented and hence cut across the traditional subjects and subject departments found in secondary schools. This specific issue has been raised many times in schools over the past few decades and is yet to be resolved, principally because educators have been unable to successfully address the problem of the overcrowded curriculum.

The new, however, utilitarian criterion suggests that content should be selected to meet students' present needs. Students need such content that deals with drug education, personal problem solving, multicultural understanding, financial awareness and so forth. While these approaches to the utilitarian criterion may cause some confusion amongst curriculum developers, and the criterion itself becomes doubtful this criterion is particularly appropriate for curricula devised for students.

(E) Learnability

As it is clear to select such content that students are able to learn, this criterion has not always been applied in the past. The learnability criterion is particularly appropriate to curricula that have to meet the needs of large numbers of students with diverse backgrounds and a wide range of ability (Young, 1989; Kennedy, 1990). In these cases, if the content is to be acquired by all students, then consideration must be made not to make the content too difficult for learners.

By contrast, there may be times when a curriculum developer is unsure just how able the learners may be Creating a curriculum for high-ability learners, the so-called gifted and talented, demands that content challenge such learners and so the learnability criterion would be used in a different manner. In the more general context, however, it is often difficult to take account of individual differences and hence aspects of content may not be learnable by some students. To lessen this problem and make content suitable to the level of the all students involved could reduce the content to the level of the lowest common denominator.

An associated aspect on the learnability criterion is that of student readiness. It is extremely difficult for curriculum developers to predetermine curriculum content that can predict learner readiness, beyond rather broad parameters. However, the classroom teacher, acting as curriculum implementer and curriculum adapter, is in an appropriate position to consider learner readiness. Thus, it is a professional judgment by the teacher, which will determine whether certain content should be included in the presentation of the curriculum based upon the readiness of the learners, to accept that content. Effective teachers have been making such Curriculum decisions for many years.

(F) Interest

The interest of learners in the curriculum's content is generally considered an

important criterion in the selection of that content by curriculum developers. However, it appears to be one of those criteria more valued in the theory than in the reality. Certainly, curriculum developers have accorded this criterion the lowest priority in practice.

The problem associated with the student interest criterion is the dilemma it causes when determining just how significant a role this criterion should play. At one extreme curriculum, developers could ignore student interest as a selection criterion. They could argue, perhaps justifiably so, that, they know what content students should learn. However, on this extreme criterion a student loses the potential interest and confidence.

Alternatively, curriculum content selected largely upon a student-interest criterion possibly suffers, from Whim, immature development, and individualistic emphasis. Students interests are of diverse nature and it becomes difficult in satisfy all the instincts of learners.

Curriculum, developers when selecting content for a curriculum should apply the criteria - significance, validity, social relevant, utility, learnability and interest. Teachers will find these criteria useful when, selecting content. For a curriculum document, particularly when decisions are required for their classroom programmes. Where possible, all criteria should be considered, although it appears that the criteria of significance, validity and utility are accorded greater priority.

1.4 Procedures of Content Selection

Smith, Stanley and Shores have discussed this issue in detail. According to them, the procedures of content selection are:

(a) judgemental
(b) experimental
(c) analytical and
(d) consensual.

1. The Judgemental Procedure

Selection of subject matter by judgement requires the curriculum worker to answer the following questions:

(i) What social and educational objectives should be accepted?

(ii) What is the existing, state of affairs in which these objectives are considered desirable and appropriate, and in which they must be realised?

(iii) What subject matter, best satisfies these objectives under the existing conditions?

The most objective selection of content by this procedure requires that the interest, knowledge and ideals of the curriculum worker rise above those of special' social vision and freedom from the restricting influences of personal perceptions and subjective thinking.

When the curriculum worker is collecting data to help determine objectives, or when he is using these data to select content with the judgmental procedure, he is not expected to make original social and historical investigations. The knowledge he needs should be drawn from scholarly work in cultural anthropology, sociology, economics, political science, history, psychology, and philosophy, etc.

In addition, the judgmental procedure advocates for extensive group discussion and deliberation, in which general points of view and personal and group opinions are critically evaluated and reconstructed in the process of agreeing upon social ideals and goals. Curriculum content based on the prejudices and rationalizations will not satisfy the conditions of the judgmental procedure. Nor will the selection of such content improve the curriculum.

The judgmental procedure is best demonstrated when the criterion of social reconstruction is the primary basis of subject matter selection. This principle makes social development a primary basis of content selection. It should not be assumed, however, that this is merely a matter of personal choice. On the contrary, the test of the judgment is the extent to which other persons accepting the same ideals, facing the same facts and possessing the same social understanding would themselves select the same subject matter or agree with its selection. Its successful implementation requires critical, informed, and judicious persons as curriculum formulators. In the hands of such people, it can be one of the most dependable methods of content selection. However, the curriculum worker cannot neglect any of its phases without the risk of serious misjudgement.

2. The Experimental Procedures

The experimental procedure of content selection tries to determine by actual testing whether or not subject matter satisfies a particular criterion. It answers such questions as: Is the subject matter interesting? Is the subject matter suitable for the adults keeping in view their needs and problems, etc? The evaluation of the subject matter must, be done under prescribed conditions and by techniques that make the procedure as valid as possible. Errors due to personal or group prejudices, misjudgement and outside conditions are to be reduced to a minimum. The experimental procedure of content selection takes the following form:

(i) Tentatively selecting subject matter in accordance with question.

(ii) Hypothesising that the tentatively selected subject matter meets conditions of the criterion (that it is interesting or useful).

(iii) Prescribing conditions for the evaluation/assessment (description of the characteristics of the children, the teacher, the classroom, the methods of teaching, the materials to be, used and other facts affecting the experiment).

(iv) Checking the results against the hypothesis to find whether the subject matter satisfies the criterion.

This procedure has not yet been used extensively, but its findings have been

generally valid. However, it is open to the objection that all factors cannot be controlled; and that its findings, are not entirely dependable. Another criticism is that the experimental procedures assume an unchanging curriculum in all the aspects related to the one under investigation. If this is not the case, the experimenter cannot know whether his results were resulted from conditions of the experiment or to some outside conditions that changed without measurement or control. However, in spite of all the criticisms this procedure remains one of the most familiar tools of selecting the subject matter.

3. The Analytical Procedure

The analytical procedure is one of the most widely known methods of content selection. It has the features like the criterion of utility. In general, it consists of an analysis of the things people do in order to discover the subject matter involved in these activities.

There are three forms of the analytical procedure; their brief description is as follows.

(i) Activity analysis

The purpose of this analysis is to discover the general activities of people of given national group or region. This analysis helps in identifying relevant activities to be selected for the curriculum.

(ii) Job analysis

This analysis is applied to vocational Perspectives. To determine what should be taught in "Professional Preparation of Teachers" course, for example, an analysis of their work would become the basis of the teacher education programmes.

(iii) Knowledge analysis

If the problem is to find what elements of written data are in general use, one procedure would be to analyse such relevant documentary materials newspapers journals. and library materials. The content to be included in a course of grammar might be determined by studying the grammatical forms found in the correspondence of people in general or of a selected group. The analytical procedure of contents selection, therefore consists of the application of certain techniques of fact-finding to the activities under investigation.

The analytical procedure can involve the use- of the following techniques:

(i) Interviewing

The interviewer requests a person on the job, or one who performs certain activities, to name the duties for which he is responsible or the ideas and skills he employs. From these

interview, and composite list of duties is drawn up.

(ii) Working on the Job

The investigator works on the job, studying the operations required and making a list of them.

(iii) Analysis of the Job or Activity by the Worker

A person who has become familiar with a job or activity is asked to list his duties or the operations he performs.

(iv) Questionnaires

The duties or operations involved in a job or activity are sought by means of an inquiry sent to the workers or person: they are in form of written question on statements on the problem.

(v) Documentary Analysis

The investigator makes an analysis of magazines, correspondence, public records and the like and tabulates the information, skills, or principles found in these documents. These findings throw light on what will be needed by persons who use these documents or who are involved in activities from which they results.

(vi) Observing the Performance of People

The curriculum worker, or a group of persons under his direction observe and tabulate their observations and findings. This may be concerned with the daily activities of people or it may involve analysis of individuals or groups in certain specific situation.

The main criticism on the analytical procedure is that it breaks operations, skills and knowledge into such small elements that the real identity of the original process or knowledge is lost.

4. The Consensual Procedure

The consensual procedure is a way of collecting people's pinions, about what they believe the curriculum should be. The results of the consensual procedure are expressed in terms of the number of persons, or percentage or persons, of a particular community or group who believe that such and such should be taught in the schools. The selection of persons whose opinions are to be sought is the first step. Such persons are usually selected because they are:

(i) Outstanding leaders from all works of life such as educators business men and workers etc.

(ii) Experts and specialists such as physicians, engineers, teachers, and artists or

(iii) Representatives of the population of a community or region.

The next phase is the development of a method for collecting opinions. Usually a questionnaire is used. In some cases, interviews or small group conferences are employed. The final stage of the consensual procedure is the tabulation and interpretation of the responses. This procedure is perhaps reliable, when it is used along with the judgemental procedure. However, it is also subject to the objection that people's replies are too often dictated by stake, unexamined prejudices and occupational basis. It is also apparent that this procedure results not in consensus but in a tabulation of votes.

1.5. Romine's Classification of Procedures

It is difficult to separate "'selection" from organization, particularly at the actual point of making choices within the overall framework of the curriculum. For that reason, the procedures dealt with above apply in varying degrees to both selection and organization of curricular experiences, suggesting means of establishing courses and other aspects of the total curriculum. A number of suggestions may be found in educational literature dealing with this task of curricular construction. Some authors discuss one or more of the procedures in terms of determining the scope of the curriculum, others in terms of the selection of curricular materials and experiences. Some of the procedures are more applicable on a broad basis, while others are more pertinent to courses of study, units, projects and related classroom activities.

Stephen Romine has discussed some procedures which can be used in selecting and organizing curricular experiences: analysis of text books surveys of opinions, the study of errors, the study of the curricula, analysis of adult activities, analysis of social functions-, and study and analysis of youth interests and needs. Although some of these have, to some degree, been discussed under the above four procedures, it would be interesting to discuss them further.

(i) The Textbook Procedure

This procedure is the most commonly employed of all for selecting and organizing curricular experiences. The content of a course or a curriculum based on this approach is determined in large part, if not entirely, by the selected textbook (or textbooks). Essentially, it assumes that the writers of texts are sufficiently expert to know what is appropriate for the youth who are required curricular experiences and use the textual materials in so doing. While such persons generally have access to such basic data and do a

real service in bringing together and organizing materials, it is doubtful whether they are capable of accurately diagnosing pupils background and needs as they are found in the thousands of classrooms over the nation. Yet in the hands of many teachers, the text serves as the prescription for all pupils, regardless of individual differences amongst them.

(ii) The survey of Opinions Procedure

This procedure has several variations and is consictered as the consensual or expert opinion procedure. Subject-matter committees, for example, make pronouncements based upon the study and opinions of the members. These decisions frequently are based upon considerable studies and deliberations and are worth careful consideration. On other occasions, the recommended ideas or practices may lose significances as unstudied opinion. The value of the procedure is greatly in restricted areas of the curriculum, as in some special subject field, or in dealing with special problems. In this procedure five groups of persons may be involved individually or collectively, (a) experts in a given field, including teachers, (b) specialists in professional education, (c) lay leaders, (d) the general public and (e) pupils. Opinions may be solicited through the committees established to study given problems.

(iii) The Study of Errors Procedure

The idea behind this procedure is that of identifying errors and shortcomings as a basis for incorporation. This is a form of analysis, which concentrates on incorrect or improper behaviour, judged in terms of given criteria of values. In the field of social studies, for example, a study can determine were not in a harmony with each other the reasons why different racial groups in a. community. Based on the findings, a series of curricular experiences should be outlined and materials selected to help promote better understanding and relationships that are more cooperative. In mathematics, a teacher may study pupil performance, determining weaknesses in several respects. Remedial work may then be implemented. For the purposes of diagnosis at the beginning of courses, a more extensive analysis may be made of both the strengths and weaknesses of pupils.

(iv) The Study of Other Curricula Procedure

Sometimes it is called as the scissors and paste" method, this procedure entails the study of curriculum guides, courses of study and similar materials prepared by other schools and agencies. The general idea is to determine from such study what others are doing which may be useful in one's own school situation. In this way a teacher, or a school, can use the substance of curricular content. A teacher, may, for example, simply look over a course of study in science that is used in another school and adopt it exactly as it is for use in his own Class. This danger must be avoided if the procedure is to serve any good purpose. In Pakistan we have prescribed content, a teacher may, however, come to know about methods of teaching as practised in other schools.

(v) The Analysis of Adult Activities Procedure

Activity analysis is not a new procedure, but it still is essential for curricular content Generally it involves the study of life activities, citizenship, vocations, and so forth to discover significant activities concerning which youth should be taught to be proficient. It is more commonly recognized and utilized, however, in relation to vocational education, wherein analysis is directed towards the identification of specific skills. Once such identification has been accomplished curriculum experiences are set up to team the skill involved.

(vi) The Social Function's Procedure

Related to the activity analysis approach, but broader in point of view, is what may be called the social functions procedure. This procedure has its basis in the concept of education as social process, and, as a primary step, it involves an analysis of the social functions which adult citizens should be capable of performing. Because it requires a broad grasp of the total environment and deals with larger areas of living. This approach is more applicable to such larger areas of the curriculum as for example, the whole so can studies programme. It may however, lead to selection that is more detailed and organization as an outgrowth of the larger planning which is done initially. This procedure entails comprehensive study of social, political, economic, and related conditions, influences and trends and these, may be approached through surveys, study of literature, research and other such methods.

(vii) The Youth Interests and Needs Procedure

Certain similarities exist between the social functions procedure and that of youth interests and needs. The former takes its scope from the functions of adult living; generally classified into a number of large areas of life problems, its sequence preferably is based upon the background, maturity, interests and needs, of boys and girls, although it may be chronological and logical. The youth interests and needs procedure involves consideration of persistent problems too, but both its scope and sequence are based upon youth, it involves the personal and social functions of boys and girls rather, than of adults. It does not avoid preparing for adult living. But its primary focus is on the here-and-now, young today and the things, which concern them.

1.6. Principles of Curriculum Organization

The principles suggested for use in selecting educational content are also useful in considering organization and may be modified slightly to become more applicable for such a purpose. The pattern of the curriculum with its sub-divisions should be planned and structured according to the bases which have been suggested as underlying the operation of the school-the purposes of education, the needs and abilities of the learners and the educative process.

In terms of these, a few general principles can be adopted to outline the job that should be done with respect to curricular organization, viz

(i) Provide for scope and sequence with flexibility

(ii) Provide a common body of educative experiences, along with, which are essential in meeting special interests and needs.

(iii) Be consonant with the manner in which learning takes place.

(iv) Implement the attainment of stated objectives, and encourage the evaluation of pupil growth and development in terms of these objectives.

(v) Consider to the individual learners and involve them in cooperative curriculum planning and active learning ventures.

(vi) Relate to the total educational picture in a way to affect favourably the learning climate in the school and in the community.

(1) Curriculum Organization?

According to Ralph Tyler, important changes in human behavior are not produced overnight. No single learning experience has a very profound influence upon the learner. Changes in ways of thinking, in fundamental habits, in major operating concepts, in attitudes, in abiding interests and the like develop slowly. It is only after months and years that we are able to see major educational objectives taking marked concrete shape. In some respects, educational experiences produce their effects in the way water dripping upon a stone wears it away. In a day or a week or a month, there is no appreciable change in the stone, but over a period of years definite erosion. is noted. Correspondingly, by the accumulation of educational experiences profound changes are brought about in the learner.

(2) Criteria for Effective Curriculum Organization: Continuity, Sequence and Integration.

As discussed by Tyler in Basic Principle of Curriculum and Instruction, there are three major criteria to be met in building an effectively organized group of learning experiences. These are continuity, sequence and integration. A summary of Tyler's discussion is given below:

(i) Continuity

It refers to the stability and consistency of the learning experiences. This refers of major curriculum elements. For example, if in social studies, the development of skills in reading social studies material is an important objective, it is necessary to see that there is a lasting opportunity for these skills to be practiced and developed. In similar way if an objective in science is to develop a meaningful concept of energy, it is important that this

concept be dealt in various parts of the science course. Continuity is thus seen to be a major factor in effective vertical organization of curricular contents.

(ii) Sequence

Keeping the learning experiences in its logical order it is called sequencing again it. It is possible for a major curriculum element to occur again and again but merely at the same level, so that there is no understanding or skill or attitude. Sequence as a criterion emphasizes the importance of having each successive experience built upon the preceding one.. For example, sequence in the development of reading skill in social studies would involve the provision of increasingly more complex social studies material, increased breadth in the operation of the skill involved in reading these materials, and increased depth of analysis so that the sixth-grade social studies programmes would not simply reiterate the reading skill involved in the fifth grade social studies programmes would not simply reiterate the reading skill involved in the fifth-grade but would go into them more broadly and deeply. Correspondingly, sequential development of a concept of energy in the natural sciences would require that each successive treatment of energy would help the, meaning of the term "energy" Sequence emphasizes not duplication but higher levels of treatment with each successive learning experience.

(iii) Integration

It is sequencing and relating learning explain in a horizontal. This refers to the horizontal relationship of curriculum experiences. The organization of these experiences should be such that they help the student increasingly to get a unified view and to unify his behaviour in relation to the elements being dealt with. For example, in developing skill, in handling quantitative problems in arithmetic, it is also important to consider the ways in which these skills can be effectively utilized in social studies, in science, in businesses and other fields so that they are not developed simply as isolated behaviours to be used in single course, but are increasingly part of the total capacities of the student to use in the varied situations of his daily life. Correspondingly, in developing concepts in the social studies, it is important to see how these ideas can be related to work going on in other subject fields so that increasingly there is unity in the students outlook, skills, attitudes and the like.

2. SELECTION OF TEACHING STRATEGIS

Learning activities may be defined as those activities offered to learners in the teaching-learning situation which are designed to enable them to require the designated content and thereby achieve the stated objectives and move broadly) the curriculum's intent. This definition incorporates all teaching or instructional strategies planned by teachers according context of the classroom or learning environment (such as independent study, fieldwork and excursions).

In the cycle of the curriculum process, learning activities are connected to content and curriculum intent. Just as content is derived from statements of aims goals, and objectives,, so curriculum developers seek to implement that content effectively through the use of appropriate learning activities. In this way the various methods not only teach the content but also help achieve the initially stated objectives.

This relationship can be explained seen well in terms of an example. In a primary school, a group of teachers acting as curriculum developers want upper-primary students to become more aware of science concepts and principles (statement of objectives). Appropriate knowledge, skills and values (statement of content) have been developed to extend the primary science syllabus and to match the level of student's cognitive development. The curriculum developers have suggested that students acquire content to understand base metals and their properties. The question now arises. What do we know about the learning process and our learners that can facilitate student acquisition of the curriculum's content?

This is a very real problem for both the curriculum developer and the classroom teacher to resolve. After all, the developer's intention is to provide the best possible guidance to teachers, who will implement this curriculum. Consequently, curriculum developers who are in close contact with current school practice are more likely to include appropriate, teaching-learning strategies in their curricula.

Thus the curriculum developers have the situational analysis and the curriculum intent stated, that a combination of interactive teaching and practical experiments would be appropriate methods for upper-primary students to learn about base metals. It may have been that demonstrations and audio-visual presentations were appropriate or under certain conditions, perhaps even an inquiry approach. The selection of appropriate teaching-learning strategies reflects the curriculum developer's professional understanding of the task at hand and the needs of students, given the directions from the earlier curriculum elements. Later in this section, a model is recommended for the selection of appropriate teaching-learning strategies,

The example above can be applied for any curriculum being developed. While in certain areas of formalised learning some methods have become commonly accepted, there may be good arguments for students learning in other ways.

2.1. Teaching-learning strategies

Teaching learning strategies are the organized teaching learning application in classroom, labs etc. The need for variety *of* teaching-learning strategies is taken up by numerous curriculum writers and educational researchers (Zais. 1976: Pratt, 1960: Joyce & Weil. 1992; Brady, 1992; Saylor Alexander & Lewis. 1981) and the literature suggests that three arguments support this position.

(1) Not all students learn equally well when the same strategies are employed: For example, some students prefer to learn through an inquiry method while others favour an expository approach.

(2) Certain teaching-learning methods are more applicable to particular situations. Lectures, for example, are not as appropriate when one is trying to develop student self-concepts as small group work or individualised tasks. No one strategy is appropriate, or can hope to be appropriate, to all learning contexts.

(3) Not a single method is superior, particularly in terms of student performance, to another in all learning situations. One may be more efficient in one situation and less effective in another. For example, lectures are very efficient use of teaching resources and may assist an educational institution to distribute resources more efficiently. But lectures may not produce effective student understanding and application of certain skills, as in a science experiment. Consequently, strategies must be matched to objectives so that the most efficient and effective one is selected-.

The following are some of the more significant teaching-learning strategies available to teachers in schools. When examining each category of method consider the advantages and disadvantages of each approach as well as the essential features involved. Furthermore, it is useful to view this variety of strategies as a problem-solving device to overcome the difficulties experienced in classroom teaching. If, for example, a teaching strategy is not working effectively, you should reflect upon whether it is appropriate to the learning context. Using the model (figure 3) outlined later in this section, you may locate a more appropriate teaching strategy. Using this model, or a similar procedure, is the way by which, curriculum developers should have selected appropriate teaching-learning strategies to meet their curriculum's intentions.

When considering a variety of teaching methods one should also distinguish between different categories, as well as within the same category in order to locate the must appropriate strategies. The main categories groupings of teachings-learning strategies are:

(1) Expository teaching
(2) Interactive teaching
(3) Small group teaching/discussion
(4) Inquiry teaching/problem solving.
(5) Individualisation
(6) Models of reality

Let us briefly examine each of-these categories of teaching-learning, strategies, in terms of their nature, examples, advantages and disadvantages. However, it should be noted that this is but a cursory analysis and is undertaken to assist the curriculum developer to be more effective in recommending an appropriate match between content and learning activities.

2.1.1. Expository teaching

Expository leading is direct approach and one way of leading. This method involves the transmission of information in a single direction from a source to learners. The source may take a variety of forms (books, television, persons), although it is commonly a teacher in a classroom. Learners arc essentially passive receivers of information in this strategy. Common examples of expository teaching are lectures, demonstrations, set reading tasks and audio-visual presentations.

The open learning university-level programmes transmitted through the television are an example of this method. Early analysis of student participation suggests that this is a significantly more popular approach to university education than was previously believed. This approach may also assist the large number of students currently unable to obtain places in our universities. In fact, the Allama lqba Open University is primarily utilising the technology of television.

Students receiving this information should be observing and listening to the best information synthesised from various sources and research and, hopefully, presented in an easily learnable way.

It is quite possible: that the person delivering the lecture, giving the demonstration or providing the audio-visual presentation is the most knowledgeable person available on the subject. In a one or two-hour presentation, a student could acquire material that would take many hours of searching and reading in the library and hence this is very efficient for the learner as well for the cumettfum provider.

Finally, through the use of Expository teaching, instructors may feel confident that they have covered the course. When they are in control of the dissemination of content they know that, at least in terms of information dissemination, the curriculum's objectives have been met and students have been exposed to the curriculum's intent. The final effectiveness of the learning, of course, rests with the learner.

However expository teaching does have many limitations as well. These methods emphasise passive learning behaviour on the part of students. Psychological studies suggest that passive methods may be relatively ineffective as the student may lack close association with the material to be learnt. Similarly, psychological experiments have also demonstrated that the retention rate of inibrnration from these sources, particularly lectures, tends to below over periods of time. And where the information is presented poorly expository teaching suffers greatly in terms of learning effectiveness.

Finally, students receive minimal immediate feedback and hence are not aware of the accuracy of their learning. Students, who participate in forms of distance education,

have shown they prefer those forms of external study that provide opportunities for quicker feedback and interaction with teachers. Recent research (Hattie, 1992) shows that feedback to learners' has one of the most powerful effects upon student learning.

Overall, expository teaching is very efficient in terms of the use of teaching resources and addressing the curriculum, but generally less effective in terms of student learning.

2.1.2 Interactive teaching

Similar in nature to expository teaching, the essential difference with interactive teaching is the deliberate encouragement of interaction between learner and teacher. This usually takes the form of question-reaction episodes inter-dispersed with expository information. Sometimes known as the discussion or question-discussion technique, it incorporates the successful features of expository teaching with interactive and feedback elements.

The advantages of interactive teaching usually overcome most of the disadvantages of expository teaching while retaining many of the advantages of the latter. Interactive teaching incorporates effective and efficient use of 'resources, provision of immediate feedback, more active learner participation and more opportunities for remedial and extension work. The immediacy of feedback assists effective learning and is further enhanced when it can be initiated by the learner and is consequently particularly powerful (Haftie, 1992). Thus the opportunity for teacher-learner interaction is the major advantage of this category" of method.

This method is extremely popular in schools for it allows classroom teachers not only to address curriculum requirements and to enhance student learning, but also to control students more effectively. Interactive teaching is also a mainstay of—schools as some strategies, such as lectures, are largely inappropriate teaching-learning strategies for some learners. For example, we would not expect year 2 students to learn about family and societal roles through lectures, though television may be an appropriate medium.

However, interactive teaching is less effective in the utilisation of resources. For example, smaller groups of students are taught in the one venue and at the same time than by the lecture method and hence more resources teachers, rooms, audio-visual machines are required for interactive teaching. If a teacher were to have several classes in the same subject and at the same level, then the interactive method requires repeat teaching. This is clearly Inefficient use of time and labour even if-preparation time is saved through repeats. Where repeat sessions do not occur then high preparation time is also a disadvantage. Consequently, interactive teaching is less efficient in terms of teaching resources, but for most educators this is more than offset by the enlianced quality of students learning and the nature of teaching-learner interactions.

2.1.3. Small-group teaching/discussion

It refers to teach the students in small groups. The principal feature of this strategy involves the division of a class into small groups, which work relatively independently to achieve a goal. In most cases this task is addressed through a group discussion procedure. Here the role of the teacher changes from one of transmitter of knowledge to coordinator of activities and guide to information and its processing. In small groups students are usually set learning tasks within the classroom context. This may involve, for example, students discussing a topical issue in social studies, resolving a problem in science or raising questions about a novel's main character. Through the interaction engendered by group discussion students acquire the skills of planning and organizing work, developing arguments, sharing knowledge, dividing tasks, adopting compromise positions and so forth. Examples of small-group teaching include group discussions, some tutorials, some seminars, 'buzz' groups, brainstorming groups and so forth.

The advantages of small-group teaching are like the interactive benefits acquired by students. The skills mentioned above the learnt through student-student and student-teacher interaction. For many students, small groups are the only school opportunity for them to acquire these skills effectively. Students have much to learn from each other and as such this strategy should not be dismissed lightly. Similarly, the ability to learn cooperatively, especially to solve problems through working in groups, is an extremely valuable skill.

On of the more significant lessons educators have learnt from the 1980s is to move away from excessively competitive, individualistic approaches to learning towards more cooperative, group-based learning approaches. Small group activities, particularly discussion groups, are an excellent way to achieve this goal. Curriculum developers should be deliberately finding ways to include this type of strategy in the curricula they are devising.

While many teachers use small-group teaching, particularly at secondary and tertiary levels in subjects such as English; social studies, history and legal studies, there is nevertheless, considerable resistance to utilising this strategy in many school subjects such as mathematics and science. Apart from requiring a changed teacher role, the strategy is invariably time consuming and appears to cover little content, particularly that which is set in the curriculum. However, more up-to-date curriculum developers have realised the value of group work both terms *of* skills and the values they engender, and incorporated such deliberately within their curriculum.

Some teachers criticized it as unproductive activities on worked if as especially when measured by the volume of noise generated. While students do not require complete silence in order to learn, and they cannot participate unless they talk to each other, more effective group work is kept task-oriented by teachers. As well, teachers need to watch that student attitudes, expressed effectively, through interaction in the small group, do not become little more than the pooling of ignorance. As well this strategy is invariably disruptive until students are experienced, somewhat haphazard in progress, often boring

for many students, frequently non-task-oriented and certainly conducive to creating classroom disruptions.

However, recent developments suggest that one thing we can learn from other countries is the ability to cooperate more effectively and to solve problems in group. Perhaps; it is suggested by many, our society, and our educational system, has promoted excessive individualism and individualistic competition at the expense of solving problems together. Business and industry are moving in that direction and now it is time for our school system to promote the virtues of cooperative group work. Teachers can face resistance from students but this largely reflects their inexperience with this teaching-learning strategy.

2.1.4. Inquiry teaching/problem solving

Generally learners are actively engaged in determining answers to questions for resolving problems in the strategy. Instead of teachers providing information for learners to digest, students are first posed a problem, a question to solve, a dilemma to resolve or an issue to address, which they learn about through the process of finding an answer. A variety of inquiry strategies are available and the inquiry approach may be used the almost any teaching-learning situation and with any subject matter. Inquiry teaching typically involves learners in four stages of activity, which may be conducted relatively independently, in pairs or in larger groups.

(1) Problem awareness: creation of doubt in, student's minds.

(2) Forming tentative hypotheses or possible solution to the problem or issue by the learner.

(3) Research and collection of date to test those hypotheses.

(4) Forming conclusions based on the evidence collected and accept or reject the hypothesis/possible solution.

A fifth stage, generalising conclusions and applying them to new data, is sometimes included in this strategy. In all of these stages, the essential feature is that the learner is actively involved in resolving the problem while the teacher serves more as a facilitator to enhance the problem-solving process.

Sometimes called enquiry teaching, discovery learning, problem solving, inductive learning, or even the scientific method. This appears to be mainly because it is a different from interactive teaching, as it involves more work for teachers. However, this should not be the case for there are numerous advantages both for learners and teachers when using inquiry/problem-solving methods.

Inquiry teaching learning is equally appropriate to a wide variety of academic subjects and themes in school. One could, for example, apply this method to social studies, English, science, mathematics and so forth. In all of these cases the role of the teacher changes to become more of facilitator and provider of resources, while the learner becomes more actively engaged in resolving the problem. And therein lies its main advantage and

disadvantage.

The main advantage of inquiry teaching is the diverse learning occurs within students. Learners participating in the inquiry/problem-solving approach are more likely to associate closely with their learning task and, by so doing, enhance the depth of their understanding. That is, learners who resolve problems for themselves retain that understanding more effectively because of the close identification with acquiring that information. In this-way students develop a range of inquiry-based skills, which may be translated to realistic situations in later life. Finally the inquiry strategy promotes logical, thinking, a positive approach to problem solving and more systematic approaches to decision-making. Where this strategy is linked with groups of learners, the problem solving approach is potentially more effective.

However inquiry teaching/problem-solving is clearly not popular in school at least with many teachers and this may be attributed to its numerous perceived disadvantages to teachers. It is an extremely time-consuming procedure when undertaken in classrooms and this places it in direct competition with other strategies attempting to address the demands of heavily knowledge-based curricula. The amount of content covered in say, science using the inquiry strategy would be but a fraction of that completed using expository or interactive strategies. Initially, it would also be more time consuming in terms of teacher preparation and student lack of familiarity with the procedure.

Furthermore, some school subjects are clearly not suited to an inquiry approach. Elements of curricula within industrial arts, home economics, physical education, science and so forth are obviously inappropriate to student experimentation if only for concern of student safety. Allowing students to inquire into the use of power saws in an industrial arts class on woodwork, for example, would clearly be inappropriate. Nevertheless, many subjects are appropriate and teachers could well give greater consideration so the inclusion of the inquiry strategy in the in the repertoire of teaching-learning strategies found in .their curricula. Curriculum developers should take a proactive position with this strategy and. as with the small group strategy, incorporate it, where appropriate, within the curriculum they are developing.

A variation of inquiry, which emphasises a team or group approach to problem solving, is known as group investigation (Sylor et al., 1981:288). Less, emphasis is placed here upon the scientific approach to data collection, although data analysis is important, while its strength lies with group cooperation and teamwork to investigate the problem. As mentioned above, this type of activity offers much for the future when extrapolated to adult, working environments.

2.1.5. Individualisation

It is an indirect approach which involve the learners to work in individually by themselves. The essential features of individualised learning are that learners complete tasks appropriate to their ability level and proceed with this learning at their own pace. The locus of responsibility for learning changes from the teacher to the learner and considerable learning is undertaken independently. Two forms of individualisation have been quite

commonly found in schools in the past (though less evident recently) and they are equally appropriate to curriculum in other educational and adult learning situations.

The Individualisation is student oriented strategy. Student learning is powerful when individualization is perceived as resolution of problems. Some of the highly structured materials are not seen as personalized although they may still be highly relevant to student needs. Unstructured individualization, however is invariably highly personalized and problem oriented and, as such very effective. In many gifted and talented programmes, for example, unstructured individualisation is both commonly used and highly desired as a means of enhancing student skills and knowledge.

Individualisation has the additional advantages of meeting student needs more appropriately and allowing for students to progress at their own pace. Such tasks many even by undertaken outside of school hours and location and student self direction may be enhanced considerably.

Many teachers resist introducing individualization into their classes as it required substantially more teacher time for organization and preparation Individualization does not have to be taken to its extreme; it certainly will involve the teacher in more initial course restructuring and lesson preparation.

Furthermore teachers have expressed some concerns about individualisation as a teaching strategy as they have less direct control over student learning particularly student exposure to content. In this situation the teacher may be unsure as to the appropriateness and breadth of the content material acquired by the learner. For extension work this may be less of a disadvantage but if teachers were to divide content into sections for individualized work concern may be evident over the students' total coverage of designated content.

Finally individualisation is different to collective work approaches as is put on individualisation of the learner.

2.1.6. Models of reality:

As alternative way to facilitate effective learning is to involve in learning situations that are as real life as possible. Sometimes it is preferably to participate in "real life" activities (such as work experience), but for many reasons this option is frequently, not available in schools. Yet we know that most students will learn certain things very effectively if we can expose them to learning activities that are as life-like as possible. In these situation we need to replicate the real world by creating models of the salient features of that world, i.e. create models of reality. Simulations, physical models, games and role playing are examples of these models. By involving students in experience that they might encounter in the real world, albert simplified for pedagogical purposes, learners have the opportunity to obtain a powerful understanding of the situation.

Examples of models of reality range from physical simulations through work models (practice leaching) through simulation games to role-playing exercises of various forms (perhaps involving students in a racial discrimination exercise so that they feel what it is like to be discriminated against.

In those models that have a high degree of reality structured within them, students learn in a situation that is as close as possible to a real life situation or context. Airlines, for example, have both flight-deck and cabin simulators that

Figure 1: Models of reality

High reality——————————————————— Abstract reality

Physical Models	**Work Models**	**Game/simulations**	**Role-playing**
Flight simulator Architectural Ship simulator	Practice teaching Articled clerkship Apprenticeship	Star power Monopoly Test cricket	Students adopt rules And act out situations To reflect: poverty job interview share trading

can replicate any known flight situation including crashes. Cabin crew and (light crew experience regular training and testing situations using the simulators so that they will know exactly what to do, particularly in cases of emergency. Many police and military forces have constructed simulation activities for combating terrorist attacks and similar emergency situations.

In schools, physical-models of reality arc relatively uncommon, cause they are very expensive to provide. One example, however, is kitchen set up in a home economics classroom. Here teachers have attempted to replicate, a household kitchen in order that students might learn to cook in as real a context as possible. Similarly, a business studies class may be set up as an office, serving counter computer station and so forth. However these are expensive options and many schools cannot afford such luxuries.

At the other extreme a classroom teacher may wish to develop empathic attitudes in students by involving learners in a role-playing exercise based on people with physical deformities. Similarly, an economics teacher might wish to familiarize students with the working of the Pakistani Stock Exchange by simulating share trading activities.

The essential procedural steps in models of reality commence with the teacher establishing a scenario (what is happening), students agreeing to abide by the rules at the model and then lengthy period of participation. All forms of simulations and role-play should also finish with a thorough debriefing session. Debriefing is a very important feature of any model of reality (especially role-play and simulations) and must be incorporated within the curriculum. And as part of the learning exercise students can be involved in reasearching the nature of the models they are using. In a simulation on share trading for example, students could research what it means to be defined as a trader, what shares are how they are traded in a modern computer system, what sorts of experience people might encounter when moves are made on the exchange and so forth.

Models of reality are very powerful learning strategies and their advantages are

numerous. They involve the learner thoroughly in the situation concerned and by attempting to deal with reality the student learners effectively and pervasively. Models of reality also provide learning situations that are often difficult to comprehend in other ways. Teaching values in a classroom learning how to cook a super tanker are difficult to achieve effusively by lecturers, discussions and so forth sponge correctly or how to steer so forth. For some forms of learning to be effective the student simply has to learn by doing.

Some schools have developed simulation games to teach less cognitively able and less cooperative students by involving them in a more experimental form of learning. Teaching the medieval system in history, for example through lectures discussions or audio-visual material is very efficient, but for many students it would be boring and dry. But to create a simulation game which re-created peasants and merchants undertaking their daily lives many hundreds of years ago is quite different learning experience. Such a simulation was created for a group of less able student, who experienced considerable difficulties learning the curriculum through expository and interactive teaching methods. But a thoughtful teacher adopted the curriculum for students to participate in a medieval simulations and the result was one of the most powerful learning experience those students had ever encountered.

Yet the disadvantages of reality models have meant that for many students in schools, models of reality are almost unknown teaching-learning strategies. Teachers often don't employ them because they are very time consuming, have questionable learner effect in terms of achieving curriculum goals require substantial additional teacher preparation and usually demand teachers change their standard classroom procedures. Consequently many students have little exposure to models of reality and when they do they often act inappropriately.

While for many training situations these strategies are essential (for example, fight-deck and cabin simulators are an essential part of airline training before crew and attendants take to the air. In schools they are perceived more as optional and certainly as expensive and hence are employed rarely. School timetables often don't allow for role-playing and simulations to occur (it is usually difficult to start and stop a simulation within 40-minute period) and many teachers are unaware how to conduct these strategies effectively consequently models of reality remain, in schools, one of the most underutilized yet effective, learning-teaching strategies available.

2.2. Other teaching-learning strategies

Some writers distinguish practice/drill as a separate strategy (Saylor et ad., 1981; Marsh, 1986), although they may be considered more a method of reinforcement than a learning strategy. Once a student has learnt something through one of the above strategies, is useful to reinforce that learning through practice/drill. The important point here is that learning is usually acquired elsewhere and the practice or drill serves to consolidate that learning in schools. Practice and drill are continuously used in subject such as maths physical education, dance and so forth. The reinforcement role they serve is important though this is usually more the responsibility of the teacher as user of a curriculum, than

the curriculum developer.

A useful way of summarizing the teaching learning strategies discussed in this unit is to place, then in perspective with each other, such as in a continuum. At one extreme of learning strategies on the continuum-below we could include learning through reality. Learning by participating in a real situation is certainly a powerful way of acquiring information, skills and values. However, it is one over which, curriculum developers and teachers generally have no control as it is beyond the domain of the educational institution concerned. While reality may be effective teacher it is rarely an efficient, systematic and thorough teacher where learning through a specifically designed curriculum can be extrapolated from one context to another.

In recent years therefore, we have witnessed a significant trend towards formalised learning situations in many institutions involved in some form of education. Take, for example, learning how to sail a yacht. One could lean through participating in real situations through trial-and-error as we call it. But does learning to sail-on a mirror or similar, centerboard dingy prepare one effectively for sailing an ocean-racing sloop? To some degree the answer is yes, but if one participated in a sailing school and learnt how to sail through learning

Figure 2: Teaching-learning strategies

Low reality

High teacher participation _____ participation Low teacher

Low leaner involvement High leaner involvement

The sailing curriculum, then one should have acquired generic and specific skills transferable to many different contexts.

2.3. Criteria for selecting learning activities

How does one select appropriate leaning activities from this diverse teaching-learning strategies to include- in a curriculum? Obviously not all activities are equally applicable to all teaching situations. Yet it is equally obvious that teachers have preferred methods, which they use constantly, almost to the exclusion of all other.

Curriculum developers have a responsibility to advocate, appropriate teaching-learning strategies for a curriculum even they believe that classroom teachers will make their own strategy selections based upon their own criteria. While teacher professionalism and autonomy should not be diminished, the curriculum developer may have very important reasons for recommending particular strategies. To achieve the desired empathic intent in a history curriculum, for example, a curriculum developer may recommend specific strategies which would achieve that intent, such as the use of simulations, role-play, small-group discussions and so forth. Using such teaching-learning

methods would help history teachers achieve the desired intent of empathic understanding of a particular historical era and Context.

Several authors have posited criteria for the purpose of selecting appropriate methods in a school situation. McNeil (1985) argued for philosophical criteria, psychological criteria, technological criteria, criteria from pressure groups and practicality as a criterion in the procedure for selecting teaching-learning strategies. While these criteria are valuable, some of them lack direct practical application for school curricula.

Zais (1976: 355-64) suggested aims, goals and objectives: foundation commitments; content: and students' experience as appropriate criteria. Brady (1992), however, argues for the selection criteria to include variety, scope, validity, appropriateness and relevance as means by which to judge learning activities.

When selecting criteria it may be useful to employ Wheeler's 'principles of learning'. Learning theory can provide an insight into students learn and therefore help match appropriate learning activities. However, there is probably more agreement amongst educators over the principles of learning than there is about learning theories. Wheeler's principles arc included below (1967: 130):

(1) Learning is an active process in which the learner must be involved.

(2) Learning proceeds more effectively if as well is being an active participant, the leaner understands what he is learning.

(3) Learning is considerably affected by individual goals, values and motives.

(4) Frequent repetition of response to a class of situations is important in learning skills.

(5) Immediate reinforcement promotes teaming; Cognitive feedback is most effective when time lapse is minimal.

(6) The wider the range of experiences presented to the learner, the more likely are generalisation and discrimination to occur.

(7) Behaviour is a function of the learner's perceptions.

(8) Similar situations may elicit different reactions from different learners.

(9) While transfer does occur, is usually much less than people think. What there is may usually be attributed to similarities between the tasks involved. Both likenesses between situations and possibilities of-transfer should be pointed out specifically.

(10) Group atmosphere affects both learning product and accrued satisfaction.

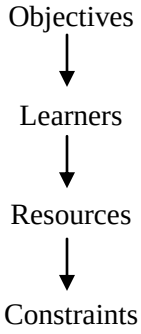
(11) Individual differences affect learning. Such differences are both biogenic and socio-cultural.

(12) All learning's are multiple. Although focus may be on one particular (desired) outcome, other learning take, place simultaneously.

With these principles in mind what is a useful procedure for selecting appropriate learning activities? “Percolating” procedure is helpful to teachers seeking to select the most effective teaching-learning strategies for curriculum: development and for the teaching of a curriculum's content their classrooms. In this model the process of percolating through the four levels is set within the broader context of the curriculum conception favoured by curriculum developers. Thus those who prefer a more academic discipline/ cognitive skills perceptive to curriculum tend to favour more expository and interactive teaching-learning strategies. While this may appear logical, curriculum developers should take cognisance of preferred learner styles, which will encourage teachers to balance the emphasis upon, strategies. Certainly curriculum developers should encourage teachers to employ a variety of teaching-learning strategies in the application of a curriculum.

There are four criteria suggested for curriculum developers to be used to serve the functions of shifting through a range of learning activities to determine the most appropriate for a curriculum. Furthermore, teachers may wish to apply the same criteria to a curriculum when applied in a specific context. These criteria are objectives, appropriateness, resources available and constraints.

Figure 3: Selection of learning activities



Objectives

Objectives are the split goals which are based the ongoing activity. The first barrier or sifting layer that curriculum developers, must pass learning, activities and through is to determine if they can achieve the stated objectives. Many learning activities are available to the curriculum developers and the teacher to select from but only a few may be appropriate to facilitate the curriculum's intention. Thus the first of the four barriers to pass through is to determine if the learning can possibly achieve the curriculum's objectives.

For example, if a science curriculum's objective requires students to understand the operation of scientific equipment such as Bunsen burner, then the interactive and expository strategies would be appropriate, although the inquiry would not. However, if the same science curriculum's objectives sought students to operate a pipette correctly and safely, then the interactive method, small-group methods, small-group method individualization would be applicable.

Similarly, if the curriculum objective seeks to emphasise cultural empathy in literature then inquiry and models of reality teaching-learning strategies would be for more appropriate than expository and interactive strategies. In the end it is important for the curriculum developer to determine exactly what the state objectives seek to achieve when deciding the most suitable learning activities for inclusion in the curriculum. In turn, this process reinforces the need for well constructed and clearly phrased statements of curriculum intent.

The major point to be made in applying the objectives criterion of the method selection algorithm is that the type of method or techniques selected by the curriculum developer is primarily dependent upon the nature of the objectives involved. As there are many teaching-learning strategies available and many appear to meet the curriculum's intentions, it is important to limit the range of potential techniques to those, which suit the objective as a first step.

Leaner appropriateness

Learners are the individuals who desire to learn and ready to change. Having limited the range of potential methods to those which satisfy the needs of the curriculum's objectives, it is important for curriculum developers to consider which of the remaining methods are consistent with the characteristics of the students involved, namely, the students" interests, abilities and levels of development. In one sense this criterion will already have been considered when the objective itself was decided upon. In applying the criterion of attainability to the selection of objectives the astute curriculum developer would have taken such obvious characteristics as age, general ability and stage of intellectual development into account when deciding on the appropriateness of the objective for the group of students in mind.

It is also possible to relate learning activities to the concepts argued by developmental psychologists such as Jean Piaget. Here some activities would be more appropriate to a particular level of learner development than others. Expository and inquiry strategies for example, may be more appropriate to the formal operations level, although for different reasons. In earlier stages, drill and repetition may be more appropriate as might individualization and small groups. After considering appropriateness of the strategies to the learner, the available pool of methods has been further reduced in number.

Resources

The things & material that facilitate learning strategies physically and psychologically are resource. Having considered the nature of the curriculum's objectives and the characteristics of the learner, curriculum developers and teachers should apply the criterion of resource availability. Some methods will require access to hardware (for example, film projectors, computers, tape-recorders) and software (for example, films, computer programmes and tapes). But do these resources exist within the institution and are they available at the required time? An objective that required students to observe the

action of the ocean waves in practice is extremely difficult for students. Clearly, the availability of needed resources is a significant criterion to be applied to the selection of methods. When each potential method is assessed in terms of this criterion, the number of viable methods remaining is likely to be reduced substantially further.

Constraints

Closely allied to the criterion of resource availability, is the notion that in all teaching-learning situations there is a number of constraints that operate to further reduce the choice of the ideal (or most appropriate) methods. Perhaps the most significant constraint in schools is time. For example, after having considered all the earlier criteria, one may have decided that an excursion or field trip is the best way to put across your objective(s). However, an excursion may take half a day and antagonise some of your colleagues, while an audio-visual presentation may have nearly the same impact in just twenty minutes and with considerably less opposition.

Another constraint might be a teacher's recognition of his/her own limitations. For example, some teachers may feel uncomfortable with the noise generated by group discussions and simulations, or the seeming chaos resulting from individualised self-pacing. This, of course, further reduces the choice of appropriate methods available for a given objective Other constraints to be considered when selecting learning activities in a school setting include:

(1) Finances available.

(2) Availability of staff (for example, for fieldwork, team teaching).

(3) School policy.

(4) Central organization.

The mode or algorithm suggests how' developers might proceed in selecting appropriate learning activities in their curricula. The model further suggests that developers have an intimate knowledge of the institution for which the curriculum is being developed in order to select methods effectively. And while the examples cited relate to schools, the selection criteria are equally appropriate to other organizations involved in Curriculum development.

Organizing, learning activities

Little research or writing is available in the curriculum field that deals with the issue of organizing or structuring learning activities. In part this is due to the strong tradition of classroom teachers determining the appropriate learning activities for their students. Teachers claim quite reasonably, that, their experience, their knowledge of the learners' abilities and the school context places them in a unique position to organize learning activities. Their professionalism, through their understanding of their students, the curriculum and the learning context, suggests to them the most appropriate organization of

learning activities.

Learning is a complex phenomenon and is influenced by various factors. Learning is acquired over an extensive period of time and hence is difficult to distinguish between influential variables in all then, the substantial number of variables and their complex interrelationships over time make it difficult to determine the precise effectiveness of single activities and how they should he organized in a curriculum. Nevertheless, three criteria proposed, by Tyler (1994) have become accepted in the field as a rule of thumb basis for organizing learning activities: They are useful for curriculum developers to consider in their planning and for teachers to incorporate in their practice. They are:

Continuity
Sequence
Integration

Continuity

Tyler (1949) defined continuity as the vertical reiteration of major curriculum elements. This means that if an objective emphasises particular processes such as inquiry skills, then these skills require repetition at various points along the curriculum in order that learners may have repeated exposure to and opportunity to practice this activity.

Sequence

This concept is an extension of continuity. Sequence requires that the activity not only be repeated, but also that it progresses from simple to complex. This hierarchical organization of learning activities may occur within a subject or theme over a period such as a year, as well as progress through many years of schooling. Thus, with reading the basic skills are acquired before complex vocabulary is added and later sophisticated comprehension is undertaken. Taba (1962:296) refers to sequencing as cumulative learning', a term which seems quite appropriate.

Integration

The third criterion suggested by Tyler refers to the horizontal relationship between learning activities, Tyler's intention was that at any point in time, the learning activities are so related that they provide a unified and integrated experience for the learner. For example, activities in English literature may involve associated mapping skills from geography as well as research skills from botany. Traditionally, educators have not performed well according to this criterion, claiming instead that the responsibility for integration lies with the learner. As well, it appears difficult to entice educators out of the shell of their subject areas and so promote integration.

3. SUMMARY

Curriculum content is the subject matter of the teaching-learning process and as such, includes the knowledge, skills and values associated with that subject. The Content select tends to emphasise cither a subject approach (knowledge] .or a process approach i (skills). The criteria for selecting effective content for a curriculum are:

1. Significance: how essential it is to the subject.
2. Validity: how accurate or true it is.
3. Relevance: whether it is worthwhile to society.
4. Utility: how useful the content is .to adult functioning.
5. Learn ability: whether the student can acquire the content.
6. Interest: whether it has intrinsic interest to learners.

Current school curricula are finite in capacity and have increasingly become overcrowded due to pressures for additional learning's. The architectonics of content includes scope and sequence, and serves the function of ordering content into systematic categories for the purpose of study.

Scope refers to the wide range of action & function of content within the curriculum at any one lime. Scope may be influenced by time constraints, a common core. special-needs integration requirements and (the total amount of content required.

Sequence is the order, in which content is presented to learners, it is influenced by principles such as simple to complex, prerequisite learning's, whole lo part, learning, chronology; increasing abstraction and spiral sequencing..

Learning and developmental theorists have increased their influence in recent decades over the scope and sequence of content. Learning activities include those specific the activities offered to learners that will allow them to understand curriculum content and so achieve the stated objectives.

A wide variety of activities, particularly teaching methods, is available and should be used appropriately. Curriculum developers should recommend the use of a variety of learning activities, which are consistent with the objectives and content.

Useful learning-teaching strategies include:

1. Expository teaching: flow of information from source to learner.
2. Interactive teaching: incorporating interchange-between source and learner.
3. Small group teaching: emphasises group participation.
4. Inquiry teaching: learners actively engage in resolving problems,
5. Individualization: completion of tasks appropriate to learner's level of ability.
6. Models of reality: involving learners in replications of the real world.

7. Reality: involves learners, outside educational institutions, learning through experience.

A model for the selection of appropriate learning activities considers in descending hierarchical order:

1. Objectives
2. The learner
3. Resource availability
4. Constraints

The organization of learning activities, unlike content, has few guidelines based on research and in the literature. Some suggestions include structuring learning activities according to continuity, sequence and integration.

4. SELF-ASSESSMENT QUESTIONS

Q. No.1: Describe the meaning and selection of curriculum content'?

Q. No.2: Describe important levels of content and their functions?

Q. No.3: What are the important general principles for selecting curricular content as well as learning experiences?

Q. Nq.4: Explain the principles of selection of learning experiences?

Q. No.5: Describe the way of selection of teaching learning methods?

Q. No.6: Differentiate between expository teaching and interactive teaching.

Q. No.7: Write the interactive teaching?

Q. Np.8: Describe the small group teaching or discussions?

Q. No. 10: Explains the inquiry teaching or problem solving methods?

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UNIT–8

CURRICULUM EVALUATION

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INTRODUCTION

Evaluation of the curriculum of any specific course deals with all the educational outcomes related to it, whereas measurement deals specifically with those characteristics that can be easily quantified. In evaluation, such techniques may have to be evolved as may help the educators to analyse and assess the merits and shortcomings of any curriculum. Further, any problems involved in implementing a curriculum successfully have to be identified and resolved in order to achieve its educational objectives.

The procedures of curriculum evaluation involve certain steps, which should, however be flexible enough to be adjusted as per exigencies arising from time to time. These procedures are useful in guiding the thinking of those who are carrying out the evaluation. The steps, which have to be evolved by careful and intensive analysis of the types of the tasks- involved. While the strategy for curriculum evaluation should be adjusted according to the particular problems and situations under consideration, certain models that will be described in this Unit, can play a very important part in the process. Curriculum evaluation should not only be a means for judging educational effectiveness, but also if applied critically and intelligently, should lead to useful decisions that can serve as a powerful force to improve the educational process as well.

In this Unit, the nature, models and aims of curriculum evaluation, educational goals and curriculum objectives, curriculum models, designing evaluation studies, methods of curriculum evaluation, problems of research *design* in curriculum evaluation, and course improvement through evaluation have been discussed at length.

OBJECTIVES

After going through this Unit, you will be able to:

1. Mention the main purposes of curriculum evaluation and the extent to which an evaluation can be made.
2. Analyse the goals and objectives of educational programmes for relating them to the evaluation.
3. Design strategies and methods for the evaluation of specific educational programmes in accordance with the needs *of* the situations.
4. Suggest means of improving educational curricula and processes in the light of the evaluation of the progranrares.

1 THE NATURE AND MODELS OF CURRICULUM EVALUATION

The evaluation of curriculum material occupies an important place as on the provision of effective, meaningful, need-based and rational curricular material depends the growth of the learners. Good materials contribute to desirable changes in the learners. It has, therefore, to be assured that the curriculum materials are of good quality. At the same time, they should be acceptable both to the learners and the teachers; They should be of practical use and fit well in the existing educational setting. This can be assured by evaluation alone. Evaluations help in modifying the curriculum to adequately meet the growing challenges.

Curriculum evaluation has become one of the current catchwords in educational parlance. It primarily refers to observations and judgments made about what actually happens in the school though judgments about what students have achieved, and what else may also be included.

Curriculum evaluation determines the worth or value of curriculum, i.e. whether the curriculum is fulfilling its purposes for which it was formulated.

Curriculum evaluation is an important step because a wrong or defective curriculum may cause serious problems; for example it may not be suitable to the needs of the individual and society and it may put heavy load on the student.

Curriculum evaluation may be studied under two distinct points of view. According to one point of view, curriculum evaluation is concerned with the measurement of the achievement of objectives. The other point of view is that curriculum evaluation is the collection and use of information to make decisions about the educational programme.

(A) The first point of view stipulates curriculum evaluation as a means for quality control in education. Curriculum evaluation starts with the act of stating the objectives of a full course or a unit of the course. This is followed by the definition of these objectives in behavioural terms. The next, stage is the development of items which aim at finding out the extent to which the new materials have which developed such behaviours as will satisfy the purposes and the objectives the curriculum developers have *in* mind.

Achievement of objectives model is an American innovation. Tyler may be called the originator of this *model* and it got a concrete shape through the efforts of Benjamin S. Bloom. This model became very popular in the world and led to the technological heights in curriculum development in the form of Programmed Learning and Self-instructional schemes. This model has been criticised on two grounds. First, it is argued that it takes into account only intended or ore-specified outcomes and ignores unintended outcomes; second, it 'allows no scope for determining the relative merit of different sets of curriculum.

(B) The second point of view envisages 'illumination' role for curriculum evaluation. Its primary objective is to provide relevant information to the decision makers so as to enable them to arrive at decisions. This implies that there is more to evaluate in the curriculum programme than its stated objectives. It highlights 'service nature' of 'curriculum evaluation'. This approach may be termed as 'decision oriented inquiry' rather than 'conclusion oriented inquiry.

Curricular programmes, in general, are evaluated by any one or a combination of the following:

(a) Self-evaluation by the participants of the curricular programme, i.e., the learners;
(b) Evaluation by the teachers;
(c) Evaluation by outside evaluators, with specified terms of reference.
(d) Follow up studies of those who have participated in the programme.

Evaluation is done at two levels: (i) Formative Level and (ii) Summative Level.

(i) Formative evaluation is conducted in 'trial' schools. Consideration is given to what actually happens in practice as the course is being implemented. As the course proceeds, an on-going or formative evaluation is carried out using, inventories, questionnaires and observational and reporting techniques to establish the following:

(ii) The formative level evaluation has two important characteristic. The curriculum developer or the writer of the curriculum himself carries but the task of material evaluation. The formative evaluation is used to improve the materials while they are prepared and developed.

(i) How far the learning prescribed has been mastered?
(ii) What are the attitudes of teachers and their pupils towards the course?

2. Sometime type of curriculum evaluation is done by persons other than the writer or the developer of the curriculum material. It takes place after the completion of the curriculum material. At this stage, attempts are made to assess what has been achieved by the pupils at the end of the course. In the summative evaluation the tests used are related to what has been finally mastered and achieved as a result of the whole course. It also attempts to establish overall attitudes of teachers and pupils to the course.

Evaluation in practice is seldom purely formative or summative, since it has usually both the objectives. Any line of distinction between the two is rather thin and not distinct.

1.1 The Aims of Curriculum Evaluation

When curricular innovations are introduced, it is an important task of educators to determine their effectiveness. An assessment of any new programme has to be making to find out if the desired outcomes are being achieved; and to what extent, if at all, it results in

significantly better learning than the existing programme. The use of evaluation techniques should enable curriculum workers to make study progress in improving the curriculum.

The major aims of curriculum evaluation may thus be summarized as follows:

(a) To determine the outcomes of a programme;
(b) To help in deciding whether to accept or reject a programme;
(c) To ascertain the need for the revision of the course content;
(d) To help in further development of the curriculum materials for continuous improvement.

Careful evaluation should, therefore, demonstrate the strengths and weaknesses in the curriculum' so that necessary changes can be made in the instructional programme.

In testing and measurement, greater emphasis, is placed upon those characteristics that are easy to quantify and, thus important outcomes such as problem solving, creativity, critical thinking, work habits, and cultural appreciations tend to be neglected. Uncritical use of testing and heavy reliance on test results has caused a widespread spurious sense of certainty about educational accomplishments. Evaluation, however, is aimed at measuring all the educational outcomes, not just those that lend themselves easily to quantification.

The purposes of evaluation for curriculum innovations include the collection of information to be used as:

(a) Feedback to the innovators for further revision of materials and methods:
(b) Input for decision-making in the course;
(c) Empirical data on behaviour changes under curriculum treatment.

Professional educators who implement the curricula should be held accountable for successful achievement of deviational outcomes. This involves

(a) Developing improved, more comprehensive measurements of pupil performance,
(b) Making, analysis of the contributions made to the pupils' performance by educators, administrators, planners, institutions and other agencies in the educational process.

Curriculum evaluation, which should assess all aspects of the programme, should give a clear picture of the educational processes and products that should not only have prognostic value as regards the success of the programme, but should also have a diagnostic value in adapting and improving the programme to the satisfaction of all concerned.

1.2 Educational Goals and Curriculum Objectives

Statement of goals and objectives of the curriculum have many valuable uses. The congruence between the intent of the curriculum and the actual educational outcomes has to be investigated. In addition to abilities, the importance of attitudes, values, interests and sentiments in the educational process has been increasingly recognized. The behavioural

approach to the statement of objectives has gained a new impetus because of its value in the educational process. Valid evaluation is facilitated by a statement of the aims and objectives of education, arranged in a hierarchy of national, regional, local, subject and course areas drawn to blocks and units and even to daily lesson.

(A) Development of Educational Goals

Philosophers and educationists' have for centuries been discussing the aims and objectives of education. How more than ever, the problem of educational goals is a top priority though still a largely unresolved problem. The main reason for this is that in spite of all our efforts, the goals produced are essentially non-functional, even when stated in behavioural terms.

The goals of education usually tend to be non-functional for the following reasons:

(ii) In the statements of objectives, too much reliance is placed on 'continued development of values and character' or 'perfecting the intrinsic powers of ever citizen' While these sound enthusiastic, these lack clarity about the goal and hence the actual educational process that needs to be adopted for achieving these goals.

(iii) There is a lack of public involvement with the educational experts and educators in the clarification of objectives. Sometimes public participation narrows down the goals, but ultimately the goals are to be controlled by them.

(iv) The goals have too often been assumed as given and concentration has been directed to how to achieve the so called assumed or given goals. Changes in goals, that form a part of the educational process, have to be created by every generation.

(v) Those at the helm of affairs tend to interpret the goals according to their perception, which makes them too hazy.

(B) Role of Educational Measurement in Defining Goals

Tests are related to the major concerns in the educational process and should provide help in the development of meaningful goals. Without proper measures of the outcomes, there would be no direction as to the goals. Without proper evaluation, would be hazy and narrowly defined. Along with the measurement of basic skills and mastery of academic subjects, academic honestly and a sense of self should be used in the discovery and development of educational goals.

(C) Consideration in Selecting Goals

The criteria for the goals of education should be developed in relation to:

(i) The values of the society as a whole and of individual members;

(ii) The historical background of the society.

(iii) The potential urge of the society to move forward and progress.

(iv) The present standards and abilities of the students: and the possibility of achieving the goals.

(v) The educational philosophies of the institutions and the extent of teach-ability of the subject-matter under study.
(vi) Different theories of learning
(vii) The measurability of the goals, if and when achieved. A goal, the achievement of which is not measurable is a vague goal.

(D) Educational Objectives and their Taxonomies

As you will recall from unit-6 objectives are more precise than goals. Broad goals are analysed and made more specific and useful as building blocks for instruction.

You will recall that the taxonomy developed by Bloom and others in USA was divided into three domains.

(i) The Cognitive Domain: It deals with thinking, knowing and problem solving.
(ii) The Affective Domain: It deals with attitudes, values, interests, appreciations and socio-emotional adjustment.
(iii) The Psychomotor Domain: It deals with manual and motor skills i.e. the activities to be performed as a result of bodily movements.

(E) **The Cognitive Domain**

The cognitive domain is divided into:
(i) The acquisition of knowledge.
(ii) The development of intellectual skills and abilities necessary to use knowledge.

More specifically, you may remember that Bloom's categories were as follows:
(i) Knowledge, recalling facts, concepts, trends generalisations, principles, etc.
(ii) Comprehension, understanding, translation, interpretation, and, extrapolation.
(iii) Application, using abstractions in particular concrete situations.
(iv) Analysis, breaking down for clarity of thinking of elements, relationships and organizational principles.
(v) Synthesis, producing a unique, communication by recognizing and rearranging the facts.
(vi) Evaluation, judging in terms of (i) internal evidence (logical) and (ii) external evidence (consistency of facts developed).

(F) The Affective Domain

As you have already read in this unit as well as in unit-6, the affective domain is concerned with values, interests etc. This domain has the following categories.

(i) Receiving (attending), awareness, willingness and selected attention.
(ii) Responding, acquiescence, willingness and satisfaction in, response.
(iii) Valuing acceptance of and preference for a value and commitment.
(iv) Organization conceptualisation as a value and organization of a value system.
(v) Characterization of a value or a value complex as a generalization.

(G) The Psychomotor Domain

This domain includes the following:
(i) Reflex movements: Functions at birth like stretching, stiffening, relaxing.
(ii) Fundamental movements such as walking, running, jumping, pulling, pushing etc.
(iii) Perceptual abilities visual and auditory discrimination, bending, bouncing, eating, writing etc.
(iv) Physical abilities such as strenuous activity, moving quickly and precisely, touching toes, stopping and starting immediately etc.
(v) Skilled movements. They include typing, skating, filing, juggling, playing musical instruments.
(vj) Non-discursive, communication: It includes behaviour ranging from facial expressions to highly sophisticated communications.

(H) Educational Objectives: a position Statement.

(i) The objectives should be apparent to educators and learners.
(ii) A sound educational system provides for occasional reassignment of immediate objectives to take advantage of special opportunities that occur.
(iii) Global objectives provide little guidance to teaching and evaluation and specific objectives most often ignore vast concerns.

1.3 Curriculum Model

(A) Stake's Countenance Model

Robert R.E. Stake, an American evaluator (1969) has attempted to explain curriculum evaluation in terms of 'antecedents' 'transactions' and 'outcomes'. The terms “antecedents' implies those aspects of the situation in which the curriculum is taught, such as aspects of the situation in which the curriculum is taught, such as the time available and the books and other sources provided. The terms 'Transactions' refers to what actually happens in lesions, including, what is done by both teachers and pupils. The term ‘outcomes’ connotes pupil's achievements, which are the effects of the curriculum on the attitudes of the subject, as well as teacher's feelings about teaching the curriculum. According to stake such an approach provides an opportunity to answer questions about 'why' the curriculum was effective or not effective in various circumstances.

This model is known as Countenance Model because different people look at the curriculum and appraise it.

Stake's evaluation model may be explained as under:

Terms	***Kinds of Information***	***Methods***
1. Antecedents	(a) Organisational background	Time Table
	(b) Resources.	Syllabus and Textbooks
	(c) Attitudes of administrators '	Interviews
	(d) Parents	
	(e) Examinations available	
	(f) Context in Curriculum	Interviews with staff
	(g) Knowledge and skills of Pupils	Planning Sessions
2.Transactions	*Teachers:*	
(In lessons)	(a) Roles adopted	Activity Records
	(b) Use of time and resource	
	(c) Contact with pupils	Observations of class
	Pupils:	
	(a) Cognitive processes	Self report by teachers
	(b) Interest and involvement	Self report by pupils
	(c) Use of time	
3. Outcomes	(a) Pupil's achievements	Test and written work
	(b) Pupil's attitudes interpretations	Questionnaires
	(c) Teacher's attitudes, Interpretations	Interviews
	(d) Effects on other parts of Institutions	

(B) Hilda Taba Model

Taba's Social Studies Model emphasises the cause and effect relationship in the curriculum process. The evaluation process in based on the assumption of a cause and effect relationship between experimental control over the study material audits effects on the achievement of the student. A systematic-variation in the study material is also used. The researcher prepares the study material in different sets, each set having some variation from the other, Material is given to different groups of people and thereafter curriculum evaluation, is done. The outcomes of the evaluation will determine the principles for developing the new programme. The material, which produces the last results, will be advocated.

(C) D.L. Stuffiebean's CIPP Model

Stuffiebean's (1969, 1971) contribution to curriculum evaluation is often referred to as the CIPP Model, stressing the need for attention to Context, Input (f) Process (p) and Product (p). The first three of these terms help to characteristics formative evaluation, while product refers to summative evaluation. All the four elements of CIPP model are necessary backgrounds tor the processing of delineating, obtaining and providing useful information for judging decision alternatives. The CIPP attempts to answer the following Basic questions:

(i) What objectives of evaluation should be accomplished?
(ii) What procedures should be followed?
(iii) Whether procedures are working properly?
(iv) Whether objectives are being achieved?

Four types of decisions are possible: decision on intended means (precedence designs), intended ends (goals), actual means (procedures) and actual ends (attainments).

(D) Michael Seriven's goal Free Model

Michael Seriven (1973) has argued that the evaluator should not look at only what is intended by the course developers since some of the important effects, pleasant or otherwise may come quite unplanned. In this model, no goals for achievement are set. It is primarily concerned with the actual effects of a programme. The checklist used in this model does not prescribe the minimum levels to be achieved by a programme. In this type of evaluation, teachers have only a limited role to play as simply they have to make the curriculum programme available for evaluation.

(E) Tyler's Objectives Model devaluation

R. Tyler propounded, the objectives Model of Evaluation in 'Basic' Principles of Curriculum' (1949). He visualised curriculum evaluation as the process of determining to what extent the programme of instruction is actually realizing the educational objectives. Curriculum evaluation is a means for 'quality control in education.

In the objectives model of curriculum, curriculum is assessed against a set of specified objectives. The model got a concrete shape through the work of B.S. Bloom who produced a classification system of objectives that covered that area of pupil-learning. He put forward his ideas in Taxonomy of educational Objectives: The classification of educational Goals. (1956).

Tyler's model involves the following steps:

(i) Formulating objectives.
(ii) Classifying the objectives.
(iii) Defining the objectives in terms of behavior.
(v) Selecting situations in which achievement of objectives will be shown.
(vi) Selecting and trying promising methods of evaluation.

(vi) Developing more improved and objective methods of evaluation.
(viii) Developing means of interpreting and using the results of various instruments.

Tyler's model emphasises instructional objectives. Important elements of this model are the specification of objectives and evaluation of student achievement in terms of these objectives. This model is likely to ignore the process of instruction. Taba's Social studies Model stresses the cause and effect relationship in the curriculum process. The key elements of this model are experimental control and systematic variation of the material to study. The curriculum, which gives better results, is advocated.

2- DESIGNING EVALUATION STUDIES

An evaluation study requires a lot of thinking as well as proper planning to get useful results. Generally, most attention has been placed on the evaluation of the content, the basic on which the curriculum development was approached. Recent trends towards more process concept of behavioural based programmes demand more adequate definition of evaluation, better related criteria for judging value and more sophisticated mechanisms for organising the procedures and made is for reporting. Every evaluation study, has peculiarities of its own but the design of the study has to be planned using relational procedures. Various strategies and models for designing evaluation studies have been tried and a summary statement of the procedures involved in some of them will now be attempted.

In organizing a curriculum evaluation, logically and intelligently, leading to decisions on the effectiveness of the programme and its possible improvement, one suggested strategy involves the following:

(a) Specification, refinement, or modification of programme goals and evaluation.,
(b) Planning for an approximate evaluation design.
(c) Selection or development of data-gathering methods.
(d) Collection of relevant data.
(e) Processing, summarizing and analysis of data.
(f) Contrasting of data and objectives.
(g) Reporting and feedback of results.

2.1 Steps in the Evaluation Process

The evaluation process should include the following steps.

(i) The involvement of all concerned in the study as facilitators of programme evaluation, such as:
    (a) Related groups, boards, parents,
    (b) Professional individuals, psychologists, teachers,
    (c) Student groups.
(ii) The formation of a cohesive model of broad goals and specific behavioural objectives, arranged in hierarchical order from general specific outcomes in respect of all the three domains of objectives.
(iii) The translation of specific objectives into a communicable form, applicable to facilitating learning; and the application of instructional strategies with content and process components.
(iv) The choice of instrumentation (test, observations, interviews, etc.) which will allow one to make inference about programme effectiveness.
(iv) Periodic observation of behaviours as valid and reliable indices.
(v) The analysis of data, given by measurers, by typical statistical methods.
(vi) The interpretation of data relevant to specific objectiveness of me programme and instructional strategies.
(viii) Recommendations culminating in further modifications and revisions of

broad goals and specific objectives to produce, substantial improvements in the programme.

2.2 Methods of Curriculum Evaluation

Some of the methods used to determine the effectiveness of an educational programme are outlined below:

(i) The cosmetic method: It deals with the apparent activities and face value of the programme. Evidence about students learning is not collected.

(ii) The cardiac method: It involves an empirical approach. The effectiveness of the programme is shown through the collection of data.

(iii) The colloquial method recommends the discussion of the finding on the programmes effectiveness by a group of people associated with it.

(iv) The curricular method: It attempts to show how the new programme could fit into the old one for improvement.

(v) The computational method: It uses the statistical analysis of the data on performance in the programme

It may be mentioned here that each of the above cited methods; has got it-own minus and plus points. It is not, therefore, advisable to recommend or adopt exclusively any one method at the cost of others for all situations and purposes. It should, therefore, mainly depend upon the relevant crucial factors, which should .help in determining the method of evaluation to be adopted.

2.3 Problems of Research Design in Curriculum Evaluation

Research projects on the effectiveness of the curriculum are carried on, generally, using a comparison between experimental and control groups. In the experimental group, the curriculum variable interacts with many other elements in the teaching process the methods of instruction; the training and indoctrination of teachers etc. A Control group is a group of subjects'(the students being included for the evaluation design) as similar as possible to the experimental group, the difference being "that they are not given the experimental treatment. A statistical comparison is then made between the achievements of the two groups to discover whether there is any significant difference between them.

As alternative to such an experimental approach to curriculum evaluation the following may be suggested:

(a) An attempt might be made to isolate characteristics of a curriculum to vary them independently. For example, teachers attitudes vary and are not under control material might be used with teacher training.

(b) Statistics could be used to suggest the process of learning as related to curriculum elements. The investigation could look for trends, ups-and-downs.

(c) Clinical studies can be done on developmental psychology problems.

(d) The curriculum variable could be redefined at the point of its effect, the interaction between the teachers and students. Observations of the process and interviews could be recorded for an assessment of teachers attitudes.

Comparison of parallel experimental and control groups may not be meaningful as this approach assumes that all the relevant variables (except the curriculum variable) have been constant. But in such complex situations, this may be difficult to achieve. Instead of taking the groups simultaneously, evaluation using longitudinal studies would consider the curriculum as a process, a succession of events changing systematically over time.

In evaluation, behavioural definitions of the objectives are demanded from the course designers. Evaluators tend to equate behavior with items in the cognitive process; however, achievement tests may be sufficient to assess the objectives of the curriculum. The study of such typical performance variables as attitudes, learning and teaching styles etc, or the development of instruments for measuring such variables are among the major technical problems of evaluation, studies. Also the new curricula are subject oriented and the evaluators"

The subject specialists often reject behaviour concept. However, the relation between the subject disciplines and the process of teaching has to be examined: and all those concerned with education should contribute not only to the improvement of a particular subject in particular circumstances but also the understanding of the process of teaching and learning.

The purpose of evaluation is to determine the success or failure of any program n achieving its objectives. In attempting to evaluate a programme a record has to be maintained for investigation regarding:

(i) The objectives of the programme
(ii) The environment provided
(iii) The transactions between the teachers and students
(iv) The students progress
(v) The side-effects of other variables
(vi) The merits and shortcomings seen from divergent viewpoints.

Depending on the interests/intents of evaluators in the concerned field, and considering the entity, standards and, anticipated decisions, some major antecedents variables, transactions between teachers and students and outcomes are listed below:

(i) Antecedents
Student characteristics
Teacher characteristics
Curricular content
Curricular context
Instructional materials
Physical plant

School organization
Community context

(ii) Transactions
Communication flow
Time allocation
Sequence of events
Reinforcement schedule
Social climate

(ii) Outcomes
Student achievement
Student attitudes
Student motor skills
Effect on teachers
Institutional effects

Such a description as given above helps to identify the many characteristics programme to be evaluated. The evaluator must choose the variables to be described and judged according to his interest and talent. As for the sources of information they may be described as teachers, administrators, parents and so on. The evaluator has to choose the relevant variables by using:

(i) Intents
(ii) Observations
(iii) Standards
(iv) Judgments

To be more specific:

(i) The intents are indicated by the different goals of the people involved
(ii) The observations refer to the perceptions of what actually happens during the implementation of the programme.
(iii) The standards depend upon the opinions of experts as to what should happen.
(iv) The judgements reflect the feelings of the people about aspects of the situation.

2.4 Course Improvement through Evaluation

In order to find out ways of improving a course to enhance learning, its effectiveness must be assessed. Data collection should include information, which will help in evaluating the difficult area of the affective outcomes of the newly developed curriculum. The most useful evaluation information is that which allows for adjustment and modification in the developmental stages of the curriculum, rather than simply examining, the end products only. This, of course, would be a deviation from the traditional testing approach.

Any tests used, should fit into the contents of the curriculum and should produce reliable and valid-scores. Test results may seem be forgotten, however, instead of being

followed up for use. Remember that the purpose of evaluation is to study the changes in the pupils following the programme and to use this information to identify where improvements should made to facilitate improved learning.

Opinions about an education programme are often biased. The gathering of data using appropriate tests produces more objective results. The achievements of standards should be measured through the use of standardized tests of all relevant areas of proficiency, while attitudes can be accessed through interviews and questionnaires.

But systematic evaluation involves more than simply administering test's and analyzing the results. Apart from measuring proficiency and attitudes; approaches to evaluation can include, for example, process studies of classroom events and follow-up studies of the later careers of the subjects. And the results of evaluation studies of evaluation, studies should be implemented to produce improvements in the curriculum.

Conclusion

Old habits of thought and techniques are poor guides for course improvement. For this, more systematic evaluation is necessary. Such evaluation should produce a description of the outcomes of a programme on a broad scale, to ascertain the changes produced during a course and the revision needed.

Even the collection of appropriate data will make little contribution, if it leads only to approval or disapproval of a programme. Evaluation is a fundamental part of curriculum development, not just an appendage. It involves collecting facts that the course developer can and will use to do a better job.

3. SUMMARY

Evaluation is concerned with making value judgements about all sorts of things in our lives. In education, evaluation usually refers to making judgments about student performance and behaviours and the use of that information to enhance both learning/teaching and the curriculum. In education, product evaluation (student performance) is distinguished from process evaluation (the learning situation).

Evaluation in curriculum is useful for providing feedback to learners, determining how well learners achieved the objectives, providing information to improve curricula, assisting learners with decision making clarifying the stated objectives, and assisting others in making decisions about students.

Evaluation can occur formatively (during the learning experience), summatively (at the end of the learning experience) or diagnose (to determine deficiencies). To make evaluative judgements, needs useful data gathered from assessment and measurement techniques. Assessment involves the interpretation of measurement data. It makes sense of the data collected on student performance. Assessment may be norm-referenced (related to other learners) or criterion/standards-referenced (related to predetermined criteria/standards).

Measurement is the collection of data, usually in quantitative terms of student performance. A range of- measurement devices is available. Useful techniques include standardised tests, teacher-made tests, oral tests, work samples, systematic observation interviews, questionnaires, checklists and rating scales, anecdotal records, sociograms and self-reports.

A curriculum evaluation algorithm involves seven stages, i.e. (1) evaluation presage, (2) task specification, (3) evaluation design, (4) data collection, (5) data analysis, (6) conclusions and recommendations, and (7) presentation of report. Curriculum materials may be evaluated by criteria such as: interest, authenticity, organization and balance, appropriateness and technical quality.

4. SELF-ASSESSMENT QUESTIONS

Q.1: Compare the domains of measures at of educational achievement and evaluation of the curriculum.

Q. 2: List the processes for which, information data are used in the evaluation of curriculum innovations?

Q. 3: Explain the value of a systematic statement of objectives of an educational programme for valid evaluation?

Q. 4: Develop a logical strategy for designing an evaluation of an educational programme?

Q. 5: List the steps involved in the evaluation process?

Q. 6: Explain the experimental research approach to finding out the effectiveness of a new curriculum?

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UNIT–9

CURRICULUM APPLICATIONS AND CHANGE

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INTRODUCTION

Change is a phenomenon, which we experience continuously. Daily, it seems, features of our society are changing. Indeed, some changes are so frequent-that we may find it difficult to keep abreast of these developments. Therefore, it is with schooling and education, substantive and frequent change appears to have become something of a norm, a continuing reality of schools. Curriculum change may be considered as a subset of educational change and as such is affected by the same type of factors affecting change within education in general.

Whether working with individuals, groups or systems the curriculum developer is sure to be involved in curriculum change. Several strategies for managing change have been identified:-

a. Empirical - rational change strategies.
b. Normative re-education strategies.
c. Power coercive strategies,
d. Process of change is elaborated into three phases.
e. Unfreezing
f. Moving
g. Re-freezing.

Social scientists have said that change probably occurs in three stages. The first stage is initimation in which ideas for change are launched and decisions are made, regarding the nature, direction and extent of change. The second stage is legitimisation, in which the sentiments on behalf of change are communicated. The third is congruence of the separate systems of values.

In organizations including school systems change occurs because of:

a. Planning by equals;
b. Indoctrination by superior officers; and
c. Outright coercion by superior officers;

In Jenkins words, "changes will occur only as the forces are modified so that the level where the forces are equal is changed". To modify the forces we may:

a. Reduce or remove restraining forces.
b. Strengthen the driving forces or add to their number.
c. Change the direction of certain forces.

This unit deals with the problems and issues occur in curriculum development. We will discuss briefly dynamics of curriculum change and its process. We will conclude with a brief over view of monitoring and curriculum evaluation.

OBJECTIVES

After studying this unit, you will be able to:

1. Interpret problems and issues in curriculum change.
2. Explain implementation and modification in curriculum development.
3. Describe dynamics of curriculum change.
4. Discuss curriculum change process.

1. PROBLEMS AND ISSUES IN CURRICULUM DEVELOPMENT

Change is the process of transforming phenomenon into something different. The past few years have witnessed the most significant and sustained period of curriculum change that we have known. Perhaps those driving curriculum change will lessen its pace, though probably not given the substantially increased politicisation of education in general and curriculum in particular, over the past decade. In order to cope effectively with curriculum change we must learn to understand more about the concepts and processes involved.

In this module, we can read curriculum implementation and curriculum change from two slightly different points of view. First, it will relate the concept of curriculum implementation that is actually a curriculum application phase in so doing it- highlights the integral nature of curriculum implementation in the entire curriculum development process.

It is important for teachers and curriculum developers to be aware of this feature of modern education, particularly in terms of the accelerated rate of curriculum change. Innovations and change generally besiege our schools, and teachers are in need of greater understanding of the concept of change and how implementation might be undertaken successfully.

If we are to seriously consider what happens after the development of some curriculum object (a written curriculum document, a curriculum project package or a set of curriculum materials) and we reflect upon the definition of curriculum, then we must take cognisance of what happens when students in classroom use that object. This involves the third phase of the model-application which, in turn, incorporates these sets of activities.

(i) Implementation of the curriculum.
(ii) Monitoring and feedback from the curriculum.
(iii) The provision of feedback data to the presage group.

For any curriculum document, materials or project to be implemented in a school or school system, change must occur. To make this change occur effectively and with minimal disruption and confusion, a plan for implementing the curriculum innovation must be devised. If not, we can expect expensive resistance to the debate that the innovation may fail to fee accepted. Similarly it is insufficient merely to devise a curriculum and hope that schools will adopt it as this is a recipe for disaster in the early stage of implementation, it is likely that modifications will be made to the curriculum. This is to be expected indeed anticipated for a curriculum cannot possibly be devised to anticipate the myriad of variables operating in schools. The degree of successful implementation will reflect to a large measure of the ability and willingness of developers to accommodation modifications to their curriculum.

The monitoring stage of the Curriculum implementation is similarly an important step in gauging the success of the curriculum activity. Feedback from the curriculum over a longer period will invariably involve the application of a-comprehensive curriculum evaluation, that is. an analysis of the effectiveness of the curriculum activity as a whole. Further feedback will be obtained from a study of product evaluation, that is, how well students have achieved the educational intentions. While the implementation of the curriculum is a short-term activity, the monitoring and feedback aspects are likely to span several years.

During the period in which the curriculum has been developed, then implemented, modified and monitored changed circumstances in education have undoubtedly occurred. Changes may well have occurred to the nature of students taking the curriculum, new educational directions may have emerged from government policy, additional resources may have become available and so forth.

To account for, and where necessary accommodate those changes, the curriculum model must be considered. This may involve major changes to the structure of the curriculum, simple cosmetic alterations or something in between. What is important, however, is that the continuation of models seen as standard procedure. As the variables that affect education are in a continual change, so -curriculum must take account of positive variations and reflect those changes. In this way, we can be assured that we have the basis for an effective, responsible curriculum document, material package and project. That way we can say that we have had contributed for change which has following steps for complete comprehension of this package.

When the curriculum worker, whether superintendent or principal, director or teacher, begins to revise or develop a curriculum programme, he is confronted by *a* host of problems and issues. It is not hard to discover the reason for this. The curriculum is the set of pupils' activities and experiences by which the educator seeks to achieve his purpose. It is, therefore, education viewed from the standpoint of means. Education moreover involves for more than the intellectual training of youth. In its fullest sense it is the shaping of character, including altitudes, habits and beliefs. Hence, to develop a curriculum is to touch at every point the complex web of moral and intellectual purposes and beliefs which ultimately define the political, economic and social arrangement of any society For this-reason over since the days of Plato and Aristotle the greatest masters of political science and of education have been insisting that to would the character of education is, in the long run to mould the character of society and individual.

In a relatively unchanging and stable society, where substantial agreement exists with respect to basic intellectual and moral beliefs, the educator can, to a marked degree, take for granted the answers to many of the crucial questions underlying the curriculum. For in such a society in primary junction of education is clear, it is simply to shape the personality and character of individual.

Straight hair, and wan lace that had seemingly never known cosmetics. “Nadia is

the kind of teacher who defeats school bond issues single-handedly, said a mutual friend. She may have been right both her appearance and her attitudes toward youngsters and subject matter were unappealing.

Nadia expected much of her pupils but less of herself. Her understanding of her subject was limited and she obviously cared little about the adolescents she taught.

I have often taught about Nadia since those days in No. l Girls High School. Where should curriculum development have begun in her classroom? With her – probably But how and under what circumstances: Just incidentally (and may be it shouldn't be so incidental) I wonder what Nadia thought of me. How did her-brash young neighbour in the classroom across the hall improve herself so that her pupils' experiences could have improved in turn?

The preceding example raises some serious questions about the process of curriculum development. Among them are these: Granted that Nadia represents "a type", are there some common ways that can be used, in helping -people everywhere improve themselves? What different approaches are necessary and desirable for use with widely individuals? Are we vain enough to believe that improvement is only for persons outside ourselves? Can curriculum leaders become so preoccupied with the intrinsic worth of-a proposed innovation that they forget that real improvement or change occurs within people including them?

1.1 Political Realities

Existing literature in curriculum development deals almost with questions of how curriculum is planned and implemented at the local level. The assumptions seem to be that professionals (teachers, administrators, and perhaps Consultants) make the curriculum that the role of parents (and often students) is crucial, and that local leadership and initiative are significant factors in curriculum quality. The result is that almost all curriculum specialists approach problems of curriculum engineering from a "rational scientific" or "human relations" point of view, seeking the resolution of conflicts because of analysis, reason, and principle. Indeed, the language used by specialists to describe the process is replete with such terms as "decision making", "planning", "research and development" and "management" (Kirst and Walker 1971, p.481). Rarely is curriculum development viewed as a policy making activity and almost never is it placed in the context of political policy making, although the "influence: of politics sometimes is acknowledge. "Political" as it is used in this section is not combined to phenomena involving government; it refers more broadly to all the processes by which conflicts among competing public policies are resolved.

Based on an extensive analysis of the sources and process of innovation in curriculum, Pellegrin (1966, p. 15) concludes:

"The greatest stimuli to changes in education originate in sources external to this field

What I have shown is that the sources of innovation lie largely outside the local community, and in most instances outside the education profession".

The statement suggests that curriculum development is not merely "influenced" by political phenomena, but that it may, itself be a political process in many ways (Kirst aad Walter 1971, p.840).

These curriculum developers have identified a number of important sources that are presently determining curriculum policy and have classified them into three broad categories.

(1) Groups that establish minimum curriculum standards.
(2) Alternative generators.
(3) Groups demanding curriculum change.

1.2 Role of the Teacher

Many people misunderstand the role of teachers in curriculum development than in any other area of the curriculum field. For example:

We see the teachers as a very active participant in the planning of the total programme of the school centre.....

Each teacher takes part in planning the total programme of the school, possibly at district and state levels and as a member of committees, councils and other groups and always as a member of faculty of a particular school (Saylor and Alexander 1947, pp.43,61).

Yet, in another sense, the teacher's role in determining- what the curriculum shall be, in fact, enormous.

Major problem is the development of deep understanding in disciplines, which are basic to implement change. To cope with the new curriculum, these questions may be asked:

(i) What specific skills are needed to be developed?
(ii) What are the most effective procedures for in-service programme?
(iii) What material is needed?

Before initiating change, it is necessary to consider the potential, attitudes, and skills of the teacher. Teachers are especially inexperienced in the process of identifying, classifying, and organizing the general principles of the contents. Moreover, new theoretical insights are also needed to link what they know about developmental growth and placing of particular learning experiences between theories of learning and their application in practical situations. This linkage will permit a clear and manageable means

of translating general ideas into practice in desired direction.

1.3 Leadership of the Profession

It was suggested that since schools and institutions are created by society to fulfill a social need, the question arises whether professional groups -should participate in the process and what role should they play. There are several reasons justifying a leadership role for the profession in curriculum change. The most obvious is the fact that members of the profession are students and authorities in the technical aspects of education. For example they have school in such areas as the psychology of learning process, the nature of the reading act techniques of effective information transmission and so on.

Another, obvious ground supporting the profession's leadership role in curriculum change is its collective expertise in the context of the cultural heritage that is the discipline of knowledge.

Since teaching is an inescapable moral enterprise, teachers may avoid students' of social and educational values and more specifically, of the foundational areas of the curriculum.

1.4 Consultants

The services of consultants in curriculum change projects have been used by school systems. The consultants are usually employed to assist administrators teachers, especially production committees at the points where they face-difficulties.

Given a physical-psychological framework within which the staff can work effectively on curriculum change, what role should the consultant play? Initially, for the consultant, who is naively called in by a school system to help with general curriculum revision", there is no alternative to an extended period of assessment. This involves deriving some understanding of where the school staff and the community "are" with regard to values, beliefs, perceptions, etc. The consultant, in other words, must "get inside" staff members' perspectives so that he can "see" the situation as they see it (Smith, Stanley and Shores, 1957. p.473).

The consultants, who understand people and how to communicate with them, will usually begin by exploring the staff s perceptual field with them.

At every stage of curriculum change, groups will be concerned with what to do how to do, and what is to be done? To guide these consultants is to visualize the situation so that careful planning may be made. Otherwise, they will not be able to tackle the following questions:

(1) What facts should be sought in the change of the present situation?
(2) How many facts should be secured by different techniques?
(3) For what purpose facts will be used?
(4) How can they be classified and interpreted for this purpose?

(5) What type of the staff needed?
(6) What type of knowledge skill and recommendation the staff will needed?

Consultants may cause a problem in the process of curriculum change, if they do not discover: the perceptual field of those with whom they have to work, get inside the perceptive of the situation and keep their own understanding of the situation in the background.

1.5 Science and Technology

It is perhaps not too much to say that every significant curriculum problem of today is rooted in one way or the other in the general problem of cultural reintegration created by the increasing advancement of science and technology. These forces are uprooting the people of Pakistani society in the same way as wet rice farming uprooted the common weeds. For a simple illustration of how society is affected by science and technology, turn back to the invention, a little over a hundred years ago, of the steam locomotive and the establishment of the first railway lines. The iron horse provided a faster mode of transportation than the stagecoach; in addition, its freight and passenger capacity was considerably greater.

It is plain that one of the ways in which invention affects social life and curriculum is by creating new jobs and wiping out others. Another is by conquering geographic distance, so that people of the nation and of the whole world are brought into a closer relationship.

1.6 Objectives of Curriculum

Educational objectives are derived from national ideology. A clear statement of objectives helps to select from vast areas of knowledge in the various disciplines which may be necessary for valid output. According to the Taba, (1962, p. 198) a platform of objectives is needed to provide a common, consistent focus for the multifarious activities we call curriculum.

It is a fact that during the last fifteen years, no issue in curriculum has received more attention than the value of objectives and the way they are stated. This is philosophical problem. The positions may be:

(1) An objective must specify the exact overt behavior.

(2) Objective must specify the behaviour or a product that indicates whether the objective has been achieved or not.
(3) Objectives are not stated.

Therefore, objectives also guide the evaluation process. What is taught and evaluated is a common discrepancy in curriculum development. The scope of evaluation is usually narrow than the scope of objectives. Selection of content and learning experiences need greater attention as discrepancies occur here. Whether the contents serve the

objectives or are implemented in the same way i.e. at functioning level. The validity and significance of the content, consistency with social realities, balance breadth and depth, learning and adaptability to your experiences and appropriateness to the needs help achieving the objectives.

Organizational Problems

Problems of organization are also important. Conflicts and misunderstanding about the nature of knowledge and how it is internalised may cause difficulties. (Taba 1962, p.291) establishes here following questions:

(1) What is meant by logic of subject matter?
(2) How can logic be reconciled with the fact that there is psychological order of learning?
(3) How can learning content be so organized that it would be possible to develop at the same time a variety of behavioural objectives, such as to acquire a way of thinking and certain relevant attitudes?

These issues may be evolved from philosophical, psychological, technical, political and practical criteria, which are not kept in view.

Another Planning

Another issue in curriculum planning is the gap between concepts for designing learning activities' and concepts for guiding teacher preparation. Paradigms for teaching are not often related to specific opportunities. Teacher training programmes do not prepare the teacher for curriculum designs they have to teach.

Curriculum Evaluation

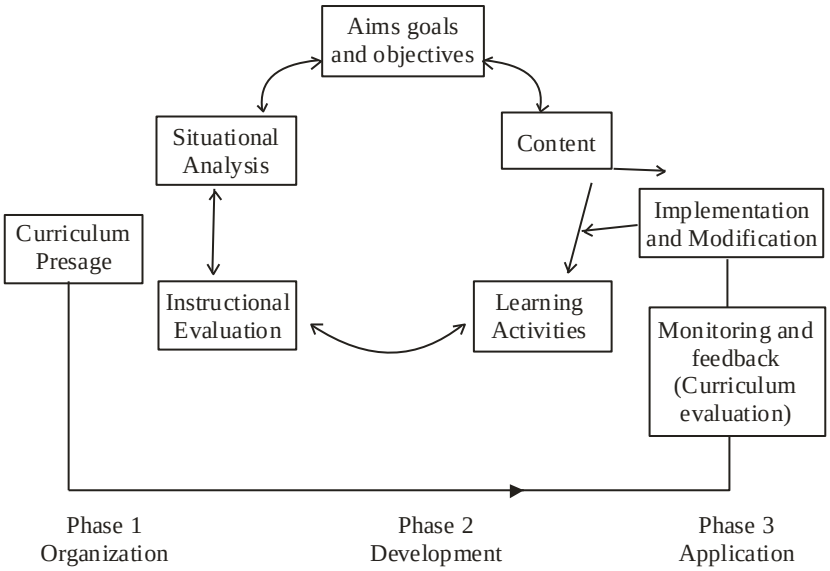
The field of evaluation is full of issues about its purpose, and how it is to be carried out. Curriculum specialists, teachers and administrators often disagree as to which techniques are to be used in evaluation. In brief, we can say that there are many problems and issues in curriculum development.

(1) Almost all curriculum specialists approached problems of curriculum development from a rational, scientific or human relations point of view, seeking the resolution of conflicts based on analysis, reason and principal.
(2) Major problem is the development of depth of understanding in disciplines, which are basic to implement change.
(3) At every stage of curriculum change teachers, consultants, leaders and science and technology advancements should be kept in mind because they have great impact on curriculum development.
(4) Educational objectives, organizations, learning activities, and evaluation processes are full of issues about their purpose and how they should be carried out.

2. IMPLEMENTATION AND MODIFICATION

An algorithmic model of curriculum development is presented as a useful way to construct a curriculum (see figure: 1). This was essentially a model for the conceptualisation of curriculum development, which also provided a set of sequential procedures for devising and developing a curriculum. Three phases are conceptualised in the model - presage, development and application. Now we are discussing the final phase. From the algorithmic perspective, the final phase (application) consists of the activities of the implementation modification monitoring, and curriculum evaluation.

Figure 1: Model of Curriculum Development



Implementation in curriculum involves changing the *status quo* by accepting and utilising a newly created curriculum or part of a curriculum. That is taking the curriculum document as devised in the development phase of the model and putting it into practice. If the curriculum is accepted and utilised successfully we say that it has become institutionalised.

In short term, any newly implemented curriculum will be expected to have trouble. To a large degree, this will reflect the effectiveness of the curriculum developers in meeting the need of the curriculum's clients. This implementation is a short-term phenomenon that attempts to integrate the new curriculum into existing practice. To some degree, the problems experienced in - implementing curricula reside with a lack of systematic implementation procedure. As noted, for any new curriculum to be implemented requires an educational institution to change. This itself is a daunting phenomenon and so a plan or strategy is required to facilitate change to take place.

The nature of change and change strategies will be dealt with more substantially. At this point, it is important for teachers to know that a new curriculum can be implemented more successfully in a school when actions are taken on the following points:

(1) A deliberate strategy is developed to implement the curriculum in the school.
(2) The curriculum is presented in terms of its favourable characteristics.
(3) Support is obtained at a systematic level for the innovation.
(4) The characteristics of the school are disposed towards the innovation.

In implementing the curriculum, it would be expected that some modification would be required. If, for example, a curriculum was developed externally to a school and was then implemented within it, one would anticipate some modifications being necessary to take account of differing local contextual factors such as the varying nature of students, differing school resources, differing teachers, parental input variations, community support, and so forth. These modifications might be minor in nature or quite substantial, depending upon how effectively the curriculum developers have undertaken their task.

What .are extremely useful for teachers in schools in recent times are the realisation, acceptance and support by central authorities of individual differences between schools. To claim, that all schools within an educational system are the same, or even substantially similar, is just nonsense. Yet, that was the very basis of systematic curriculum developers, such as those in a centralised organization to take into account all variable relating to different schools, particularly as some in one school may be counterproductive to others in another. Similarly, educators and the communities cannot expect schooling outcomes to be the same.

For example like newly constructed history curriculum disseminated by a central curriculum development and assessment agency. The assumption made by those agencies is that all schools will use the uniform curriculum document the same, as their students are essentially the same. Yet, we all know that this is not the case. One school may have a very advanced approach to teaching history, ample teaching learning resources and students able to cope with the abstract, learning involved. Another school may resist history, possess few resources and have students struggling to understand English as a language let alone the subtleties of history. Clearly these schools cannot, nor should they have to, adverse the history curriculum in identical ways.

Even if a curriculum was initially developed within a school, or by a cluster of local schools, one would expect some adjustments to be made as the curriculum was taken from theory to practice through the implementation process. To a significant degree, then, the success of a curriculum's implementation will reflect the willingness and ability of developers to accommodate changes to their curriculum. This may require changes to curriculum intent, content, learning activities or evaluation procedures and, in turn, may require developers to revise their situational analysis.

This sub process, within the final phase of the curriculum development, model, is likely to occur in the short term and may require regular reflection to ensure that the curriculum is achieving its purpose within a school's context. Consequently, the modification of a curriculum-may occur every year or two, though these changes are likely to be minor m nature.

The essential feature of this first part of the model's application phase is that developers have a useful role to play in the implementation of their curricula within schools. As developers, they can explain the working of the curriculum in detail. As developers, are in a better place to make modifications, if required, to the curriculum. Moreover, as developers involved in implementation they have a stake in making their curriculum work successfully. While these are short-term considerations, they are important to the long-term success of the curriculum.

(1) Implementation in curriculum involves changing the status by accepting and utilising a newly created curriculum or part of a curriculum.

(2) In implementing the curriculum, some modifications are necessary to take account of differing local contextual factors such as the varying nature of students, differing school resources, differing teachers, parental input variations community support and so forth.

(3) To a significant degree, the success of a curriculum's implementation will reflect the willingness and ability of developers to accommodate changes to curriculum content, intent, learning activities and evaluation.

(4) Curriculum developers have many useful roles in the implementation of their curricula within the school.

3. MONITORING AND CURRICULUM EVALUATION

As a rule of Thumb, over the long term, a curriculum will require continual monitoring and the provision of substantive feedback to curriculum developers every five to seven years. A decade or so ago we would have said that curricula-would require substantial revision every seven to ten years, but at this, time of substantial educational and societal change, a seven year period may be the outer parameter.

As the task of curriculum development is such an extensive, resource demanding activity, it is obvious that constant development or redevelopment-of curricula is impracticable. However, it is equally obvious that societal and educational conditions are changing constantly and these will need to be reflected in the school curriculum. How are these positions to be resolved?

The long-term aspect of the final phase recommends constant, monitoring of the curriculum. This is undertaken as part of the regular activities of classroom teachers through the collection of feedback data on student performance. In addition, some group is needed to undertake the task *of* regularly and systematically monitoring the curriculum. If those implementing and using the curriculum are amenable to also pooling their feedback data, the process of monitoring the curriculum can be achieved relatively painlessly.

As the situation changes with time (due to changed societal circumstances, different types of students, new technology, additional content information and so forth); so the monitoring of the curriculum should report the direction for needed changes. On an annual basis, this feedback can be forwarded to the original curriculum development group for consideration and for making relatively minor changes to the curriculum document.

Experience suggests that constant monitoring of curricula, while desirable, is rarely achieved consistently at school level. After several years of operation, it is highly probable that a curriculum is no longer as appropriate-as previously and substantial adjustments may be required. After several years of tinkering with minor changes to the curriculum, it is vivid that a substantial revision is required. At this point, a comprehensive curriculum evaluation is an essential task to be undertaken.

An approach for conducting comprehensive curriculum evaluation has been mentioned and one such method can be seen in Print -(1990) Curriculum Review of Social Studies and Social Science Education. Many other ways of conducting curriculum evaluations are available as may be seen in the literature (Kemmis & Stake. 1988.- Stufflebeam, et al. 1972, Parlett & Hamilton, 1972;, Davis. 1981: Eisner. 1979).

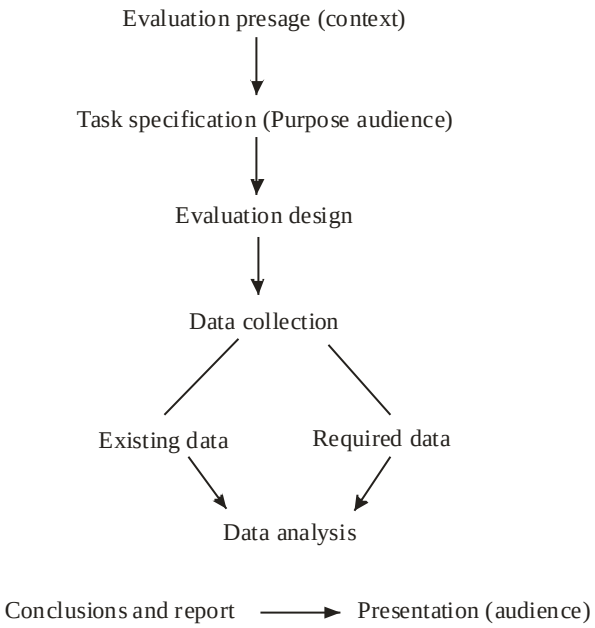


Figure 2: Curriculum Evaluation Algorithm

The essential stages in conducting a curriculum evaluation is outlined in figure 2. The algorithm suggests a seven-stage sequential procedure for evaluating a curriculum (such as a school's curriculum or year 11 and 12 history) or a component of a curriculum (such as years 5 and 6 of the primary mathematics syllabus). The same procedure is appropriate for evaluating a component part of a curriculum such as unit of work (e.g. coastal landforms in year 11 geography) or a sequence of teaching units as well as for an entire curriculum.

(1) A curriculum will require continual monitoring and the provision of substantive feedback to curriculurn developers every five to seven years.

(2) Constant monitoring of the curriculum is undertaken as part of the regular activities of classroom teachers through the collection of feedback data on student performance.

(3) As the situation changes with time, the monitoring of the curriculum should report the direction for needed changes.

(4) With minor changes to the curriculum, a substantial revision is required and for that a comprehensive curriculum evaluation is essential.

4. DYNAMICS OF CURRICULUM CHANGE

Curriculum change invariably reflects change in the society and education in general. Hence most of what we deal within curriculum change within the school content addresses ways of implementing change effectively. A specific curriculum innovation may lead society in changing in a particular direction, but usually curriculum change reflects societal change.

Consequently, when we consider curriculum change we need to include both the content of that change as well as the process by which that change comes about. By content, we mean knowledge, skills, concepts, understandings, values and so forth associated with the material concerned, such as a new history syllabus or a primary mathematics syllabus. The change process refers to means by which teachers will be introduced to that content and how they will be convinced to adopt and implement it. This requires the use of appropriate change strategies to convince teachers of the need for the change, the value of participating in the change and importance of developing appropriate perceptions, beliefs and actions that accord with that change.

This is not an easy or straightforward task, nor should it be. An important beginning is for curriculum developers, indeed, all participants in the educational process, be aware of the generic forces of curriculum change It they are to survive effectively. As the fast pace of curriculum change continues apparently unabated, driven by the increased politicisation of curriculum, so those involved in developing and implementing curricula will need to know more about the nature of change and how to deal with it purposefully. To that end, the following change, process will examine the steps of change.

(1) In curriculum change, we need to consider both content of the change as well as the process.

(2) The change process requires the use of appropriate change strategies.

(3) All participants in the educational process be aware of the generic forces of curriculum change.

5. CURRICULUM CHANGE PROCESS

In schools today teachers encounter at least three iayers of curriculum change - national, state and school levels. Most recently, teachers have been exposed to embryonic changes resulting from a "national curriculum", a set of national curriculum statements designed to provide direction for curriculum across the states and territories (Kennedy, 1989:1991). Such curriculum changes are just beginning to impact upon classroom teachers and although the effect currently is negligible, the degree of that impact is likely to substantially increase over the next few years.

A second and currently more profound degree -of impact in curriculum change is the state, level of curriculum development. Each state has a curriculum and assessment agency, which is responsible for devising, implementing and assessing certain system level curricula, usually restricted to secondary schools. The state department of education has responsibility for curricula in-primary, schools. The process of developing and implementing curricula creates significant curriculum change as a baseline, let alone a range, of other curriculum, initiatives at the state level (such as gender curriculum issues, gifted and talented education, multicultural education and so forth).

(2) The School Level Curriculum

The third level of curriculum change results from initiatives at the school level (or perhaps a group of schools), usually resulting from attempts to address the needs of diverse students. Contextual factors, such as language background, numeric skills, employment opportunities, ethnic background and so forth, require teachers in schools to reflect upon their curriculum offerings in order to make appropriate changes or to develop more suitable curricula. Consequently, teachers must consider another layer of curriculum change. However, it should be noted that little evidence exists of significant curriculum initiatives being undertaken by groups or clusters of schools (as in regions, districts, or clusters) despite schools being encouraged to seek an individual identity through the creation of a school vision and mission statements.

If teachers and curriculum developers are in a position to understand the nature of the change process, they are also better served to take appropriate action. In particular, teachers can relate to the various problems thrown up by change as it affects their school. Typical questions posed in schools by teachers include:

(1) Do we need this change?
(2) Why has it arisen?
(3) How will it affect us?
(4) What ways can we best support/oppose it?
(5) Will it improve the quality of learning?
(6) How can we ensure the continuance of the change?

Curriculum change, it has been said, reflects the broader change in education and the social system at large. However, how can that change be made to be taken abroad

effectively in educational institutions?

There has been extensive research in this area and most curriculum writers agree that three (or four as stated here) basic phases of curriculum change process may be distinguished (Fullon, 1982, 1987; Miles et al, 1987; Smith and Lovat,/ 1991, Print 1988).

The figure 3 depicts the readily apparent nature of the interaction between the four phases. While these phases have been separated for analysis purposes, in reality they will merge imperceptibly into each other and thus depict more of a continuous process. However, what constitutes these found phases and how do they interact with each other?

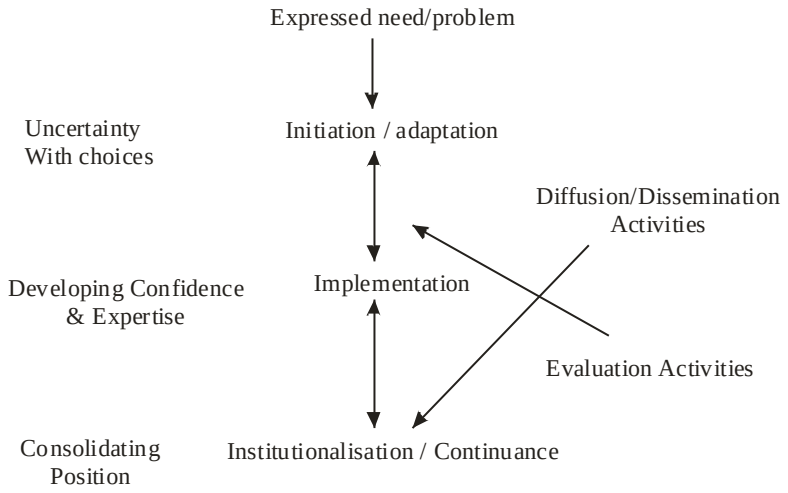


Figure 3: Educational Change Process

Source: After Fullan, 1982, 1987; Miles, 1987; Print, 1988.

5.1 Need

The beginnings of the curriculum change process lies in expression of concern, dissatisfaction or need with the current curriculum or curriculum practices. Consequently, a demand or expressed need is created for a variation in the existing situation. This expressed need is created from a variety of resources such as teachers, students, parents administrators; employers, educational systems or some combination of these sources.

Change will not occur without this need being present, although not all individuals perceive the same need for change, while others are not aware of the need for change at all. At times persuasive methods to promote change are necessary if change is to succeed. In recent years, for example, all educational systems within Pakistan have experienced curriculum change. In many cases, the need for change was perceived by a small group of powerful decision makers, who then sought to convince others that the need and direction

of proposed curriculum change was indeed appropriate.

Acceptance of the need for curriculum change by a substantial majority of those involved. If classroom, teachers, for example, do not recognise and accept the need for a particular curriculum innovation they will become resistant to it and hence place the success of the curriculum change in jeopardy. Thus, the first phase of successful curriculum change may involve those initiating the change in the process of convincing change participants as to its value. To facilitate that process the characteristics of innovations relative advantage, compatibility, complexity, and communication should be employed purposively.

In the field of school curriculum there appears, nevertheless, particularly in recent years, to be a constant demand for variation. This demand may be generated generally in society or from within the educational system as suggested earlier by Skilback (1984: 54-7). Certainly educational system and political inspired curriculum changes have been experienced throughout Pakistan and as we move into the early 1990 we may be experiencing the, most significant curriculum change - the move towards a national curriculum.

If you believe that change is necessary in schools, you might pose the following questions to yourself.

(1) Are you aware of societal pressures indicating the need for change in schools?
(2) What is it about your institution's curriculum that requires change?
(3) In answering these questions, we have a starting point for the change process.

5.2 Adoption

Adoption means the deliberate acceptance of an innovation to resolve an: expressed need. After seeking information to resolve the problem, alternatives may be examined and this process serves as a step towards consolidation of change. From amongst the many alternatives available, an attempted resolution or innovation is decided and accepted. At this stage, however, the innovation is by no means secure with in the institution.

Dissemination or diffusion of information about the proposed change occurs with varying degrees of success within educational institutions. Significant works in the literature on educational change (Fullan, 1982, 1987, Huberman & Miles, 1984; Miles et al., 1988) suggest a number of factors that codetermine adoption rates within institutions. Among the more important factors in accounting for the successful adoption of the change are:

(1) Access to information of decision makers.
(2) Alternative innovations in competition.
(3) Central administrative support for a particular change.

(4) Shared vision of the need for change.
(5) Good quality innovation/change programme.
(6) Funding availability to support implementation.
(7) Purposeful role and effectiveness of change agent.
(8) Significant and sustained community pressures.
(9) Government position on the proposed change.

These factors will have a significant impact upon how willingly the change is adopted by teachers in schools.

5.3 Implementation

The implementation phase may be considered as a continuum, stretching from the adoption of an innovation until its complete acceptance or institutionalisation. In education, implementation begins with the initial attempts by educators to effect the innovation into various schools. As this phase progresses, participants usually develop confidence and expertise with the innovation and so it has a greater chance of success. This is a crucial phase in the change process, for although the innovation has been adopted, few participants or even change agents would guarantee successful implementation to the institutionalisation phase. Indeed, the innovation will succeed or fail in large measure by how effectively it is implemented.

It may however, be for a variety of reasons that this phase is not successful and hence the innovation long-wishes. Alternatively, perhaps the rate of acceptance is slow and the subsequent usage rate is low. We should be aware of innovations that have failed to gain substantial acceptance at the implementation phase

The innovation could be implemented very successfully. There are several factors, which research (Fullan, 1982, 1987; Huberman & Miles, 1984; Miles et, al, 1987) has indicated and which will influence the effectiveness of the implernentation phase and hence the rate at which the innovation is accepted and used. These may be categorised as:

(1) Strategies for enhancing change.
(2) Characteristics of the innovation.
(3) Internal nature of the organization.
(4) Characteristics of the broader societal context.

The implementation phase of any change process will be enhance if the internal nature of the receiving organization and the characteristics of the social context are conductive to change.

The third factor above concerns with the internal nature of the organization undergoing change. Institutions that have an internal means to facilitate change will be in a favourable position to make the implementation phase more effective. Take, for example, a school experiencing difficulty-matching students needs, with existing curricula. At a systematic level, an educational system, which has effective lines of communication,

strong leadership, adequate support services, significant administrative commitment to the innovation and so forth, then successful implementation can occur.

Similarly, if society is desirous of particular change, or is being pushed towards a changed direction and there is political support for the change, then effective implementation in the school system will be enhanced. Perhaps society perceives it is time for a change and hence is relatively supportive of curriculum initiatives, which will produce substantive change.

Implementation, the third stage in the change process, shows that the innovation is gaining strength but its transformation into the fourth phase cannot yet be guaranteed. In fact, the implementation phase is a testing procedure, which will determine the innovation's ultimate fate. Meanwhile, an innovation exists somewhere along with an implementation continuum ranging from non-implementation to complete acceptance.

5.4 Institutionalisation/Contrivance

This phase must take time and change cannot be considered to have occurred successfully until institutionalisation is evident. Indeed, many innovations appear to succeed in the earlier stages only to flounder when exposed to the broader context for which they were intended.

Many innovations in curriculum receive artificial support in the form of finance, consultant, and administrative favour and so forth during the early stages of implementation. A decisive test for the success of an innovation is to gauge the effectiveness of the innovation on the removal of those crutches.

Facilitating the achievement of institutionalisation (and by mean the phase of implementation as well) is a major goal of those, who would bring about planned educational change. To enhance the achievement of that goal within educational organizations, in relation to curriculum change, teachers and curriculum developers need to consider the following factors (Fullan, 1982, 1987; Miles, etal, 1988):

(1) Strong administrative commitments.
(2) Positive pressure and support from within the school.
(3) Support activities (in-service, local facilitators)
(4) Resource allocation to found change.
(5) Removal of competing practices.
(6) Believable evaluation of innovation.
(7) Linked with current classroom practices-.

By contrast, schools that are lacking in support and. which are essentially destabilised by high staff turnover, inadequate leadership support, low levels of resources, poor support services and constant student mobility, have difficulty in institutionalising change.

The four phases of the change process may be summarized in terms of a matrix which considers the phases in relation to the typical sorts of activities conducted, factors to enhance that phase and the typical time taken in schools to achieve that phase.

Table Curriculum Change Matrix

Phase	Typical Activities	Enhancing Factors	Time-line
Need felt problem, deciding to start,	Perceived heed. examine similar contexts, needs assessment.	Shared vision. active advocate. local agendas, and similar societal needs	Short-medium
Adoption	Launching process. acceptance. building commitment, front-end training for key people.	All the above factors. plus quality innovation, fit local setting, perceived ' value.	Relatively short after-initiation phase.
Implementation	Design action plan, setting goals, maintaining. active commitment, developing confidence and expertise.	Strong leadership, commitment, shared control, clear direction, rewards for participants, peer support, quality innovation.	Short-long (Perhaps incomplete)
Institutionalisation	Building strength. evaluating, changing. organization, integrating into structures, building networks.	Strong leadership, institutional support, resource support, in-service staff, remove competing activities evaluation.	Medium-long, depending upon effect of implementation.

6. SUMMARY

The final phase of the model of curriculum development outlined in this unit is application. This involves implementing and modifying the curriculum in the short-term as well as long-term monitoring and curriculum evaluation. “With an examination of the final phase of the curriculum algorithm application, it is now complete.

Change is a phenomenon, which we experience continuously. Curriculum change may be considered as a subset of educational, change and, as such, is affected by the same type of factors affecting change within education in general like political realities, role of the teachers, and leadership of the profession, consultants, science and technology.

When the curriculum developers begin to revise or develop a curriculum programme, he is confronted by a host of problem and issues of objectives; contents, and learning activities and involves:

(1) Initiation: on initial concern.
(2) Adoption: acceptance but not security.
(3) Implementation: effecting change into practice.
(4) Institutionalisation: consolidation of the change into practice.

Innovations that are more readily adopted and implemented are characterised by a high degree of relative advantage, compatibility, effective communication, trial ability and low complexity.

Curriculum developers must be aware of change and how to facilitate it, as this is a feature of a successfully implemented curriculum

7. SELF-ASSESSMENT QUESTIONS

Q.1: Critically analyse the concept of change in curriculum development?

Q.2: There is a relationship of curriculum application and change, how?

Q.3: What actions are taken when a new curriculum is implemented more successfully in a school?

Q.4: Discuss monitoring and curriculum evaluation?

Q.5: All participants in the educational process have to be aware of the dynamics of curriculum change. Support the statement?

Q.6: As a teacher or curriculum developer, what are the factors you need to consider in relation to curriculum change.

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